Math 62CM: Multilinear algebra, differential forms, Stokes' theorem and applications

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Contents

Part I

From Linear functions to Differential 1-forms

Chapter 1

Vector space and its dual

1.1 The dual space V^*

Let V be a real vector space of finite dimension n. Choosing a basis $v_1, \ldots, v_n$ defines an *isomorphism* $V \to \mathbb{R}^n$,

$$v = x_1 v_1 + \cdots + x_n v_n \mapsto X = (x_1, \ldots, x_n).$$

However, it is important to remember that this isomorphism depends on the choice of a basis.

A function $\ell : V \to \mathbb{R}$ is called *linear* if $\ell(\lambda v) = \lambda \ell(v)$ and $\ell(v_1 + v_2) = \ell(v_1) + \ell(v_2)$, for any $v, v_1, v_2 \in V,\ \lambda \in \mathbb{R}$. It follows that

$$\ell\left(\sum_1^k \lambda_j v_j\right) = \sum_1^k \lambda_j \ell(v_j).$$

The set of all linear functions on V will be denoted by V^*.

Proposition 1.1. *V^* is a vector space of the same dimension as V.*

Proof. One can add linear functions and multiply them by real numbers:

$$\begin{aligned}(\ell_1 + \ell_2)(x) &= \ell_1(x) + \ell_2(x)\\ (\lambda\ell)(x) &= \lambda\ell(x) \quad \text{for } \ell, \ell_1, \ell_2 \in V^*,\ x \in V,\ \lambda \in \mathbb{R}\end{aligned}$$

It is straightforward to check that all axioms of a vector space are satisfied for V^*. Let us now check that $\dim V = \dim V^*$.

Choose a basis $v_1 \dots v_n$ of V. For any $x \in V$ let $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ be its coordinates in the basis $v_1 \dots v_n$. It is important to observe that *each of the coordinates $x_1, \dots, x_n$ can be viewed as a linear function on V*. Indeed,

1) the coordinates of the sum of two vectors are equal to the sum of the corresponding coordinates;

2) when a vector is multiplied by a number, its coordinates are being multiplied by the same number.

Thus $x_1, \dots, x_n$ are vectors from the space V^*. Let us show now that they form a basis of V^*. Indeed, any linear function $l \in V^*$ can be written in the form $l(x) = a_1x_1 + \dots + a_nx_n$ which means that l is a linear combination of $x_1 \dots x_n$ with coefficients $a_1, \dots, a_n$. Thus $x_1, \dots, x_n$ generate V^*. On the other hand, if $a_1x + \dots + a_nx_n$ is the 0-function, then all the coefficients must be equal to 0; i.e. functions $x_1, \dots, x_n$ are linearly independent. Hence $x_1, \dots, x_n$ form a basis of V and therefore, $\dim V^* = n = \dim V$. ■

The space V^* is called *dual* to V and the basis $x_1, \dots, x_n$ *dual* to $v_1 \dots v_n$.[1]

Exercise 1.2. *Prove the converse: given any basis $\ell_1, \dots, \ell_n$ of V^* we can construct a dual basis $w_1, \dots, w_n$ of V so that the functions $\ell_1, \dots, \ell_n$ serve as coordinate functions for this basis.*

Recall that vector spaces of the same dimension are isomorphic. For instance, if we fix bases in both spaces, we can map vectors of the first basis into the corresponding vectors of the second basis, and extend this map by linearity to an isomorphism between the spaces. In particular, by sending a basis $S = \{v_1, \dots, v_n\}$ of a space V into the dual basis $S^* := \{x_1, \dots, x_n\}$ of the dual space V^* we can establish an isomorphism $i_S : V \to V^*$. However, this isomorphism *is not canonical*, i.e. it depends on the choice of the basis $v_1, \dots, v_n$.

[1]It is sometimes customary to denote dual bases in V and V^* by the same letters but using lower indices for V and upper indices for V^*, e.g. $v_1, \dots, v_n$ and $v^1, \dots, v^n$. However, in these notes we do not follow this convention.

If V is a Euclidean space, i.e. a space with a scalar product $\langle x, y\rangle$, then this allows us to define another isomorphism $V \to V^*$ without choosing any basis. This isomorphism associates with a vector $v \in V$ a linear function $l_v(x) = \langle v, x\rangle$. We will denote the corresponding map $V \to V^*$ by $\mathcal{D}$. Thus we have $\mathcal{D}(v) = l_v$ for any vector $v \in V$.

Exercise 1.3. *Prove that $\mathcal{D} : V \to V^*$ is an isomorphism. Show that $\mathcal{D} = i_S$ for any* orthonormal *basis S.*

The isomorphism $\mathcal{D}$ is independent of a choice of a basis, but still it is not completely canonical because it depends on a choice of a scalar product. However, when talking about *Euclidean spaces*, this isomorphism is canonical.

Remark 1.4. The definition of the dual space V^* also works in the infinite-dimensional case.

Exercise 1.5. Show that both maps i_S and $\mathcal{D}$ are injective in the infinite dimensional case as well.

Remark 1.6. However V is infinite-dimensional then neither of the above maps is surjective. In fact, the operation of taking the dual alway increases the dimension of the space in the sense of its cardinality. Given an infinite-dimensional space V, its basis is a set $\{v_\alpha\}_{\alpha\in A}$ of its vectors $v_\alpha \in V$ indexed by a set A, such that any *finite* subset of this set is linearly independent, and any vector in V can be presented as a (unique) *finite* linear combination of vectors from $\{v_\alpha\}_{\alpha\in A}$. One can show any vector space admits a basis, and the cardinality of any two bases of V is the same (i.e. they can be enumerated by the same set A). The cardinality of A is called the dimension of V, and denoted dim V. It turned out that if V is infinite-dimensional then one always has dim $V^* > \dim V$.

For instance, consider the space of infinite sequences $(x_1, x_2, \dots)$ such that only finite number of terms are not equal to zero. This space is usually denoted by $\mathbb{R}^\infty$. The space $\mathbb{R}^\infty$ has a countable standard basis $e_1, e_2, \dots,$ where e_j is a vector in V such that $x_j = 1$ and $x_i = 0$ for $i \neq j$, so its dimension is equal to the cardinality of the set $\mathbb{N}$ of natural numbers. The dual space $(\mathbb{R}^\infty)^*$ is the space $\mathbb{R}^{\mathbb{N}}$ of *arbitrary* sequences $(x_1, x_2, \dots)$ without any extra conditions. Indeed, a linear function on any space is determined by its values on the vectors of a basis, and any combination of values is possible. The One can show that $\mathbb{R}^{\mathbb{N}}$ has no countable basis, and in fact, any basis of $\mathbb{R}^{\mathbb{N}}$ must have a cardinality of a continuum, i.e. it can be indexed by real numbers.

This pattern persists, and the dimension of the double dual space even larger,

$$\dim \mathbb{R}^\infty < \dim(\mathbb{R}^\infty)^* = \dim \mathbb{R}^{\mathbb{N}} < \dim(\mathbb{R}^{\mathbb{N}})^* = \dim((\mathbb{R}^\infty)^*)^* < \dots \ .$$

1.2 Canonical isomorphism between $(V^*)^*$ and V

Proposition 1.7. *If V is finite-dimensional that the double dual space $(V^*)^*$, dual to the dual space V^* of V, is canonically isomorphic in the finite-dimensional case to V.*

The word *canonically* means that the isomorphism is independent of any additional choices.

Proof. When we write $f(x)$ we usually mean that the function f is fixed but the argument x can vary. However, we can also take the opposite point of view, that x is fixed but f can vary. Hence, we can view the point x as a function on the vector space of functions.

If $x \in V$ and $f \in V^*$ then the above argument allows us to consider vectors of the space V as linear functions on the dual space V^*. Thus we can define a map $I : V \to V^{**}$ by the formula

$$x \mapsto I(x) \in (V^*)^*, \quad \text{where} \quad I(x)(l) = l(x) \quad \text{for any} \quad l \in V^*. \qquad \blacksquare$$

Exercise 1.8. Prove that if V is finite-dimensional then I is an isomorphism. Moreover, prove that if S and S^* are dual bases in V and V^* then the composition of isomorphisms $i_S : V \to V^*$ and $i_{S^*} : V^* \to (V^*)^*$ coincides with I. Show that in the infinite-dimensional case the map $I : V \to (V^*)^*$ is still injective. However, as it is explained above in Remark 1.6 in the infinite-dimensional case it is never surjective.

1.3 The map $\mathcal{A}^*$

Given a map $\mathcal{A} : V \to W$ one can define a *dual map* $\mathcal{A}^* : W^* \to V^*$ as follows. For any linear function $\ell \in W^*$ we define the function $\mathcal{A}^*(\ell) \in V^*$ by the formula $\mathcal{A}^*(\ell)(x) = \ell(A(x))$, $x \in V$. In other words, $\mathcal{A}^*(\ell) = \ell \circ \mathcal{A}$.[2]

[2]In fact, the above formula makes sense in much more general situation. Given any map $\Phi : X \to Y$ between two sets X and Y the formula $\Phi^*(h) = h \circ \Phi$ defines a map $\Phi^* : F(Y) \to F(X)$ between the spaces of functions on Y and X. Notice that this map goes in the direction opposite to the direction of the map Φ.

Given bases $\mathcal{B}_v = \{v_1, \dots, v_n\}$ and $\mathcal{B}_w = \{w_1, \dots, w_k\}$ in the vector spaces V and W one can associate with the map $\mathcal{A}$ a matrix $A = M_{\mathcal{B}_v\mathcal{B}_w}(\mathcal{A})$. Its columns are coordinates of the vectors $\mathcal{A}(v_j), j = 1, \dots, n$, in the basis $\mathcal{B}_w$. Dual spaces V^* and W^* have dual bases $X = \{x_1, \dots x_n\}$ and $Y = \{y_1, \dots, y_k\}$ which consist of coordinate functions corresponding to the basis $\mathcal{B}_v$ and $\mathcal{B}_w$. Let us denote by A^* the matrix of the dual map $\mathcal{A}^*$ with respect to the bases Y and X, i. e. $A^* = M_{YX}(\mathcal{A}^*)$.

Proposition 1.9. *The matrices A and A^* are transpose to each other, i.e. $A^* = A^T$.*

Proof. Recall the definition of the matrix of a linear map. We should take vectors of the basis $Y = \{y_1, \dots, y_k\}$, apply to them the map $\mathcal{A}^*$, expand the images in the basis $X = \{x_1, \dots x_n\}$ and write the components of these vectors as *columns* of the matrix A^*. Set $\tilde{y}_i = \mathcal{A}^*(y_i)$, $i = 1, \dots, k$. For any vector $u = \sum\limits_{j=i}^{n} u_j v_j \in V$, we have $\tilde{y}_i(u) = y_i(\mathcal{A}(u))$. The coordinates of the vector $\mathcal{A}(u)$ in the basis $w_1, \dots, w_k$ may be obtained by multiplying the matrix A by the column $\begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}$. Hence,

$$\tilde{y}_i(u) = y_i(\mathcal{A}(u)) = \sum_{j=1}^{n} a_{ij} u_j.$$

But we also have

$$\sum_{j=1}^{n} a_{ij} x_j(u) = \sum_{j=1}^{n} a_{ij} u_j.$$

Hence, the linear function $\tilde{y}_i \in V^*$ has an expansion $\sum\limits_{j=1}^{n} a_{ij} x_j$ in the basis $X = \{x_1, \dots x_n\}$ of the space V^*. Hence the i-th column of the matrix A^* equals $\begin{pmatrix} a_{i1} \\ \vdots \\ a_{in} \end{pmatrix}$, so that the whole matrix A^* has the form

$$A^* = \begin{pmatrix} a_{11} & \cdots & a_{k1} \\ \cdots & \dots & \cdots \\ a_{1n} & \cdots & a_{kn} \end{pmatrix} = A^T .$$

$\blacksquare$

Exercise 1.10. *Given a linear map $\mathcal{A} : V \to W$ with a matrix A, find $\mathcal{A}^*(y_i)$.*

Answer. The map $\mathcal{A}^*$ sends the coordinate function y_i on W to the function $\sum\limits_{j=1}^{n} a_{ij}x_j$ on V, i.e. to its expression in coordinates x_i.

Proposition 1.11. *Consider linear maps*

$$U \xrightarrow{\mathcal{A}} V \xrightarrow{\mathcal{B}} W\ .$$

Then $(\mathcal{B} \circ \mathcal{A})^* = \mathcal{A}^* \circ \mathcal{B}^*$.

Proof. For any linear function $\ell \in W^*$ we have

$$(\mathcal{B} \circ \mathcal{A})^*(\ell)(x) = \ell(\mathcal{B}(\mathcal{A}(x)) = \mathcal{A}^*(\mathcal{B}^*(\ell))(x)$$

for any $x \in U$. ■

Exercise 1.12. *Suppose that V is a Euclidean space and $\mathcal{A}$ is a linear map $V \to V$. Prove that for any two vectors $X, Y \in V$ we have*

$$\langle \mathcal{A}(X), Y \rangle = \langle X, \mathcal{D}^{-1} \circ \mathcal{A}^* \circ \mathcal{D}(Y) \rangle. \tag{1.3.1}$$

Solution. By definition of the operator $\mathcal{D}$ we have

$$\langle X, \mathcal{D}^{-1}(Z) \rangle = Z(X)$$

for any vector $Z \in V^*$. Applying this to $Z = \mathcal{A}^* \circ \mathcal{D}(Y)$ we see that the right-hand side of (1.3.1) is equal to to $\mathcal{A}^* \circ \mathcal{D}(Y)(X)$. On the other hand, the left-hand side can be rewritten as $\mathcal{D}(Y)(\mathcal{A}(X))$. But $\mathcal{A}^* \circ \mathcal{D}(Y)(X) = \mathcal{D}(Y)(\mathcal{A}(X))$. ■

Let us recall that if V is an Euclidean space, then operator $\mathcal{B} : V \to V$ is called *adjoint* to $\mathcal{A} : V \to V$ if for any two vectors $X, Y \in V$ one has

$$\langle \mathcal{A}(X), Y \rangle = \langle X, \mathcal{B}(Y) \rangle.$$

The adjoint operator always exist and unique. It is denoted by $\mathcal{A}^\star$. Clearly, $(\mathcal{A}^\star)^\star = \mathcal{A}$. In any *orthonormal* basis the matrices of an operator and its adjoint are transpose to each other. An operator $\mathcal{A} : V \to V$ is called *self-adjoint* if $\mathcal{A}^\star = \mathcal{A}$, or equivalently, if for any two vectors $X, Y \in V$ one has

$$\langle \mathcal{A}(X), Y \rangle = \langle X, \mathcal{A}(Y) \rangle.$$

The statement of Exercise 1.12 can be otherwise expressed by saying that the adjoint operator $\mathcal{A}^\star$ is equal to $\mathcal{D}^{-1} \circ \mathcal{A}^* \circ \mathcal{D} : V \to V$. In particular, an operator $\mathcal{A} : V \to V$ is self-adjoint if and only if $\mathcal{A}^* \circ \mathcal{D} = \mathcal{D} \circ \mathcal{A}$.

Remark 1.13. As it follows from Proposition 1.9 and Exercise 1.12 the matrix of a self-adjoint operator in any *orthonormal* basis is symmetric. This is not true if the basis is not orthonormal.

Chapter 2

Topological preliminaries

2.1 Elements of topology in a vector space

We recall in this section some basic topological notions in a finite-dimensional vector space and elements of the theory of continuous functions. The proofs of most statements are straightforward and we omit them.

Suppose V is a Euclidean space. The notation

$$B_r(p) := \{x \in V; ||x - p|| < r\},\ D_r(p) := \{x \in V; ||x - p|| \leq r\} \text{ and } S_r(p) := \{x \in V; ||x - p|| = r\}$$

stand for open, closed balls and the sphere of radius r centered at a point $p \in V$.

Open and closed sets

A set $U \subset V$ is called *open* if for any $x \in U$ there exists $\epsilon > 0$ such that $B_\epsilon(x) \subset U$.

A set $A \subset V$ is called *closed* if its complement $V \setminus A$ is open. Equivalently,

Lemma 2.1. *The set A is closed if and only if for any sequence $x_n \in A$, $n = 1, 2, \ldots$ which converges to a point $a \in V$, the limit point a belongs to A.*

Exercise 2.2. *Prove that though we defined open sets using a Euclidean structure on the vector space V the definition of open and closed sets is independent of this choice.*

Points which appear as limits of sequences of points $x_n \in A$ are called *limit points* of A.

There are only two subsets of V which are simultaneously open and closed: V and $\varnothing$.

Lemma 2.3. *1. For any family $U_\lambda, \lambda \in \Lambda$ of open sets the union $\bigcup\limits_{\lambda\in\Lambda} U_\lambda$ is open.*

2. For any family $A_\lambda, \lambda \in \Lambda$ of closed sets the intersection $\bigcap\limits_{\lambda\in\Lambda} A_\lambda$ is closed.

3. The union $\bigcup\limits_1^n A_i$ of a finitely many closed sets is closed.

4. The intersection $\bigcap\limits_1^n U_i$ of a finitely many open sets is open.

By a *neighborhood* of a point $p \in V$ we understand *any open set $U \ni p$.*

Given any subset $X \subset V$ a point $p \in V$ is called

- *interior* point for A if there is a neighborhood $U \ni p$ such that $U \subset A$;

- *boundary* point if it is not an interior point neither for A nor for its complement $V \setminus A$.

We emphasize that a boundary point of A may or may not belong to A. Equivalently, a point $p \in V$ is a boundary point of A if it is *a limit point both for A and $V \setminus A$.*

The set of all interior points of A is called the *interior of* A and denoted $\operatorname{Int} A$. The set of all boundary points of A is called the *boundary* of A and denoted ∂A. The union of all limit points of A is called the *closure* of A and denoted $\overline{A}$.

Lemma 2.4. *1. We have $\overline{A} = A \cup \partial A$, $\operatorname{Int} A = A \setminus \partial A$.*

2. $\overline{A}$ is equal to the intersection of all closed sets containing A.

3. $\operatorname{Int} U$ is the union of all open sets contained in A.

Given a subset $X \subset V$

- a subset $Y \subset X$ is called *relatively open* in X if there exists an open set $U \subset V$ such that $Y = X \cap U$.

- a subset $Y \subset X$ is called *relatively closed* in X if there exists a closed set $A \subset V$ such that $Y = X \cap A$.

One also calls relatively and open and closed subsets of X just *open* and *closed in X*.

2.2 Everywhere and nowhere dense sets

A *closed* set A is called nowhere dense if $\operatorname{Int} A = \varnothing$. For instance any finite set is nowhere dense. Any linear subspace $L \subset V$ is nowhere dense in V if $\dim L < \dim V$. Here is a more interesting example of a nowhere dense set.

Fix some number $\epsilon < 1$. For any interval $\Delta = [a,b]$ we denote by Δ_ϵ the *open* interval centered at the point $c = \frac{a+b}{2}$, the middle point of Δ, of the total length equal to $\epsilon(b-a)$. We denote by $C(\Delta) := \Delta \setminus \Delta_\epsilon$. Thus $C(\Delta)$ consists of two disjoint smaller closed intervals. Let $I = [0,1]$. Take $C(I) = I_1 \cup I_2$. Take again $C(I_1) \cup C(I_2)$ then again apply the operation C to four new closed intervals. Continue the process, and take the intersection of all sets arising on all steps of this construction. The resulted closed set $K_\epsilon \subset I$ is nowhere dense. It is called a *Cantor set*.

A subset $B \subset A$ is called *everywhere dense* in A if $\overline{B} \supset A$. For instance the the set $\mathbf{Q} \cap I$ of rational points in the interval $I = [0,1]$ is everywhere dense in I.

2.3 Compactness and connectedness

A set $A \subset V$ is called *compact* if one of the following equivalent conditions is satisfied:

COMP1. A is closed and bounded.

COMP2. from any infinite sequence of points $x_n \in A$ one can choose a subsequence x_{n_k} converging to a point $a \in A$.

COMP3. from any family $U_\lambda, \lambda \in \Lambda$ of open sets covering A, i.e. $\bigcup\limits_{\lambda \in \Lambda} U_\lambda \supset A$, one can choose finitely many sets $U_{\lambda_1}, \dots, U_{\lambda_k}$ which cover A, i.e. $\bigcup\limits_1^k U_{\lambda_k} \supset A$.

The equivalence of these definitions is a combination of theorems of Bolzano-Weierstrass and Émile Borel.

A set A is called *path-connected* if for any two points $a_0, a_1 \in A$ there is a continuous path $\gamma : [0,1] \to A$ such that $\gamma(0) = a_0$ and $\gamma(1) = a_1$.

A set A is called *connected* if one cannot present A as a union $A = A_1 \cup A_2$ such that $A_1 \cap A_2 = \varnothing$, $A_1, A_2 \neq \varnothing$ and both A_1 and A_2 are simultaneously relatively closed and open in A.

Lemma 2.5. *Any path-connected set is connected.*

Proof. Suppose that A is disconnected. Then it can be presented as a union $A = A_0 \cup A_1$ of two non-empty relatively open (and hence relatively closed) subsets. Consider the function $\phi : A \to \mathbb{R}$ defined by the formula

$$\phi(x) = \begin{cases} 0, & x \in A_0, \\ 1, & x \in A_1. \end{cases}$$

We claim that the function ϕ is is continuous. Indeed, For each $i = 0, 1$ and any point $a \in A_i$ there exists $\epsilon > 0$ such that $B_\epsilon(x) \cap A \subset A_i$. Hence the function ϕ is constant on $B_\epsilon(x) \cap A$, and hence continuous at the point x. Now take points $x_0 \in A_0$ and $x_1 \in A_1$ and connect them by a path $\gamma : [0, 1] \to A$ (this path exists because A is path-connected). Consider the function $\psi := \phi \circ \gamma : [0, 1] \to \mathbb{R}$. This function is continuous (as a composition of two continuous maps). Furthermore, $\psi(0) = 0, \psi(1) = 1$. Hence, by an intermediate value theorem of Cauchy the function ψ must take all values in the interval $[0, 1]$. But this is a contradiction because by construction the function ψ takes no other values except 0 and 1. ■

Lemma 2.6. *Any open connected subset $U \subset \mathbb{R}^n$ is path connected.*

Proof. Take any point $a \in U$. Denote by C_a the set of all points in U which can be connected with a by a path. We need to prove that $C_a = U$.

First, we note that C_a is open. Indeed, if $b \in C_a$ then using openness of U we can find $\epsilon > 0$ such the ball $B_\epsilon(b) \in U$. Any point of $c \in B_\epsilon(b)$ can be connected by a straight interval $I_{bc} \subset B_\epsilon(b)$ with b, and hence it can be connected by a path with a, i.e. $c \in C_a$. Thus $B_\epsilon(b) \subset C_a$, and hence C_a is open. Similarly we prove that the complement $U \setminus C_a$ is open. Indeed, take $b \notin C_a$. As above, there exists an open ball $B_\epsilon(b) \subset U$. Then $B_\epsilon(b) \subset U \setminus C_a$. Indeed, if it were possible to connect a point $c \in B_\epsilon(b)$ with a by a path, then the same would be true for b, because b and c are connected by the interval I_{bc}. Thus, we have $U = C_a \cup (U \setminus C_a)$, both sets C_a and $U \setminus C_a$ are open and C_a is non-empty. Hence, $U \setminus C_a$ is empty in view of connectedness of U. Thus, $C_a = U$, i.e. U is path-connected. ■

Exercise 2.7. Show that a closure of a connected set is connected.

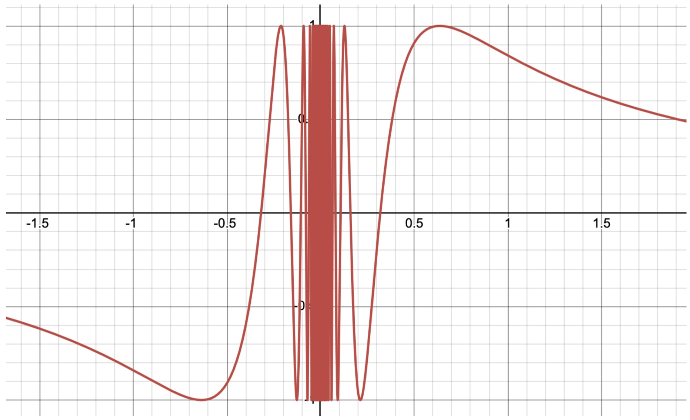


Figure 2.1: The closure of the graph $y = \sin\frac{1}{x}, x \neq 0$ is a connected but not path connected set.

Exercise 2.8. In general a connected set need not to be path-connected, e.g. the closure of the graph of the function $\sin\frac{1}{x}, x \in \mathbb{R} \setminus 0$. Prove it. Note that even the closure of the positive part of the graph $\{y = \sin\frac{1}{x}, x > 0\}$ is not path connected, and in particular the analog of the statement in Exercise 2.7 does not hold for path connected sets.

Exercise 2.9. Prove that any non-empty connected (= path-connected) open subset of $\mathbb{R}$ is equal to an interval (a, b) (we allow here $a = -\infty$ and $b = \infty$). If one drops the condition of openness, then one needs to add a closed and semi-closed intervals and a point.

Remark 2.10. One of the corollaries of this exercise is that in $\mathbb{R}$ any connected set is path-connected.

Solution. Let $A \subset \mathbb{R}$ be a non-empty connected subset. Let $a < b$ be two points of A. Suppose that a point $c \in (a, b)$ does not belong to A. Then we can write $A = A_0 \cup A_1$, where $A_0 = A \cap (-\infty, 0), A_1 = A \cap (0, \infty)$. Both sets A_0 and A_1 are relatively open and non-empty, which contradicts connectedness of A. Hence if two points a and b, $a < b$, are in A, then the whole interval $[a, b]$ is also contained in A. Denote $m := \inf A$ and $M := \sup A$ (we assume that $m = -\infty$ if A is unbounded from below and $M = +\infty$ if A is unbounded from above). Then the above argument shows that the open interval (m, M) is contained in A. Thus, there could be 5 cases:

- $m, M \notin A$; in this case $A = (m, M)$;
- $m \in A, M \notin A$; in this case $A = [m, M)$;

- $m \notin A, M \in A$; in this case $A = (m, M]$;
- $m, M \in A$ and $m < M$; in this case $A = [m, M]$;
- $m, M \in A$ and $m = M$; in this case A consists of one point.

■

2.4 Connected and path-connected components

Lemma 2.11. *Let A_λ, $\lambda \in \Lambda$ be any family of connected (resp. path-connected) subsets of a vector space V. Suppose $\bigcap\limits_{\lambda\in\Lambda} A_\lambda \neq \varnothing$. Then $\bigcup\limits_{\lambda\in\Lambda} A_\lambda$ is also connected (resp. path-connected)*

Proof. Pick a point $a \in \bigcap\limits_{\lambda\in\Lambda} A_\lambda$. Consider first the case when A_λ are path connected. Pick a point $a \in \bigcap\limits_{\lambda\in\Lambda} A_\lambda$. Then a can be connected by path with all points in A_λ for any points in $\lambda \in \Lambda$. Hence, all points of A_λ and $A_{\lambda'}$ can be connected with each other for any $\lambda, \lambda' \in \Lambda$.

Suppose now that A_λ are connected. Denote $A := \bigcup\limits_{\lambda\in\Lambda} A_\lambda$. Suppose A can be presented as a union $A = U \cup U'$ of disjoint relatively open subsets, *where we denoted by U the set which contains the point $a \in \bigcap\limits_{\lambda\in\Lambda} A_\lambda$*. Then for each $\lambda \in \Lambda$ the intersections $U_\lambda := U \cap A_\lambda$ and $U'_\lambda := U' \cap A_\lambda$ are relatively open in A_λ. We have $A_\lambda = U_\lambda \cup U'_\lambda$. By assumption, $U_\lambda \ni a$, and hence $U_\lambda \neq \varnothing$. Hence, connectedness of A_λ implies that $U'_\lambda = \varnothing$. But then $U' = \bigcup\limits_{\lambda\in\Lambda} U'_\lambda = \varnothing$, and therefore A is connected.

■

Given any set $A \subset V$ and a point $a \in A$ the *connected component* (resp. *path-connected component $C_a \subset A$* of the point $a \in A$ is the union of all connected (resp. path-connected) subsets of A which contains the point a. Due to Lemma 2.11 the (path-)connected component C_a is itself (path-)connected, and hence it is *the biggest (path-)connected subset of A which contains the point a.* The path-connected component of a can be equivalently defined as the set of all points of A one can connect with a by a path in A.

Note that (path-)connected components of different points either coincide or do not intersect, and hence the set A can be presented as a disjoint union of (path-)-connected components.

Lemma 2.6 shows that for open sets in a vector space V the notions of connected and path-connected components coincide, and due to Exercise 2.9 the same is true for any subset in $\mathbb{R}$. In particular, any open set $U \subset \mathbb{R}$ can be presented as a union of disjoint open intervals, which are its connected (= path-connected) components. Note that the number of these intervals can be infinite, but always countable.

2.5 Continuous maps and functions

Let V, W be two Euclidean spaces and A is a subset of V. A map $f : A \to W$ is called *continuous* if one of the three equivalent properties hold:

1. For any $\epsilon > 0$ and any point $x \in A$ there exists $\delta > 0$ such that $f(B_\delta(x) \cap A) \subset B_\epsilon(f(x))$.
2. If for a sequence $x_n \in A$ there exists $\lim x_n = x \in A$ then the sequence $f(x_n) \in W$ converges to $f(x)$.
3. For any open set $U \subset W$ the pre-image $f^{-1}(U)$ is relatively open in A.
4. For any closed set $B \subset W$ the pre-image $f^{-1}(B)$ is relatively closed in A.

Let us verify equivalence of 3 and 4. For any open set $U \subset W$ its complement $B = W \setminus U$ is closed and we have $f^{-1}(U) = A \setminus f^{-1}(B)$. Hence, if $f^{-1}(U)$ is relatively open, i.e. $f^{-1}(U) = U' \cap A$ for an open set $U' \subset V$, then $f^{-1}(B) = A \cap (V \setminus U')$, i.e. $f^{-1}(B)$ is relatively closed. The converse is similar.

Let us deduce 1 from 3. The ball $B_\epsilon(f(x))$ is open. Hence $f^{-1}(B_\epsilon(f(x)))$ is relatively open in A. Hence, there exists $\delta > 0$ such that $B_\delta(x) \cap A \subset f^{-1}(B_\epsilon(f(x)))$, i.e. $f(B_\delta(x) \cap A) \subset B_\epsilon(f(x))$. We leave the converse and the equivalence of definition 2 to the reader.

Remark 2.12. Consider a map $f : A \to W$ and denote $B := f(A)$. Then definition 3 can be equivalently stated as follows:

3′. If $U \subset B$ is relatively open in B then its pre-image $f^{-1}(U)$ is relatively open in A.

Definition 4 can be reformulated in a similar way.

Indeed, we have $U = U' \cap A$ for an open set $U' \subset W$, while $f^{-1}(U) = f^{-1}(U')$.

The following theorem summarize properties of continuous maps.

Theorem 2.13. *Let $f : A \to W$ be a continuous map. Then*

1. *if A is compact then $f(A)$ is compact;*
2. *if A is connected then $f(A)$ is connected;*
3. *if A is path connected then $f(A)$ is path-connected.*

Proof. 1. Take any infinite sequence $y_n \in f(A)$. Then there exist points $x_n \in A$ such that $y_n = f(x_n), n = 1, \dots$. Then there exists a converging subsequence $x_{n_k} \to a \in A$. Then by continuity $\lim\limits_{k\to\infty} f(x_{n_k}) = f(a) \in f(A)$, i.e. $f(A)$ is compact.

2. Suppose that $f(A)$ can be presented as a union $B_1 \cup B_2$ of simultaneously relatively open and closed disjoint non-empty sets. Then $f^{-1}(B_1), f^{-1}(B_2) \subset A$ are simultaneously relatively open and closed in A, disjoint and non-empty. We also have $f^{-1}(B_1) \cup f^{-1}(B_2) = f^{-1}(B_1 \cup B_2) = f^{-1}(f(A)) = A$. Hence A is disconnected which is a contradiction.

3. Take any two points $y_0, y_1 \in f(A)$. Then there exist $x_0, x_1 \in A$ such that $f(x_0) = y_0, f(x_1) = y_1$. But A is path-connected. Hence the points x_0, x_1 can be connected by a path $\gamma : [0,1] \to A$. Then the path $f \circ \gamma : [0,1] \to f(A)$ connects y_0 and y_1, i.e. $f(A)$ is path-connected. ■

Note that in the case $W = \mathbb{R}$ Theorem 2.13.1 is just the Weierstrass theorem: a continuous function f on a compact set $A \subset V$ is bounded and achieves its maximal and minimal values.

We finish this section by a theorem of Georg Cantor about uniform continuity.

Theorem 2.14. *Let A be compact and $f : A \to W$ a continuous map. Then for any $\epsilon > 0$ there exists $\delta > 0$ such that for any $x \in A$ we have $f(B_\delta(x)) \subset B_\epsilon(f(x))$.*

Proof. Choose $\epsilon > 0$. By continuity of f for every point $x \in A$ there exists $\delta(x) > 0$ such that

$$f(B_{\delta(x)}(x)) \subset B_{\frac{\epsilon}{2}}(f(x)).$$

We need to prove that $\inf\limits_{x\in A} \delta(x) > 0$. Note that for any point in $y \in B_{\frac{\delta(x)}{2}}(x)$ we have $B_{\frac{\delta(x)}{2}}(y) \subset B_{\delta(x)}(x)$, and hence

$$f(B_{\frac{\delta(x)}{2}}(y)) \subset f(B_{\delta(x)}(x)) \subset B_{\frac{\epsilon}{2}}(f(x)) \subset B_\epsilon(f(y)),$$

because $f(x) \in B_{\frac{\epsilon}{2}}(y)$.

By compactness, from the covering $\bigcup\limits_{x\in A} B_{\frac{\delta(x)}{2}}(x)$ we can choose a finite number of balls $B_{\frac{\delta(x_j)}{2}}(x_j), j = 1,\dots,N$ which still cover A. Then $\delta = \min\limits_{k} \frac{\delta(x_j)}{2}$ satisfy the condition of the theorem, i.e. $f(B_\delta(x)) \subset B_\epsilon(f(x))$ for any $x \in A$.

2.6 Homeomorphism

Given two subsets $A, B \subset V$ of a vector space V we say that a map $f : A \to B$ is a *homeomorphism* if it is 1-1 (i.e. bijective), continuous, and the inverse map $f^{-1} : B \to A$ is also a homeomorphism.

Equivalently, one can say that $f : A \to B$ is a homeomorphism if (and only if) it is 1-1 and if images and pre-images of open sets in one of the subsets are open in the other one, i.e. if it induces 1-1 map between open sets in A and open sets in B.

Theorem 2.13 implies

Corollary 2.15. *The property of compactness, connectedness and path connectedness are preserved by a homeomorphism.*

Exercise 2.16. *1. All open intervals $(a, b) \subset \mathbb{R}$, $a < b$, where we allow a to be $-\infty$ and b to be $+\infty$, are homeomorphic to $\mathbb{R}$.*

2. The circle $S := \{x_1^2 + x_2^2 = 1\} \subset \mathbb{R}^2$ is not *homeomorphic to $\mathbb{R}$, but the punctured circle $S \setminus (0, 1)$ is homeomorphic to $\mathbb{R}$.*

3. The set $\mathbb{R}^2 \setminus \{x_2 = 0, x_1 \geq 0\}$ is homeomeorphic to $\mathbb{R}^2$.

Theorem 2.17. *Suppose A is compact and $f : A \to B$ is a continuous 1-1 map. Then $f^{-1} : B \to A$ is also continuous, i.e. f is a homeomorphism.*

Proof. We need to show the following claim: for any sequence $a_j \in A, j = 1,\dots$ such that $f(a_j) \underset{j\to\infty}{\to} b \in B$ the sequence $\{a_j\}$ converges to a point in A. In view of compactness of A we can choose a subsequence $\{a_{j_k}\}$ which converges to a point $a \in A$. Let us verify that the whole sequence $\{a_j\}$ also converges to the same point. Indeed, otherwise we could choose for some $\epsilon > 0$ a subsequence of points a_{j_m} such that $||a_{j_m} - a|| \geq \epsilon$. Passing, if necessary to a subsequence of this sequence we conclude that $a_{j_m} \to \widetilde{a} \in A$ and $\widetilde{a} \neq a$. But then by continuity we get $f(a_{j_m}) \to f(\widetilde{a}) \in B$. But them map f is 1-1. Hence, $f(\widetilde{a}) \neq b$, which is a contradiction. ■

Exercise 2.18. Give an example of a non-compact A for which a continuous 1-1 map $f : A \to B$ has a discontinuous inverse.

Chapter 3

Vector fields and differential 1-forms

The main idea of the differential calculus is that it is useful to associate with a non-linear object a *family of linear objects* which approximate it near near the corresponding points. In Physics context, these families are usually referred as *fields.* In this chapter we will illustrate this idea by three important examples: vector fields, differential 1-forms and plane fields.

Given a vector space V we will denote by V_x the vector space V with the origin translated to the point $x \in V$. There is a projection $\pi_x : V_x \to V$, $\pi_x(v) = x + v$ which maps the origin of V_x to the point x. The map π_x is *not linear* but *affine*, i.e. a linear map plus a constant. The linear map $v \in V_x \mapsto \pi_x(v) - x \in V$ is called the *parallel transport* of V_x to V. One can think of V_x as a tangent space to V at the point x, and sometimes use the notation T_xV.

Given any domain $U \subset V$ the collection $\{T_xV, x \in U\}$ of tangent spaces at points of U is called the *tangent bundle* to U and denoted by TU. We denote by π the projection $TU \to U$ given by $\pi(x, h) = x, h \in T_xU$ Individual tangent spaces $T_xV, x \in U$ (which we will also denote sometimes by T_xU are fibers $\pi^{-1}(x)$ of this projection. The parallel transport defines a 1-1 map $TU \to U \times V$.

Though the parallel transport allows one to identify spaces V and T_xU, it will be important for us to think about them as different spaces.

3.1 Vector fields and differential 1-forms

Definition 3.1. 1. A *vector field* v on a domain $U \subset V$ is a function which associates with each point $x \in U$ a vector $v_x \in V_x$. Equivalently, one can say that a vector field v on U is a map $U \to TU$ such that $\pi \circ v(x) = x$ for each $x \in U$, i.e. it maps every point $x \in U$ to a fiber r $\pi^{-1}(x) = T_x U$ over this point.
2. A *differential 1-form* α on a domain $U \subset V$ is a function which associates with each point $x \in U$ a vector $v(x) \in V_x^*$, i.e a linear function $\alpha_x : V_x \to \mathbb{R}$. Equivalently, one can say that a differential form on U is a function $\alpha : TU \to \mathbb{R}$ which is linear on each fiber $\pi^{-1}(x) = T_x U$.

Thus, differential 1-form is a function of two vector arguments: $x \in U$ and $h \in V_x$. It depends on h linearly, and on x in an arbitrary (but usually continuous) way.

In any vector space V a vector $v \in V = V_0$ can be parallel transported to all points $x \in V$, thus defining a constant vector field $\mathbf{v}$ on V. In particular, if $v_1, \dots, v_n$ is a basis in V then the vector fields $\mathbf{v}_1, \dots, \mathbf{v}_n$ define a basis $\mathbf{v}_1(x), \dots, \mathbf{v}_n(x) \in V_x$ for any $x \in V$. For different points x these bases differ by a parallel transport.

Like vectors, vector fields can be added: $(v + w)(x) = v(x) + w(x)$. But while vectors can be multiplied only by numbers, vector fields can be multiplied by functions: given a vector field v on $U \subset V$ and a function $f : U \to \mathbb{R}$ we define the vector field fv by the formula $(fv)(x) = f(x)v(x)$.

Given any vector field w on a domain $U \subset V$ we can express the vector $w(x) \in V_x$ through the basis $\mathbf{v}_1(x), \dots, \mathbf{v}_n(x) \in V_x$:

$$w(x) = w_1(x)\,\mathbf{v}_1(x) + \dots + w_n(x)\,\mathbf{v}_n(x),$$

and hence present w in the form $\sum\limits_1^n w_j\,\mathbf{v}_j$.

Therefore, defining a vector field on U is the same as defining n functions $w_1, \dots, w_n : U \to \mathbb{R}$, i.e. defining a map $(v_1, \dots, v_n) : U \to \mathbb{R}^n$. Thus, when a basis of V is fixed, the difference between a map $U \to \mathbb{R}^n$ and a vector field on U is just a matter of geometric interpretation:
- when we speak about a *vector field* v we view $v(x)$ as a vector in V_x, i.e. originated at the point $x \in U$;
- when we speak about a map $v : U \to \mathbb{R}^n$ we view $v(x)$ as a point of the space V, or as a vector originated at $\mathbf{0} \in V$.

3.2 Differential of a function

Let $f : U \to \mathbb{R}$ be a function on a domain $U \subset V$ in a vector space V. The function f is called *differentiable* at a point $x \in U$ if there exists a linear function $l : V_x \to \mathbb{R}$ such that

$$f(x+h) - f(x) = l(h) + o(||h||)$$

for any sufficiently small vector h, where the notation $o(t)$ stands for any function such that $\frac{o(t)}{t} \underset{t\to 0}{\to} 0$. The linear function l is called the *differential* of the function f at the point x and is denoted by $d_x f$. In other words, f is differentiable at $x \in U$ if for any $h \in V_x$ there exists a limit

$$l(h) = \lim_{t\to 0} \frac{f(x+th) - f(x)}{t},$$

and the limit $l(h)$ *linearly* depends on h. The value $l(h) = d_x f(h)$ is called the *directional derivative* of f at the point x in the direction h. The function f is called differentiable on the whole domain U if it is differentiable at each point of U.

Simply speaking, the differentiability of a function means that at a small scale near a point x the function behaves approximately like a linear function, called the differential of the function at the point x. However these linear functions vary from point to point, and thus the family $\{d_x f\}_{x\in U}$ of all these linear functions, is a differential 1-form. We call it the *differential* of the function f, and denote it by df (without a reference to a particular point x).

Let us summarize the above discussion. Let $f : U \to \mathbb{R}$ be a differentiable function. Then for each point $x \in U$ there exists a linear function $d_x f : V_x \to \mathbb{R}$, the differential of f at the point x, defined by the formula

$$d_x f(h) = \lim_{t\to 0} \frac{f(x+th) - f(x)}{t}, x \in U, h \in V_x.$$

The family $df = \{d_x f\}_{x\in U}$ is a differential 1-form.

Let $v_1, \dots, v_n$ be a basis in V, and $\mathbf{v}_1, \dots, \mathbf{v}_n$ the corresponding constant vector fields obtained by parallel transporting the basis $v_1, \dots, v_n$. Then we have

$$\begin{aligned} d_x f(\mathbf{v}_i(x)) &= \lim_{t\to 0} \frac{f(x_1, \dots, x_{i-1}, x_i + t, x_{i+1}, \dots, x_n) - f(x_1, \dots, x_i, \dots, x_n)}{t} \\ &= \frac{\partial f}{\partial x_i}(x),\ x \in U,\ i = 1, \dots, n, \end{aligned}$$

where $x_1, \dots, x_n$ are coordinates with respect to the chosen basis $v_1, \dots, v_n$.

Therefore, for $h = \sum\limits_1^n h_j \, \mathbf{v}_j(x)$ we get by linearity

$$d_x f(h) = \sum_1^n \frac{\partial f}{\partial x_j}(x) h_j.$$

Notice that if f is a linear function,

$$f(x) = a_1 x_1 + \dots + a_n x_n ,$$

then for each $x \in V$ we have

$$d_x f(h) = a_1 h_1 + \dots + a_n h_n \ , h = (h_1, \dots, h_n) \in V_x \, .$$

Thus the differential of a linear function f at any point $x \in V$ coincides with this function, parallel transported to the space V_x. This observation, in particular, can be applied to linear coordinate functions $x_1, \dots, x_n$ with respect to a chosen basis of V. We will refer to their differentials $dx_1, \dots, dx_n$ as *basic differential 1-forms.* Thus we have

$$d_a x_j(h) = h_j, \ \text{ for any } a \in V, h \in V_x.$$

As we will see, the class of differential 1-forms is much larger than the class of differentials of functions.

Similar to vector fields, differential 1-forms can be added and multiplied by functions. Let us choose a basis $v_1, \dots, v_n$ in V, and let $x_1, \dots, x_n$ be the corresponding coordinate functions. Their differentials $d_a x_1, \dots, d_a x_n$ at any point $a \in \mathbb{R}^n$ form a basis of the dual space V_a^*. It follows that *any differential 1-form α on U can be presented as*

$$\alpha = f_1 dx_1 + \dots + f_n dx_n, \tag{3.2.1}$$

for some functions $f_1, \dots, f_n : U \to \mathbb{R}$. This expression explains the origin of the adjective *differential.*

Exercise 3.2. Prove that the presentation (3.2.1) for a differential 1-form α is unique (for a fixed coordinate system).

Lemma 3.3. *For a differentiable function $f : U \to \mathbb{R}$ we have*

$$df = \sum_1^n \frac{\partial f}{\partial x_k} dx_k.$$

Proof. For any vector $h = (h_1, \dots, h_n) \in V_x$ we have

$$d_x f(h) = \sum_1^n \frac{\partial f}{\partial x_k}(x) h_k.$$

On the other hand,

$$\left(\sum_1^n \frac{\partial f}{\partial x_k} dx_k \right)_x (h) = \sum_1^n \frac{\partial f}{\partial x_k}(x) dx_k(h) = \sum_1^n \frac{\partial f}{\partial x_k}(x) h_k = d_x f(h).$$

■

Example 3.4. Differential 1-forms on the line. Consider an interval $U = (a, b) \subset \mathbb{R}$. As $\mathbb{R}$ has a canonical basis 1, there is a canonical coordinate x on it, and its differential dx is a basic differential 1-form. For any point $a \in \mathbb{R}$ and any vector $h \in \mathbb{R}_a$ we have $d_a x(h) = h$. Any other differential 1-form α on U can be written as $\alpha = g(x)dx$, and if $\alpha = df$ for some differentiable function $f : (a, b) \to \mathbb{R}$ then $df = f'(x)dx$. Note that this formula means that the two differential forms df and dx are proportional, and $f'(x)$ is the proportionality coefficient which depends on the point x. Hence, Leibniz's notation for the derivative $f'(x) = \frac{df}{dx}$ makes sense as a ratio of two differential forms on the line.

Proposition 3.5. Leibniz rule. *Let $f, g : U \to \mathbb{R}$ be two smooth functions. Then*

$$d(fg) = f dg + g df.$$

.

This follows from the well-known rule for the derivative of a product of two functions.

We recall that existence of partial derivatives at a point $a \in U$ does not guarantee the differentiability of f at the point a.

Exercise 3.6. Prove that if partial derivatives exist in a neighborhood of a and *continuous* at the point a then f is differentiable at this point.

3.3 C^k-smooth functions

The functions whose first partial derivatives are continuous in U are called C^1*-smooth*, or sometimes just smooth. Equivalently, we can say that f is smooth if the differential $d_x f$ continuously depends on the point $x \in U$.

More generally, for $k \geq 1$ a function $f : U \to \mathbb{R}$ is called C^k-smooth if all its partial derivatives up to order k are continuous in U. The space of C^k-smooth functions is denoted by $C^k(U)$. We will also use the notation $C^0(U)$ and $C^\infty(U)$ which stands, respectively, for the spaces of continuous functions and functions with continuous derivatives of all orders. In this notes we will often speak of smooth functions without specifying the class of smoothness, *assuming that functions have as many continuous derivatives as necessary to justify our computations.*

Differential 1-forms and vector fields are called C^k-smooth if their coordinate functions are C^k-smooth. It is straightforward to check that this property is independent of the choice of coordinates.

Remark 3.7. Sometimes we will need to consider smooth maps, functions, vectors fields, differential forms, etc. defined on a closed subset A of a vector space V. We will always mean by that the these objects are defined on some open neighborhood $U \supset A$. It will not be important for us how exactly these objects are extended to U. However, to make sense of differentiability we need to assume that they are extended. In fact, one can define what differentiability means without any extension, but this would go beyond the goals of these lecture notes. Moreover, a theorem of Hassler Whitney asserts that any function smooth on a closed subset $A \subset V$ can be extended to a smooth function defined on the whole space V.

3.4 Gradient vector field

If V is an Euclidean space, i.e. a vector space with an inner product $\langle\, , \rangle$, then there exists a canonical isomorphism $\mathcal{D} : V \to V^*$, defined by the formula $\mathcal{D}(v)(x) = \langle v, x\rangle$ for $v, x \in V$. Of course, $\mathcal{D}$ defines an isomorphism $V_x \to V_x^*$ for each $x \in V$. Set

$$\nabla f(x) = \mathcal{D}^{-1}(d_x f).$$

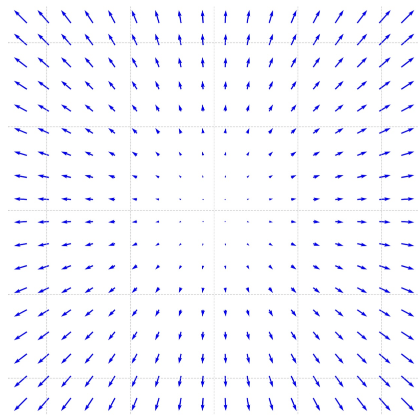


Figure 3.1: Gradient vector field of a function $f(x,y)=x^2+y^2$.

The vector $\nabla f(x)$ is called the *gradient* of the function f at the point $x\in U$. We will also use the notation $\mathrm{grad} f(x)$.

By definition we have

$$\langle \nabla f(x), h\rangle = d_x f(h) \qquad \text{for any vector} \quad h\in V\,.$$

If $||h||=1$ then $d_xf(h)=\langle\nabla f(x),h\rangle=||\nabla f(x)||\cos\varphi$, where φ is the angle between the vectors $\nabla f(x)$ and h. In particular, the directional derivative $d_xf(h)$ has its maximal value when $\varphi=0$. Thus the direction of the gradient is the direction of the maximal growth of the function and the length of the gradient equals this maximal value.

As in the case of a differential, the gradient varies from point to point, and the family of vectors $\{\nabla f(x)\}_{x\in U}$ forms the *gradient vector field* ∇f.

Coordinate expression of the gradient vector field. Suppose that $(V,\langle\ ,\ \rangle)$ is a Euclidean vector space. Choose a (not necessarily orthonormal) basis $v_1,\dots,v_n$. Let us find the coordinate description of the gradient vector field ∇f, i.e. find the coefficients a_j in the expansion $\nabla f(x)=\sum\limits_1^n a_i(x)\frac{\partial}{\partial x_i}$. By definition we have

$$\langle\nabla f(x),h\rangle = d_xf(h)=\sum_1^n\frac{\partial f}{\partial x_j}(x)h_j \tag{3.4.1}$$

for any vector $h\in V_x$ with coordinates $(h_1,\dots,h_n)$ in the basis $v_1,\dots,v_n$ parallel transported to V_x. Let us denote $g_{ij}=\langle v_i,v_j\rangle$. Thus $G=(g_{ij})$ is a symmetric $n\times n$ matrix, which is called the

Gram matrix of the basis $v_1, \dots, v_n$. Then the equation (3.4.1) can be rewritten as

$$\sum_{i,j=1}^{n} g_{ji} a_i h_j = \sum_{1}^{n} \frac{\partial f}{\partial x_j}(x) h_j.$$

Because h_j are arbitrarily numbers it implies that the coefficients with h_j in the right and left sides should coincide for all $j = 1, \dots, n$. Hence we get the following system of linear equations:

$$\sum_{i=1}^{n} g_{ij} a_i = \frac{\partial f}{\partial x_j}(x), \quad j = 1, \dots, n, \tag{3.4.2}$$

or in matrix form

$$G \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x_1}(x) \\ \vdots \\ \frac{\partial f}{\partial x_n}(x) \end{pmatrix},$$

and thus

$$\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = G^{-1} \begin{pmatrix} \frac{\partial f}{\partial x_1}(x) \\ \vdots \\ \frac{\partial f}{\partial x_n}(x) \end{pmatrix}, \tag{3.4.3}$$

i.e.

$$\nabla f = \sum_{i,j=1}^{n} g^{ij} \frac{\partial f}{\partial x_i}(x) \frac{\partial}{\partial x_j}, \tag{3.4.4}$$

where we denote by g^{ij} the entries of the inverse matrix $G^{-1} = (g_{ij})^{-1}$

If the basis $v_1, \dots, v_n$ is orthonormal then G is the unit matrix, and thus in this case

$$\nabla f = \sum_{1}^{n} \frac{\partial f}{\partial x_j}(x) \frac{\partial}{\partial x_j}, \tag{3.4.5}$$

i.e. ∇f has coordinates $(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n})$. However, simple expression (3.4.5) for the gradient holds **only in the orthonormal basis**. In the general case one has a more complicated expression (3.4.4).

3.5 Vector fields as first order partial differential operators

Vector fields naturally arise in a context of Physics, Mechanics, Hydrodynamics, etc. as force, velocity and other physical fields. A gradient vector field ∇f of a function f provides us with an example of a vector field, but as we shall see, gradient vector fields form only a small, very special class of vector fields.

Let $C^\infty(U)$ denote the *vector space of infinitely differentiable functions* on a domain $U \subset V$. Let v be a C^∞-smooth vector field on V. We associate with v a linear operator

$$D_v : C^\infty(U) \to C^\infty(U),$$

given by the formula

$$D_v(f) = df(v),\ f \in C^\infty(U).$$

In other words, we compute at any point $x \in U$ the directional derivative of f in the direction of the vector $v(x)$.

Clearly, the operator D_v is linear: $D_v(af+bg) = aD_v(f)+bD_v(g)$ for any functions $f, g \in C^\infty(U)$ and any real numbers $a, b \in \mathbb{R}$. It also satisfies the *Leibniz rule*:

$$D_v(fg) = D_v(f)g + fD_v(g).$$

Let $v_1, \dots, v_n$ be a basis of V, and $\mathbf{v}_1, \dots, \mathbf{v}_n$ the corresponding vector fields. Let $x_1, \dots, x_n$ be the coordinate functions in the basis $v_1, \dots, v_n$. Note that $(D_{\mathbf{v}_j} f)(x) = \frac{\partial f}{\partial x_j}(x)$, i.e. the operator $D_{\mathbf{v}_j}$ is the operator $\frac{\partial}{\partial x_j}$ of taking j-th partial derivative. More generally, if $w = \sum\limits_1^n c_k \mathbf{v}_k$, where $c_1, \dots, c_k$ are functions on U, then

$$(D_w f)(x) = \sum_1^n c_k(x) \frac{\partial f}{\partial x_k}(x),$$

and hence, we can present the operator D_w as $D_w = \sum\limits_1^n c_k \frac{\partial}{\partial x_k}$.

It is customary not to distinguish between a vector field w and the operator D_w, and therefore use the notation $\frac{\partial}{\partial x_k}$ or ∂_{x_k} for a basic vector field $\mathbf{v}_k$, and the notation $\sum\limits_1^n c_k \frac{\partial}{\partial x_k}$ or $\sum\limits_1^n c_k \partial_{x_k}$ for a vector field $w = \sum\limits_1^k c_k \mathbf{v}_k$. Let us stress the point that the coefficients c_k here are *functions* on U, and not constants.

3.6 Closed and exact differential 1-forms

A differential 1-form α on a domain $U \subset V$ is called *exact* if there exists a C^1-smooth function $g : U \to \mathbb{R}$ such that $\alpha = dg$. The function g is called a *primitive* of the exact form α. We note that the primitive is defined uniquely *up to adding a constant*:

$$dg = d\widetilde{g} \quad \Rightarrow \widetilde{g} = g + C.$$

The following lemma provides a *necessary* condition for a closed form to be exact.

Lemma 3.8. *If a differential $\alpha = \sum\limits_1^n f_j dx_j$ with C^1-smooth coefficients $f_1, \dots, f_n : U \to \mathbb{R}$ is exact then for any $i, j \in \{1, \dots, n\}$ and any point $x \in U$ we have*

$$\frac{\partial f_i}{\partial x_j}(x) = \frac{\partial f_j}{\partial x_i}(x). \tag{3.6.1}$$

Proof. If $\alpha = dg$ then $f_i = \frac{\partial g}{\partial x_i}$ and $\frac{\partial f_i}{\partial x_j} = \frac{\partial^2 g}{\partial x_i \partial x_j}$, so the formula (3.6.1) is just the mixed derivatives equality $\frac{\partial^2 g}{\partial x_i \partial x_j} = \frac{\partial^2 g}{\partial x_j \partial x_i}$. ■

For instance, the form $x_1 dx_2$ on $\mathbb{R}^2$ is not exact.

A differential form $\alpha = \sum\limits_1^n f_j dx_j$ with C^1-smooth coefficients $f_j : U \to \mathbb{R}$ is called *closed* if it satisfies condition (3.6.1).

This definition has a defect that it is coordinate dependent. It is not a priori clear that if this condition is satisfied in one coordinate system then it is satisfied in any other. We will later show that it is independent.

Exercise 3.9. Which of the following forms are closed and which are exact?

$$\alpha_1 = xdy, \quad \alpha_2 = \frac{1}{2}(xdy - ydx), \quad \alpha_3 = \alpha_1 - \alpha_2, \quad \alpha_4 = \frac{\alpha_2}{x^2 + y^2}.$$

The forms α_1, α_2 and α_3 are defined on $\mathbb{R}^2$. The form α_4 is defined on $\mathbb{R}^2 \setminus 0$.

Answer. α_1 and α_2 are not closed, α_3 is exact, α_4 is closed but not exact. The non-exactness of α_4 is proven in Example 3.13.1 below.

3.7 Smooth maps and their differentials

Let V, W be two vector spaces of arbitrary (not, necessarily, equal) dimensions and $U \subset V$ be an open domain in V.

Recall that a map $f : U \to W$ is called *differentiable* if for each $x \in U$ there exists a linear map

$$l : V_x \to W_{f(x)}$$

such that

$$l(h) = \lim_{t \to 0} \frac{f(x + th) - f(x)}{t}$$

for any $h \in V_x$. In other words,

$$f(x + th) - f(x) = tl(h) + o(t), \text{ where } \frac{o(t)}{t} \underset{t \to 0}{\to} 0.$$

The map l is denoted by $d_x f$ and is called *the differential* of the map f at the point $x \in U$. Thus, $d_x f$ is a linear map $V_x \to W_{f(x)}$.

Example 3.10. Consider a map $f : \mathbb{R}^2 \to \mathbb{R}^2$ given by the formula

$$f(x_1, x_2) = (x_1^3 \sin \frac{\pi x_1 x_2}{2}, x_1 e^{\sin \pi x_2}).$$

Compute its differential at the point $a = (1, 2)$.

Solution. We have

$$\begin{aligned} f(1 + h_1, 2 + h_2) &= ((1 + 3h_1 + \dots) \sin(\pi + \pi h_1 + \frac{\pi h_2}{2} + \dots), (1 + h_1) e^{\sin(2\pi + 2h_2)} \\ &= \left(-(1 + 3h_1 + \dots)(\pi h_1 + \frac{\pi h_2}{2} + \dots), (1 + h_1)(1 + 2h_2 + \dots) \right) = \\ &= (-\pi h_1 - \frac{\pi h_2}{2} + \dots, h_1 + 2h_2 + \dots), \end{aligned}$$

where dots denote terms $o(||h||)$.

Hence, $d_{(1,2)} f$ is a linear map $d_{(1,2)} f : \mathbb{R}^2_{(1,2)} \to \mathbb{R}^2_{(0,1)}$ given by the formula

$$d_{(1,2)} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} = \begin{pmatrix} -\pi & -\pi/2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix}.$$

Of course, $\begin{pmatrix} -\pi & -\pi/2 \\ 1 & 2 \end{pmatrix}$ is the Jacobi matrix $J_{(1,2)} f = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(a) & \frac{\partial f_1}{\partial x_2}(a) \\ \frac{\partial f_2}{\partial x_1}(a) & \frac{\partial f_2}{\partial x_2}(a) \end{pmatrix}$.

The space $W_{f(x)}$ can be identified with W via a parallel transport, and hence sometimes it is convenient to think about the differential as a map $V_x \to W$. In particular, in the case of a *linear function.* i.e. when $W = \mathbb{R}$ it is customary to do that, and hence we defined earlier in Section 3.2 the differential of a function $f : U \to \mathbb{R}$ at a point $x \in U$ as a linear function $V_x \to \mathbb{R}$, i.e. an element of V_x^*, rather than a linear map $V_x \to W_{f(x)}$.

Let us pick bases in V and W and let $(x_1, \dots, x_k)$ and $(y_1, \dots, y_n)$ be the corresponding coordinate functions. Then each of the spaces V_x and W_y, $x \in V$, $y \in W$ inherits a basis obtained by parallel transport of the bases of V and W. In terms of these bases, the differential $d_x f$ is given by the Jacobi matrix

$$\begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_k} \\ \dots & \dots & \dots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_k} \end{pmatrix}$$

In other words, given a vector $h \in V_x$ with coordinates $\begin{pmatrix} h_1 \\ \dots \\ h_k \end{pmatrix}$ the column of coordinates of the vector $d_x f(h)$ is given by the formula

$$d_x f(h) = J_x(f)h = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_k} \\ \dots & \dots & \dots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_k} \end{pmatrix} \begin{pmatrix} h_1 \\ \cdot \\ h_k \end{pmatrix}.$$

As we already pointed out, in what follows we will consider only sufficiently smooth maps, i.e. we assume that all maps and their coordinate functions are differentiable as many times as we need it.

If a map $f : U \to U' \subset W$ is differentiable at every point of U then the collection of differentials $\{d_x f,\ x \in U\}$ can be viewed as a map $df : TU \to TU'$. Note that we have the following *commuting* diagram:

$$\begin{array}{ccc} TU & \xrightarrow{df} & TU' \\ \pi \downarrow & & \downarrow \pi \\ U & \xrightarrow{f} & U' \end{array}$$

Here the word "commuting" means that the compositions of arrows with the same first and last space is the same, i.e in this case that $\pi \circ df = f \circ \pi$. We can also say that df is a *fiberwise linear map*, i.e. it maps the fiber $\pi^{-1}(x) = T_x U$ over a point $x \in U$ linearly to the fiber $\pi^{-1}(f(x)) = T_{f(x)}U'$ over the point $f(x) \in U'$.

3.8 Operator f^*

Let U be a domain in a vector space V, U' a domain in a vector space W and $f : U \to U'$ a smooth map. Then for any point $x \in U$ we have a linear map

$$d_x f : V_x \to W_{f(x)}$$

Let α be a differential 1-form on U'.

Definition 3.11. *The pull-back form $f^*\alpha$ on U is defined by the formula*

$$(f^*\alpha)_x = (d_x f)^* \alpha_{f(x)}, x \in U.$$

In other words, for any point $x \in U$ and a vector $h \in V_x$ we have $(f^*\alpha)_x(h) = \alpha_{f(x)}(d_x f(h))$.

Yet equivalently, we can say that $f^*\alpha = \alpha \circ df$, where we view the differential α as the function $TU' \to \mathbb{R}$, the differential form $f^*\alpha$ as a function $TU \to \mathbb{R}$ and the differential df as a map $TU \to TU'$.

Lemma 3.12. *Let $f : U \to U'$ a smooth map.*

1. *For any two forms α, β on U' we have $f^*(\alpha + \beta) = f^*\alpha + f^*\beta$.*

2. *For any smooth function $\phi : U' \to \mathbb{R}$ we have $f^*(\phi\alpha) = (\phi \circ f) f^*\alpha$..*

3. *For any smooth function $g : U' \to \mathbb{R}$ we have*

$$f^* dg = d(g \circ f).$$

Proof. 1. We have

$$(f^*(\alpha + \beta))_x = (d_x f)^*(\alpha_{f(x)} + \beta_{f(x)}) = (d_x f)^*(\alpha_{f(x)}) + (d_x f)^*(\beta_{f(x)}) = (f^*\alpha)_x + (f^*\beta)_x.$$

2. Similarly,

$$(f^*(\phi\alpha))_x(h) = (\phi\alpha)_{f(x)}(d_x f(h)) = \phi(f(x))\alpha_{f(x)}(d_x f(h)) = \phi(f(x))(f^*\alpha)_x(h).$$

3. This is just the chain rule: for any point $x \in U$ and a vector $h \in V_x$ we have

$$(f^* dg)_x(h) = d_{f(x)} g(d_x f(h)) = d_x(g \circ f)(h).$$

■

Example 3.13. 1. In $V = \mathbb{R}^2$ with Cartesian coordinates denoted (r, ϕ) consider a domain $U := \{r > 0, 0 < \phi < 2\pi\}$, and in $W = \mathbb{R}^2$ with Cartesian coordinates denoted (x, y) consider a domain $U' = \mathbb{R}^2 \setminus 0$. Let $\alpha = \frac{xdy-ydx}{x^2+y^2}$ be a differential form on $U' = \mathbb{R}^2 \setminus 0$. Consider a map $P : U \to U'$ given by the formula $P(r, \phi) = (r\cos\phi, r\sin\phi)$. Then

$$
\begin{aligned}
P^*\alpha &= \frac{r\cos\phi d(r\sin\phi) - r\sin\phi d(r\cos\phi)}{(r\cos\phi)^2 + (r\sin\phi)^2} \\
&= \frac{1}{r^2}\left(r\cos\phi(\sin\phi\, dr + r\cos\phi\, d\phi) - r\sin\phi(\cos\phi\, dr - r\sin\phi\, d\phi)\right) \\
&= \frac{1}{r^2}(r^2((\cos\phi)^2 + (\sin\phi)^2)d\phi = d\phi.
\end{aligned}
$$

As we will see in the next section, the map P can be viewed as introducing polar coordinates r, ϕ in $\mathbb{R}^2$, and the above computation can be equivalently interpreted as expressing the form α in polar coordinates. More precisely, away from the origin the two 1-forms, dr and $d\phi$ are linearly independent, and hence any differential 1-form can be expressed in the form $g(r, \phi)dr + h(r, \phi)d\phi$. As the above computation shows, we have $g = 0$ and $h = 1$.

Note that as a corollary of the above computation we conclude that the form $\alpha = \frac{xdy-ydx}{x^2+y^2}$ on $\mathbb{R}^2 \setminus 0$ is closed. We also can see that it is not exact. Indeed, if $df = \alpha$ then $f = \phi + \text{const}$, but ϕ is not a continuous function on $\mathbb{R}^2 \setminus 0$.

2. Let $\alpha = f_1 dx_1 + \ldots f_n dx_n$ be a differential 1-form in a domain $U \subset V$. Consider a (smooth) path $\gamma = (\gamma_1, \ldots, \gamma_n) : (a, b) \to U$. Then

$$\gamma^*\alpha = \left(f_1(\gamma(t))\gamma_1'(t) + \cdots + f_n(\gamma(t))\gamma_n'(t)\right) dt.$$

3. Take a differential form $\alpha = dz - ydx$ in $\mathbb{R}^3$. Given two smooth functions $f, g : \mathbb{R} \to \mathbb{R}^3$ define a map $F : \mathbb{R} \to \mathbb{R}^3$ by the formula

$$F(t) = (t, g(t), f(t)), \quad t \in \mathbb{R}.$$

Then

$$F^*\alpha = df - g(t)dt = (f'(t) - g(t))dt.$$

Hence, $F^*\alpha = 0$ if and only if $g = f'$.

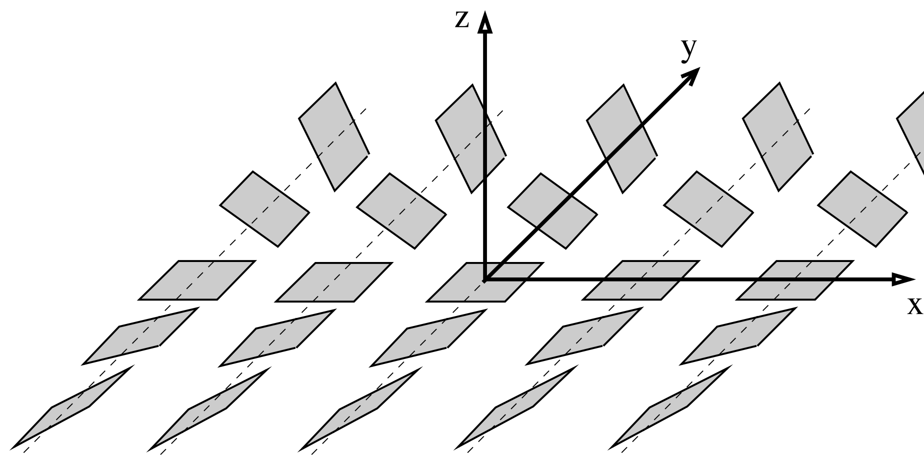


Figure 3.2: Contact structure

3.9 Hyperplane fields and Pfaffian equations

Given a non-zero linear function ℓ on an n-dimensional vector space V the equation $\ell = 0$ defines a *hyperplane*, i.e. an $(n-1)$-dimensional subspace $\{v \in V; \ell(v) = 0\} \subset V$.

Suppose we are given a differential 1-form λ on a domain $U \subset V$. Suppose that $\lambda_x \neq 0$ for each $x \in U$. Then one can consider a *hyperplane field* $\xi := \{\lambda = 0\}$, where $\xi_x := \{v \in V_x;\ \lambda_x(v) = 0\}$. One says that ξ is given by a *Pfaffian* (in honor of a German mathematician Johann Friedrich Pfaff (1765–1825)) equation

$$\lambda = 0. \tag{3.9.1}$$

Example 3.14. Let $V = \mathbb{R}^3$ with coordinates (x, y, z)

1. Let $\lambda = dz$. Then $\xi = \{dz = 0\}$ is the horizontal plane field which is equal to $\mathrm{Span}(\frac{\partial}{\partial x}, \frac{\partial}{\partial y})$.

2. Let $\lambda = dz + xdy + ydx$. Then the Pfaffian equation $dz - xdy - ydx = 0$, or equivalently, $d(z - \frac{1}{2}(x^2 + y^2)) = 0$ defines a family of planes tangent to parabolloids $\{z = \frac{1}{2}(x^2 + y^2) + C\}$.

3. Let $\lambda = dz - ydx$. Then the plane field $\xi := \{dz - ydx = 0\}$ is shown on Fig. 3.9. This plane is *non-integrable* in the following sense: there are no surfaces in $\mathbb{R}^3$ tangent to ξ. Indeed, the plane field ξ is transverse to the z-direction. Hence, any surface S, which is tangent to it in a neighborhood of a point $a = (x_0, y_0, z_0)$, can be presented as a graph $\{z = f(x, y)\}$ of a

function defined in a neighborhood U of the point (x_0, y_0) and which satisfies $f(x_0, y_0) = z_0$. (See Implicit Function Theorem in the next section). Consider a map $F : U \to \mathbb{R}^3$ given by the formula $F(x, y) = (x, y, f(x, y))$. Then $F^*\alpha = df - ydx$. But $F^*\alpha = 0$. Indeed, for any point $a \in U$ and any vector $T \in \mathbb{R}^2_a$ we have $(F^*\alpha)_a(T) = \alpha(d_aF(T)) = 0$ because $d_aF(T) \in \xi = \{\alpha = 0\}$. But this means that the form ydx is exact, but it is not even closed.

The plane field ξ is called a *contact structure*. It plays an important role in symplectic and contact geometry, which is, in turn, the geometric language for Mechanics and Geometric Optics. In the next section we give an example of a problem in Mechanics where a contact structure naturally arises.

3.10 Skating and contact structures

Configuration space of a mechanical system is a space of all its possible positions. For instance, one needs 3 coordinates (x, y, z) to describe a position of a particle in the 3-dimensional space $\mathbb{R}^3$, so the configuration space of a free particle is $\mathbb{R}^3$. If the particle is connected by a rod of length r to the origin, then the coordinates are *constrained* by the equation

$$x^2 + y^2 + z^2 = r^2,$$

i.e. the configuration space becomes the sphere $S_r := \{x^2 + y^2 + z^2 = r^2\} \subset \mathbb{R}^3$.

One says that a mechanical system has k *degrees of freedom* if its configuration space is k-dimensional. So the free particle in $\mathbb{R}^3$ has three degrees of freedom, and constrained one only two.

Viewing r as a control parameter which can be changed, we can still consider $\mathbb{R}^3$ as the configuration space of the system, but the motion is *constrained* to the family of concentric spheres S_r.

We can express this constraint by stating that the velocity vector $(\dot{x}, \dot{y}, \dot{z})$ of the particle should be *orthogonal* to radius-vector (x, y, z), i.e. satisfy the condition $x\dot{x} + y\dot{y} + z\dot{z} = 0$.

In other words, the velocity vector of a moving point in the configuration space should be *tangent to the plane field defined by the Pfaffian equation*

$$xdx + ydy + zdz = 0.$$

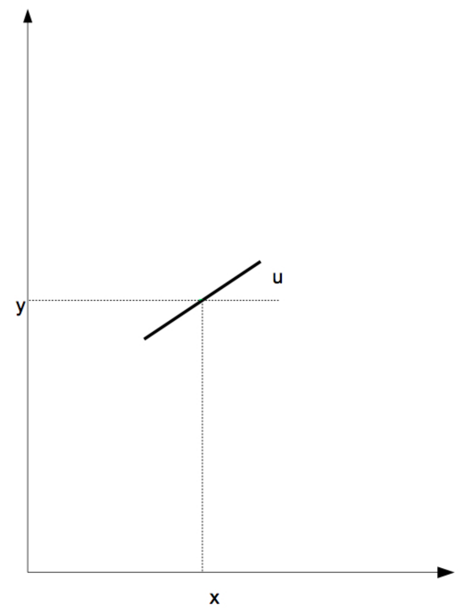


Figure 3.3: Configuration space of the skating problem

Constraints of this type are called *holonomic.*

Skating is an example of a system with *non-holonomic constraints.* An ice skate position is determined by 3 coordinates: (x, y) - the points of contact, and a circular coordinate $u \in [0, 2\pi]$ - the direction of the skate. Hence, the configuration space is $\mathbb{R}^2 \times S^1$.

The skating equation is that the velocity vector $v = (\dot{x}, \dot{y})$ is proportional to $(\cos u, \sin u)$, i.e $-\dot{x}\sin u + \dot{y}\sin u = 0$. There are no constraints on $\dot{u}$.

In other words, the trajectory $(x(t), y(t), u(t))$ of the skate is *tangent to the plane field*

$$\xi = \{-\sin u\, dx + \cos u\, dy = 0\}.$$

Such constraint is called in *non-holonomic.*

The plane field ξ is another example of a *contact structure.* The curves tangent to it are called em Legendrian. Thus, trajectory of a skate is a Legendrian curve in the configuration space

Contact plane field *completely non-integrable*: as we already had seen, there are no surfaces tangent to it, and unlike the holonomic case *one can travel between any two points along a Legendrian curve.*

Chapter 4

Inverse and Implicit Function Theorems

4.1 Diffeomorphisms and Curvilinear Coordinate Systems

Diffeomorphisms

Let $U \subset V$ and $U' \subset V'$ be two domains in vector spaces V and V' of the same dimension n. A C^k-smooth map $f : U \to U'$ is called a *C^k-diffeomorphism*, $k \geq 1$, if it is 1-1 and the inverse map $f^{-1} : U' \to U$ is also $\mathbb{C}^k$- smooth. If $k = 0$, i.e. f and f^{-1} are assumed to be only continuous, then f is called a *homeomorphism*. As we will see in the next section, as a corollary of the Inverse Function Theorem 4.3, if f is 1-1 and its differential $d_x f : V_x \to V'_{f(x)}$ is invertible for each $x \in U$ then f^{-1} is automatically C^k-smooth, and hence f is a C^k-diffeomorphism. Below we drop C^k from the notation, and just talk about diffeomorphisms assuming that they are as smooth as necessary. We will specify the smoothness class k in some cases when this is important.

Exercise 4.1. 1. Show that any smooth function $h : \mathbb{R} \to \mathbb{R}$ with with $\lim\limits_{x \to \pm\infty} h(x) = \pm\infty$ and $h' > 0$ s a diffeomorphism.

2. Is the map $\mathbb{R} \to \mathbb{R}$ given by the formula $x \to x^3$ a diffeomorphism?

3. Show that the map P defined in Example 3.13.1 is a diffeomorphism of the domain

$$\{r > 0, -\pi < \phi < \pi\} \subset \mathbb{R}^2$$

(we consider here (r, ϕ) as Cartesian coordinates in $\mathbb{R}^2$) onto the domain

$$U'' = \mathbb{R}^2 \setminus \{y = 0, x \leq 0\}.$$

Verify that the inverse map $Q = P^{-1}$ is given by the formula $Q(x, y) = \left(\sqrt{x^2 + y^2}, \arctan \frac{y}{x}\right)$.

Curvilinear coordinates

A (non-linear or curvilinear) *coordinate system* on a domain U in an n-dimensional space V is a diffeomorphism $f = (f_1, \ldots, f_n) : U \to U' \subset \mathbb{R}^n$.

Thus a *coordinate map* f associates n coordinates $y_1 = f_1(x), \ldots, y_n = f_n(x)$ with each point $x \in U$. The inverse map $f^{-1} : U' \to U$ is called a *parameterization*. If one already has another set of coordinates $x_1 \ldots x_n$ on U, then the coordinate map f expresses new coordinates $y_1 \ldots y_n$ through the old ones, while the parametrization map expresses the old coordinates through the new ones.

Example 4.2. Consider $\mathbb{R}^3$ with Cartesian coordinates (r, ϕ, θ), and take a domain

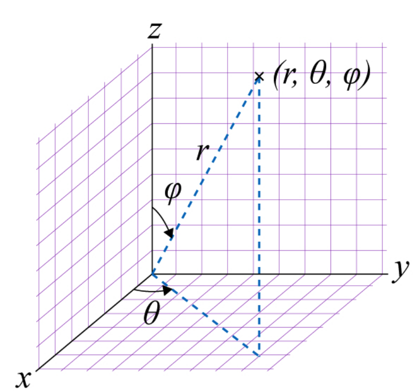
$$U := \{r > 0, 0 < \phi < \pi, 0 < \theta < 2\pi\}.$$

Consider a map $S : U \to \mathbb{R}^3$, where the target $\mathbb{R}^3$ is endowed with Cartesian coordinates (x, y, z), given by the formula $S(r, \phi, \theta) = (x, y, z)$, where

$$x = r \sin \varphi \cos \theta,$$

$$y = r \sin \varphi \sin \theta,$$

$$z = r \cos \varphi \,.$$



Then S is a diffeomorphism of U onto the domain $U' = \mathbb{R}^3 \setminus \{y = 0, x \geq 0\}$. This is a parameterization map for the *spherical coordinate system.*

Lemma 3.12 applied to a *parameterization* map f can be interpreted by saying the differential of a function has the same expression in the new coordinates and in the old ones.

Consider a domain $U \subset V$ with (possibly curvilinear) coordinates $(u_1, \dots, u_n)$. Given a point $c \in U$ with coordinates $(c_1, \dots, c_n)$, the curve $\Gamma_c^j := \{u_i = c_i, i = 1, \dots, n; i \neq j\}$ is called the *j-th coordinate line* through the point c. For instance, coordinate lines for polar coordinates in $\mathbb{R}^2$ are concentric circles and rays, while coordinate lines for spherical coordinates in $\mathbb{R}^3$ are rays from the origin, and latitudes and meridians on concentric spheres.

4.2 Inverse function theorem

Theorem 4.3. *Let U be an open set in $\mathbb{R}^n$ which contains the origin 0. Let $f : U \to \mathbb{R}^n$ be a C^k-smooth map such that $f(0) = 0$ and $d_0 f : \mathbb{R}_0^n \to \mathbb{R}_0^n$ is an invertible linear map (i.e.* $\det d_0 f \neq 0$*. Then there are neighborhoods $U_1, U_2 \ni 0$, $U_1 \subset U$ such that $f|_{U_1}$ is a diffeomorphism $U_1 \to U_2$.*

Proof. Note that if the statement holds for maps f, g then it also holds for $f \circ g$ and and for f^{-1}. The statement obviously holds for the case when f is a linear map. Hence, it is sufficient to consider the case $d_0 f = \mathrm{Id}$. Indeed, we can reduce to this case by taking a composition $f \circ (d_0 f)^{-1}$. Hence, we can write

$$f(x) = x + \phi(x),\ x \in U, \quad \text{where} \;\; d_0 \phi = 0. \tag{4.2.1}$$

Let us first show that

Lemma 4.4. *For any $\epsilon > 0$ there exists $\delta > 0$ such that for any $x, y \in B_\delta(0)$ we have*

$$(1 - \epsilon)||x - y|| \leq ||f(x) - f(y)|| \leq (1 + \epsilon)||x - y||. \tag{4.2.2}$$

Proof of Lemma 4.4. A *norm* of a linear map $A : \mathbb{R}^n \to \mathbb{R}^n$ is defined as $||A|| = \sup\limits_{||x||=1} ||Ax||$. We then have $\|Ax|\| \leq \|A\|\|x\|$ for any $x \in \mathbb{R}^n$.

Exercise 4.5. Prove that if (a_{ij}) is a matrix of A then $||A||^2 \leq \sum\limits_{i,j=1}^{n} a_{ij}^2$.

As the map ϕ is C^1 and $d_0\phi = 0$ we can choose for any $\epsilon > 0$ a $\delta > 0$ such that $||d_z\phi|| < \epsilon$ for any $u \in B_\delta(0)$. It follows that for a sufficiently small δ we have $||\phi(x) - \phi(y)|| \leq \epsilon||x - y||$ for any $x, y \in B_\delta(0)$, see the proof of Lemma 14.7. Applying (4.2.1) we get $\|f(x) - f(y)\| = \|(x - y) + (\phi(x) - \phi(y))\|$, and hence, by (4.2.2) we get

$$(1-\epsilon)\|x-y\| \leq \|x-y\| - \|\phi(x)-\phi(y)\| \leq \|f(x)-f(y)\| \leq \|x-y\| + \|\phi(x)-\phi(y)\| \leq (1+\epsilon)\|x-y\|.$$

■

We are continuing our proof of Theorem 4.3. Let us choose $\delta > 0$ provided by Lemma 4.4 for $\epsilon = \frac{1}{2}$. Inequality (4.2.2) then implies that the map f is *injective* on $B_\delta(0)$. Indeed, if $f(x) - f(y)$ then inequality (4.2.2) implies that $x = y$.

Let us also show that $f(B_\delta(0)) \supset B_{\delta/8}(0)$. Given $y \in B_{\delta/8}(0)$ consider the function $\theta_y : B_\delta(0) \to \mathbb{R}$ given by the formula $\theta_y(x) = \|f(x) - y\|^2$.

As the function ϕ_y is continuous of the closed ball $\overline{B_\delta(0)}$ it achieves a minimum there at some point $x_0 \in \overline{B_\delta(0)}$. Inequality (4.2.2) implies that

$$\min_{||x||=\delta} \phi_y(x) \geq \frac{3}{8}, \text{ while } \max_{||x||=\delta/8} \phi_y(x) \leq \frac{5}{16} < \frac{3}{8}.$$

Hence, $x_0 \notin \partial B_\delta(0)$, i.e. it is an interior point of $B_\delta(0)$, which means it is a critical point of ϕ_y, i.e. $d_{x_0}\phi_y = 0$. Computing partial derivatives of the function ϕ_y we get

$$0 = \frac{\partial \phi_y}{\partial x_j}(x_0) = \left\langle \frac{\partial f}{\partial x_j}(x_0), f(x_0) - y \right\rangle.$$

By assumption, $\text{Span}\left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right) = \mathbb{R}^n_{x_0}$, and therefore, $f(x_0) = y$. Denote $U := f^{-1}(B_{\delta/8}(0))$ and $U' =: B_{\delta/8}(0)$. We have U is open as the pre-image of an open set under a continuous map, and $U \ni 0$. Hence, $f|_U := U \to B_{\delta/8}(0)$ is a 1-1 map. To verify that f is a diffeomorphism we need to check that the inverse map $g := f^{-1} : U' \to U$ is C^1-smooth. Note that by construction $g(0) = 0$. Let us rewrite the inequality (4.2.2) in terms of g:

$$\frac{1}{1+\epsilon}||x - y|| \leq ||g(x) - g(y)|| \leq \frac{1}{1-\epsilon}||x - y||. \tag{4.2.3}$$

This immediately implies that g is Lipshitz, and hence, continuous. Setting $y = 0$ and using $g(y) = 0$ we also get that

$$\frac{1}{1+\epsilon} \leq \frac{||g(x)||}{||x||} \leq \frac{1}{1-\epsilon},$$

and hence, $\frac{||g(x)||}{||x||} \underset{x\to 0}{\to} 1$.

Let us now check that g is of class C^1. We rewrite (4.2.1) as $x - g(x) = \phi(g(x))$. To verify the differentiability at 0 we need to check that $\lim\limits_{x\to 0} \frac{||\phi(g(x))||}{||x||} = 0$. Recall that by continuity of g we have $\lim\limits_{x\to 0} g(x) = 0$. Hence, by definition of ϕ we have $\lim\limits_{x\to 0} \frac{||\phi(g(x))||}{||g(x)||} = 0$. Therefore,

$$\lim_{x\to 0} \frac{||\phi(g(x))||}{||x||} = \lim_{x\to 0} \frac{||\phi(g(x))||}{||g(x)||} \lim_{x\to 0} \frac{||g(x)||}{||x||} = 0.$$

This proves that that g is differentiable at 0. But the same argument applies to any point x where $d_{g(x)}f \neq 0$, which is true for all points in U. The fact that $d_x f$ continuously depends on x follows from the chain rule: $d_x g = (d_{g(x)}f)^{-1}$. This concludes the proof of the inverse function theorem. ■

Remark 4.6. One can show that $g = f^{-1}$ has the same class of smoothness as f, i.e. if f is C^k, then g is C^k and f is C^∞ then g is C^∞.

4.3 Implicit function theorem

Consider $\mathbb{R}^n$ with coordinates $(x_1, \dots, x_n)$, $\mathbb{R}^m$ with coordinates $(y_1, \dots, y_m)$ and $\mathbb{R}^{n+m} = \mathbb{R}^n \times \mathbb{R}^m$ with coordinates $(x, y) = (x_1, \dots, x_n, y_1, \dots, y_m)$. Given a map $f : \mathbb{R}^{n+m} \to \mathbb{R}^m$ we denote by f_y the Jacobi matrix

$$f_y := \begin{pmatrix} \frac{\partial f_1}{\partial y_1} & \dots & \frac{\partial f_1}{\partial y_m} \\ \dots & \dots & \dots \\ \frac{\partial f_m}{\partial y_1} & \dots & \frac{\partial f_m}{\partial y_m} \end{pmatrix}$$

Theorem 4.7. *Let U be a neighborhood of 0 in $\mathbb{R}^{n+m}$, U' a neighborhood of 0 in $\mathbb{R}^m$. Let $f : U \to U'$ be a C^1-map such that $f(0) = 0$ and $\det(f_y) \neq 0$. Then there exists a neighborhood $U'' \ni 0$ in $\mathbb{R}^n$, a neighborhood $\widetilde{U} \subset U$ of 0 in $\mathbb{R}^{n+m}$ and a C^1-smooth map $h : U'' \to \widetilde{U}$ such*

$$\widetilde{U} \cap \{f(x, y) = 0\} = \{y = h(x);\ x \in U''\}.$$

Proof. Consider a map $F : U \to \mathbb{R}^n \times U' \subset \mathbb{R}^{n+m}$ given by the formula $F(x, y) = (x, f(x, y))$, $(x, y) \in U$. Note that $F(0) = 0$ and $d_0 F : \mathbb{R}^{n+m}_0 \to \mathbb{R}^{n+m}_0$ is invertible. Hence, we can apply Theorem 4.3 to conclude that F is a diffeomorphism on a smaller neighborhood $U_1 \ni 0, U_1 \subset U$ onto a neighborhood $U_2 \ni 0$ in $\mathbb{R}^{n+m}$. Let $g = F^{-1}$. We have $g(x, y) = (g_1(x, y), g_2(x, y)) \in \mathbb{R}^n \times \mathbb{R}^m$ and

$$F(g_1(x, y), g_2(x, y)) = (g_1(x, y), f(g_1(x, y), g_2(x, y))) = (x, y),$$

$g_1(x,y) = x$ and $f(x, g_2(x,y)) = y$. Denote $h(x) := g_2(x,0)$. Then we have

$$f(x, h(x)) = 0$$

for x in a sufficiently small neighborhood $U'' \ni 0, U'' \subset \mathbb{R}^n$. Conversely if $(x,y) \in U_1$ and $f(x,y) = 0$ then $y = g_2(x,0)$. Let a neighborhood $U'' \ni 0$, $U'' \subset \mathbb{R}^n$ be so small that $g_2(x,0) \in U_1$ for $x \in U''$ then we get

$$\{f(x,y) = 0\} \cap U_1 \cap \{x \in U_1\} = \{y = h(x), x \in U''\}.$$

■

Exercise 4.8. *In notation of Theorem 4.7 denote*

$$f_x := \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \dots & \dots & \dots \\ \frac{\partial f_m}{\partial x_1} & \dots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}, \quad h_x := \begin{pmatrix} \frac{\partial h_1}{\partial x_1} & \dots & \frac{\partial h_1}{\partial x_n} \\ \dots & \dots & \dots \\ \frac{\partial h_m}{\partial x_1} & \dots & \frac{\partial h_m}{\partial x_n} \end{pmatrix}.$$

Use the chain rule to show that

$$h_x(x) = -(f_y(x, h(x)))^{-1} f_x(x, h(x)), x \in U''.$$

Chapter 5

Integration of differential 1-forms

5.1 One-dimensional Riemann integral for functions and differential 1-forms

A *partition* $\mathcal{P}$ of an interval $[a,b]$ is a finite sequence $a = t_0 < t_1 < \dots < t_N = b$. We will denote by $T_j, j = 0,\dots,N$ the vector $t_{j+1} - t_j \in \mathbb{R}_{t_j}$ and by Δ_j the interval $[t_j, t_{j+1}]$. The length $t_{j+1} - t_j = ||T_j||$ of the interval Δ_j will be denoted by δ_j. The number $\max\limits_{j=1,\dots,N} \delta_j$ is called the *fineness* or the *size* of the partition $\mathcal{P}$ and will be denoted by $\delta(\mathcal{P})$.

Let us first recall the definition of (Riemann) integral of a *bounded* function of one variable. Given a function $f : [a,b] \to \mathbb{R}$ we will form a *lower* and *upper* integral sums corresponding to the partition $\mathcal{P}$:

$$\begin{aligned} L(f;\mathcal{P}) &= \sum_0^{N-1} (\inf_{[t_j,t_{j+1}]} f)(t_{j+1} - t_j)\,, \\ U(f;\mathcal{P}) &= \sum_0^{N-1} (\sup_{[t_j,t_{j+1}]} f)(t_{j+1} - t_j)\,, \end{aligned} \tag{5.1.1}$$

The function is called *Riemann integrable* if

$$\sup_{\mathcal{P}} L(f;\mathcal{P}) = \inf_{\mathcal{P}} U(f;\mathcal{P}),$$

and in this case this number is called the (Riemann) *integral of the function f over the interval* $[a,b]$. The integrability of f can be equivalently reformulated as follows. Let us choose a set $C =$

$\{c_1, \dots, c_{N-1}\}$, $c_j \in \Delta_j$, and consider an integral sum

$$I(f; \mathcal{P}, C) = \sum_{0}^{N-1} f(c_j)(t_{j+1} - t_j),\ c_j \in \Delta_j. \tag{5.1.2}$$

Then the function f is integrable if there exists a limit $\lim\limits_{\delta(\mathcal{P}) \to 0} I(f, \mathcal{P}, C)$. In this case this limit is equal to the integral of f over the interval $[a, b]$. Let us emphasize that if we already know that the function is integrable, then to compute the integral one can choose *any* sequence of integral sum, provided that their fineness goes to 0. In particular, sometimes it is convenient to choose $c_j = t_j$, and in this case we will write $I(f; \mathcal{P})$ instead of $I(f; \mathcal{P}, C)$.

The integral has different notations. It can be denoted sometimes by $\int\limits_{[a,b]} f$, but the most common notation for this integral is $\int\limits_a^b f(x)dx$. This notation hints that we are integrating here the differential form $f(x)dx$ rather than a function f. Indeed, given a differential form $\alpha = f(x)dx$ we have $f(c_j)(t_{j+1} - t_j) = \alpha_{c_j}(T_j)$,[1] and hence

$$I(\alpha; \mathcal{P}, C) = I(f; \mathcal{P}, C) = \sum_{0}^{N-1} \alpha_{c_j}(T_j),\ c_j \in \Delta_j. \tag{5.1.3}$$

We say that a differential 1-form α is *integrable* if there exists a limit $\lim\limits_{\delta(\mathcal{P}) \to 0} I(\alpha, \mathcal{P}, C)$, which is called in this case the *integral of the differential* 1*-form* α *over the oriented interval* $[a, b]$ and will be denoted by $\int\limits_{\overrightarrow{[a,b]}} \alpha$, or simply $\int\limits_a^b \alpha$. By definition, we say that $\int\limits_{\overleftarrow{[a,b]}} \alpha = -\int\limits_{\overrightarrow{[a,b]}} \alpha$. This agrees with the definition $\int\limits_{\overleftarrow{[a,b]}} \alpha = \lim\limits_{\delta(\mathcal{P}) \to 0} \sum\limits_{1}^{N-1} \alpha_{c_j}(-T_j)$, and with the standard calculus rule $\int\limits_b^a f(x)dx = -\int\limits_a^b f(x)dx$.

Let us recall that a map $\phi : [a, b] \to [c, d]$ is called a *diffeomorphism* if it is smooth and has a smooth inverse map $\phi^{-1} : [c, d] \to [a, b]$. This is equivalent to one of the following:

- $\phi(a) = c; \phi(b) = d$ and $\phi' > 0$ everywhere on $[a, b]$. In this case we say that ϕ preserves orientation.

- $\phi(a) = d; \phi(b) = c$ and $\phi' < 0$ everywhere on $[a, b]$. In this case we say that ϕ reverses orientation.

[1] Here we parallel transported the vector T_j from the point t_j to the point $c_j \in [t_j, t_{j+1}]$.

Theorem 5.1. *Let $\phi : [a,b] \to [c,d]$ be a diffeomorphism. Then if a 1-form $\alpha = f(x)dx$ is integrable over $[c,d]$ then its pull-back $f^*\alpha$ is integrable over $[a,b]$, and we have*

$$\int\limits_{\overrightarrow{[a,b]}} \phi^*\alpha = \int\limits_{\overrightarrow{[c,d]}} \alpha, \tag{5.1.4}$$

if ϕ preserves the orientation and

$$\int\limits_{\overrightarrow{[a,b]}} \phi^*\alpha = \int\limits_{\overleftarrow{[c,d]}} \alpha = -\int\limits_{\overrightarrow{[c,d]}} \alpha,$$

if ϕ reverses the orientation.

Remark 5.2. We will show later a stronger result:

$$\int\limits_{\overrightarrow{[a,b]}} \phi^*\alpha = \int\limits_{\overrightarrow{[c,d]}} \alpha$$

for *any* $\phi : [a,b] \to [c,d]$ with $\phi(a) = c, \phi(b) = d$, which is not necessarily a diffeomorphism.

Proof. We consider only the orientation preserving case, and leave the orientation reversing one to the reader. Choose any partition $\mathcal{P} = \{a = t_0 < \cdots < t_{N-1} < t_N = b\}$ of the interval $[a,b]$ and choose any set $C = \{c_0, \ldots, c_{N-1}\}$ such that $c_j \in \Delta_j$. Then the points $\widetilde{t}_j = \phi(t_j) \in [c,d]$, $j = 0, \ldots N$ form a partition of $[c,d]$. Denote this partition by $\widetilde{\mathcal{P}}$, and denote $\widetilde{\Delta}_j := [\widetilde{t}_j, \widetilde{t}_{j+1}] \subset [c,d]$, $\widetilde{c}_j = \phi(c_j)$, $\widetilde{C} = \phi(C) = \{\widetilde{c}_0, \ldots, \widetilde{c}_{N-1}\}$. Then we have

$$\begin{aligned} I(\phi^*\alpha, \mathcal{P}, C) = \sum_0^{N-1} \phi^*\alpha_{c_j}(T_j) = \sum_0^{N-1} \alpha_{\widetilde{c}_j}(d\phi(T_j)) = \\ \sum_0^{N-1} \alpha_{\widetilde{c}_j}(\phi'(c_j)\delta_j). \end{aligned} \tag{5.1.5}$$

Recall that according to the mean value theorem there exists a point $d_j \in \Delta_j$, such that

$$\widetilde{T}_j = \widetilde{t}_{j+1} - \widetilde{t}_j = \phi(t_{j+1}) - \phi(t_j) = \phi'(d_j)(t_{j+1} - t_j).$$

Note also that the function ϕ' is *uniformly continuous*, i.e. for any $\epsilon > 0$ there exists $\delta > 0$ such that for any $t, t' \in [a,b]$ such that $|t - t'| < \delta$ we have $|\phi'(t) - \phi'(t')| < \epsilon$. Besides, the function ϕ'

is bounded above and below by some positive constants: $m < \phi' < M$. Hence $m\delta_j < \widetilde{\delta}_j < M\delta_j$ for all $j = 1, \dots, N-1$. Hence, if $\delta(\mathcal{P}) < \delta$ then we have

$$\left|I(\phi^*\alpha, \mathcal{P}, C) - I(\alpha; \widetilde{\mathcal{P}}, \widetilde{C})\right| = \left|\sum_0^{N-1} \alpha_{\widetilde{c}_j}\big((\phi'(c_j) - \phi'(d_j))\delta_j\big)\right| =$$

$$\leq \frac{\epsilon}{m}\left|\sum_1^{N-1} f(\widetilde{c}_j)\widetilde{\delta}_j\right| = \frac{\epsilon}{m}\left|I(\alpha; \widetilde{\mathcal{P}}, \widetilde{C})\right|. \tag{5.1.6}$$

When $\delta(\mathcal{P}) \to 0$ we have $\widetilde{\delta}(\mathcal{P}) \to 0$, and hence by assumption $I(\alpha; \widetilde{\mathcal{P}}, \widetilde{C}) \to \int\limits_c^d \alpha$, but this implies that $I(\phi^*\alpha, \mathcal{P}, C) - I(\alpha; \widetilde{\mathcal{P}}, \widetilde{C}) \to 0$, and thus $\phi^*\alpha$ is integrable over $[a, b]$ and

$$\int\limits_a^b \phi^*\alpha = \lim_{\delta(\mathcal{P})\to 0} I(\phi^*\alpha, \mathcal{P}, C) = \lim_{\delta(\widetilde{\mathcal{P}})\to 0} I(\alpha, \widetilde{\mathcal{P}}, \widetilde{C}) = \int\limits_c^d \alpha.$$

■

If we write $\alpha = f(x)dx$, then $\phi^*\alpha = f(\phi(t))\phi'(t)dt$ and the formula (5.1) takes a familiar form of the change of variables formula from the 1-variable calculus:

$$\int\limits_c^d f(x)dx = \int\limits_a^b f(\phi(t))\phi'(t)dt.$$

5.2 Integration of differential 1-forms along curves

Curves as paths

A *path*, or *parametrically given curve* in a domain U in a vector space V is a map $\gamma : [a, b] \to U$. We will assume in what follows that all considered paths are differentiable.

Given a differential 1-form α in U we define the *integral of α over γ* by the formula

$$\int\limits_\gamma \alpha = \int\limits_{[a,b]} \gamma^*\alpha.$$

If γ is a *loop*, i.e. $\gamma(a) = \gamma(b)$ then one often write $\oint\limits_\gamma \alpha$ instead of $\int\limits_\gamma \alpha$.

Example 5.3. Consider the form $\alpha = dz - ydx + xdy$ on $\mathbb{R}^3$. Let $\gamma : [0, 2\pi] \to \mathbb{R}^3$ be a *helix* given by parametric equations $x = R\cos t, y = R\sin t, z = Ct$. Then

$$\int\limits_{\gamma} \alpha = \int\limits_{0}^{2\pi} (Cdt + R^2(\sin^2 tdt + \cos^2 tdt)) = \int\limits_{0}^{2\pi} (C + R^2)dt = 2\pi(C + R^2).$$

Note that $\int\limits_{\gamma} \alpha = 0$ when $C = -R^2$. One can observe that in this case the curve γ is tangent to the plane field ξ given by the Pfaffian equation $\alpha = 0$.

Proposition 5.4. *Let a path $\widetilde{\gamma}$ be obtained from $\gamma : [a, b] \to U$ by a reparametrization, i.e. $\widetilde{\gamma} = \gamma \circ \phi$, where $\phi : [c, d] \to [a, b]$ is an orientation preserving diffeomorphism. Then $\int\limits_{\widetilde{\gamma}} \alpha = \int\limits_{\gamma} \alpha$.*

Indeed, applying Theorem 5.1 we get

$$\int\limits_{\widetilde{\gamma}} \alpha = \int\limits_{c}^{d} \widetilde{\gamma}^* \alpha = \int\limits_{c}^{d} \phi^*(\gamma^* \alpha) \int\limits_{a}^{b} \gamma^* \alpha = \int\limits_{\gamma} \alpha.$$

A vector $\gamma'(t) \in V_{\gamma(t)}$ is called the *velocity* vector of the path γ.

Curves as 1-dimensional submanifolds

A subset $\Gamma \subset U$ is called a *1-dimensional submanifold* of U if for any point $x \in \Gamma$ there is a neighborhood $U_x \subset U$ and a diffeomorphism $\Phi_x : U_x \to \Omega_x \subset \mathbb{R}^n$, such that $\Phi_x(x) = 0 \in \mathbb{R}^n$ and $\Phi_x(\Gamma \cap U_x)$ either coincides with $\{x_2 = \ldots x_n = 0\} \cap \Omega_x$, or with $\{x_2 = \ldots x_n = 0,\ x_1 \geq 0\} \cap \Omega_x$. In the latter case the point x is called a *boundary* point of Γ. In the former case it is called an *interior* point of Γ.

A 1-dimensional submanifold is called *closed* if it is compact and has no boundary. An example of a closed 1-dimensional manifold is the circle $S^1 = \{x^2 + y^2 = 1\} \subset \mathbb{R}^2$.

WARNING. The word *closed* is used here in a different sense than when one speaks about closed *subsets*. For instance, a circle in $\mathbb{R}^2$ is both, a closed subset and a closed 1-dimensional submanifold, while a closed interval is a closed subset but not a closed submanifold: it has 2 boundary points. An open interval in $\mathbb{R}$ (or any $\mathbb{R}^n$) is a submanifold without boundary but it is not closed because it is not compact. A line in a vector space is a 1-dimensional submanifold which is a closed subset of the ambient vector space. However, it is not compact, and hence not a closed submanifold.

Proposition 5.5. *1. Suppose that a path $\gamma : [a,b] \to U$ is an embedding. This means that $\gamma'(t) \neq 0$ for all $t \in [a,b]$ and $\gamma(t) \neq \gamma(t')$ if $t \neq t'$.*[2] *Then $\Gamma = \gamma([a,b])$ is 1-dimensional compact submanifold with boundary.*

2. *Suppose $\Gamma \subset U$ is given by equations $F_1 = 0, \dots, F_{n-1} = 0$ where $F_1, \dots, F_{n-1} : U \to \mathbb{R}$ are smooth functions such that for each point $x \in \Gamma$ the differential $d_x F_1, \dots, d_x F_{n-1}$ are linearly independent. Then Γ is a 1-dimensional submanifold of U.*

3. *Any compact connected 1-dimensional submanifold $\Gamma \subset U$ can be parameterized either by an embedding $\gamma : [a,b] \to \Gamma \hookrightarrow U$ if it has non-empty boundary, or by an embedding $\gamma : S^1 \to \Gamma \hookrightarrow U$ if it is closed.*

Proof. 1. Take a point $c \in [a,b]$. By assumption $\gamma'(c) \neq 0$. Let us choose an affine coordinate system $(y_1, \dots, y_n)$ in V centered at the point $C = \gamma(c)$ such that the vector $\gamma'(c) \in V_C$ coincide with the first basic vector. In these coordinates the map γ can be written as $(\gamma_1, \dots, \gamma_n)$ where $\gamma_1'(c) = 1$, $\gamma_j'(c) = 0$ for $j > 1$ and $\gamma_j(c) = 0$ for all $j = 1, \dots, n$. By the inverse function theorem the function γ_1 is a diffeomorphism of a neighborhood of c onto a neighborhood of 0 in $\mathbb{R}$ (if c is one of the end points of the interval, then it is a diffeomorphism onto the corresponding one-sided neighborhood of 0). Let σ be the inverse function defined on the interval Δ equal to $(-\delta, \delta)$, $[0, \delta)$ and $(-\delta, 0]$, respectively, depending on whether c is an interior point, $c = a$ or $c = b$, so that $\gamma_1(\sigma(u)) = u$ for any $u \in \Delta$. Denote $\widetilde{\Delta} = \sigma(\Delta) \subset [a,b]$. Then $\gamma(\widetilde{\Delta}) \subset U$ can be given by the equations:

$$y_2 = \widetilde{\gamma}_2(u) := \gamma_2(\sigma(y_1)), \ \dots, \ y_n = \widetilde{\gamma}_n(u) := \gamma_n(\sigma(y_1)); \ y_1 \in \Delta.$$

Let us denote

$$\theta = \theta(\delta) := \max_{j=2,\dots,n} \max_{u \in \Delta} |\widetilde{\gamma}_j(u)|.$$

Denote

$$P_\delta := \{|y_1| \leq \delta, |y_j| \leq \theta(\delta)\}.$$

We have $\gamma(\widetilde{\Delta}) \subset P_\delta$.

We will show now that for a sufficiently small δ we have $\gamma([a,b]) \cap P_\delta = \gamma(\widetilde{\Delta})$. For every point $t \in [a,b] \setminus \operatorname{Int} \widetilde{\Delta}$ denote $d(t) = ||\gamma(t) - \gamma(c)||$. Recall now the condition that $\gamma(t) \neq \gamma(t')$

[2] If only the former property is satisfied then γ is called an *immersion*.

for $t \neq t'$. Hence $d(t) > 0$ for all $t \in [a,b] \setminus \operatorname{Int}\widetilde{\Delta}$. The function $d(t)$ is continuous and hence achieve the minimum value on the compact set $[a,b] \setminus \operatorname{Int}\widetilde{\Delta}$. Denote $d := \min\limits_{t \in [a,b]\setminus \operatorname{Int}\widetilde{\Delta}} d(t) > 0$. Chose $\delta' < \min(d, \delta)$ and such that $\theta(\delta') = \max\limits_{j=2,\dots,n} \max\limits_{|u-c|<\delta'} |\widetilde{\gamma}_j(u)| < d$. Let $\Delta' = \widetilde{\Delta} \cap \{|u| \leq \delta'\}|$. Then

$$\gamma([a,b]) \cap P_{\delta'} = \{y_2 = \widetilde{\gamma}_2(u),\ \dots,\ y_n = \widetilde{\gamma}_n(u);\ y_1 \in \Delta'\}.$$

2. Take a point $c \in \Gamma$. The linear independent 1-forms $d_cF_1, \dots, d_cF_{n-1} \in V_c^*$ can be completed by a 1-form $l \in V_c^*$ to a basis of V_c^*. We can choose an affine coordinate system in V with c as its origin and such that the function x_n coincides with l. Then the Jacobian matrix of the functions $F_1, \dots, F_{n-1}, x_n$ is non-degenerate at $c = 0$, and hence by the inverse function theorem the map $F = (F_1, \dots, F_{n-1}, x_n) : V \to \mathbb{R}^n$ is invertible in the neighborhood of $c = 0$, and hence these functions can be chosen as new curvilinear coordinates $y_1 = F_1, \dots, y_{n-1} = F_{n-1}, y_n = x_n$ near the point $c = 0$. In these coordinates the curve Γ is given near c by the equations $y_1 = \dots = y_{n-1} = 0$.

3. We postpone the proof of this part till Section 15 below. ■

In the case where Γ is closed we will usually parameterize it by a path $\gamma : [a,b] \to \Gamma \subset U$ with $\gamma(a) = \gamma(b)$. For instance, we parameterize the circle $S^1 = \{x^2 + y^2 = 1\} \subset \mathbb{R}^2$ by a path $[0, 2\pi] \mapsto (\cos t, \sin t)$. Such γ, of course, cannot be an embedding, but we will require that $\gamma'(t) \neq 0$ and that for $t \neq t'$ we have $\gamma(t) \neq \gamma(t')$ *unless* one of these points is a and the other one is b. We will refer to 1-dimensional submanifolds simply as *curves*, respectively closed, with boundary etc.

Given a curve Γ its *tangent line* at a point $x \in \Gamma$ is a subspace of V_x generated by the velocity vector $\gamma'(t)$ for any local parameterization $\gamma : [a,b] \to \Gamma$ with $\gamma(t) = x$. If Γ is given implicitly, as in 5.5.2, then the tangent line is defined in V_x by the system of linear equations $d_xF_1 = 0, \dots, d_xF_{n-1} = 0$.

Orientation of a curve Γ is the continuously depending on points orientation of all its tangent lines. If the curve is given as a path $\gamma : [a,b] \to \Gamma \subset U$ such that $\gamma'(t) \neq 0$ for all $t \in [a,b]$ than it is canonically oriented. Indeed, the orientation of its tangent line l_x at a point $x = \gamma(t) \in \Gamma$ is defined by the velocity vector $\gamma'(t) \in l_x$.

It turns out that one can define an integral of a differential form α over an oriented compact curve directly without referring to its parameterization. For simplicity we will restrict our discussion to the case when the form α is continuous.

Let Γ be a compact connected oriented curve. A *partition* of Γ is a sequence of points $\mathcal{P} = \{z_0, z_1, \dots, z_N\}$ ordered according to the orientation of the curve and such that the boundary points of the curve (if they exist) are included into this sequence. If Γ is closed we assume that $z_N = z_0$. The *fineness* $\delta(\mathcal{P})$ of $\mathcal{P}$ is by definition is $\max\limits_{j=0,\dots,N-1} \operatorname{dist}(z_j, z_{j+1})$ (we assume here that V a Euclidean space).

Definition 5.6. *Let α be a differential 1-form and Γ a compact connected oriented curve. Let $\mathcal{P} = \{z_0, \dots, z_N\}$ be its partition. Then we define*

$$\int\limits_{\Gamma} \alpha = \lim_{\delta(\mathcal{P}) \to 0} I(\alpha, \mathcal{P}),$$

where $I(\alpha, \mathcal{P}) = \sum\limits_{0}^{N-1} \alpha_{z_j}(Z_j)$, $\;Z_j = z_{j+1} - z_j \in V_{z_j}$.

When Γ is a closed submanifold then one sometimes uses the notation $\oint\limits_{\Gamma} \alpha$ instead of $\int\limits_{\Gamma} \alpha$.

Proposition 5.7. *If one chooses a parameterization $\gamma : [a, b] \to \Gamma$ which respects the given orientation of Γ then*

$$\int\limits_{\gamma} \alpha = \int\limits_{a}^{b} \gamma^* \alpha = \int\limits_{\Gamma} \alpha.$$

Proof. Indeed, let $\widetilde{\mathcal{P}} = \{t_0, \dots, t_N\}$ be a partition of $[a, b]$ such that $\gamma(t_j) = z_j$, $j = 0, \dots, N$.

$$I(\gamma^* \alpha, \widetilde{\mathcal{P}}) = \sum_{1}^{N-1} \gamma^* \alpha_{t_j}(T_j) = \sum_{1}^{N-1} \alpha_{z_j}(U_j),$$

where $T_j := t_{j+1} - t_j \in \mathbb{R}_{t_j}$, and $U_j := d_{t_j}\gamma(T_j) \in V_{z_j}$ is a tangent vector to Γ at the point z_j. Let us evaluate the difference $U_j - Z_j$. Choosing some Cartesian coordinates in V we denote by $\gamma_1, \dots, \gamma_n$ the coordinate functions of the path γ. Then using the mean value theorem for each of the coordinate functions we get $\gamma_i(t_{j+1}) - \gamma_i(t_j) = \gamma_i'(c_j^i)\delta_j$ for some $c_j^i \in \Delta_j$, $i = 1, \dots, n;\ j = 0, \dots, N-1$. Thus

$$Z_j = \gamma(t_{j+1}) - \gamma(t_j) = (\gamma_1'(c_j^1), \dots, \gamma_i'(c_j^n))\delta_j.$$

On the other hand, $\;U_j = d_{t_j}\gamma(T_j)) \in V_{z_j} = \gamma'(t_j)\delta_j$. Hence,

$$||Z_j - U_j|| = \delta_j \sqrt{\sum_{1}^{n} (\gamma_i'(c_j^i) - \gamma_i'(t_j))^2}.$$

Note that if $\delta(\mathcal{P}) \to 0$ then we also have $\delta(\widetilde{\mathcal{P}}) \to 0$, and hence using smoothness of the path γ we conclude that for any $\epsilon > 0$ there exists $\delta > 0$ such that $||Z_j - U_j|| < \epsilon\delta_j$ for all $j = 1, \dots, N$. Thus

$$\sum_1^{N-1} \alpha_{z_j}(\widetilde{T}_j) - \sum_1^{N-1} \alpha_{z_j}(Z_j) \underset{\delta(\mathcal{P})}{\to} 0,$$

and therefore

$$\int_a^b \gamma^*\alpha = \lim_{\delta(\widetilde{\mathcal{P}})\to 0} I(\gamma^*\alpha, \widetilde{\mathcal{P}}) = \lim_{\delta(\mathcal{P})\to 0} I(\alpha, \mathcal{P}) = \int_\Gamma \alpha.$$

■

5.3 Integrals of closed and exact differential 1-forms

Theorem 5.8. *Let $\alpha = df$ be an exact 1-form in a domain $U \subset V$. Then for any path $\gamma : [a, b] \to U$ which connects points $A = \gamma(a)$ and $B = \gamma(b)$ we have*

$$\int_\gamma \alpha = f(B) - f(A).$$

In particular, if γ is a loop then $\oint\limits_\gamma \alpha = 0$.

Similarly for an oriented curve $\Gamma \subset U$ with boundary $\partial\Gamma = B - A$ we have

$$\int_\Gamma \alpha = f(B) - f(A).$$

Proof. We have $\int\limits_\gamma df = \int\limits_a^b \gamma^* df = \int\limits_a^b d(f \circ \gamma) = f(\gamma(b)) - f(\gamma(a)) = f(B) - f(A)$. ■

It turns out that closed forms are *locally exact*. A domain $U \subset V$ is called *star-shaped* with respect to a point $a \in V$ if with any point $x \in U$ it contains the whole interval $I_{a,x}$ connecting a and x, i.e. $I_{a,x} = \{a + t(x - a);\, t \in [0, 1]\}$. In particular, any convex domain is star-shaped.

Proposition 5.9. *Let α be a closed 1-form on a star-shaped domain $U \subset V$. Then it is exact.*

Proof. Define a function $F : U \to \mathbb{R}$ by the formula

$$F(x) = \int_{\overrightarrow{I_{a,x}}} \alpha, \quad x \in U,$$

where the intervals $I_{a,x}$ are oriented from 0 to x.

We claim that $dF = \alpha$. Let us identify V with the $\mathbb{R}^n$ choosing a as the origin $a = 0$. Then α can be written as $\alpha = \sum\limits_1^n P_k(x)dx_k$, and $I_{0,x}$ can be parameterized by

$$t \mapsto tx,\ t \in [0,1].$$

Hence,

$$F(x) = \int\limits_{\overrightarrow{I_{0,x}}} \alpha = \int\limits_0^1 \sum_1^n P_k(tx)x_k dt. \tag{5.3.1}$$

Differentiating the integral over x_j as parameters, we get

$$\frac{\partial F}{\partial x_j} = \int\limits_0^1 \sum_{k=1}^n tx_k \frac{\partial P_k}{\partial x_j}(tx)dt + \int\limits_0^1 P_j(tx)dt.$$

But the closedness of α means that $\frac{\partial P_k}{\partial x_j} = \frac{\partial P_j}{\partial x_k}$, and using this we can further write

$$\frac{\partial F}{\partial x_j} = \int\limits_0^1 \left(\sum_{k=1}^n tx_k \frac{\partial P_j}{\partial x_k}(tx) + P_j(tx)\right) dt = \int\limits_0^1 \frac{d\left(tP_j(tx)\right)}{dt} dt = P_j(x)$$

Thus

$$dF = \sum_{j=1}^n \frac{\partial F}{\partial x_j} dx_j = \sum_{j=1}^n P_j(x)dx_j = \alpha$$

Corollary 5.10. *The differential* 1*-form* $\alpha = \frac{xdy - ydx}{x^2+y^2}$ *is closed but not exact.*

Proof. We already had seen that the form α is closed. Take the unit circle $S^1 := \{x^2 + y^2 = 1\}$ oriented counter-clockwise, and compute $\oint\limits_{S^1} \alpha$. Using polar coordinates we recall that $\alpha = d\phi$, and parameterize the circle by $\phi \mapsto (1,\phi)$, $\phi \in [-\pi,\pi]$. Then $\oint\limits_{S^1} \alpha = \int\limits_{-\pi}^{\pi} d\phi = 2\pi \neq 0$. Hence, α is not exact because according to Theorem 5.8 the integral of an exact form over a loop is equal to 0. ■

Corollary 5.11. *Let $f : U \to U'$ be a smooth map and α is a closed differential 1-form on U'. Then the pull-back $f^*\alpha$ is a closed differential 1-form on U. In particular, the condition that a form is closed is independent of a choice of a (curvilinear) coordinates.*

Proof. For the form α to be closed it is the same, according to according to Proposition 5.9, as to be locally exact but the pullback of an exact form is exact: $f^*dh = d(h \circ f)$, and hence, $f^*\alpha$ is also locally exact, and thus closed. ■

Remark 5.12. On $\mathbb{R}$, or any interval $[a, b] \subset \mathbb{R}$ *any* 1*-form is exact, and hence, closed.* Indeed, $\alpha = f(x)dx$. Hence, for $F(x) = \int_a^x f(t)dt$ we have $dF = F'(x)dx = f(x)dx = \alpha$.

5.4 Homotopic paths

The word "homotopy" means *continuous deformation.* Consider two continuous paths $\gamma_0, \gamma_1 : [a, b] \to U \subset V$ be two paths in a vector space V with the same end-points:

$$\gamma_0(a) = \gamma_1(a) = A \in U, \gamma_0(b) = \gamma_1(b) = B \in U.$$

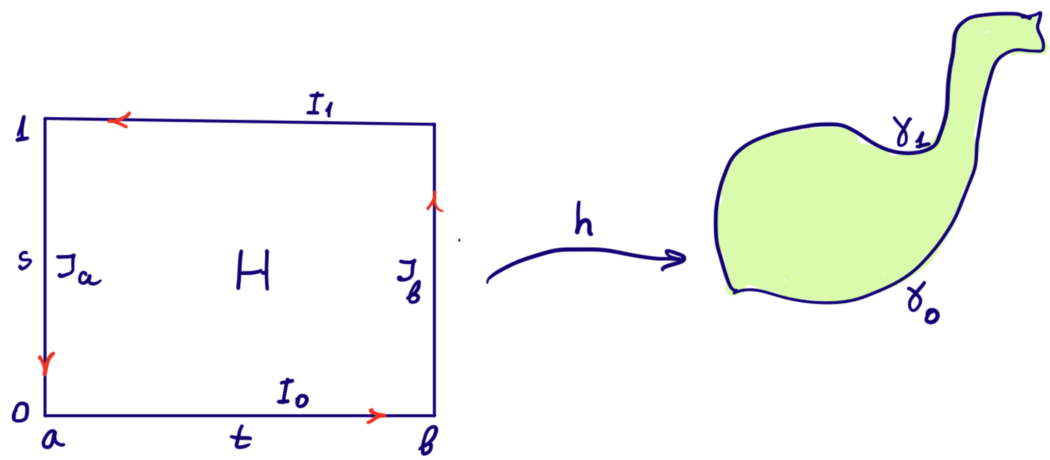
The paths γ_0, γ_1 are called *homotopic* with fixed ends if there exists a continuous map $h : [a, b] \times [0, 1] \to U$ such that

$$h(t, 0) = \gamma_0(t),\ h(t, 1) = \gamma_1(t) \text{ for all } t \in [a, b];$$
$$h(a, s) = A,\ h(b, s) = B \text{ for all } s \in [0, 1];$$

Denote by $\gamma_s : [a, b] \to U,\ s \in [0, 1]$ the path $\gamma_s(t) = h(t, s)$. Then when the parameter s varies from 0 to 1 the path γ_s is continuously deforming from γ_0 to γ_1 keeping the end points $\gamma_s(a)$ and $\gamma_s(b)$ fixed.

Proposition 5.13. *Let α be a closed differential* 1*-form in U. Suppose two paths $\gamma_0, \gamma_1 : [a, b] \to U \subset V$ are homotopic with fixed ends $A = \gamma_0(a) = \gamma_1(a)$ and $B = \gamma_0(b) = \gamma_1(b)$. Then*

$$\int_{\gamma_0} \alpha = \int_{\gamma_1} \alpha.$$



Proof. Denote by H the rectangle $[a,b]\times[0,1]$ and by ∂H its boundary oriented counter-clockwise. We view H is a subset $\{a \le t \le b, 0 \le s \le 1\}$ in $\mathbb{R}^2$ with coordinates t,s. Let $h : H \to U$ be a homotopy between γ_0 and γ_1, i.e. $h(t,0)=\gamma_0(t), h(t,1)=\gamma_1(t)$ for $t\in[a,b]$ and $h(a,s)=A, h(b,s)=B$ for all $s\in[0,1]$. According to Corollary 5.11 the form $h^*\alpha$ is closed. Note that the rectangle $H\subset\mathbb{R}^2$ is star-shaped. Hence, according to Proposition 5.9 the form $h^*\alpha$ on H is exact, and therefore, $\int\limits_{\partial H} h^*\alpha = 0$.

Denote $I_0 := \{s=0, a\le t\le b\}$, $I_1 := \{s=1, a\le t\le b\}$, $J_a := \{0\le s\le 1, t=a\}$, $J_b := \{0\le s\le 1, t=b\}$. We have $\partial H = I_0\cup I_1\cup J_a\cup J_b$. If we orient the intervals I_0, I_1, J_a, J_b as the counter-clockwise boundary of ∂H we get

$$\begin{aligned}\int\limits_{\partial H} h^*\alpha &= \int\limits_{I_0} h^*\alpha + \int\limits_{J_b} h^*\alpha + \int\limits_{I_1} h^*\alpha + \int\limits_{J_a} h^*\alpha \\ &= \int\limits_{\gamma_0}\alpha - \int\limits_{\gamma_1}\alpha,\end{aligned}$$

because $\int\limits_{J_a} h^*\alpha = \int\limits_{J_b} h^*\alpha = 0$.

Thus,

$$\int\limits_{\gamma_0}\alpha = \int\limits_{\gamma_1}\alpha,$$

as required. ∎

Remark 5.14. The above proof has a small gap. While the homotopy was defined as a continuous map, the integrals were defined only over smooth paths. One remedy would be just to restrict the notion of homotopy to be smooth. However, as we will see later, it is not necessary: existence of a continuous homotopy implies existence of a smooth one.

As we already had seen, the integral of a 1-form over a path remains unchanged, if we reparametrize the path by an orientation preserving diffeomorphism. The following corollary of Proposition 5.13 shows that the reparametrization need not even be a diffeomorphism as long as it matches the end points in the correct order.

Corollary 5.15. *Let β be any differential 1-form on a domain $U \subset V$. Let $\gamma : [a,b] \to U$ be a smooth path. Consider any smooth function $\phi : [c,d] \to [a,b]$ such that $\phi(s) = a, \phi(d) = b$. Then $\int\limits_{\gamma} \beta = \int\limits_{\gamma\circ\phi} \beta$.*

Proof. Denote $\alpha := \gamma^*\beta$. Then, by definition B $\int\limits_{\gamma} \beta = \int\limits_{a}^{b} \alpha$ and $\int\limits_{\gamma\circ\phi} \beta = \int\limits_{c}^{d} \phi^*\alpha$. The map $\phi : [c,d] \to [a,b]$ is a path in $[a,b]$ with end points a and b, and $\int\limits_{c}^{d} \phi^*\alpha = \int\limits_{\phi} \alpha$. The identity map $j : [a,b] \to [a,b]$ is another paths with the same end points, and we have $\int\limits_{a}^{b} \alpha = \int\limits_{j} \alpha$. But the path ϕ and j are homotopic with fixed ends. Indeed, the required homotopy $h(t,s)$ can be defined by the formula $h(t,s) = (1-s)j(t) + s\phi(t)$. Applying Proposition 5.13 we conclude that

$$\int\limits_{\gamma} \beta = \int\limits_{j} \alpha = \int\limits_{\phi} \alpha = \int\limits_{\gamma\circ\phi} \beta.$$

■

5.5 Simply connected domains

A subset $U \subset V$ is called *simply connected*, or 1-connected if it is path-connected and any two paths with the same ends are homotopic (via a homotopy with fixed ends).

Exercise 5.16. 1. Prove that any vector space V, or more generally, any star-shaped domain $U \subset V$, is simply connected.

2. Show that $\mathbb{R}^2 \setminus 0$ *is not* simply connected.

Theorem 5.17. *Any closed differential* 1*-form in a simply connected domain is exact.*

Proof. Let α be a closed differential 1-form. Choose a reference point $a \in U$. By assumption, U is path-connected. Hence, any other point x can be connected to A by a path $\gamma_x : [0,1] \to U$, i.e. $\gamma_x(0) = a,\ \gamma_x(1) = x$. Let us define the function $F : U \to \mathbb{R}$ by the formula $F(x) = \int\limits_{\gamma_x} \alpha$. Note that due to the simply-connectedness of the domain U, any $\delta : [0,1] \to U$ connecting b and x is homotopic to γ_x relative its ends, and hence according to Proposition 5.13 we have $\int\limits_{\gamma_x} \alpha = \int\limits_{\delta} \alpha$. Thus the above definition of the function F is independent of the choice of paths γ_x. We claim that the function F is differentiable and $dF = \alpha$. Take any point $a \in U$. By Proposition 5.9 the form α is exact in a ball $B_\epsilon(a) \subset U$. Let G be a primitive of α in $B_\epsilon(a)$. Then for any point $x \in B_\epsilon(a)$ we have $G(x) = G(a) + \int\limits_{\delta_x} \alpha$, where δ_x is any path connecting a with x in $B_\epsilon(a)$. Let σ be a path obtained by concatenating the paths γ_a from A to a and the path δ_x from a to x. Then

$$F(x) = \int\limits_{\gamma_x} \alpha = \int\limits_{\sigma} \alpha = \int\limits_{\gamma_a} \alpha + \int\limits_{\delta_x} \alpha = \int\limits_{\gamma_a} \alpha + G(x) - G(a).$$

Thus the function $F|_{B_\epsilon(a)}$ differs from G by a constant. Therefore, it is differentiable and $dF = dG = \alpha$. ■

Part II

Multilinear Algebra

Chapter 6

Multilinear and bilinear functions and quadratic forms

6.1 Multilinear functions

A function $l(X_1, X_2, \ldots, X_k)$ of k vector arguments $X_1, \ldots, X_k \in V$ (i.e. a function $l : \underbrace{V \times \cdots \times V}_{k} \to \mathbb{R}$) is called *k-linear* (or multilinear) if it is linear with respect to each argument when all other arguments are fixed. We say *bilinear* instead of 2-linear. We will also consider 0-linear functions, which are just *constants*.

Multilinear functions are also called *tensors*. Sometimes, one may also say a "k-linear *form*", or simply k-form instead of a "k-linear functions". However, we will reserve the term k-form for a skew-symmetric tensors which will be defined in Section 7.3 below.

If one fixes a basis $v_1 \ldots v_n$ in the space V then with each bilinear function $f(X, Y)$ one can associate a square $n \times n$ matrix as follows. Set $a_{ij} = f(v_i, v_j)$. Then $A = (a_{ij})_{i,j=1,\ldots,n}$ is called *the matrix of the function f in the basis $v_1, \ldots, v_n$*. For any 2 vectors

$$X = \sum_1^n x_i v_i,\; Y = \sum_1^n y_j v_j$$

we have

$$f(X, Y) = f\left(\sum_{i=1}^n x_i v_i, \sum_{j=1}^n y_j v_j\right) = \sum_{i,j=1}^n x_i y_j f(v_i, v_j) = \sum_{i,j=1}^n a_{ij} x_i y_j = X^T A Y\,.$$

Exercise 6.1. *How does the matrix of a bilinear function depend on the choice of a basis?*

Answer. The matrices A and $\tilde{A}$ of the bilinear form $f(x,y)$ in the bases $v_1,\ldots,v_n$ and $\tilde{v}_1,\ldots,\tilde{v}_n$ are related by the formula $\tilde{A} = C^T AC$, where C is the matrix of transition from the basis $v_1 \ldots v_n$ to the basis $\tilde{v}_1 \ldots \tilde{v}_n$, i.e the matrix whose columns are the coordinates of the basis $\tilde{v}_1 \ldots \tilde{v}_n$ in the basis $v_1 \ldots v_n$.

Similarly, with a k-linear function $f(X_1,\ldots,X_k)$ on V and a basis $v_1,\ldots,v_n$ one can associate a "k-dimensional" matrix

$$A = \{a_{i_1 i_2 \ldots i_k};\ \ 1 \le i_1,\ldots,i_k \le n\},$$

where

$$a_{i_1 i_2 \ldots i_k} = f(v_{i_1},\ldots,v_{i_k})\,.$$

If $X_i = \sum\limits_{j=1}^{n} x_{ij} v_j,\ \ i = 1,\ldots,k$, then

$$f(X_1,\ldots,X_k) = \sum_{i_1,i_2,\ldots i_k=1}^{n} a_{i_1 i_2 \ldots i_k} x_{1i_1} x_{2i_2} \ldots x_{ki_k}\ ,$$

see Proposition 7.1 below.

6.2 More about bilinear functions

Let $f : V \times V \to \mathbb{R}$ be a bilinear function. We say that a vector $x \in V$ is a null-vector for f if $f(x,y) = 0$ for any $y \in V$. The set of all null-vectors forms a subspace of V denoted by $\operatorname{Ker} f$, which is called the *kernel* or the *null-space* of the bilinear function f. Thus,

$$\operatorname{Ker} f = \{x \in V; f(x,y) = 0 \ \text{ for any } y \in V.\}$$

The difference $\dim V - \dim \operatorname{Ker} f$ is called the *rank* of the bilinear function f. The function f is called *non-degenerate* if $\operatorname{rank} f = \dim V$, i.e. if its kernel is equal to $\{0\}$.

Lemma 6.2. *The rank of a bilinear form is equal to the rank of its matrix in any basis of V. In particular, a bilinear form is non-degenerate if and only if its matrix is non-degenerate.*

Proof. Let A be the matrix of f in a basis $v_1, \ldots, v_n$ of V. Let $X, Y \in \mathbb{R}^n$ denote columns of coordinates of vectors $x, y \in V$. Then

$$f(x,y) = X^T AY = (X^T AY)^T = Y^T A^T X = \langle Y, A^T X \rangle,$$

where $\langle , \rangle$ denote the standard scalar product in $\mathbb{R}^n$. We have

$$x \in \operatorname{Ker} f \iff A^T X \perp Y \text{ for any } Y \in \mathbb{R}^n \iff A^T X = 0.$$

In other words $x \in \operatorname{Ker} f$ if and only if the column vector $X \in \mathbb{R}^n$ is in the space of solutions of the homogeneous system of linear equations $A^T X = 0$ with the matrix A^T, and thus $\dim \operatorname{Ker} f$ is the dimension of the space of solutions of the system $A^T X = 0$. On the other hand, this dimension is equal to $n - \operatorname{rank} A^T$. But $\operatorname{rank} A^T = \operatorname{rank} A$, and therefore

$$\operatorname{rank} f = n - \dim \operatorname{Ker} f = n - (n - \operatorname{rank} A) = \operatorname{rank} A.$$

■

6.3 Symmetric bilinear functions and quadratic forms

A function $Q : V \to \mathbb{R}$ on a vector space V is called *quadratic* if there exists a bilinear function $f(X, Y)$ such that

$$Q(X) = f(X, X), \; X \in V. \tag{6.3.1}$$

One also uses the term *quadratic form.* The bilinear function $f(X, Y)$ is not uniquely determined by the equation (6.3.1). For instance, all the bilinear functions x_1y_2, x_2y_1 and $\frac{1}{2}(x_1y_2 + x_2y_1)$ on $\mathbb{R}^2$ define the same quadratic form x_1x_2.

On the other hand, there is a 1-1 correspondence between quadratic form and *symmetric* bilinear functions. A bilinear function $f(X, Y)$ is called symmetric if $f(X, Y) = f(Y, X)$ for all $X, Y \in V$.

Lemma 6.3. *Given a quadratic form $Q : V \to \mathbb{R}$ there exists a unique symmetric bilinear form $f(X, Y)$ such that $Q(X) = f(X, X), X \in V$.*

Proof. If $Q(X) = f(X, X)$ for a symmetric f then

$$\begin{aligned} Q(X+Y) &= f(X+Y, X+Y) = f(X,X) + f(X,Y) + f(Y,X) + f(Y,Y) \\ &= Q(X) + 2f(X,Y) + Q(Y), \end{aligned}$$

and hence

$$f(X,Y) = \frac{1}{2}\big(Q(X+Y) - Q(X) - Q(Y)\big). \tag{6.3.2}$$

I leave it as an exercise to check that the formula (6.3.2) always defines a symmetric bilinear function. ∎

Let $S = \{v_1, \dots, v_n\}$ is a basis of V. The matrix $A = (a_{ij})$ of a symmetric bilinear form $f(X,Y)$ in the basis S is called also the matrix of the corresponding quadratic form $Q(X) = f(X,X)$. This matrix is symmetric, and

$$Q(X) = \sum_{ij} a_{ij}x_ix_j = a_{11}x_1^2 + \dots + a_{nn}x_n^2 + 2\sum_{i<j} a_{ij}x_ix_j.$$

Thus the matrix A is diagonal if and only if the quadratic form Q is the sum of squares (with coefficients). Let us recall that if one changes the basis S to a basis $\widetilde{S} = \{\widetilde{v}_1, \dots, \widetilde{v}_n\}$ then the matrix of a bilinear form f changes to $\widetilde{C} = C^TAC$.

Exercise 6.4. **(Sylvester's inertia law)** Prove that there is always a basis $\widetilde{S} = \{\widetilde{v}_1, \dots, \widetilde{v}_n\}$ in which a quadratic form Q is reduced to a sum of squares. The number of positive, negative and zero coefficients with the squares is independent of the choice of the basis.

Thus in some coordinate system a quadratic form can always be written as

$$-\sum_1^k x_i^2 + \sum_{k+1}^{k+l} x_j^2,\ k+l \le n.$$

The number k of negative squares is called the *negative index* or simply the *index* of the quadratic form Q, the total number $k+l$ of non-zero squares is called the *rank* of the form. It coincides with the rank of the matrix of the form in *any* basis. A bilinear (and quadratic) form is called *non-degenerate* if its rank is maximal possible, i.e. equal to n. For a non-degenerate quadratic for Q the difference $l-k$ between the number of positive and negative squares is called the *signature*.

A quadratic form Q is called *positive definite* if $Q(X) \ge 0$ and if $Q(X) = 0$ then $X = 0$. Equivalently, one can say that a form is positive definite if it is non-degenerate and its negative index is equal to 0.

Chapter 7

Tensor and exterior products

7.1 Tensor product

Given a k-linear function ϕ and a l-linear function ψ, one can form a $(k+l)$-linear function, which will be denoted by $\phi \otimes \psi$ and called *the tensor product* of the functions ϕ and ψ. By definition

$$\phi \otimes \psi(X_1, \dots, X_k, X_{k+1}, \dots, X_{k+l}) := \phi(X_1, \dots, X_k) \cdot \psi(X_{k+1}, \dots, X_{k+l}). \tag{7.1.1}$$

For instance, the tensor product of two linear functions l_1 and l_2 is a bilinear function $l_1 \otimes l_2$ defined by the formula

$$l_1 \otimes l_2(U, V) = l_1(U) l_2(V). \tag{7.1.2}$$

Let $v_1 \dots v_n$ be a basis in V and $x_1, \dots, x_n$ a dual basis in V^*, i.e. $x_1, \dots, x_n$ are coordinates of a vector with respect to the basis $v_1, \dots, v_n$.

The tensor product $x_i \otimes x_j$ is a bilinear function $x_i \otimes x_j(Y, Z) = y_i z_j$. Thus a bilinear function f with a matrix A can be written as a linear combination of the functions $x_i \otimes x_j$ as follows:

$$f = \sum_{i,j=1}^{n} a_{ij}\, x_i \otimes x_j,$$

where a_{ij} is the matrix of the form f in the basis $v_1 \dots v_n$. Similarly any k-linear function f with a "k-dimensional" matrix $A = \{a_{i_1 i_2 \dots i_k}\}$ can be written (see 7.1 below) as a linear combination of functions

$$x_{i_1} \otimes x_{i_2} \otimes \dots \otimes x_{i_k},\ 1 \le i_1, i_2, \dots, i_k \le n\,.$$

Namely, we have

$$f = \sum_{i_1, i_2, \dots i_k = 1}^{n} a_{i_1 i_2 \dots i_k} x_{i_1} \otimes x_{i_2} \otimes \cdots \otimes x_{i_k} \,.$$

7.2 Spaces of multilinear functions

All k-linear functions, or *k-tensors*, on a given n-dimensional vector space V themselves form a vector space, which will be denoted by $V^{*\otimes k}$. The space $V^{*\otimes 1}$ is, of course, just the dual space V^*.

Proposition 7.1. *Let $v_1, \dots v_n$ be a basis of V, and $x_1, \dots, x_k$ be the dual basis of V^* formed by coordinate functions with respect to the basis V. Then n^k k-linear functions $x_{i_1} \otimes \cdots \otimes x_{i_k}$, $1 \le i_1, \dots, i_k \le n$, form a basis of the space $V^{*\otimes k}$.*

Proof. Take a k-linear function F from $V^{*\otimes k}$ and evaluate it on vectors $v_{i_1}, \dots, v_{i_k}$:

$$F(v_{i_1}, \dots, v_{i_k}) = a_{i_1 \dots i_k} \,.$$

We claim that we have

$$F = \sum_{1 \le i_1, \dots, i_k \le n} a_{i_1 \dots i_k} x_{i_1} \otimes \cdots \otimes x_{i_k} \,.$$

Indeed, the functions on the both sides of this equality being evaluated on any set of k basic vectors $v_{i_1}, \dots, v_{i_k}$, give the same value $a_{i_1 \dots i_k}$. The same argument shows that if $\sum\limits_{1 \le i_1, \dots, i_k \le n} a_{i_1 \dots i_k} x_{i_1} \otimes \cdots \otimes x_{i_k} = 0$, then all coefficients $a_{i_1 \dots i_k}$ should be equal to 0. Hence the functions $x_{i_1} \otimes \cdots \otimes x_{i_k}$, $1 \le i_1, \dots, i_k \le n$, are linearly independent, and therefore form a basis of the space $V^{*\otimes k}$ ■

7.3 Symmetric and skew-symmetric tensors

A multilinear function (tensor) is called *symmetric* if it remains unchanged under the transposition of any two of its arguments:

$$f(X_1, \dots, X_i, \dots, X_j, \dots, X_k) = f(X_1, \dots, X_j, \dots, X_i, \dots, X_k)$$

Equivalently, one can say that a k-tensor f is symmetric if

$$f(X_{i_1}, \dots, X_{i_k}) = f(X_1, \dots, X_k)$$

for any permutation $i_1, \dots, i_k$ of indices $1, \dots, k$.

Exercise 7.2. *Show that a bilinear function $f(X, Y)$ is symmetric if and only if its matrix (in any basis) is symmetric.*

Notice that the tensor product of two symmetric tensors usually *is not* symmetric.

Example 7.3. *Any linear function is (trivially) symmetric. However, the tensor product of two functions $l_1 \otimes l_2$ is not a symmetric bilinear function unless l_1 is proportional to l_2. On the other hand, the function $l_1 \otimes l_2 + l_2 \otimes l_1$ is symmetric.*

A tensor is called *skew-symmetric (or anti-symmetric)* if it changes its sign when one transposes any two of its arguments:

$$f(X_1, \ldots, X_i, \ldots, X_j, \ldots, X_k) = -f(X_1, \ldots, X_j, \ldots, X_i, \ldots, X_k).$$

Equivalently, one can say that a k-tensor f is anti-symmetric if

$$f(X_{i_1}, \ldots, X_{i_k}) = (-1)^{\mathrm{inv}(i_1 \ldots i_k)} f(X_1, \ldots, X_k)$$

for any permutation $i_1, \ldots, i_k$ of indices $1, \ldots, k$, where $\mathrm{inv}(i_1 \ldots i_k)$ is the number of inversions in the permutation $i_1, \ldots, i_k$. Recall that two indices i_k, i_l form an *inversion* if $k < l$ but $i_k > i_l$.

The matrix A of a bilinear skew-symmetric function is skew-symmetric, i.e.

$$A^T = -A.$$

Example 7.4. *The determinant* $\det(X_1, \ldots, X_n)$ *(considered as a function of columns $X_1, \ldots, X_n$ of a matrix) is a skew-symmetric n-linear function.*

Exercise 7.5. *Prove that any n-linear skew-symmetric function on $\mathbb{R}^n$ is proportional to the determinant.*

Linear functions are trivially anti-symmetric (as well as symmetric).

As in the symmetric case, the tensor product of two skew-symmetric functions is not skew-symmetric. We will define below in Section 7.5 a new product, called an *exterior product* of skew-symmetric functions, which will again be a skew-symmetric function.

7.4 Symmetrization and anti-symmetrization

The following constructions allow us to create symmetric or anti-symmetric tensors from arbitrary tensors. Let $f(X_1, \dots, X_k)$ be a k-tensor. Set

$$f^{\text{sym}}(X_1, \dots, X_k) := \sum_{(i_1 \dots i_k)} f(X_{i_1}, \dots, X_{i_k})$$

and

$$f^{\text{asym}}(X_1, \dots, X_k) := \sum_{(i_1 \dots i_k)} (-1)^{\text{inv}(i_1,\dots,i_k)} f(X_{i_1}, \dots, X_{i_k})$$

where the sums are taken over all permutations $i_1, \dots, i_k$ of indices $1, \dots, k$. The tensors f^{sym} and f^{asym} are called, respectively, *symmetrization* and *anti-symmetrization* of the tensor f. It is now easy to see that

Proposition 7.6. *The function f^{sym} is symmetric. The function f^{asym} is skew-symmetric. If f is symmetric then $f^{\text{sym}} = k!f$ and $f^{\text{asym}} = 0$. Similarly, if f is anti-symmetric then $f^{\text{asym}} = k!f$, $f^{\text{sym}} = 0$.*

Exercise 7.7. *Let $x_1, \dots, x_n$ be coordinate function in $\mathbb{R}^n$. Find $(x_1 \otimes x_2 \otimes \dots \otimes x_n)^{asym}$.*

Answer. The determinant.

7.5 Exterior product

For our purposes skew-symmetric functions will be more important. Thus we will concentrate on studying operations on them.

Skew-symmetric k-linear functions are also called *exterior k-forms.* Let ϕ be an exterior k-form and ψ an exterior l-form. We define an exterior $(k+l)$-form $\psi \wedge \psi$, *the exterior product of ϕ and ψ*, as

$$\phi \wedge \psi := \frac{1}{k!l!} (\phi \otimes \psi)^{\text{asym}}.$$

In other words,

$$\phi\wedge\psi(X_1,\dots,X_k,X_{k+1},\dots,X_{k+l})=\frac{1}{k!l!}\sum_{i_1,\dots i_{k+l}}(-1)^{\mathrm{inv}(i_1,\dots,i_{k+l})}\phi(X_{i_1}\dots,X_{i_k})\psi(X_{i_{k+1}},\dots,X_{i_{k+l}}),$$

where the sum is taken over all permutations of indices $1,\dots,k+l$.

Note that because the anti-symmetrization of an anti-symmetric k-tensor amounts to its multiplication by $k!$, we can also write

$$\phi\wedge\psi(X_1,\dots,X_{k+l})=\sum_{i_1<\dots<i_k,i_{k+1}<\dots<i_{k+l}}(-1)^{\mathrm{inv}(i_1,\dots,i_{k+l})}\phi(X_{i_1},\dots,X_{i_k})\psi(X_{i_{k+1}},\dots,X_{i_{k+l}}),$$

where the sum is taken over all permutations $i_1,\dots,i_{k+l}$ of indices $1,\dots,k+l$.

Exercise 7.8. *The exterior product operation has the following properties:*

- *For any exterior k-form ϕ and exterior l-form ψ we have $\phi\wedge\psi=(-1)^{kl}\psi\wedge\phi$.*
- *Exterior product is linear with respect to each factor:*

$$\begin{aligned}(\phi_1+\phi_2)\wedge\psi &= \phi_1\wedge\psi+\phi_2\wedge\psi\\ (\lambda\phi)\wedge\psi &= \lambda(\phi\wedge\psi)\end{aligned}$$

  for k-forms ϕ,ϕ_1,ϕ_2, l-form ψ and a real number λ.

- *Exterior product is associative: $(\phi\wedge\psi)\wedge\omega=\phi\wedge(\psi\wedge\omega)$.*

First two properties are fairly obvious. To prove associativity one can check that both sides of the equality $(\phi\wedge\psi)\wedge\omega=\phi\wedge(\psi\wedge\omega)$ are equal to

$$\frac{1}{k!l!m!}(\phi\otimes\psi\otimes\omega)^{\mathrm{asym}}.$$

In particular, if ϕ,ψ and ω are 1-forms, i.e. if $k=l=m=1$ then

$$\psi\wedge\phi\wedge\omega=(\phi\otimes\psi\otimes\omega)^{\mathrm{asym}}.$$

This formula can be generalized for computing the exterior product of any number of 1-forms:

$$\phi_1\wedge\dots\wedge\phi_k=(\phi_1\otimes\dots\otimes\phi_k)^{\mathrm{asym}}. \tag{7.5.1}$$

Example 7.9. $x_1 \wedge x_2 = x_1 \otimes x_2 - x_2 \otimes x_1$. For 2 vectors, $U = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}, V = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$, we have

$$x_1 \wedge x_2(U,V) = u_1 v_2 - u_2 v_1 = \begin{vmatrix} u_1 & v_1 \\ u_2 & v_2 \end{vmatrix}.$$

For 3 vectors U, V, W we have

$$x_1 \wedge x_2 \wedge x_3(U,V,W) =$$

$$= x_1 \wedge x_2(U,V)x_3(W) + x_1 \wedge x_2(V,W)x_3(U) + x_1 \wedge x_2(W,U)x_3(V) =$$

$$(u_1v_2 - u_2v_1)w_3 + (v_1w_2 - v_2w_1)u_3 + (w_1u_2 - w_2u_1)v_3 = \begin{vmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{vmatrix}.$$

The last equality is just the expansion formula of the determinant according to the last row.

Proposition 7.10. *Any exterior* 2*-form f can be written as*

$$f = \sum_{1 \le i < j \le n} a_{ij}\, x_i \wedge x_j$$

Proof. We had seen above that any bilinear form can be written as $f = \sum\limits_{ij} a_{i,j}\, x_i \otimes x_j$. If f is skew-symmetric then the matrix $A = (a_{ij})$ is skew-symmetric, i.e. $a_{ii} = 0$, $a_{ij} = -a_{ji}$ for $i \neq j$.

Thus, $f = \sum\limits_{1 \le i<j \le n} a_{ij}(x_i \otimes x_j - x_j \otimes x_i) = \sum\limits_{1 \le i<j \le n} a_{ij}\, x_i \wedge x_j$. ■

Exercise 7.11. *Prove that any exterior k-form f can be written as*

$$f = \sum_{1 \le i, < \ldots < i_k \le n} a_{i_1 \ldots i_k}\, x_{i_1} \wedge x_{i_2} \wedge \ldots x_{i_k}.$$

The following proposition can be proven by induction over k, similar to what has been done in Example 7.9 for the case $k = 3$.

Proposition 7.12. *For any k 1-forms $l_1, \dots, l_k$ and k vectors $X_1, \dots, X_k$ we have*

$$l_1 \wedge \dots \wedge l_k(X_1, \dots, X_k) = \begin{vmatrix} l_1(X_1) & \dots & l_1(X_k) \\ \dots & \dots & \dots \\ l_k(X_1) & \dots & l_k(X_k) \end{vmatrix} . \tag{7.5.2}$$

Corollary 7.13. *The 1-forms $l_1, \dots, l_k$ are linearly dependent as vectors of V^* if and only if $l_1 \wedge \ldots \wedge l_k = 0$. In particular, $l_1 \wedge \ldots \wedge l_k = 0$ if $k > n = \dim V$.*

Proof. If $l_1, \dots, l_k$ are dependent then for any vectors $X_1, \dots, X_k \in V$ the rows of the determinant in the equation (7.12) are linearly dependent. Therefore, this determinant is equal to 0, and hence $l_1 \wedge \ldots \wedge l_k = 0$. In particular, when $k > n$ then the forms $l_1, \dots, l_k$ are dependent (because $\dim V^* = \dim \mathrm{V} = n$).

On the other hand, if $l_1, \dots, l_k$ are linearly independent, then the vectors $l_1, \dots, l_k \in V^*$ can be completed to form a basis $l_1, \dots, l_k, l_{k+1}, \dots, l_n$ of V^*. According to Exercise 1.2 there exists a basis $w_1, \dots, w_n$ of V that is dual to the basis $l_1, \dots, l_n$ of V^*. In other words, $l_1, \dots, l_n$ can be viewed as coordinate functions with respect to the basis $w_1, \dots, w_n$. In particular, we have $l_i(w_j) = 0$ if $i \neq j$ and $l_i(w_i) = 1$ for all $i, j = 1, \dots, n$. Hence we have

$$l_1 \wedge \dots \wedge l_k(w_1, \dots, w_k) = \begin{vmatrix} l_1(w_1) & \dots & l_1(w_k) \\ \dots & \dots & \dots \\ l_k(w_1) & \dots & l_k(w_k) \end{vmatrix} = \begin{vmatrix} 1 & \dots & 0 \\ \dots & 1 & \dots \\ 0 & \dots & 1 \end{vmatrix} = 1 ,$$

i.e. $l_1 \wedge \dots \wedge l_k \neq 0$. ■

Proposition 7.12 can be also deduced from formula (7.5.1).

Corollary 7.13 and Exercise 7.11 imply that there are no non-zero k-forms on an n-dimensional space for $k > n$.

7.6 Spaces of symmetric and skew-symmetric tensors

As was mentioned above, k-tensors on a vector space V form a vector space under the operation of addition of functions and multiplication by real numbers. We denoted this space by $V^{*\otimes k}$. Symmetric and skew-symmetric tensors form subspaces of this space $V^{*\otimes k}$, which we denote, respectively, by $S^k(V^*)$ and $\Lambda^k(V^*)$. In particular, we have

$$V^* = S^1(V^*) = \Lambda^1(V^*)$$

.

Exercise 7.14. *What is the dimension of the spaces $S^k(V^*)$ and $\Lambda^k(V^*)$?*

Answer.

$$\dim \Lambda^k(V^*) = \left(\begin{array}{c} n \\ k \end{array} \right) = \frac{n!}{k!(n-k)!}$$

$$\dim S^k(V^*) = \frac{(n+k-1)!}{k!(n-1)!}.$$

The basis of $\Lambda^k(V^*)$ is formed by exterior k-forms $x_{i_1} \wedge x_{i_2} \wedge \ldots \wedge x_{i_k}, 1 \le i_1 < i_2 < \ldots < i_k \le n$.

The basis of $S^k(V^*)$ is formed by symmetric forms

$$\left(x_1^{\otimes i_1} \otimes x_2^{\otimes i_2} \ldots x_n^{\otimes i_n} \right)^{\mathrm{sym}}, \quad i_j \ge 0, \ \sum_{j=1}^{n} i_j = k.$$

7.7 Operator $\mathcal{A}^*$ on spaces of tensors

For any linear operator $\mathcal{A} : V \to W$ we introduced above in Section 1.3 the notion of a dual linear operator $\mathcal{A}^* : W^* \to V^*$. Namely $\mathcal{A}^*(l) = l \circ \mathcal{A}$ for any element $l \in V^*$, which is just a linear function on V. In this section we extend this construction to k-tensors for $k \ge 1$, i.e. we will define a map $\mathcal{A}^* : W^{*\otimes k} \to V^{*\otimes k}$.

Given a k-tensor $\phi \in W^{*\otimes k}$ and k vectors $X_1, \ldots, X_k \in V$ we define

$$\mathcal{A}^*(\phi)(X_1, \ldots, X_k) = \phi\left(\mathcal{A}(X_1), \ldots, \mathcal{A}(X_k)\right).$$

Note that if ϕ is symmetric, or anti-symmetric, so is $\mathcal{A}^*(\phi)$. Hence, the map $\mathcal{A}^*$ also induces the maps $S^k(W^*) \to S^k(V^*)$ and $\Lambda^k(W^*) \to \Lambda^k(V^*)$. We will keep the same notation $\mathcal{A}^*$ for both of these maps as well.

Exercise 7.15. *Suppose V is provided with a basis $v_1,\dots,v_n$ and $x_{i_1}\otimes\dots\otimes x_{i_k}$, $1\le i_1,\dots,i_k\le n$, is the corresponding basis of the space $V^{*\otimes k}$. Suppose that the map $\mathcal{A}:V\to V$ has a matrix $A=(a_{ij})$ in the basis $v_1,\dots,v_n$. Find the matrix of the map $\mathcal{A}^*:V^{*\otimes k}\to V^{*\otimes k}$ in the basis $x_{i_1}\otimes\dots\otimes x_{i_k}$.*

Proposition 7.16. *Let $\mathcal{A}:V\to W$ be a linear map. Then*

1. *$\mathcal{A}^*(\phi\otimes\psi)=\mathcal{A}^*(\phi)\otimes\mathcal{A}^*(\psi)$ for any $\phi\in W^{*\otimes k},\psi\in W^{*\otimes l}$;*

2. *$\mathcal{A}^*(\phi^{\text{asym}})=(\mathcal{A}^*(\phi))^{\text{asym}}$, $\mathcal{A}^*(\phi^{\text{sym}})=(\mathcal{A}^*(\phi))^{\text{sym}}$;*

3. *$\mathcal{A}^*(\phi\wedge\psi)=\mathcal{A}^*(\phi)\wedge\mathcal{A}^*(\psi)$ for any $\phi\in\Lambda^k(W^*),\psi\in\Lambda^l(W^*)$.*

If $\mathcal{B}:W\to U$ is another linear map then $(\mathcal{B}\circ\mathcal{A})^*=\mathcal{A}^*\circ\mathcal{B}^*$.

Proof.

1. Take any $k+l$ vectors $X_1,\dots,X_{k+l}\in V$. Then by definition of the operator $\mathcal{A}^*$ we have

$$\begin{aligned}\mathcal{A}^*(\phi\otimes\psi)(X_1,\dots,X_{k+l})=&\phi\otimes\psi(\mathcal{A}(X_1),\dots,\mathcal{A}(X_{n+k})=\\ &\phi(\mathcal{A}(X_1),\dots,\mathcal{A}(X_k)\psi(\mathcal{A}(X_{k+1}),\dots,\mathcal{A}(X_{n+k}))=\\ &\mathcal{A}^*(\phi)(X_1,\dots,X_k)\mathcal{A}^*(\psi)(X_{k+1},\dots,X_{k+n})=\\ &\mathcal{A}^*(\phi)\otimes\mathcal{A}^*(\psi)(X_1,\dots,X_{k+l}).\end{aligned}$$

2. Given k vectors $X_1,\dots,X_k\in V$ we get

$$\mathcal{A}^*(\phi^{asym})(X_1,\dots,X_k)=\phi^{asym}(\mathcal{A}(X_1),\dots,\mathcal{A}(X_k))=\sum_{(i_1\dots i_k)}(-1)^{\text{inv}(i_1,\dots,i_k)}\phi(\mathcal{A}(X_{i_1}),\dots,\mathcal{A}(X_{i_k}))=$$
$$\sum_{(i_1\dots i_k)}(-1)^{\text{inv}(i_1,\dots,i_k)}\mathcal{A}^*(\phi)(X_{i_1},\dots,X_{i_k})=(\mathcal{A}^*(\phi))^{asym}(X_1,\dots,X_k),$$

where the sum is taken over all permutations $i_1,\dots,i_k$ of indices $1,\dots,k$. Similarly one proves that $\mathcal{A}^*(\phi^{\text{sym}})=(\mathcal{A}^*(\phi))^{\text{sym}}$.

3. $\mathcal{A}^*(\phi\wedge\psi)=\frac{1}{k!l!}\mathcal{A}^*\left((\phi\otimes\psi)^{asym})\right)=\frac{1}{k!l!}\left(\mathcal{A}^*(\phi\otimes\psi)\right)^{asym}=\mathcal{A}^*(\phi)\wedge\mathcal{A}^*(\psi)$.

The last statement of Proposition 7.16 is straightforward and its proof is left to the reader.

Let us now discuss how to compute $\mathcal{A}^*(\phi)$ in coordinates. Let us fix bases $v_1, \dots, v_m$ and $w_1, \dots, w_n$ in spaces V and W. Let $x_1, \dots, x_m$ and $y_1, \dots, y_n$ be coordinates and

$$A = \begin{pmatrix} a_{11} & \dots & a_{1m} \\ \vdots & \vdots & \vdots \\ a_{n1} & \dots & a_{nm} \end{pmatrix}$$

be the matrix of a linear map $\mathcal{A} : V \to W$ in these bases. Note that the map $\mathcal{A}$ in these coordinates is given by n linear coordinate functions:

$$\begin{aligned} l_1(x_1, \dots, x_m) &= a_{11}x_1 + a_{12}x_2 + \dots + a_{1m}x_m \\ l_2(x_1, \dots, x_m) &= a_{21}x_1 + a_{22}x_2 + \dots + a_{2m}x_m \\ &\dots \\ l_n(x_1, \dots, x_k) &= a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nm}x_n \end{aligned}$$

We have already computed in Section 1.3 that $\mathcal{A}^*(y_k) = l_k = \sum\limits_{j=1}^{m} a_{kj}x_j$, $k = 1, \dots, n$. Indeed, the coefficients of the function $l_k = \mathcal{A}^*(y_k)$ form the k-th column of the transpose matrix A^T. Hence, using Proposition 7.16 we compute:

$$\mathcal{A}^*(y_{j_1} \otimes \dots \otimes y_{j_k}) = l_{j_1} \otimes \dots \otimes l_{j_k}$$

and

$$\mathcal{A}^*(y_{j_1} \wedge \dots \wedge y_{j_k}) = l_{j_1} \wedge \dots \wedge l_{j_k}.$$

Consider now the case when $V = W$, $n = m$, and we use the same basis $v_1, \dots, v_n$ in the source and target spaces.

Proposition 7.17.

$$\mathcal{A}^*(x_1 \wedge \dots \wedge x_n) = (\det A)\, x_1 \wedge \dots \wedge x_n.$$

Note that the determinant $\det A$ is independent of the choice of the basis. Indeed, the matrix of a linear map changes to a similar matrix $C^{-1}AC$ in a different basis, and $\det C^{-1}AC = \det A$. Hence, we can write $\det \mathcal{A}$ instead of $\det A$, i.e. attribute the determinant to the linear operator $\mathcal{A}$ rather than to its matrix A.

Proof. We have

$$\mathcal{A}^*(x_1 \wedge \dots \wedge x_n) = l_1 \wedge \dots \wedge l_n = \sum_{i_1=1}^{n} a_{1i_1} x_{i_1} \wedge \dots \wedge \sum_{i_n=1}^{n} a_{ni_n} x_{i_n} =$$
$$\sum_{i_1,\dots,i_n=1}^{n} a_{1i_1} \dots a_{ni_n} x_{i_1} \wedge \dots \wedge x_{i_n}.$$

Note that the in the latter sum all terms with repeating indices vanish, and hence we can replace this sum by a sum over all permutations of indices $1, \dots, n$. Thus, we can continue

$$\mathcal{A}^*(x_1 \wedge \dots \wedge x_n) = \sum_{i_1,\dots,i_n} a_{1i_1} \dots a_{ni_n} x_{i_1} \wedge \dots \wedge x_{i_n} =$$
$$\left(\sum_{i_1,\dots,i_n} (-1)^{\mathrm{inv}(i_1,\dots,i_n)} a_{1i_1} \dots a_{ni_n} \right) x_1 \wedge \dots \wedge x_n = \det A \, x_1 \wedge \dots \wedge x_n.$$

Exercise 7.18. *Apply the equality*

$$\mathcal{A}^*(x_1 \wedge \dots \wedge x_k \wedge x_{k+1} \wedge \dots \wedge x_n) = \mathcal{A}^*(x_1 \wedge \dots \wedge x_k) \wedge \mathcal{A}^*(x_{k+1} \wedge \dots \wedge x_n)$$

for a map $\mathcal{A} : \mathbb{R}^n \to \mathbb{R}^n$ to deduce the formula for expansion of a determinant according to its first k rows:

$$\det A = \sum_{i_1 < \dots < i_k,\; j_1 < \dots < j_{n-k};\; i_m \neq j_l} (-1)^{\mathrm{inv}(i_1,\dots,j_{n-k})} \begin{vmatrix} a_{1,i_1} & \dots & a_{1,i_k} \\ \vdots & \vdots & \vdots \\ a_{k,i_1} & \dots & a_{k,i_k} \end{vmatrix} \begin{vmatrix} a_{k+1,j_1} & \dots & a_{k+1,j_{n-k}} \\ \vdots & \vdots & \vdots \\ a_{n,j_1} & \dots & a_{n,j_{n-k}} \end{vmatrix}.$$

Chapter 8

More about tensors

The notation $V^{*\otimes k}$ for the space of k-tensors hints that there should be an operation of *tensor product of spaces*, so that this space could be understood as the k-th tensor power of the space V^*. Such an operation indeed exists, and it leads to a more general definition of tensors, which we discuss below in this section. We begin with a very important notion of a quotient space.

8.1 Quotient space

Let V be a vector space and $L \subset V$ be its linear subspace. Given a vector $a \in V$ let us denote by L_a the affine subspace $a + L = \{a + x;\ x \in L\}$. Note that two affine subspaces L_a and L_b, $a, b \in V$ coincide if $a - b \in L$ and are disjoint if $a - b \notin L$. Consider the set, denoted by V/L of all affine subspaces parallel to L. The set can be made into a vector space by defining the operations by the formulas

$$L_a + L_b := L_{a+b},\ \ \lambda L_a := L_{\lambda a};\ a, b \in V,\ \lambda \in \mathbb{R}$$

If a' and b' are other vectors in V such that $a' - a, b' - b \in L$ then $(a' + b') - (a + b) \in L$ and $\lambda a' - \lambda a \in L$, and hence $L_{a'+b'} = L_{a+b},\ L_{\lambda a'} = Ł_{\lambda a}$.

The vector space V/L is called the *quotient space* of V by L.

In other words, we can say that V/L is obtained from V by identifying vectors which differ by a vector in L. The operations in V then naturally descend to the operations on the quotient space.

If $N \subset V$ is any linear subspace such that $\dim L + \dim N = \dim V$ and $L \cap N = \{0\}$ then the map $N \to V/L$ given by the formula $x \in N \mapsto L_x \in V/L$ is an isomorphism (why?). In particular, $\dim(V/L) = \dim V - \dim L = \operatorname{codim}_V L$.

If V is a Euclidean space then we can choose as N the orthogonal complement $L^\perp$ of L in V, and thus L/V is isomorphic to $L^\perp$. The advantage of the quotient construction is that it is *canonical*, i.e. depends only on V and L, while $L^\perp$ depends on a choice of the Euclidean structure (scalar product) on V.

Note that there exists a canonical projection $\pi_L : V \to V/L$ defined by the formula $x \mapsto L_x$, $x \in V$. Clearly, π_L is a linear surjective map.

Exercise 8.1. 1. *Prove that* $\operatorname{Ker} \pi_L = L$.

2. *Show that given a surjective linear map* $F : V \to W$*, there exists an isomorphism*

$$\overline{F} : V/\operatorname{Ker} F \to W, \ \ \textit{such that} \ \ F = \overline{F} \circ \pi_{\operatorname{Ker} F}.$$

8.2 Direct sum of vector spaces

Recall that the Cartesian product $X \times Y$ of two sets X and Y is just the set of pairs (x, y), $x \in X, y \in Y$. When $X = V, Y = W$ are vector spaces, then their direct product $V \times W$ has itself a vector space structure, where we define

$$(v_1, w_1) + (v_2, w_2) = (v_1 + v_2, w_1 + w_2) \ \text{ for } v_1, v_2 \in V, w_1, w_2 \in W, \text{ and}$$

$$\lambda(v, w) = (\lambda v, \lambda w) \ \text{ for } v \in V, w \in W, \lambda \in \mathbb{R}.$$

The resulted vector space is usually cold the *direct sum* of vector spaces v and W, and is usually denoted by $V \oplus W$ instead of $V \times W$ (though the notation $V \times W$ is also acceptable). Note that V, W can be viewed as subspaces $V \oplus 0 := \{(v, 0);\ v \in V\}$ and $0 \oplus W := \{(0, w);\ w \in W\}$ of $V \oplus W$

Suppose now that V and W are linear subspaces of a vector space H. Then there is a unique homomorphism $j : V \oplus W \to H$, which coincides with the inclusion on V and W:

$$j(v, w) := v + w.$$

We have

$$\operatorname{Im} j = \operatorname{Span}(V, W), \ \ \text{and } \operatorname{Ker} j = \{(v, v);\ v \in V \cap W\}.$$

In particular, j is an isomorphism if (and only if) $V \cap W = \{0\}$ and $\mathrm{Span}(V, W) = H$. In that case one says that H is a direct sum of its subspaces V and W, and (slightly abusing the notation) writes $H = V \oplus W$.

8.3 Tensor product of vector spaces

Given two vector spaces V and W. let us denote by VW the vector space generated by all vectors of $V \oplus W$, i.e. all pairs (v, w), $v \in V, w \in W$. In other words a vector in VW is a (formal) linear combination $\sum\limits_1^k \lambda_j(v_j, w_j)$, $\lambda_j \in \mathbb{R}$, $v_j \in V, w_j \in W$. The space VW is a huge infinite-dimensional space. Consider its subspace L, the span of vectors

$$\lambda(v,w) - (\lambda v, w), \lambda(v,w) - (v, \lambda w), (v_1 + v_2, w) - (v_1, w) - (v_2, w), (v, w_1 + w_2) - (v, w_1) - (v, w_2),$$

where v, v_1, v_2 run through all vectors in V, w, w_1, w_2 run through all vectors of W, and λ through all real numbers.

By definition, the quotient space VW/L is called the *tensor product* of spaces V and W and is denoted by $V \otimes W$. For any two vectors $v \in V$ and $w \in W$ we denote by

$$v \otimes w := \pi_L(v, w) \in V \otimes W. \tag{8.3.1}$$

We will call $v \otimes w$ the tensor product of vectors v and w. Note that by definition $v \otimes w$ is a bilinear operation, i.e.

$$\lambda(v \otimes w) = (\lambda v) \otimes w = v \otimes \lambda w,$$
$$(v_1 + v_2) \otimes w = v_1 \otimes w + v_2, \otimes w,\ v \otimes (w_1 + w_2) = v \otimes w_1 + v \otimes w_2),$$

for any $v, v_1, v_2 \in V$, $w, w_1, w_2 \in W$, $\lambda \in \mathbb{R}$.

Let $v_1, \dots, v_n$ be a basis of V, $w_1, \dots, w_m$ be a basis of W. Consider vectors

$$v = \sum_1^n x_i v_i,\ w = \sum_1^m y_j w_j.$$

Then

$$v \otimes w = \sum_{i=1}^n \sum_{j=1^m} x_i y_j v_i \otimes w_j.$$

Hence, vectors $v_i \otimes w_j \in V \otimes W, i = 1, \dots, n, j = 1, \dots, m$ generate $V \otimes W$.

Exercise 8.2. *Prove that the vectors $v_i \otimes w_j \in V \otimes W, i = 1, \ldots, n, j = 1, \ldots, m$ are linearly independent, and hence form a basis of $V \otimes W$.*

Hence, $V \otimes W$ is a finite-dimensional vector space and $\dim(V \otimes W) = mn$.

It is straightforward to check that the operation $V \otimes W$ is associative, i.e.$V_1 \otimes (V_2 \otimes V_3)$ is canonically isomorphic to $(V_1 \otimes V_2) \otimes V_3$. Hence, we can drop the parenthesis and write $V_1 \otimes \cdots \otimes V_k$ for the k-multiple tensor product, and $V^{\otimes k}$ for the k-tensor degree.

8.4 Tensor square and bilinear functions

For two linear functions $l_1, l_2 \in V^*$ we defined two tensor products, which we both denoted by $l_1 \otimes l_2$, one by formula (7.1.2) and the other one by formula (8.3.1). As we show in the next proposition, the two operations coincide. But to avoid a confusion let us temporarily write $l_1 \otimes_1 l_2$ for the first operation defined by (7.1.2) and $l_1 \otimes_2 l_2$ for the second one defined by (8.3.1).

Proposition 8.3. *There exists a canonical isomorphism space $V^* \otimes V^*$ is canonically isomorphic to the space $V^{*\otimes 2}$ of bilinear functions. Under this isomorphism the two tensor product operations $l_1 \otimes_1 l_2$ and $l_1 \otimes_2 l_2$ coincide.*

Proof. The required isomorphism $T : V^* \otimes T^* \to V^{*\otimes 2}$ is defined by the formula

$$T(l_1 \otimes_2 l_2)(X, Y) = l_1(X) l_2(Y) = l_1 \otimes_1 l_2)(X, Y), \; X, Y \in V, l_1, l_2 \in V^*.$$

We leave it to the reader as an exercise to check that the map T is a well defined isomorphism. ■

Hence, from now on we can drop the subscripts $\otimes_1$ and $\otimes_2$ from the notation of the tensor product and identify $V^{*\otimes k}$ with the k-th tensor power $(V^*)^{\otimes k}$ of the dual space V^*.

The elements of the space $V^{\otimes k}$ are also called *tensors*, and when one need to consider both tensors from $V^{\otimes k}$ and from $(V^*)^{\otimes k} = V^{*\otimes k}$ the former are called *covariant* and the latter *contravariant* tensors. Their behavior differ under linear maps. If we are given a linear map $\mathcal{A} : V \to W$, it induces a map

$$\mathcal{A}^{\otimes k} : V^{\otimes k} \to W^{\otimes k}$$

by the formula

$$\mathcal{A}^{\otimes k}(X_1 \otimes \cdots \otimes X_k) = \mathcal{A}(X_1) \otimes \cdots \otimes \mathcal{A}(X_k).$$

On the other hand, $\mathcal{A}$ induces a dual map $\mathcal{A}^*: W^* \to V^*$, and hence a map

$$(\mathcal{A}^*)^{\otimes k}: (W^*)^{\otimes k} \to (V^*)^{\otimes k}$$

which is the same as the map $W^{*\otimes k} \to V^{*\otimes k}$ which we defined in Section 7.7 and denoted simply by $\mathcal{A}^*$.

One can also talk about tensors which are *k times covariant and l time contravariant* by taking tensor products of k copies of the space V and l copies of the space V^*.

The notions of symmetric and skew-symmetric tensors can be generalized in a similar way by associating with any vector space V its *symmetric* k-th power $S^k(V)$, $k = 0, \dots,$ and its a *skew-symmetric (= exterior)* k-th power $\Lambda^k(V), k = 0, \dots, n = \dim V$.

Chapter 9

Orientation and Volume

9.1 Orientation

We say that two bases $v_1, \dots, v_k$ and $w_1, \dots, w_k$ of a vector space V define *the same orientation* of V if the matrix of transition from one of these bases to the other has a positive determinant. Clearly, if we have 3 bases, and the first and the second define the same orientation, and the second and the third define the same orientation then the first and the third also define the same orientation. Thus, one can subdivide the set of all bases of V into two classes:

- all bases in each of these classes define the same orientation;

- two bases chosen from different classes define opposite orientation of the space.

To orient the space simply means to choose one of these two classes of bases.

There is no way to say which orientation is "positive", or which is "negative"—it is a question of convention. For instance, the so-called counter-clockwise orientation of the plane depends from which side we look at the plane. The positive orientation of our physical 3-space is a physical, not mathematical, notion.

Suppose we are given two oriented spaces V, W of the same dimension. An invertible linear map (= isomorphism) $\mathcal{A} : V \to W$ is called *orientation preserving* if it maps a basis which defines the given orientation of V to a basis which defines the given orientation of W.

Any non-zero exterior n-form η on V induces an orientation of the space V. Indeed, the preferred set of bases is characterized by the property $\eta(v_1, \dots, v_n) > 0$.

9.2 Orthogonal transformations

Let V be a Euclidean vector space. Recall that a linear operator $\mathcal{U} : V \to V$ is called *orthogonal* if it preserves the scalar product, i.e. if

$$\langle \mathcal{U}(X), \mathcal{U}(Y) \rangle = \langle X, Y \rangle, \tag{9.2.1}$$

for any vectors $X, Y \in V$. Recall that we have

$$\langle \mathcal{U}(X), \mathcal{U}(Y) \rangle = \langle X, \mathcal{U}^\star (\mathcal{U}(Y)) \rangle,$$

where $\mathcal{U}^\star : V \to V$ is the adjoint operator to $\mathcal{U}$, see Section 1.3 above.

Hence, the orthogonality of an operator $\mathcal{U}$ is equivalent to the identity $\mathcal{U}^\star \circ \mathcal{U} = \mathrm{Id}$, or $\mathcal{U}^\star = \mathcal{U}^{-1}$. Here we denoted by Id the identity operator, i.e. $\mathrm{Id}(X) = X$ for any $X \in V$.

Let us recall, see Exercise 1.12, that the adjoint operator $U^\star$ is related to the dual operator $U^* : V^* \to V^*$ by the formula

$$U^\star = \mathcal{D}^{-1} \circ U^* \circ \mathcal{D}.$$

Hence, for an orthogonal operator $\mathcal{U}$, we have $\mathcal{D}^{-1} \circ U^* \circ \mathcal{D} = \mathcal{U}^{-1}$, i.e.

$$\mathcal{U}^* \circ \mathcal{D} = \mathcal{D} \circ \mathcal{U}^{-1}. \tag{9.2.2}$$

Let $v_1, \dots, v_n$ be an orthonormal basis in V and U be the matrix of $\mathcal{U}$ in this basis. The matrix of the adjoint operator in an orthonormal basis is the transpose of the matrix of this operator. Hence, the equation $\mathcal{U}^* \circ \mathcal{U} = \mathrm{Id}$ translates into the equation $U^T U = E$, or equivalently $UU^T = E$, or $U^{-1} = U^T$ for its matrix. Matrices, which satisfy this equation are called *orthogonal*. If we write

$$U = \begin{pmatrix} u_{11} & \dots & u_{1n} \\ \dots & \dots & \dots \\ u_{n1} & \dots & u_{nn} \end{pmatrix},$$

then the equation $U^T U = E$ can be rewritten as

$$\sum_i u_{ki} u_{ji} = \begin{cases} 1, & \text{if } k = j, \\ 0, & \text{if } k \neq j, \end{cases}.$$

Similarly, the equation $UU^T = E$ can be rewritten as

$$\sum_i u_{ik} u_{ij} = \begin{cases} 1, & \text{if } k = j, \\ 0, & \text{if } k \neq j, \end{cases}.$$

The above identities mean that columns (and rows) of an orthogonal matrix U form an orthonormal basis of $\mathbb{R}^n$ with respect to the dot-product.

In particular, we have

$$1 = \det(U^T U) = \det(U^T) \det U = (\det U)^2,$$

and hence $\det U = \pm 1$. In other words, the determinant of any orthogonal matrix is equal ± 1. We can also say that the determinant of an orthogonal *operator* is equal to ± 1 because the determinant of the matrix of an operator is independent of the choice of a basis. Orthogonal transformations with $\det = 1$ preserve the orientation of the space, while those with $\det = -1$ reverse it.

Composition of two orthogonal transformations, or the inverse of an orthogonal transformation is again an orthogonal transformation. The set of all orthogonal transformations of an n-dimensional Euclidean space is denoted by $O(n)$. Orientation preserving orthogonal transformations sometimes called *special*, and the set of special orthogonal transformations is denoted by $SO(n)$. For instance $O(1)$ consists of two elements and $SO(1)$ of one: $O(1) = \{1, -1\}$, $SO(1) = \{1\}$. $SO(2)$ consists of rotations of the plane, while $O(2)$ consists of rotations and reflections with respect to lines.

9.3 Determinant and Volume

We begin by recalling some facts from Linear Algebra. Let V be an n-dimensional Euclidean space with an inner product $\langle\ ,\ \rangle$. Given a linear subspace $L \subset V$ and a point $x \in V$, the *projection* $\operatorname{proj}_L(x)$ is a vector $y \in L$ which is uniquely characterized by the property $x - y \perp L$, i.e. $\langle x - y, z \rangle = 0$ for any $z \in L$. The length $||x - \operatorname{proj}_L(x)||$ is called the *distance* from x to L; we denote it by $\operatorname{dist}(x, L)$.

Let $U_1, \ldots, U_k \in V$ be linearly independent vectors. The k-dimensional *parallelepiped spanned by vectors* $U_1, \ldots, U_k$ is, by definition, the set

$$P(U_1, \ldots, U_k) = \left\{ \sum_1^k \lambda_j U_j;\ 0 \le \lambda_1, \ldots, \lambda_k \le 1 \right\} \subset \mathrm{Span}(U_1, \ldots, U_k).$$

Given a k-dimensional parallelepiped $P = P(U_1, \ldots, U_k)$ we will define its k-dimensional *volume* by the formula

$$\mathrm{Vol} P = ||U_1|| \mathrm{dist}(U_2, \mathrm{Span}(U_1)) \mathrm{dist}(U_3, \mathrm{Span}(U_1, U_2)) \ldots \mathrm{dist}(U_k, \mathrm{Span}(U_1, \ldots, U_{k-1})). \quad (9.3.1)$$

Of course we can write $\mathrm{dist}(U_1, 0)$ instead of $||U_1||$. This definition agrees with the definition of the area of a parallelogram, or the volume of a 3-dimensional parallelepiped in the elementary geometry.

Proposition 9.1. *Let $v_1, \ldots, v_n$ be an orthonormal basis in V. Given n vectors $U_1, \ldots, U_n$ let us denote by U the matrix whose columns are coordinates of these vectors in the basis $v_1, \ldots, v_n$:*

$$U := \begin{pmatrix} u_{11} & \ldots & u_{1n} \\ & \vdots & \\ u_{n1} & \ldots & u_{nn} \end{pmatrix}$$

Then

$$\mathrm{Vol}\, P(U_1, \ldots, U_n) = |\det U|.$$

Proof. If the vectors $U_1, \ldots, U_n$ are linearly dependent then $\mathrm{Vol}\, P(U_1, \ldots, U_n) = \det U = 0$. Suppose now that the vectors $U_1, \ldots, U_n$ are linearly independent, i.e. form a basis. Consider first the case where this basis is orthonormal. Then the matrix U is orthogonal. i.e. $UU^T = E$, and hence $\det U = \pm 1$. But in this case $\mathrm{Vol}\, P(U_1, \ldots, U_n) = 1$, and hence $\mathrm{Vol}\, P(U_1, \ldots, U_n) = |\det U|$.

Now let the basis $U_1, \ldots, U_n$ be arbitrary. Let us apply to it the Gram-Schmidt orthonormalization process. Recall that this process consists of the following steps. First, we normalize the vector U_1, then subtract from U_2 its projection to $\mathrm{Span}(U_1)$, Next, we normalize the new vector U_2, then subtract from U_3 its projection to $\mathrm{Span}(U_1, U_2)$, and so on. At the end of this process we obtain an orthonormal basis. It remains to notice that each of these steps affected $\mathrm{Vol}\, P(U_1, \ldots, U_n)$ and $|\det U|$ in a similar way. Indeed, when we multiplied the vectors by a positive number, both the volume and the determinant were multiplied by the same number. When we subtracted from a vector U_k its projection to $\mathrm{Span}(U_1, \dots, U_{k-1})$, this affected neither the volume nor the determinant. ■

Corollary 9.2. *1. Let $x_1, \dots, x_n$ be a Cartesian coordinate system.*[1] *Then*

$$\mathrm{Vol}\, P(U_1, \dots, U_n) = |x_1 \wedge \dots x_n(U_1, \dots, U_n)|.$$

2. Let $\mathcal{A} : V \to V$ be a linear map. Then

$$\mathrm{Vol}\, P(\mathcal{A}(U_1), \dots, \mathcal{A}(U_n)) = |\det \mathcal{A}| \mathrm{Vol}\, P(U_1, \dots, U_n).$$

Proof.

1. According to 7.12, $x_1 \wedge \dots x_n(U_1, \dots, U_n) = \det U$.

2. $x_1 \wedge \dots x_n(\mathcal{A}(U_1), \dots, \mathcal{A}(U_n)) = \mathcal{A}^*(x_1 \wedge \dots \wedge x_n)(U_1, \dots, U_n) = \det \mathcal{A} x_1 \wedge \dots x_n(U_1, \dots, U_n)$. ■

In view of Proposition 9.1 and the first part of Corollary 9.2 the value

$$x_1 \wedge \dots x_n(U_1, \dots, U_n) = \det U$$

is called sometimes the *signed volume* of the parallelepiped $P(U_1, \dots, U_n)$. It is positive when the basis $U_1, \dots, U_n$ defines the given orientation of the space V, and it is negative otherwise.

Note that $x_1 \wedge \dots x_k(U_1, \dots, U_k)$ for $0 \le k \le n$ is the signed k-dimensional volume of the orthogonal projection of the parallelepiped $P(U_1, \dots, U_k)$ to the coordinate subspace $\{x_{k+1} = \dots = x_n = 0\}$.

For instance, let ω be the 2-form $x_1 \wedge x_2 + x_3 \wedge x_4$ on $\mathbb{R}^4$. Then for any two vectors $U_1, U_2 \in \mathbb{R}^4$ the value $\omega(U_1, U_2)$ is the sum of signed areas of projections of the parallelogram $P(U_1, U_2)$ to the coordinate planes spanned by the two first and two last basic vectors.

9.4 Volume and Gram matrix

In this section we will compute the $\mathrm{Vol}_k P(v_1, \dots, v_k)$ in the case when the number k of vectors is less than the dimension n of the space.

[1] i.e. a coordinate system with respect to an orthonormal basis

Let V be an Euclidean space. Given vectors $v_1, \dots, v_k \in V$ we can form a $k \times k$-matrix

$$G(v_1, \dots, v_k) = \begin{pmatrix} \langle v_1, v_1 \rangle & \dots & \langle v_1, v_k \rangle \\ \dots & \dots & \dots \\ \langle v_k, v_1 \rangle & \dots & \langle v_k, v_k \rangle \end{pmatrix}, \tag{9.4.1}$$

which is called the *Gram matrix* of vectors $v_1, \dots, v_k$.

Suppose we are given Cartesian coordinate system in V and let us form a matrix C whose columns are coordinates of vectors $v_1, \dots, v_k$. Thus the matrix C has n rows and k columns. Then

$$G(v_1, \dots, v_k) = C^T C,$$

because in Cartesian coordinates the scalar product looks like the dot-product.

We also point out that if $k = n$ and vectors $v_1, \dots, v_n$ form a basis of V, then $G(v_1, \dots, v_k)$ is just the matrix of the bilinear function $\langle X, Y \rangle$ in the basis $v_1, \dots, v_n$. It is important to point out that while the matrix C depends on the choice of the basis, the matrix G does not.

Proposition 9.3. *Given any k vectors $v_1, \dots, v_k$ in an Euclidean space V the volume $\mathrm{Vol}_k P(v_1, \dots, v_k)$ can be computed by the formula*

$$\mathrm{Vol}_k P(v_1, \dots, v_k)^2 = \det G(v_1, \dots, v_k) = \det C^T C, \tag{9.4.2}$$

where $G(v_1, \dots, v_k)$ is the Gram matrix and C is the matrix whose columns are coordinates of vectors $v_1, \dots, v_k$ in some orthonormal basis.

Proof. Suppose first that $k = n$. Then according to Proposition 9.1 we have

$$\mathrm{Vol}_k P(v_1, \dots, v_k) = |\det C|.$$

But $\det C^T C = (\det C)^2$, and the claim follows.

Denote $L := \mathrm{Span}(v_1, \dots, v_k)$, $G := G(v_1, \dots, v_k)$. We complement $v_1, \dots, v_k$ by an orthonormal basis $v_{k+1}, \dots, v_n$ of $L^\perp$. Then $G(v_1, \dots, v_n)$ is a block $n \times n$ matrix $\begin{pmatrix} G & 0 \\ 0 & E \end{pmatrix}$. Hence,

$$\det G(v_1, \dots, v_n) = \det G(v_1, \dots, v_k).$$

Therefore,

$$
\begin{aligned}
(\mathrm{Vol}_k P(v_1, \ldots, v_k))^2 &= (\mathrm{Vol}_n P(v_1, \ldots, v_n))^2 \\
&= \det G(v_1, \ldots, v_n) = \det G(v_1, \ldots, v_k).
\end{aligned}
$$

■

Remark 9.4. *Note that* $\det G(v_1, \ldots, v_k) \geq 0$ *and* $\det G(v_1, \ldots, v_k) = 0$ *if and only if the vectors* $v_1, \ldots, v_k$ *are linearly dependent.*

Chapter 10

Dualities

10.1 Duality between k-forms and $(n-k)$-forms on a n-dimensional Euclidean space V

Let V be an n-dimensional vector space. As we have seen above, the space $\Lambda^k(V^*)$ of k-forms, and the space $\Lambda^{n-k}(V^*)$ of $(n-k)$-forms have the same dimension $\frac{n!}{k!(n-k)!}$; these spaces are therefore isomorphic. Suppose that V is an *oriented Euclidean* space, i.e. it is supplied with an orientation and an inner product $\langle \, , \, \rangle$. It turns out that in this case there is a canonical way to establish this isomorphism which will be denoted by

$$\star : \Lambda^k(V^*) \to \Lambda^{n-k}(V^*)\,.$$

Definition 10.1. *Let α be a k-form. Then given any vectors $U_1, \dots, U_{n-k}$, the value $\star\alpha(U_1, \dots, U_{n-k})$ can be computed as follows. If $U_1, \dots, U_{n-k}$ are linearly dependent then $\star\alpha(U_1, \dots, U_{n-k}) = 0$. Otherwise, let $S^\perp$ denote the orthogonal complement to the space $S = \mathrm{Span}(U_1, \dots, U_{n-k})$. Choose a basis $Z_1, \dots, Z_k$ of $S^\perp$ such that*

$$\mathrm{Vol}_k(Z_1, \dots, Z_k) = \mathrm{Vol}_{n-k}(U_1, \dots, U_{n-k})$$

and the basis $Z_1, \dots, Z_k, U_1, \dots, U_{n-k}$ defines the given orientation of the space V. Then

$$\star\alpha(U_1, \dots, U_{n-k}) = \alpha(Z_1, \dots, Z_k). \tag{10.1.1}$$

Let us first show that

Lemma 10.2. *$\star\alpha$ is a $(n-k)$-form, i.e. $\star\alpha$ is skew-symmetric and multilinear.*

Proof. To verify that $\star\alpha$ is skew-symmetric we note that for any $1 \le i < j \le n-q$ the bases

$$Z_1, Z_2, \dots, Z_k, U_1, \dots, U_i, \dots, U_j, \dots, U_{n-k}$$

and

$$-Z_1, Z_2, \dots, Z_k, U_1, \dots, U_j, \dots, U_i, \dots, U_{n-k}$$

define the same orientation of the space V, and hence

$$\begin{aligned} &\star\,\alpha(U_1, \dots, U_j, \dots, U_i, \dots, U_{n-k}) = \alpha(-Z_1, Z_2, \dots, Z_k) \\ = &-\alpha(Z_1, Z_2, \dots, Z_k) = -\star\,\alpha(U_1, \dots, U_i, \dots, U_j, \dots, U_{n-k}). \end{aligned}$$

Hence, in order to check the *multi*-linearity it is sufficient to prove the linearity of α with respect to the first argument only. It is also clear that

$$\star\alpha(\lambda U_1, \dots, U_{n-k}) = \star\lambda\alpha(U_1, \dots, U_{n-k}). \tag{10.1.2}$$

Indeed, multiplication by $\lambda \neq 0$ does not change the span of the vectors $U_1, \dots, U_{n-q}$, and hence if $\star\alpha(U_1, \dots, U_{n-k}) = \alpha(Z_1, \dots, Z_k)$ then $\star\alpha(\lambda U_1, \dots, U_{n-k}) = \alpha(\lambda Z_1, \dots, Z_k) = \lambda\alpha(Z_1, \dots, Z_k)$.

Thus it remains to check that

$$\star\alpha(U_1 + \widetilde{U}_1, U_2, \dots, U_{n-k}) = \star\alpha(U_1, U_2, \dots, U_{n-k}) + \star\alpha(\widetilde{U}_1, U_2, \dots, U_{n-k})).$$

Let us denote $L := \mathrm{Span}(U_2, \dots, U_{n-k})$ and observe that $\mathrm{proj}_L(U_1 + \widetilde{U}_1) = \mathrm{proj}_L(U_1) + \mathrm{proj}_L(\widetilde{U}_1)$. Denote $N := U_1 - \mathrm{proj}_L(U_1)$ and $\widetilde{N} := \widetilde{U}_1 - \mathrm{proj}_L(\widetilde{U}_1)$. The vectors N and $\widetilde{N}$ are normal components of U_1 and $\widetilde{U}_1$ with respect to the subspace L, and the vector $N + \widetilde{N}$ is the normal component of $U_1 + \widetilde{U}_1$ with respect to L. Hence, we have

(i) $\star\,\alpha(U_1, \dots, U_{n-k}) = \star\alpha(N, \dots, U_{n-k})$,

(ii) $\star\,\alpha(\widetilde{U}_1, \dots, U_{n-k}) = \star\alpha(\widetilde{N}, \dots, U_{n-k})$, and

(iii) $\star\,\alpha(U_1 + \widetilde{U}_1, \dots, U_{n-k}) = \star\alpha(N + \widetilde{N}, \dots, U_{n-k})$.

Indeed, in each of these three cases,

- vectors on both side of the equality span the same space;

- the parallelepiped which they generate have the same volume, and

- the orientation which these vectors define together with a basis of the complementary space remains unchanged.

Hence, it is sufficient to prove that

$$\star\alpha(N+\widetilde{N},U_2,\dots,U_{n-k})=\star\alpha(N,U_2,\dots,U_{n-k})+\star\alpha(\widetilde{N},U_2,\dots,U_{n-k}). \tag{10.1.3}$$

If the vectors N and $\widetilde{N}$ are linearly dependent, i.e. one of them is a multiple of the other, then (10.1.3) follows from (10.1.2).

Suppose now that N and $\widetilde{N}$ are linearly independent. Let $L^\perp$ denote the orthogonal complement of $L=\operatorname{Span}(U_2,\dots,U_{n-k})$. Then $\dim L^\perp=k+1$ and we have $N,\widetilde{N}\in L^\perp$. Let us denote by M the plane in $L^\perp$ spanned by the vectors N and $\widetilde{N}$, and by $M^\perp$ its orthogonal complement in $L^\perp$. Note that $\dim M^\perp=k-1$.

Choose any orientation of M so that we can talk about counter-clockwise rotation of this plane. Let $Y,\widetilde{Y}\in M$ be vectors obtained by rotating N and $\widetilde{N}$ in M counter-clockwise by the angle $\frac{\pi}{2}$. Then $Y+\widetilde{Y}$ can be obtained by rotating $N+\widetilde{N}$ in M counter-clockwise by the same angle $\frac{\pi}{2}$. Let us choose in $M^\perp$ a basis $Z_2,\dots,Z_k$ such that

$$\mathrm{Vol}_{k-1}P(Z_2,\dots,Z_k)=\mathrm{Vol}_{n-k-1}P(U_2,\dots,U_{n-k}).$$

Note that the orthogonal complements to $\operatorname{Span}(N,U_2,\dots,U_{n-k})$, $\operatorname{Span}(\widetilde{N},U_2,\dots,U_{n-k})$, and to $\operatorname{Span}(N+\widetilde{N},U_2,\dots,U_{n-k})$ in V coincide, respectively, with the orthogonal complements to the the vectors $N,\widetilde{N}$ and to $N+\widetilde{N}$ in $L^\perp$. In other words, we have

$$\begin{gathered}
(\operatorname{Span}(N,U_2,\dots,U_{n-k}))_V^\perp=\operatorname{Span}(Y,Z_2,\dots,Z_k),\\
\Big(\operatorname{Span}(\widetilde{N},U_2,\dots,U_{n-k})\Big)_V^\perp=\operatorname{Span}(\widetilde{Y},Z_2,\dots,Z_k)\ \text{ and}\\
\Big(\operatorname{Span}(N+\widetilde{N},U_2,\dots,U_{n-k})\Big)_V^\perp=\operatorname{Span}(Y+\widetilde{Y},Z_2,\dots,Z_k).
\end{gathered}$$

Next, we observe that

$$\mathrm{Vol}_{n-k}P(N,U_2,\dots,U_{n-k})=\mathrm{Vol}_kP(Y,Z_2,\dots,Z_k),$$

$$\mathrm{Vol}_{n-k} P(\widetilde{N}, U_2, \dots, U_{n-k}) = \mathrm{Vol}_k P(\widetilde{Y}, Z_2, \dots, Z_k) \text{ and}$$

$$\mathrm{Vol}_{n-k} P(N + \widetilde{N}, U_2, \dots, U_{n-k}) = \mathrm{Vol}_k P(Y + \widetilde{Y}, Z_2, \dots, Z_k).$$

Consider the following 3 bases of V:

$$Y, Z_2, \dots, Z_k, N, U_2, \dots, U_{n-k},$$

$$\widetilde{Y}, Z_2, \dots, Z_k, \widetilde{N}, U_2, \dots, U_{n-k},$$

$$Y + \widetilde{Y}, Z_2, \dots, Z_k, N + \widetilde{N}, U_2, \dots, U_{n-k},$$

and observe that all three of them define the same a priori given orientation of V. Thus, by definition of the operator $\star$ we have:

$$\begin{aligned}
&\star\, \alpha(N + \widetilde{N}, U_2, \dots, U_{n-k}) = \alpha(Y + \widetilde{Y}, Z_2, \dots, Z_k) \\
&= \alpha(Y, Z_2, \dots, Z_k) + \alpha(\widetilde{Y}, Z_2, \dots, Z_k) = \star\alpha(N, U_2, \dots, U_{n-k}) + \star\alpha(\widetilde{N}, U_2, \dots, U_{n-k}).
\end{aligned}$$

This completes the proof that $\star\alpha$ is an $(n-k)$-form. ■

Thus the map $\alpha \mapsto \star\alpha$ defines a map $\star : \Lambda^k(V^*) \to \Lambda^{n-k}(V^*)$. Clearly, this map is linear. In order to check that $\star$ is an isomorphism let us choose an orthonormal basis in V and consider the coordinates $x_1, \dots, x_n \in V^*$ corresponding to that basis.

Let us recall that the forms $x_{i_1} \wedge x_{i_2} \wedge \dots \wedge x_{i_k}$, $1 \le i_1 < i_2 < \dots < i_k \le n$, form a basis of the space $\Lambda^k(V^*)$.

Lemma 10.3.

$$\star x_{i_1} \wedge x_{i_2} \wedge \dots \wedge x_{i_k} = (-1)^{\mathrm{inv}(i_1, \dots, i_k, j_1, \dots, j_{n-k})} x_{j_1} \wedge x_{j_2} \wedge \dots \wedge x_{j_{n-k}}\,, \tag{10.1.4}$$

where $j_1 < \dots < j_{n-k}$ is the set of indices, complementary to $i_1, \dots, i_k$.

Proof. Evaluating $\star(x_{i_1} \wedge \dots \wedge x_{i_k})$ on basic vectors $v_{j_1}, \dots, v_{j_{n-k}}$ where $1 \le j_1 < \dots < j_{n-q} \le n$, we get 0 unless all the indices $j_1, \dots, j_{n-k}$ are all different from $i_1, \dots, i_k$, while in the latter case we get

$$\star(x_{i_1} \wedge \dots \wedge x_{i_k})(v_{j_1}, \dots, v_{j_{n-k}}) = (-1)^{\mathrm{inv}(i_1, \dots, i_k, j_1, \dots, j_{n-k})}.$$

Hence,

$$\star x_{i_1} \wedge x_{i_2} \wedge \cdots \wedge x_{i_k} = (-1)^{\mathrm{inv}(i_1,\dots,i_k,j_1,\dots,j_{n-k})} x_{j_1} \wedge x_{j_2} \wedge \cdots \wedge x_{j_{n-k}}.$$

■

Thus $\star$ establishes a 1-1 correspondence between the bases of the spaces $\Lambda^k(V^*)$ and the space $\Lambda^{n-k}(V^*)$, and hence it is an isomorphism. Note that by linearity for any form

$$\alpha = \sum_{1\le i_1<\dots<i_k\le n} a_{i_1\dots i_k} x_{i_1} \wedge \cdots \wedge x_{i_k})$$

we have

$$\star\alpha = \sum_{1\le i_1<\dots<i_k\le n} a_{i_1\dots i_k} \star (x_{i_1} \wedge \cdots \wedge x_{i_k}) .$$

Examples.

1. $\star C = Cx_1 \wedge \cdots \wedge x_n$; in other words the isomorphism $\star$ acts on constants (= 0-forms) by multiplying them by the volume form.

2. In $\mathbb{R}^3$ we have

$$\star x_1 = x_2 \wedge x_3 \,, \star x_2 = -x_1 \wedge x_3 = x_3 \wedge x_1 \,, \star x_3 = x_1 \wedge x_2 \,,$$

$$\star(x_1 \wedge x_2) = x_3 \,, \star(x_3 \wedge x_1) = x_2 \,, \star(x_2 \wedge x_3) = x_1 \,.$$

3. More generally, given a 1-form $l = a_1x_1 + \cdots + a_nx_n$ we have

$$\star l = a_1 x_2 \wedge \cdots \wedge x_n - a_2 x_1 \wedge x_3 \wedge \cdots \wedge x_n + \cdots + (-1)^{n-1} a_n x_1 \wedge \cdots \wedge x_{n-1} \,.$$

In particular for $n = 3$ we have

$$\star(a_1x_1 + a_2x_2 + a_3x_3) = a_1 x_2 \wedge x_3 + a_2 x_3 \wedge x_1 + a_3 x_1 \wedge x_2 \,.$$

Proposition 10.4.

$$\star^2 = (-1)^{k(n-k)}\mathrm{Id}, \quad i.e. \quad \star(\star\omega) = (-1)^{k(n-k)}\omega \quad \textit{for any } k\textit{-form} \quad \omega \,.$$

In particular, if dimension $n = \dim V$ *is odd then* $*^2 = \mathrm{Id}$*. If* n *is even and* ω *is a* k*-form then* $\star(\star\omega) = \omega$ *if* k *is even, and* $\star(\star\omega) = -\omega$ *if* k *is odd.*

Proof. It is sufficient to verify the equality

$$\star(\star\omega) = (-1)^{k(n-k)}\omega$$

for the case when ω is a basic form, i.e.

$$\omega = x_{i_1} \wedge \dots \wedge x_{i_k}\,, \ \ 1 \le i_1 < \dots < i_k \le n\,.$$

We have

$$\star(x_{i_1} \wedge \dots \wedge x_{i_k}) = (-1)^{\mathrm{inv}(i_1,\dots,i_k,j_1,\dots,j_{n-k})} x_{j_1} \wedge x_{j_2} \wedge \dots \wedge x_{j_{n-k}}$$

and

$$\star(x_{j_1} \wedge x_{j_2} \wedge \dots \wedge x_{j_{n-k}}) = (-1)^{\mathrm{inv}(j_1,\dots,j_{n-k},i_1,\dots,i_k)} x_{i_1} \wedge \dots \wedge x_{i_k}\,.$$

But the permutations $i_1 \dots i_k j_1 \dots j_{n-k}$ and $j_1 \dots j_{n-k} i_1 \dots i_k$ differ by $k(n-k)$ transpositions of pairs of its elements. Hence, we get

$$(-1)^{\mathrm{inv}(i_1,\dots,i_k,j_1,\dots,j_{n-k})} = (-1)^{k(n-k)}(-1)^{\mathrm{inv}(j_1,\dots,j_{n-k},i_1,\dots,i_k)}\,,$$

and, therefore,

$$\begin{aligned}
\star\big(\star(x_{i_1} \wedge \dots \wedge x_{i_k})\big) &= \star\big((-1)^{\mathrm{inv}(i_1,\dots,i_k,j_1,\dots,j_{n-k})} x_{j_1} \wedge x_{j_2} \wedge \dots \wedge x_{j_{n-k}}\big) \\
&= (-1)^{\mathrm{inv}(i_1,\dots,i_k,j_1,\dots,j_{n-k})} \star (x_{j_1} \wedge x_{j_2} \wedge \dots \wedge x_{j_{n-k}}) \\
&= (-1)^{\mathrm{inv}(i_1,\dots,i_k,j_1,\dots,j_{n-k}) + \mathrm{inv}(j_1,\dots,j_{n-k},i_1,\dots,i_k)} x_{i_1} \wedge \dots \wedge x_{i_k} \\
&= (-1)^{k(n-k)} x_{i_1} \wedge \dots \wedge x_{i_k}\,.
\end{aligned}$$

■

Exercise 10.5. *(a) For any special orthogonal operator $\mathcal{A}$ the operators $\mathcal{A}^*$ and $\star$ commute, i.e.*

$$\mathcal{A}^* \circ \star = \star \circ \mathcal{A}^*.$$

(b) Let A be an orthogonal matrix of order n with $\det A = 1$*. Prove that for any $k \in \{1, \dots, n\}$ the absolute value of each k-minor M of A is equal to the absolute value of its complementary minor of order $(n-k)$. (Hint: Apply (a) to the form $x_{i_1} \wedge \dots \wedge x_{i_k}$).*

(c) Let V be an oriented 3-dimensional Euclidean space. Prove that for any two vectors $X, Y \in V$, their cross-product can be written in the form

$$X \times Y = \mathcal{D}^{-1}\left(\star(\mathcal{D}(X) \wedge \mathcal{D}(Y))\right).$$

10.2 Euclidean structure on the space of exterior forms

Suppose that the space V is oriented and Euclidean, i.e. it is endowed with an inner product $\langle\, , \rangle$ and an orientation.

Given two forms $\alpha, \beta \in \Lambda^k(V^*)$, $k = 0, \dots, n$, let us define

$$\langle\langle \alpha, \beta \rangle\rangle = \star(\alpha \wedge \star\beta).$$

Note that $\alpha \wedge \star\beta$ is an n-form for every k, and hence, $\langle\langle \alpha, \beta \rangle\rangle$ is a 0-form, i.e. a real number.

Proposition 10.6. *1. The operation $\langle\langle\, , \rangle\rangle$ defines an inner product on $\Lambda^k(V^*)$ for each $k = 0, \dots, n$.*

2. If $\mathcal{A} : V \to V$ is a special orthogonal operator then the operator $\mathcal{A}^* : \Lambda^k(V^*) \to \Lambda^k(V^*)$ is orthogonal with respect to the inner product $\langle\langle\, , \rangle\rangle$.

Proof. 1. We need to check that $\langle\langle \alpha, \beta \rangle\rangle$ is a symmetric bilinear function on $\Lambda^k(V^*)$ and $\langle\langle \alpha, \alpha \rangle\rangle > 0$ unless $\alpha = 0$. Bilinearity is straightforward. Hence, it is sufficient to verify the remaining properties for basic vectors $\alpha = x_{i_1} \wedge \dots \wedge x_{i_k}, \beta = x_{j_1} \wedge \dots \wedge x_{j_k}$, where $1 \le i_1 < \dots < i_k \le n$, $1 \le j_1 < \dots < j_k \le n$. Here $(x_1, \dots, x_n)$ is any Carterian coordinate system in V which defines its given orientation.

Note that $\langle\langle \alpha, \beta \rangle\rangle = 0 = \langle\langle \beta, \alpha \rangle\rangle$ unless $i_m = j_m$ for all $m = 1, \dots, k$, and in the the latter case we have $\alpha = \beta$. Furthermore, we have

$$\langle\langle \alpha, \alpha \rangle\rangle = \star(\alpha \wedge \star\alpha) = \star(x_1 \wedge \dots \wedge x_n) = 1 > 0.$$

2. The inner product $\langle\langle\, , \rangle\rangle$ is defined only in terms of the Euclidean structure and the orientation of V. Hence, for any special orthogonal operator $\mathcal{A}$ (which preserves these structures) the induced operator $\mathcal{A}^* : \Lambda^k(V^*) \to \Lambda^k(V^*)$ preserves the inner product $\langle\langle\, , \rangle\rangle$. ∎

Note that we also proved that the basis of k-forms $x_{i_1} \wedge \cdots \wedge x_{i_k}$, $1 \le i_1 < \cdots < i_k \le n$, is orthonormal with respect to the scalar product $\langle\langle , \rangle\rangle$. Hence, we get

Corollary 10.7. *Suppose that a k-form α can be written in Cartesian coordinates as*

$$\alpha = \sum_{1 \le i_1 < \cdots < i_k \le n} a_{i_1 \dots i_k} x_{i_1} \wedge \cdots \wedge x_{i_k}.$$

Then

$$|| \alpha ||^2 = \langle\langle \alpha, \alpha \rangle\rangle = \sum_{1 \le i_1 < \cdots < i_k \le n} a^2_{i_1 \dots i_k}.$$

Corollary 10.8. *For any two exterior k-forms we have*

$$\star\alpha \wedge \beta = (-1)^{k(n-k)} \alpha \wedge \star\beta.$$

Exercise 10.9. *1. Show that for any k-forms we have*

$$\langle\langle \alpha, \beta \rangle\rangle = \langle\langle \star\alpha, \star\beta \rangle\rangle.$$

2. Show that if α, β are 1-forms on an Euclidean space V. Then

$$\langle\langle \alpha, \beta \rangle\rangle = \langle \mathcal{D}^{-1}(\alpha), \mathcal{D}^{-1}(\beta) \rangle,$$

i.e the scalar product $\langle\langle\ ,\ \rangle\rangle$ on V^* is the push-forward by $\mathcal{D}$ of the scalar product $\langle\ ,\ \rangle$ on V.

Proposition 10.10. *Let V be a Euclidean n-dimensional space. Choose an orthonormal basis $e_1, \dots, e_n$ in V. Then for any vectors $Z_1 = (z_{11}, \dots, z_{n1}), \dots, Z_k = (z_{1k}, \dots, z_{nk}) \in V$ we have*

$$(\mathrm{Vol}_k P(Z_1, \dots, Z_k))^2 = \sum_{1 \le i_1 < \cdots < i_k \le n} Z^2_{i_1, \dots, i_k}, \tag{10.2.1}$$

where

$$Z_{i_1, \dots, i_k} = \begin{vmatrix} z_{i_1 1} & \dots & z_{i_1 k} \\ \dots & \dots & \dots \\ z_{i_k 1} & \dots & z_{i_k k} \end{vmatrix}.$$

Proof. Consider linear functions $l_j = \mathcal{D}(Z_j) = \sum\limits_{i=1}^{n} z_{ij} x_i \in V^*$, $j = 1, \dots, k$. Then

$$l_1 \wedge \dots \wedge l_k = \sum_{i_1=1}^{n} z_{i_1 1} x_{i_1} \wedge \dots \wedge \sum_{i_k=1}^{n} z_{i_k k} x_{i_k} =$$
$$\sum_{i_1,\dots,i_k} z_{i_1 1} \dots z_{i_k k} x_{i_1} \wedge \dots \wedge x_{i_k} = \sum_{1 \le i_1 < \dots < i_k \le n} Z_{i_1,\dots,i_k} x_{i_1} \wedge \dots x_{i_k}. \qquad (10.2.2)$$

Hence, according to Proposition 10.7 we have

$$|| \, l_1 \wedge \dots \wedge l_k \, ||^2 = \sum_{1 \le i_1 < \dots < i_k \le n} Z^2_{i_1,\dots,i_k}, \qquad (10.2.3)$$

Thus, Proposition 10.2.1 follows from the following lemma.

Lemma 10.11. *For any k vectors $v_1, \dots, v_k$ we have*

$$\mathrm{Vol}_k P(v_1, \dots, v_k) = ||l_1 \wedge \dots \wedge l_k||,$$

where $l_j = \mathcal{D}(v_i)$, $i = 1, \dots, k$.

Proof of Lemma 10.11. We can assume that the vectors $v_1, \dots, v_k$ are linearly independent. Otherwise, both sides of the required identity are equal to 0. Let $f_1, \dots, f_k$ be an orthonormal basis of the k-dimensional subspace $L = \mathrm{Span}(v_1, \dots, v_k) \subset V$. Then both quantities, $\mathrm{Vol}_k(v_1, \dots, v_k)$ and $||l_1 \wedge \dots \wedge l_k||$ are equal to the absolute value of the determinant of the $k \times k$ matrix formed by coordinates of $v_1, \dots, v_k$ in the basis $f_1, \dots, f_k$, and hence, $\mathrm{Vol}_k P(v_1, \dots, v_k) = ||l_1 \wedge \dots \wedge l_k||$. This proves Lemma 10.11, and with it Proposition 10.7. ■

We recall that an alternative formula for computing $\mathrm{Vol}_k P(Z_1, \dots, Z_k)$ was given earlier in Proposition 9.3.

10.3 Contraction

Let V be a vector space and $\phi \in \Lambda^k(V^*)$ a k-form. Given a vector $v \in V$ define a $(k-1)$-form $\psi = v \lrcorner \phi$ by the formula

$$\psi(X_1, \dots, X_{k-1}) = \phi(v, X_1, \dots, X_{k-1})$$

for any vectors $X_1, \dots, X_{k-1} \in V$. We say that the form ψ is obtained by a *contraction* of ϕ with the vector v. This operation is also often called an *interior product* of v with ϕ and denoted by $i(v)\phi$ instead of $v \lrcorner \phi$. In these notes we will not use this notation.

Proposition 10.12. *Contraction* $\lrcorner$ *is a bilinear operation, i.e.*

$$
\begin{aligned}
(v_1 + v_2) \lrcorner \phi &= v_1 \lrcorner \phi + v_2 \lrcorner \phi \\
(\lambda v) \lrcorner \phi &= \lambda (v \lrcorner \phi) \\
v \lrcorner (\phi_1 + \phi_2) &= v \lrcorner \phi_1 + v \lrcorner \phi_2 \\
v \lrcorner (\lambda \phi) &= \lambda (v \lrcorner \phi).
\end{aligned}
$$

Here $v, v_1, v_2 \in V$; $\phi, \phi_1, \phi_2 \in \Lambda^k(V^*)$; $\lambda \in \mathbb{R}$.

The proof is straightforward.

Let ϕ be a non-zero n-form. Then we have

Proposition 10.13. *The map* $\lrcorner : V \to \Lambda^{n-1}(V^*)$, *defined by the formula* $\lrcorner(v) = v \lrcorner \phi$ *is an isomorphism between the vector spaces* V *and* $\Lambda^{n-1}(V^*)$.

Proof. Take a basis $v_1, \dots, v_n$. Let $x_1, \dots, x_n \in V^*$ be the dual basis, i.e. the corresponding coordinate system. Then $\phi = a x_1 \wedge \ldots \wedge x_n$, where $a \neq 0$. To simplify the notation let us assume that $a = 1$, so that

$$\phi = x_1 \wedge \ldots \wedge x_n.$$

Let us compute the images $v_i \lrcorner \phi$, $i = 1, \dots, k$ of the basic vectors. Let us write

$$v_i \lrcorner \phi = \sum_1^n a_j x_1 \wedge \cdots \wedge x_{j-1} \wedge x_{j+1} \wedge \cdots \wedge x_n.$$

Then

$$
\begin{aligned}
& v_l \lrcorner \phi(v_1, \dots, v_{l-1}, v_{l+1}, \dots, v_n) \\
&= \sum_1^n a_j x_1 \wedge \cdots \wedge x_{j-1} \wedge x_{j+1} \wedge \cdots \wedge x_n(v_1, \dots, v_{l-1}, v_{l+1}, \dots, v_n) = (-1)^{l-1} a_l, \quad (10.3.1)
\end{aligned}
$$

but on the other hand,

$$\begin{aligned}
&v_i \lrcorner\, \phi(v_1,\dots,v_{l-1},v_{l+1},\dots,v_n) = \phi(v_i,v_1,\dots,v_{l-1},v_{l+1},\dots,v_n)\\
&= (-1)^{i-1}\phi(v_1,\dots,v_{l-1},v_i,v_{l+1},\dots,v_n) = \begin{cases} (-1)^{i-1} & , \ l=i\,;\\ 0 & , \ \text{otherwise}\end{cases}
\end{aligned} \tag{10.3.2}$$

Thus,

$$v_i \lrcorner\, \phi = (-1)^{i-1} x_1 \wedge \dots \wedge x_{i-1} \wedge x_{i+1} \wedge \dots \wedge x_n.$$

Hence, the map $\lrcorner$ sends a basis of V^* into a basis of $\Lambda^{n-1}(V^*)$, and therefore it is an isomorphism.

■

Take a vector $v = \sum\limits_1^n a_j v_j$. Then we have

$$v \lrcorner\, (x_1 \wedge \dots \wedge x_n) = \sum_1^n (-1)^{i-1} a_i x_1 \wedge \dots \wedge x_{i-1} \wedge x_{i+1} \wedge \dots \wedge x_n. \tag{10.3.3}$$

This formula can be interpreted as the formula of expansion of a determinant according to the first column (or the first row). Indeed, for any vectors $U_1,\dots,U_{n-1}$ we have

$$v \lrcorner\, \phi(U_1,\dots,U_{n-1}) = \det(v,U_1,\dots,U_{n-1}) = \begin{vmatrix} a_1 & u_{1,1} & \dots & u_{1,n-1}\\ \dots & \dots & \dots & \dots\\ a_n & u_{n,1} & \dots & u_{n,n-1}\end{vmatrix},$$

where $\begin{pmatrix} u_{1,i}\\ \vdots\\ u_{n,i}\end{pmatrix}$ are coordinates of the vector $U_i \in V$ in the basis $v_1,\dots,v_n$. On the other hand,

$$\begin{aligned}
v \lrcorner\, \phi(U_1,\dots,U_{n-1}) &= \sum_1^n (-1)^{i-1} a_i x_1 \wedge \dots \wedge x_{i-1} \wedge x_{i+1} \wedge \dots \wedge x_n(U_1,\dots,U_{n-1})\\
&= a_1 \begin{vmatrix} u_{2,1} & \dots & u_{2,n-1}\\ u_{3,1} & \dots & u_{3,n-1}\\ \dots & \dots & \dots\\ u_{n,1} & \dots & u_{n,n-1}\end{vmatrix} + \dots + (-1)^{n-1} a_n \begin{vmatrix} u_{1,1} & \dots & u_{1,n-1}\\ u_{2,1} & \dots & u_{2,n-1}\\ \dots & \dots & \dots\\ u_{n-1,1} & \dots & u_{n-1,n-1}\end{vmatrix}.
\end{aligned} \tag{10.3.4}$$

Suppose that $\dim V = 3$. Then the formula (10.3.3) can be rewritten as

$$v \lrcorner (x_1 \wedge x_2 \wedge x_3) = a_1 x_2 \wedge x_3 + a_2 x_3 \wedge x_1 + a_3 x_1 \wedge x_2,$$

where $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ are coordinates of the vector V. Let us describe the geometric meaning of the operation $\lrcorner$. Set $\omega = v \lrcorner (x_1 \wedge x_2 \wedge x_3)$. Then $\omega(U_1, U_2)$ is the volume of the parallelogram defined by the vectors U_1, U_2 and v. Let ν be the unit normal vector to the plane $L(U_1, U_2) \subset V$. Then we have

$$\omega(U_1, U_2) = \text{ Area } P(U_1, U_2) \cdot \langle v, \nu \rangle .$$

If we interpret v as the velocity of a fluid flow in the space V then $\omega(U_1, U_2)$ is just an amount of fluid flown through the parallelogram Π generated by vectors U_1 and U_2 for the unit time. It is called *the flux* of v through the parallelogram Π.

Let us return back to the case $\dim V = n$.

Exercise 10.14. *Let*

$$\alpha = x_{i_1} \wedge \dots \wedge x_{i_k}, \quad 1 \le i_1 < \dots < i_k \le n$$

and $v = (a_1, \dots, a_n)$. *Show that*

$$v \lrcorner \alpha = \sum_{j=1}^{k} (-1)^{j+1} a_{i_j} x_{i_1} \wedge \dots x_{i_{j-1}} \wedge x_{i_{j+1}} \wedge \dots x_{i_k} .$$

The next proposition establishes a relation between the isomorphisms $\star, \lrcorner$ and $\mathcal{D}$.

Proposition 10.15. *Let V be a Euclidean space, and $x_1, \dots, x_n$ be coordinates in an orthonormal basis. Then for any vector $v \in V$ we have*

$$\star \mathcal{D} v = v \lrcorner (x_1 \wedge \dots \wedge x_n) .$$

Proof. Let $v = (a_1, \dots, a_n)$. Then $\mathcal{D} v = a_1 x_1 + \dots + a_n x_n$ and

$$\star \mathcal{D} v = a_1 x_2 \wedge \dots \wedge x_n - a_2 x_1 \wedge x_3 \wedge \dots \wedge x_n + \dots + (-1)^{n-1} a_n x_1 \wedge \dots \wedge x_{n-1} .$$

But according to Proposition 10.13 the $(n-1)$-form $v \lrcorner (x_1 \wedge \dots \wedge x_n)$ is defined by the same formula. ■

We finish this section by the proposition which shows how the contraction operation interacts with the exterior product.

Proposition 10.16. *Let α, β be forms of order k and l, respectively, and v a vector. Then*

$$v \lrcorner (\alpha \wedge \beta) = (v \lrcorner \alpha) \wedge \beta + (-1)^k \alpha \wedge (v \lrcorner \beta).$$

Proof. Note that given any indices $k_1, \dots k_m$ (not necessarily ordered) we have $e_i \lrcorner x_{k_1} \wedge \dots \wedge x_{k_m} = 0$ if $i \notin \{k_1, \dots, k_m\}$ and $e_i \lrcorner x_{k_1} \wedge \dots \wedge x_{k_m} = (-1)^J x_{k_1} \wedge \dots \overset{i}{\vee} \dots \wedge x_{k_m}$, where $J = \operatorname{inv}(i; k_1, \dots, k_m)$ is the number of variables ahead of x_i.

By linearity it is sufficient to consider the case when v, α, β are basic vector and forms, i.e.

$$v = e_i, \alpha = x_{i_1} \wedge \dots \wedge x_{i_k},\ \beta = x_{j_1} \dots x_{j_l}.$$

We have $v \lrcorner \alpha \neq 0$ if and only if the index i is among the indices $i_1, \dots, i_k$. In that case

$$v \lrcorner \alpha = (-1)^J x_{i_1} \wedge \dots \overset{i}{\vee} \dots \wedge x_{i_k}$$

and if $v \lrcorner \beta \neq 0$ then

$$v \lrcorner \beta = (-1)^{J'} x_{j_1} \wedge \dots \overset{i}{\vee} \dots \wedge x_{j_l},$$

where $J = \operatorname{inv}(i; i_1, \dots, i_k)$, $J' = \operatorname{inv}(i; j_1, \dots, j_l)$.

If it enters α bot not β then

$$v \lrcorner (\alpha \wedge \beta) = (-1)^J x_{i_1} \wedge \dots \overset{i}{\vee} \dots \wedge x_{i_k} \wedge x_{j_1} \wedge \dots \wedge x_{j_l} = (v \lrcorner \alpha) \wedge \beta,$$

while $\alpha \wedge (v \lrcorner \beta) = 0$.

Similarly, if it enters β but not α then

$$v \lrcorner (\alpha \wedge \beta) = (-1)^{J'+m} x_{i_1} \wedge \dots \wedge x_{i_k} \wedge x_{j_1} \wedge \dots \overset{i}{\vee} \dots \wedge x_{j_l} = (-1)^k \alpha \wedge (v \lrcorner \beta),$$

while $v \lrcorner \alpha \wedge \beta = 0$. Hence, in both these cases the formula holds.

If x_i enters both products then $\alpha \wedge \beta = 0$, and hence $v \lrcorner (\alpha \wedge \beta) = 0$.

On the other hand,

$$\begin{aligned}(v \lrcorner \alpha) \wedge \beta + (-1)^k \alpha \wedge (v \lrcorner \beta) &= (-1)^J x_{i_1} \wedge \dots \overset{i}{\vee} \dots \wedge x_{i_k} \wedge x_{j_1} \dots \wedge x_{j_l} \\ + (-1)^{k+J'} x_{i_1} \wedge \dots \wedge x_{i_k} \wedge x_{j_1} \wedge \dots \overset{i}{\vee} \dots \wedge x_{j_l} &= 0,\end{aligned}$$

because the products $x_{i_1} \wedge \dots \overset{i}{\vee} \dots \wedge x_{i_k} \wedge x_{j_1} \dots \wedge x_{j_l}$ and $x_{i_1} \wedge \dots \wedge x_{i_k} \wedge x_{j_1} \wedge \dots \overset{i}{\vee} \dots \wedge x_{j_l}$ differ only in the position of x_i. In the first product it is at the $(k + J')$-s position, and in the second at $(J+1)$-st position. Hence, the difference in signs is $(-1)^{J+J'+k+1}$, which leads to the required cancellation.

$\blacksquare$

Chapter 11

Complex vector spaces

11.1 Complex numbers

The space $\mathbb{R}^2$ can be endowed with an associative and commutative multiplication operation. This operation is uniquely determined by three properties:

- it is a bilinear operation;
- the vector $(1, 0)$ is the unit;
- the vector $(0, 1)$ satisfies $(0, 1)^2 = (-1, 0)$.

The vector $(0, 1)$ is usually denoted by i, and we will simply write 1 instead of the vector $(1, 0)$. Hence, any point $(a, b) \in \mathbb{R}^2$ can be written as $a + bi$, where $a, b \in \mathbb{R}$, and the product of $a + bi$ and $c + di$ is given by the formula

$$(a + bi)(c + di) = ac - bd + (ad + bc)i.$$

The plane $\mathbb{R}^2$ endowed with this multiplication is denoted by $\mathbb{C}$ and called the set of *complex numbers*. The real line generated by 1 is called the *real axis*, the line generated by i is called the *imaginary axis*. The set of real numbers $\mathbb{R}$ can be viewed as embedded into $\mathbb{C}$ as the real axis. Given a complex number $z = x + iy$, the numbers x and y are called its *real* and *imaginary* parts, respectively, and denoted by $\mathrm{Re}z$ and $\mathrm{Im}z$, so that $z = \mathrm{Re}z + i\mathrm{Im}z$.

For any non-zero complex number $z = a + bi$ there exists an inverse z^{-1} such that $z^{-1}z = 1$. Indeed, we can set

$$z^{-1} = \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}i.$$

The commutativity, associativity and existence of the inverse is easy to check, but it should not be taken for granted: it is impossible to define a similar operation any $\mathbb{R}^n$ for $n > 2$.

Given $z = a + bi \in \mathbb{C}$ its conjugate is defined as $\bar{z} = a - bi$. The conjugation operation $z \mapsto \bar{z}$ is the reflection of $\mathbb{C}$ with respect to the real axis $\mathbb{R} \subset \mathbb{C}$. Note that

$$\mathrm{Re} z = \frac{1}{2}(z + \bar{z}), \;\; \mathrm{Im} z = \frac{1}{2i}(z - \bar{z}).$$

Let us introduce the polar coordinates (r, ϕ) in $\mathbb{R}^2 = \mathbb{C}$. Then a complex number $z = x + yi$ can be written as $r\cos\phi + ir\sin\phi = r(\cos\phi + i\sin\phi)$. This form of writing a complex number is called, sometimes, t*trigonometric*. The number $r = \sqrt{x^2 + y^2}$ is called the *modulus* of z and denoted by $|z|$ and ϕ is called the *argument* of ϕ and denoted by $\arg z$. Note that the argument is defined only $\mod 2\pi$. The value of the argument in $[0, 2\pi)$ is sometimes called the *principal value* of the argument. When z is real than its modulus $|z|$ is just the absolute value. We also not that $|z| = \sqrt{z\bar{z}}$.

An important role plays the *triangle inequality*

$$\big|\,|z_1| - |z_2|\,\big| \le |z_1 + z_2| \le |z_1| + |z_2|.$$

Exponential function of a complex variable

Recall that the exponential function e^x has a Taylor expansion

$$e^x = \sum_0^\infty \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots.$$

We then define for a *complex* argument z the exponential function by the same formula

$$e^z := 1 + z + \frac{z^2}{2!} + \dots + \frac{z^n}{n!} + \dots.$$

One can check that this power series *absolutely converging* for all z and satisfies the formula

$$e^{z_1 + z_2} = e^{z_1} e^{z_2}.$$

In particular, we have

$$e^{iy} = 1 + iy - \frac{y^2}{2!} - i\frac{y^3}{3!} + \frac{y^4}{4!} + \dots + \dots \tag{11.1.1}$$

$$= \sum_{k=0}^{\infty}(-1)^k\frac{y^{2k}}{2k!} + i\sum_{k=0}^{\infty}(-1)^k\frac{y^{2k+1}}{(2k+1)!}. \tag{11.1.2}$$

But $\sum\limits_{k=0}^{\infty}(-1)^k\frac{y^{2k}}{2k!} = \cos y$ and $\sum\limits_{k=0}^{\infty}(-1)^k\frac{y^{2k+1}}{(2k+1)!} = \sin y$, and hence we get Euler's formula

$$e^{iy} = \cos y + i\sin y,$$

and furthermore,

$$e^{x+iy} = e^x e^{iy} = e^x(\cos y + i\sin y),$$

i.e. $|e^{x+iy}| = e^x, \arg(e^z) = y$.

Hence, any complex number $z = r(\cos\phi + i\sin\phi)$ can be rewritten in the form $z = re^{i\phi}$. This is called the *exponential form* of the complex number z.

Note that

$$\left(e^{i\phi}\right)^n = e^{in\phi},$$

and hence if $z = re^{i\phi}$ then $z^n = r^n e^{in\phi} = r^n(\cos n\phi + i\sin n\phi)$.

Note that the operation $z \mapsto iz$ is the rotation of $\mathbb{C}$ counterclockwise by the angle $\frac{\pi}{2}$. More generally a multiplication operation $z \mapsto zw$, where $w = \rho e^{i\theta}$ is the composition of a rotation by the angle θ and a radial dilatation (homothety) in ρ times.

Exercise 11.1.

1. Compute $\sum\limits_{0}^{n}\cos k\theta$ and $\sum\limits_{1}^{n}\sin k\theta$.
2. Compute $1 + \binom{n}{4} + \binom{n}{8} + \binom{n}{12} + \dots$.

11.2 Complex vector space

In a real vector space one knows how to multiply a vector by a real number. In a complex vector space there is defined an operation of multiplication by a complex number. Example is the space $\mathbb{C}^n$ whose vectors are n-tuples $z = (z_1, \dots, z_n)$ of complex numbers, and multiplication by any complex number $\lambda = \alpha + i\beta$ is defined component-wise: $\lambda(z_1, \dots, z_n) = (\lambda z_1, \dots, \lambda z_n)$. Complex vector space can be viewed as an upgrade of a real vector space, or better to say as a real vector space with an additional structure.

In order to make a real vector space V into a complex vector space, one just needs to define how to multiply a vector by i. This operation must be a linear map $\mathcal{J} : V \to V$ which should satisfy the condition $\mathcal{J}^2 = -\mathrm{Id}$, i.e $\mathcal{J}(\mathcal{J}(v)) = i(iv) = -v$.

Example 11.2. *Consider $\mathbb{R}^{2n}$ with coordinates $(x_1, y_1, \dots, x_n, y_n)$. Consider a $2n \times 2n$-matrix*

$$J = \begin{pmatrix} 0 & -1 & 0 & 0 & & & \\ 1 & 0 & 0 & 0 & & & \\ 0 & 0 & 0 & -1 & & & \\ 0 & 0 & 1 & 0 & & & \\ & & & & \dots & & \\ & & & & & 0 & -1 \\ & & & & & 1 & 0 \end{pmatrix}$$

Then $J^2 = -I$. Consider a linear operator $\mathcal{J} : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ with this matrix, i.e. $\mathcal{J}(Z) = JZ$ for any vector $Z \in \mathbb{R}^{2n}$ which we view as a column-vector. Then $\mathcal{J}^2 = -\mathrm{Id}$, and hence we can define on $\mathbb{R}^{2n}$ a complex structure (i.e.the multiplication by i by the formula

$$iZ = \mathcal{J}(Z),\ Z \in \mathbb{R}^{2n}.$$

This complex vector space is canonically isomorphic to $\mathbb{C}^n$, where we identify the real vector $(x_1, y_1, \dots, x_n, y_n) \in \mathbb{R}^{2n}$ with a complex vector $(z_1 = x_1 + iy_1, \dots, z_n x_n + iy_n)$.

On the other hand, any complex vector space can be viewed as a real vector space. In order to do that we just need to "forget" how to multiply by i. This procedure is called the *realification* of a complex vector space. For example, the realification of $\mathbb{C}^n$ is $\mathbb{R}^{2n}$. Sometimes to emphasize the realification operation we will denote the realification of a complex vector space V by $V_{\mathbb{R}}$. As the sets these to objects coincide.

Given a complex vector space V we can define linear combinations $\sum \lambda_i v_i$, where $\lambda_i \in \mathbb{C}$ are complex numbers, and thus similarly to the real case talk about really dependent, really independent vectors. Given vectors $v_1, \dots v_n \in V$ we define its *complex span* $\mathrm{Span}_{\mathbb{C}}(V_1, \dots, v_n)$ by the formula

$$\mathrm{Span}_{\mathbb{C}}(v_1, \dots, v_n) = \left\{ \sum_1^n \lambda_i v_i,\ \lambda_i \in \mathbb{C}. \right\}$$

. A *basis* of a complex vector space is a system of complex linear independent vectors $v_1, \ldots, v_n$ such that $\mathrm{Span}_{\mathbb{C}}(v_1, \ldots, v_n) = V$. The number of vectors in a complex basis is called the *complex dimension* of V and denoted $\dim_{\mathbb{C}} V$.

For instance $\dim \mathbb{C}^n = n$. On the other hand, its realification $\mathbb{R}^{2n}$ has real dimension $2n$. In particular, $\mathbb{C}$ is a complex vector space of dimension 1, and therefore it is called a complex *line* rather than a plane.

Exercise 11.3. *Let $v_1, \ldots, v_n$ be a complex basis of a complex vector space V. Find the real basis of its realification $V_{\mathbb{R}}$.*

Answer. $v_1, iv_1, v_2, iv_2, \ldots, v_n, iv_n$.

There is another important operation which associates with a real vector space V of real dimension n a complex vector space $V_{\mathbb{C}}$ of complex dimension n. It is done in a way similar to how we made complex numbers out of real numbers. As a real vector space the space $V_{\mathbb{C}}$ is just the direct sum $V \oplus V = \{(v, w);\ v, w \in V\}$. This is a real space of dimension $2n$. We then make $V \oplus V$ into a complex vector space by defining the multiplication by i by the formula:

$$i(v, w) = (-w, v).$$

We will write vector $(v, 0)$ simply by v and $(0, v) = i(v, 0)$ by iv. Hence, every vector of $V_{\mathbb{C}}$ can be written as $v + iw$, where $v, w \in V$. If $v_1, \ldots, v_n$ is a real basis of V, then the same vectors form a complex basis of $V_{\mathbb{C}}$.

Complexification of a real vector space

Given a real space V one can associate with it a complex vector space $V_{\mathbb{C}}$, called the *complexification* of V, as follows. It is made of vectors of V in exactly the same way as complex numbers made of reals. Namely, $V_{\mathbb{C}}$ consists of expressions $X + iY$, where $X, Y \in V$. We define a multiplication of a complex number $a + ib$ by a vector $X + iY$ by a formula

$$(a + ib)(X + iY) = aX - bY + i(aY + bX).$$

For instance, $\mathbb{C}^n$ is canonically isomorphic to $(\mathbb{R}^n)_{\mathbb{C}}$, because any vector $Z = (z_1 = x_1 + iy_1, \ldots, z_n = x_n + iy_n) \in \mathbb{C}^n$ can be uniquely written as $Z = X + iY$, where $X = (x_1, \ldots, x_n), Y =$

$(y_1, \ldots, y_n) \in \mathbb{R}^n$. Note that the *realification* of the space $V_\mathbb{C}$, i.e. the space $(V_\mathbb{C})_\mathbb{R}$ is canonically is just $V \oplus V = \{(X, Y),\ X, Y \in \mathbb{R}^n\}$.

If $v_1, \ldots, v_n$ is a basis of a V over $\mathbb{R}$, then the same vectors form a basis of the complexified space $V_\mathbb{C}$ over $\mathbb{C}$. Thus $\dim_\mathbb{R} V = \dim_\mathbb{C} V_\mathbb{C}$.

11.3 Complex linear maps

Complex linear maps and their realifications

Given two complex vector spaces V, W a map $\mathcal{A} : V \to W$ is called *complex linear* (or $\mathbb{C}$-linear) if $\mathcal{A}(X+Y) = \mathcal{A}(X) + \mathcal{A}(Y)$ and $\mathcal{A}(\lambda X) = \lambda \mathcal{A}(X)$ for any vectors $X, Y \in V$ and any *complex* number $\lambda \in \mathbb{C}$. Thus complex linearity is stronger condition than the real linearity. The difference is in the additional requirement that $\mathcal{A}(iX) = i\mathcal{A}(X)$. In other words, the operator $\mathcal{A}$ must *commute with the operation of multiplication by* i.

Any linear map $\mathcal{A} : \mathbb{C} \to \mathbb{C}$ is a multiplication by a complex number $a = c + id$. If we view $\mathbb{C}$ as $\mathbb{R}^2$ and right the real matrix of this map in the standard basis $1, i$ we get the matrix $\begin{pmatrix} c & -d \\ d & c \end{pmatrix}$. Indeed, $\mathcal{A}(1) = a = c + di$ and $\mathcal{A}(i) = ai = -d + ci$, so the first column of the matrix is equal to $\begin{pmatrix} c \\ d \end{pmatrix}$ and the second one is equal to $\begin{pmatrix} -d \\ c \end{pmatrix}$.

If we have bases $v_1, \ldots, v_n$ of V and $w_1, \ldots w_m$ of W then one can associate with $\mathcal{A}$ an $m \times n$ *complex* matrix A by the same rule as in the real case.

Recall (see Exercise 11.3) that vectors $v_1, v_1' = iv_1, v_2, v_2' = iv_2, \ldots, v_n, v_n' = iv_n$ and $w_1, w_1' = iw_1, w_2, w_2' = iw_2, \ldots w_m, w_m' = iw_m)$ form real bases of the realifications $V_\mathbb{R}$ and $W_\mathbb{R}$ of the spaces V and W. if

$$A = \begin{pmatrix} a_{11} & \ldots & a_{1n} \\ & \ldots & \\ a_{m1} & \ldots & a_{mn} \end{pmatrix}$$

is the complex matrix of $\mathcal{A}$ then the real matrix $A_\mathbb{R}$ of the map $\mathcal{A}$ is the real basis has order $2n \times 2n$ and is obtained from A by replacing each complex element $a_{kl} = c_{kl} + id_{kl}$ by a 2×2 matrix $\begin{pmatrix} c_{kl} & -d_{kl} \\ d_{kl} & c_{kl} \end{pmatrix}$.

Exercise 11.4. *Prove that*

$$\det A_{\mathbb{R}} = |\det A|^2.$$

Complexification of real linear maps

Given a real linear map $\mathcal{A} : V \to W$ one can define a complex linear map $\mathcal{A}_{\mathbb{C}} : V_{\mathbb{C}} \to W_{\mathbb{C}}$ by the formula

$$\mathcal{A}_{\mathbb{C}}(v + iw) = \mathcal{A}(v) + i\mathcal{A}(w).$$

. If A is the matrix of $\mathcal{A}$ in a basis $v_1, \dots, v_n$ then $\mathcal{A}_{\mathbb{C}}$ has the same matrix in the same basis viewed as a complex basis of $V_{\mathbb{C}}$. The operator $\mathcal{A}_{\mathbb{C}}$ is called the *complexification* of the operator $\mathcal{A}$.

In particular, one can consider $\mathbb{C}$-linear functions $V \to \mathbb{C}$ on a complex vector space V. Complex coordinates $z_1, \dots, z_n$ in a complex basis are examples of $\mathbb{C}$-linear functions, and any other $\mathbb{C}$-linear function on V has a form $c_1 z_1 + \dots c_n z_n$, where $c_1, \dots, c_n \in \mathbb{C}$ are complex numbers.

Complex-valued $\mathbb{R}$-linear functions

It is sometimes useful to consider also $\mathbb{C}$-valued $\mathbb{R}$-linear functions on a complex vector space V, i.e. $\mathbb{R}$-linear maps $V \to \mathbb{C}$ (i.e. a linear map $V_{\mathbb{R}} \to \mathbb{R}^2$). Such a $\mathbb{C}$-valued function has the form $\lambda = \alpha + i\beta$, where α, β are usual real linear functions. For instance the function $\bar{z}$ on $\mathbb{C}$ is a $\mathbb{C}$-valued $\mathbb{R}$-linear function which is not $\mathbb{C}$-linear.

If $z_1, \dots, z_n$ are complex coordinates on a complex vector space V then any $\mathbb{R}$-linear complex-valued function can be written as $\sum\limits_1^n a_i z_i + b_i \bar{z}_i$, where $a_i, b_i \in \mathbb{C}$ are complex numbers.

We can furthermore consider complex-valued tensors and, in particular complex-valued exterior forms. A $\mathbb{C}$-valued k-form λ can be written as $\alpha + i\beta$ where α and β are usual $\mathbb{R}$-valued k-forms. For instance, we can consider on $\mathbb{C}^n$ the 2-form $\omega = \frac{i}{2}\sum\limits_1^n z_k \wedge \bar{z}_k$. It can be rewritten as $\omega = \frac{i}{2}\sum\limits_1^n (x_k + iy_k) \wedge (x_k - iy_k) = \sum\limits_1^n x_k \wedge y_k$.

Part III

Calculus of differential forms

Chapter 12

Definition and properties of differential k-forms

12.1 Differential k-forms

Similarly to vector fields and differential 1-forms we can consider *fields of exterior k-forms*, $k \geq 0$, i.e. functions on $U \subset V$ which associate to each point $x \in U$ a k-form from $\Lambda^k(V_x^*)$. These fields of exterior k-forms are called *differential k-forms.*

Thus the relation between k-forms and differential k-forms is exactly the same as the relation between vectors and vector-fields, and linear functions and differential 1-forms.

A differential 0-form f on U associates with each point $x \in U$ a 0-form on V_x, i. e. a number $f(x) \in \mathbb{R}$. Thus differential 0-forms on U are just functions $U \to \mathbb{R}$.

All operations which we defined for exterior forms can be defined as pointwise operations on differential forms. In particular,

- given a differential k-form α and a differential ℓ-form β we define a differential $(k+l)$-form $\alpha \wedge \beta$ as $(\alpha \wedge \beta)_x = \alpha_x \wedge \beta_x$;

- given differential k-forms $\alpha_1, \dots, \alpha_m$ on a domain U and functions $f_1, \dots, f_m : U \to \mathbb{R}$ we can form a linear combination of $\alpha_1, \dots, \alpha_m$ with functional coefficients $f_1, \dots, f_m$:

$$(f_1\alpha_1 + \dots f_m\alpha_m)_x = f_1(x)(\alpha_1)_x + \dots + f_m(x)(\alpha_m)_x;$$

- given a differential k-form α we define $\star\alpha$ by $(\star\alpha)_x = \star(\alpha_x)$;

- given a differential k-form α and a vector field v on a domain $U \subset V$ we define the contraction $v \lrcorner \alpha$ as a differential $(k-1)$-form which satisfies $(v \lrcorner \alpha)_x = v(x) \lrcorner \alpha_x$.

12.2 Coordinate description of differential forms

In a similar way as any exterior k-form can be expressed as a linear combination of basic forms $x_{i_1} \wedge \dots \wedge x_{i_k} <\ i_1 < \dots < i_k$, any differential k form on a domain $U \subset V$ in an n-dimensional space V can be written as

$$\alpha = \sum_{1 \le i_1 < \dots < i_k \le n} f_{i_1 \dots i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k},$$

where $f_{i_1 \dots i_k} : U \to \mathbb{R}$ are *functions* on U.

For instance, any differential 2-form $\mathbb{R}$ on a 3-dimensional space can be written as

$$\omega = b_1(x) dx_2 \wedge dx_3 + b_2(x) dx_3 \wedge dx_1 + b_3(x) dx_1 \wedge dx_2$$

where b_1, b_2, and b_3 are functions on V. Any differential 3-form Ω on a 3-dimensional space V has the form

$$\Omega = c(x) dx_1 \wedge dx_2 \wedge dx_3$$

for a function c on V.

12.3 Operator f^*

Let U be a domain in a vector space V and $f : U \to W$ a smooth map. Then the differential df defines a *family* of linear maps

$$d_x f : T_x U \to T_{f(x)}(W), x \in V.$$

Equivalently we can say that the differential df is a *fiberwise linear map* $df : TU \to TW$ between the tangent bundles TU and TW.

A differential k-form ω on W defines an exterior k-form ω_y on the space $W_y = T_y W$ for each $y \in W$.

We define the differential k-form $f^*\omega$ on U by the formula

$$(f^*\omega)|_{V_x} = (d_x f)^*(\omega|_{W_{f(x)}}).$$

Here the notation $\omega|_{W_y}$ stands for the exterior k-form defined by the differential form ω on the space W_y.

In other words, for any k vectors, $H_1, \dots, H_k \in V_x$ we have

$$f^*\omega(H_1, \dots, H_k) = \omega(d_x f(H_1), \dots, d_x f(H_k)).$$

We say that the differential form $f^*\omega$ is *induced from* ω by the map f, or that $f^*\omega$ is the *pull-back of* ω *by* f.

Example 12.1. Let $\Omega = h(x)dx_1 \wedge \dots \wedge dx_n$. Then formula (9.2) implies

$$f^*\Omega = h \circ f \det Df dx_1 \wedge \dots \wedge dx_n.$$

Here

$$\det Df = \begin{vmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \dots & \dots & \dots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{vmatrix}$$

is the determinant of the Jacobian matrix of $f = (f_1, \dots, f_n)$.

Similarly to Proposition 1.11 we get

Proposition 12.2. *Given* 2 *maps*

$$U_1 \xrightarrow{f} U_2 \xrightarrow{g} U_3$$

and a differential k *form* ω *on* U_3 *we have*

$$(g \circ f)^*(\omega) = f^*(g^*\omega)\,.$$

An important special case of the pull-back operator f^* is the *restriction* operator. Namely Let $L \subset V$ be an affine subspace. Let $j : L \hookrightarrow V$ be the inclusion map. Then given a differential k-form α on a domain $U \subset V$ we can consider the form $j^*\alpha$ on the domain $U' := L \cap U$. This form is called the *restriction* of the form α to U' and it is usually denoted by $\alpha|_{U'}$. Thus the restricted form $\alpha|_{U'}$ is the same form α but viewed as a function of a point $a \in U'$ and vectors $T_1, \ldots, T_k \in L_a$, rather than in V_a.

12.4 Coordinate description of the operator f^*

Consider first the linear case. Let $\mathcal{A}$ be a linear map $V \to W$ and $\omega \in \Lambda^p(W^*)$. Let us fix coordinate systems $x_1, \ldots, x_k$ in V and $y_1, \ldots, y_n$ in W. If A is the matrix of the map $\mathcal{A}$ then we already have seen in Section 7.7 that

$$\mathcal{A}^* y_j = l_j(x_1, \ldots, x_k) = a_{j1}x_1 + a_{j2}x_2 + \ldots + a_{jk}x_k, \quad j = 1, \ldots, n,$$

and that for any exterior k-form

$$\omega = \sum_{1 \le i_1 < \ldots < i_p \le n} A_{i_1, \ldots, i_p}\, y_{i_1} \wedge \ldots \wedge y_{i_p}$$

we have

$$\mathcal{A}^*\omega = \sum_{1 \le i, < \ldots < i_p \le n} A_{i_1 \ldots i_p} l_{i_1} \wedge \ldots \wedge l_{i_p}.$$

Now consider the non-linear situation. Let ω be a differential p-form on a domain $U' \subset W$. Thus it can be written in the form

$$\omega = \sum A_{i_1 \ldots i_p}(y) dy_{i_1} \wedge \ldots dy_{i_p}$$

for some functions $A_{i_1 \ldots i_p}$ on U' .

Let U be a domain in V and $f : U \to U'$ a smooth map.

Proposition 12.3. *$f^*\omega = \sum A_{i_1, \ldots, i_p}(f(x)) df_{i_1} \wedge \ldots \wedge df_{i_p}$, where $f_1, \ldots, f_n$ are coordinate functions of the map f.*

Proof. For each point $x \in U$ we have, by definition,

$$f^*\omega|_{V_x} = (d_x f)^*(\omega|_{W_{fx}})$$

But the coordinate functions of the linear map $d_x f$ are just the differentials $d_x f_i$ of the coordinate functions of the map f. Hence the desired formula follows from the linear case proven in the previous proposition. ■

12.5 Examples

1. Consider the domain $U = \{r > 0, 0 \le \varphi < 2\pi\}$ on the plane $V = \mathbb{R}^2$ with Cartesian coordinates (r, φ). Let $W = \mathbb{R}^2$ be another copy of $\mathbb{R}^2$ with Cartesian coordinates (x, y). Consider a map $P : V \to W$ given by the formula

$$P(r, \varphi) = (r \cos \varphi, r \sin \varphi).$$

This map introduces (r, φ) as *polar coordinates* on the plane W. Set $\omega = dx \wedge dy$. It is called *the area form* on W. Then

$$P^*\omega = d(r \cos \varphi) \wedge d(r \sin \varphi) = (\cos \varphi dr + rd(\cos \varphi)) \wedge (\sin \varphi dr + rd(\sin \varphi) =$$

$$(\cos \varphi dr - r \sin \varphi d\varphi) \wedge (\sin \varphi dr + r \cos \varphi d\varphi) =$$

$$\cos \varphi \cdot \sin \varphi dr \wedge dr - r \sin^2 \varphi d\varphi \wedge dr + r \cos^2 \varphi dr \wedge d\varphi - r^2 \sin \varphi \cos \varphi d\varphi \wedge d\varphi =$$

$$r \cos^2 \varphi dr \wedge d\varphi + r \sin^2 dr \wedge d\varphi = rdr \wedge d\varphi.$$

2. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a smooth function and the map $F : \mathbb{R}^2 \to \mathbb{R}^3$ be given by the formula

$$F(x, y) = (x, y, f(x, y))$$

Let

$$\omega = P(x, y, z)dy \wedge dz + Q(x, y, z)dz \wedge dx + R(x, y, z)dx \wedge dy$$

be a differential 2-form on $\mathbb{R}^3$. Then

$$\begin{aligned}
F^*\omega &= P(x,y,f(x,y))dy \wedge df + \\
&+ Q(x,y,f(x,y))df \wedge dx + R(x,y,f(x,y))dx \wedge dy \\
&= P(x,y,f(x,y))dy \wedge (f_x dx + f_y dy) + \\
&+ Q(x,y,f(x,y))(f_x dx + f_y dy) \wedge dx + \\
&+ R(x,y,f(x,y))dx \wedge dy = \\
&= (R(x,y,f(x,y)) - P(x,y,f(x,y))f_x - Q(x,y,f(x,y))f_y)dx \wedge dy
\end{aligned}$$

where f_x, f_y are partial derivatives of f.

3. If $p > k$ then the pull-back $f^*\omega$ of a p-form ω on U to a k-dimensional space V is equal to 0.

12.6 Riemannian metric

We already discussed vector fields, line and and plane fields, differential 1-forms (i.e. fields of linear functions), and differential k-forms (i.e. fields of skew-symmetric k-linear functions) for $k > 1$. One can also consider other tensor fields, e.g. fields of not necessarily skew-symmetric functions, as well as fields of other objects, e.g. of quadratic forms. In this section we consider as an example fields of *symmetric bilinear functions*, and the tightly related fields of *quadratic forms*. We restrict the discussion to the 2-dimensional case.

A *Riemann* (or *Riemannian*) metric on a domain $U \subset \mathbb{R}^2$ is a field of scalar products on planes $\mathbb{R}^2_a$. Defining a scalar product is the same as defining a positive definite quadratic form. Hence, a Riemannian metric can be viewed as a field of positive definite quadratic forms $g_a : \mathbb{R}^2_a \to \mathbb{R}$. Recall that the differential 1-forms dx and dy are fields of linear functions on $\mathbb{R}^2_a$. Hence, $dx^2, dy^2, dxdy$ are fields of quadratic forms, and any field of quadratic forms on U can be written as a combination $E(x,y)dx^2 + 2F(x,y)dxdy + G(x,y)dy^2$, where $E(x,y), F(x,y), G(x,y)$ are functions on U (the coefficient 2 in front of F we put in order that the matrix of the quadratic form could be written as $\begin{pmatrix} E & F \\ F & G \end{pmatrix}$. Positive definiteness is equivalent to the conditions

$$E > 0, EG - F^2 > 0.$$

Equivalently, a Riemannian metric can be viewed as a field of symmetric positive definite bilinear forms, or 2-tensors. Viewed this way it can be written as

$$E(x,y)dx \otimes dx + F(x,y)dx \otimes dy + F(x,y)dx \otimes dy + G(x,y)dy \otimes dy.$$

Writing it in this form usually is not as convenient.

For any vector $T \in \mathbb{R}^2_a$ its *norm* $||T||_g$ with respect to the metric g is equal to $\sqrt{g_a(T)}$. We will omit the subscript g from the notation $||T||_g$ when the choice of g is clear from the context.

Example 12.4. 1. The differential quadratic form $dx^2 + dy^2$ is the standard Euclidean metric on $\mathbb{R}^2$.
2. The Riemannian metric $\frac{dx^2+dy^2}{y}$ on the upper-half plane $\mathbb{R}^2_+ = \{y > 0\} \subset \mathbb{R}^2$ is called the *Lobachevsky metric.*

Exercise 12.5. Let $e_x, e_y \in \mathbb{R}^2_a$ be the vectors $(1,0)$ and $(0,1)$. Show that

$$||e_x|| = \sqrt{E(a)}, ||e_y|| = \sqrt{G(a)},\ \cos\alpha = \frac{F(a)}{\sqrt{E(a)G(a)}},$$

where α is the angle between the vectors e_x and e_y, and

$$\text{Area}P(e_x, e_y) = \sqrt{E(a)G(a) - (F(a))^2},$$

where $P(e_x, e_y)$ is the parallelogram, spanned by e_x and e_y.

The differential 2-form $\sqrt{EG - F^2}dx \wedge dy$ is called the *area form* associated to the Riemannian metric g.

A Riemannian metric g on U allows us to compute the length of a curve. Given a path $\phi : [a,b] \to U$ we define the length of this path by integrating its velocity:

$$l(\phi) = \int\limits_a^b \Big|\Big|\frac{d\phi}{dt}(t)\Big|\Big|_g dt.$$

Thanks to the change of variable formula the length of the path is independent of the parameterization and orientation of the path (provided that we run through each point of the path only once).

Furthermore, given a Riemannian metric we can define the distance function between points $x, y \in U$ by the formula

$$d(x,y) = \inf_{\phi} l(\phi),$$

where the infimum is taken over all paths connecting x and y.

Let $U, \widetilde{U} \subset \mathbb{R}^2$ be two domains endowed with Riemannian metrics $g = Edx^2 + 2Fdxdy + Gdy^2$ on U and $\widetilde{g} = \widetilde{E}dx^2 + 2\widetilde{F}dxdy + \widetilde{G}dy^2$ on $\widetilde{U}$. A diffeomorphism (i.e. a smooth 1-1 map with a smooth inverse) $f : U \to \widetilde{U}$, is called an *isometry* if $f^*\widetilde{g} = g$, i.e. for any vector $T \in \mathbb{R}^2_a,\ \ a \in U$ we have

$$g(T) = \widetilde{g}(df(T))$$

.

For instance any linear orthogonal map $A : \mathbb{R}^2 \to \mathbb{R}^2$ (i.e. a rotation or a reflection), as well as any translation $z \mapsto z + c$ is clearly an isometry of a Euclidean metric $dx^2 + dy^2$ on $\mathbb{R}^2$. It is a surprising fact that the converse is also true.

Lemma 12.6. *Any isometry*

$$f : (\mathbb{R}^2, dx^2 + dy^2) \to (\mathbb{R}^2, dx^2 + dy^2)$$

is a composition of a linear orthogonal transformation with a translation.

Chapter 13

Exterior differential

13.1 Coordinate definition of the exterior differential

Let us denote by $\Omega^k(U)$ the space of all differential k-forms on U. When we will need to specify the class of smoothness of coefficients of the form we will use the notation $\Omega^k_l(U)$ for the space of differential k-forms with C^l-smooth coefficients, i.e. which depend C^l-smoothly on a point of U.

We will define a map

$$d : \Omega^k(U) \to \Omega^{k+1}(U),$$

or more precisely

$$d : \Omega^k_l(U) \to \Omega^{k+1}_{l-1}(U),$$

(thus assuming that $l \geq 1$), which is called the *exterior differential.* In the current form it was introduced by Élie Cartan, but essentially it was known already to Henri Poincaré.

We first define it in coordinates and then prove that the result is independent of the choice of the coordinate system. We will clarify later (see Lemma 16.7) the meaning of the exterior differential by showing how to define it without introducing any coordinates.

Let us fix a coordinate system $x_1, \dots, x_n$ in $V \supset U$. A differential k-form $\omega \in \Omega^k(U)$ can be written as

$$\omega = \sum_{i_1 < \dots < i_k} a_{i_1 \dots, i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k}$$

where $a_{i_1\dots i_k}$ are functions on the domain U. Define

$$d\omega := \sum_{i_1<\dots<i_k} da_{i_1\dots i_k} \wedge dx_{i_1} \wedge \dots \wedge dx_{i_k}.$$

Examples. 1. Let $\omega \in \Omega^1(U)$, i.e. $\omega = \sum_{i=1}^{n} a_i dx_i$. Then

$$d\omega = \sum_{i=1}^{n} da_i \wedge dx_i = \sum_{i=1}^{n} \left(\sum_{j=1}^{n} \frac{\partial a_i}{\partial x_j} dx_j \right) \wedge dx_i = \sum_{1\le i<j\le n} \left(\frac{\partial a_j}{\partial x_i} - \frac{\partial a_i}{\partial x_j} \right) dx_i \wedge dx_j.$$

For instance, when $n = 2$ we have

$$d(a_1dx_1 + a_2dx_2) = \left(\frac{\partial a_2}{\partial x_1} - \frac{\partial a_1}{\partial x_2} \right) dx_1 \wedge dx_2$$

For $n = 3$, we get

$$d(a_1dx_1+a_2dx_2+a_3dx_3) = \left(\frac{\partial a_3}{\partial x_2} - \frac{\partial a_2}{\partial x_3} \right) dx_2\wedge dx_3+\left(\frac{\partial a_1}{\partial x_3} - \frac{\partial a_3}{\partial x_1} \right) dx_3\wedge dx_1+\left(\frac{\partial a_2}{\partial x_1} - \frac{\partial a_1}{\partial x_2} \right) dx_1\wedge dx_2.$$

2. Let $n = 3$ and $\omega \in \Omega^2(U)$. Then

$$\omega = a_1dx_2 \wedge dx_3 + a_2dx_3 \wedge dx_1 + a_3dx_1 \wedge dx_2$$

and

$$\begin{aligned} d\omega &= da_1 \wedge dx_2 \wedge dx_3 + da_2 \wedge dx_3 \wedge dx_1 + da_3 \wedge dx_1 \wedge dx_2 \\ &= \left(\frac{\partial a_1}{\partial x_1} + \frac{\partial a_2}{\partial x_2} + \frac{\partial a_3}{\partial x_3} \right) dx_1 \wedge dx_2 \wedge dx_3 \end{aligned}$$

3. For 0-forms, i.e. functions the exterior differential coincides with the usual differential of a function.

13.2 Properties of the operator d

Proposition 13.1. *For any two differential forms, $\alpha \in \Omega^k(U)$, $\beta \in \Omega^l(U)$ we have*

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta.$$

Proof. We have

$$\begin{aligned}
\alpha &= \sum_{i_1 < \ldots < i_k} a_{i_1 \ldots i_k} dx_{i_1} \wedge \ldots \wedge dx_{i_k} \\
\beta &= \sum_{j_1 < \ldots < j_l} b_{j_i \ldots j_l} dx_{j_1} \wedge \ldots \wedge dx_{j_l} \\
\alpha \wedge \beta &= \left(\sum_{i_1 < \ldots < i_k} a_{i_1 \ldots i_k} dx_{i_1} \wedge \ldots \wedge dx_{i_k} \right) \wedge \left(\sum_{j_i < \ldots < j_l} b_{j_1 \ldots j_l} dx_{j_1} \wedge \ldots \wedge dx_{j_l} \right) \\
&= \sum_{\substack{i_1 < \ldots < i_k \\ j_1 < \ldots < j_l}} a_{i_1 \ldots i_k} b_{j_1 \ldots j_l} dx_{i_l} \wedge \ldots \wedge dx_{i_k} \wedge dx_{j_l} \wedge \ldots \wedge dx_{j_l}
\end{aligned}$$

$$
\begin{aligned}
d(\alpha\wedge\beta) &= \sum_{\substack{i_1<\ldots<i_k\\ j_1<\ldots<j_l}} (b_{j_1\ldots j_l} da_{i_1\ldots i_k} + a_{i_1\ldots i_k} db_{j_1\ldots j_l}) \wedge dx_{i_1}\wedge\ldots\wedge dx_{i_k}\wedge dx_{j_1}\wedge\ldots\wedge dx_{j_l} \\
&= \sum_{\substack{i_1<\ldots<i_k\\ j_1<\ldots<j_l}} b_{j_1\ldots j_l} da_{i_1\ldots i_k}\wedge dx_{i_1}\wedge\ldots\wedge dx_{i_k}\wedge dx_{j_l}\wedge\ldots\wedge dx_{j_l} \\
&+ \sum_{\substack{i_1<\ldots<i_k\\ j_1<\ldots<j_l}} a_{i_1\ldots i_k} db_{j_1\ldots j_l}\wedge dx_{i_1}\wedge\ldots\wedge dx_{i_k}\wedge dx_{j_1}\wedge\ldots\wedge dx_{j_l} \\
&= \left(\sum_{i_1<\ldots<i_k} da_{i_1\ldots i_k}\wedge dx_{i_1\ldots}\wedge dx_{i_k}\right)\wedge\left(\sum b_{j_1\ldots j_l} dx_{j_1}\wedge\ldots\wedge dx_{j_l}\right) \\
&+ (-1)^k\left(\sum_{i_1<\ldots<i_k} a_{i_1\ldots i_k} dx_{i_1}\wedge\ldots\wedge dx_{i_k}\right)\wedge\left(\sum_{j_i<\ldots<j_l} db_{j_1\ldots j_l}\wedge dx_{j_1}\wedge\ldots\wedge dx_{j_l}\right) \\
&= d\alpha\wedge\beta + (-1)^k\alpha\wedge d\beta.
\end{aligned}
$$

Notice that the sign $(-1)^k$ appeared because we had to make k transposition to move $db_{j_1\ldots j_l}$ to its place. ■

Proposition 13.2. *For any differential k-form ω we have*

$$dd\omega = 0.$$

Proof. Let $\omega = \sum_{i_1<\ldots<i_k} a_{i_1\ldots i_k} dx_{i_1}\wedge\ldots\wedge dx_{i_k}$. Then we have

$$d\omega = \sum_{i_1<\ldots<i_k} da_{i_1\ldots i_k}\wedge dx_{i_1}\wedge\ldots\wedge dx_{i_k}.$$

Applying Proposition 13.1 we get

$$
\begin{aligned}
dd\omega = \sum_{i_1<\ldots<i_k} & dda_{i_1\ldots i_k}\wedge dx_{i_1}\wedge\ldots\wedge dx_{i_k} - da_{i_1\ldots i_k}\wedge ddx_{i_1}\wedge\ldots\wedge dx_{i_k} + \ldots \\
& + (-1)^k da_{i_1\ldots i_k}\wedge dx_{i_1}\wedge\ldots\wedge ddx_{i_k}.
\end{aligned}
$$

But $ddf = 0$ for any function f as was shown above. Hence all terms in this sum are equal to 0, i.e. $ddw = 0$. ■

Definition. A k-form ω is called *closed* if $d\omega = 0$. It is called *exact* if there exists a $(k-1)$-form θ such that $d\theta = \omega$. The form θ is called the *primitive* of the form ω.

If $\omega = \sum a_i dx_i$ then $d\omega = \sum\limits_{i<j} \left(\frac{\partial a_j}{\partial x_i} - \frac{\partial a_j}{\partial x_j}\right) dx_i \wedge dx_j$. Hence,

$$d\omega = 0 \quad \Leftrightarrow \quad \frac{\partial a_j}{\partial x_i} = \frac{\partial a_j}{\partial x_j} \text{ for all } i, j.$$

Thus the *new and old definition of closed 1-forms coincide.*

The statement $dd\omega = 0$ can be reformulated as follows:

Corollary 13.3. *Every exact form is closed.*

The converse is not true in general, as we already had seen for a differential 1-form $\omega = \frac{xdy - ydx}{x^2+y^2}$ on the punctured plane $U = \mathbb{R}^2 \setminus 0$. We will see later similar examples of closed but not exact k-forms for $k > 1$.

Proposition 13.4. *Operators f^* and d commute, i.e. for any differential k-form $\omega \in \Omega^k(W)$, and a smooth map $f : U \to W$ we have*

$$df^*\omega = f^*d\omega$$

.

Proof. Suppose first that $k = 0$, i.e. ω is a function $\varphi : W \to \mathbb{R}$. Then $f^*\varphi = \varphi \circ f$. Then $d(\varphi \circ f) = f^*d\varphi$. Indeed, for any point $x \in U$ and a vector $X \in V_x$ we have

$$d(\varphi \circ f)(X) = d\varphi(d_x f(X))$$

(chain rule)

But $d\varphi(d_x f(X)) = f^*(d\varphi(X))$.

Consider now the case of arbitrary k-form ω,

$$\omega = \sum_{i_1 < \ldots < i_k} a_{i_1 \ldots i_k} dx_{i_1} \wedge \ldots \wedge dx_{i_k}.$$

Then

$$f^*\omega = \sum_{i_1<\ldots<i_k} a_{i_1\ldots i_k} \circ f df_{i_1} \wedge \ldots \wedge df_{i_k}$$

where $f_1, \ldots, f_n$ are coordinate functions of the map f. Using the previous theorem and taking into account that $d(df_i) = 0$, we get

$$d(f^*\omega) = \sum_{i_1<\ldots<i_k} d(a_{i_1\ldots i_k} \circ f) \wedge df_{i_1} \wedge \ldots \wedge df_{i_k}.$$

On the other hand

$$d\omega = \sum_{i_1<\ldots<i_k} da_{i_1\ldots i_k} \wedge dx_{i_1} \wedge \ldots \wedge dx_{i_k}$$

and therefore

$$f^*dw = \sum_{i_1<\ldots<i_k} f^*(da_{i_1\ldots i_k}) \wedge df_{i_1} \wedge \ldots \wedge df_{i_k}.$$

But according to what is proven above, we have

$$f^*da_{i_1\ldots i_k} = d(a_{i_1\ldots i_k} \circ f)$$

Thus,

$$f^*d\omega = \sum_{i_1<\ldots<i_k} d(a_{i_1\ldots i_k} \circ f) \wedge df_{i_1} \wedge \ldots \wedge df_{i_k} = df^*\omega$$

∎

The above theorem shows, in particular, that the definition of the exterior differential is independent of the choice of the coordinates. Moreover, one can even use non-linear (curvilinear) coordinate systems, like polar coordinates on the plane.

13.3 Geometric definition of the exterior differential

We will show later (see Lemma 16.7) that one can give another equivalent definition of the operator d without using any coordinates at all. But as a first step we give below an equivalent definition of the exterior differential which is manifestly invariant of a choice of *affine* coordinates.

Given a point $a \in V$ and vectors $h_1, \dots, h_k \in V_a$ we will denote by $P_a(h_1, \dots, h_k)$ the k-dimensional parallelepiped

$$\left\{ a + \frac{1}{2}\sum_{1}^{k} u_j h_j;\ |u_1|, \dots, |u_k| \le 1 \right\}$$

centered at the point a with its sides parallel to the vectors $h_1, \dots, h_k$. For instance, for $k = 1$ the parallelepiped $P_a(h)$ is an interval centered at a of length $|h|$.

As it was explained above in Section 13.1 for a 0-form $f \in \Omega^0(U)$, i.e. a function $f : U \to \mathbb{R}$ its exterior differential is just its usual differential, and hence it can be defined by the formula

$$(df)_a(h) = d_a f(h) = \lim_{t\to 0} \frac{1}{t}\left(f\left(a + \frac{th}{2}\right) - f\left(a - \frac{th}{2}\right)\right), t \in \mathbb{R}, h \in V_a, a \in U$$

Proposition 13.5. *Let $\alpha \in \Omega_1^{k-1}(U)$ be a differential $(k-1)$-form on a domain $U \subset \mathbb{R}^n$. Then for any point $a \in U$ and any vectors $h_1, \dots, h_k \in V_a$ we have*

$$\begin{aligned}(d\alpha)_a(h_1, \dots, h_k) &= \lim_{t\to 0} \frac{1}{t^k}\sum_{j=1}^{k}(-1)^{j-1}\left(\alpha_{a+\frac{th_j}{2}}(th_1, \dots, \overset{j}{\vee}, \dots, th_k) - \alpha_{a-\frac{th_j}{2}}(th_1, \dots, \overset{j}{\vee}, \dots, th_k)\right)\\ &= \lim_{t\to 0} \frac{1}{t}\sum_{j=1}^{k}(-1)^{j-1}\left(\alpha_{a+\frac{th_j}{2}}(h_1, \dots, \overset{j}{\vee}, \dots, h_k) - \alpha_{a-\frac{th_j}{2}}(h_1, \dots, \overset{j}{\vee}, \dots, h_k)\right)\end{aligned} \tag{13.3.1}$$

For instance, for a 1-form $\alpha \in \Omega^1(U)$, we have

$$\begin{aligned}(d\alpha)_a(h_1, h_2) &:= \lim_{t\to 0} \frac{1}{t^2}\left(\alpha_{a+\frac{th_1}{2}}(th_2) + \alpha_{a-\frac{th_1}{2}}(-th_2) + \alpha_{a+\frac{th_2}{2}}(-th_1) + \alpha_{a-\frac{th_2}{2}}(th_1)\right)\\ &= \lim_{t\to 0} \frac{1}{t}\left(\alpha_{a+\frac{th_1}{2}}(h_2) - \alpha_{a-\frac{th_1}{2}}(h_2) - \alpha_{a+\frac{th_2}{2}}(h_1) + \alpha_{a-\frac{th_2}{2}}(h_1)\right)\end{aligned}$$

Proof. First we observe that for a C^1-smooth form α the limit in the right-hand side of formula (13.3.1) exists and define a k-form (i.e. for each a it is a multilinear skew-symmetric function of vectors $h_1, \dots, h_k \in \mathbb{R}^n_a$). We can also assume that the vectors $h_1, \dots, h_k$ are linearly independent. Otherwise, both definitions give 0. Denote $L_a = \mathrm{Span}(h_1, \dots, h_k) \subset V_a$ and consider the restriction $\alpha' := \alpha|_{L\cap U}$. Let $(y_1, \dots, y_k)$ be coordinates in L_a corresponding to the basis $h_1, \dots, h_k$. Then we can write $\alpha' := \sum_{i=1}^{k} P_i dy_1 \wedge \dots \overset{i}{\vee} \dots \wedge dy_k$ and thus

$$\begin{aligned}
&\lim_{t\to 0} \frac{1}{t} \sum_{j=1}^{k} (-1)^{j-1} \left(\alpha_{a+\frac{th_j}{2}}(h_1, \dots, \overset{j}{\vee}, \dots, h_k) - \alpha_{a-\frac{th_j}{2}}(h_1, \dots, \overset{j}{\vee}, \dots, h_k) \right) \\
&= \sum_{j=1}^{k} \sum_{i=1}^{k} (-1)^{j-1} \lim_{t\to 0} \frac{P_i(a + \frac{th_j}{2}) - P_i(a - \frac{th_j}{2})}{t} dy_1 \wedge \dots \overset{i}{\vee} \dots \wedge dy_k(h_1, \dots, \overset{j}{\vee}, \dots, h_k) \qquad (13.3.2) \\
&= \sum_{j=1}^{k} \sum_{i=1}^{k} (-1)^{j-1} \frac{\partial P_i}{\partial y_j}(a) dy_1 \wedge \dots \overset{i}{\vee} \dots \wedge dy_k(h_1, \dots, \overset{j}{\vee}, \dots, h_k).
\end{aligned}$$

But

$$dy_1 \wedge \dots \overset{i}{\vee} \dots \wedge dy_k(h_1, \dots, \overset{j}{\vee}, \dots, h_k) = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j. \end{cases}$$

Hence, the expression in (13.3.2) is equal to $\sum_{1}^{k} (-1)^{i-1} \frac{\partial P_i}{\partial y_i}(a)$. But

$$\begin{aligned}
d\alpha' &= \sum_{i=1}^{k} dP_i \wedge dy_1 \wedge \dots \overset{i}{\vee} \dots \wedge dy_k \\
&= \sum_{i=1}^{k} \left(\sum_{j=1}^{k} \frac{\partial P_i}{\partial y_j} dy_j \right) \wedge dy_1 \wedge \dots \overset{i}{\vee} \dots \wedge dy_k \\
&= \sum_{i=1}^{k} (-1)^{i-1} \frac{\partial P_i}{\partial y_i} dy_1 \wedge \dots \wedge dy_i \wedge \dots \wedge dy_k,
\end{aligned}$$

and therefore

$$(d\alpha')_a(h_1, \dots, h_k) = \sum_{1}^{k} (-1)^{i-1} \frac{\partial P_i}{\partial y_i}(a),$$

which matches the expression in (13.3.2). Hence, the two definitions of the exterior differential coincide.

13.4 Closed differential k-forms are locally exact

We will prove in this section a generalization of Proposition 5.9 to closed differential k-forms of arbitrary order k.

Theorem 13.6 (*Poincaré's lemma*)**.** *Let $U \subset V$ be a star-shaped domain in a m-dimensional space, and $\alpha \in \Omega^k(U)$, $1 \le k \le m$ be a closed differential k-form in U. Then α is exact, i.e. there exists a differential $(k-1)$-form $\beta \in \Omega^{k-1}(U)$ such that $d\beta = \alpha$.*

In particularly, we get

Corollary 13.7. *Any closed differential k-form is locally exact.*

The following lemma is the key to the proof.

Lemma 13.8. *Let $U \subset V$ be any domain (not necessarily star-shaped). Consider the product $\widehat{U} := U \times [0,1]$. Let $j_0, j_1 : U \to \widehat{U}$ be the inclusion maps $j_0(x) = (x,0) \in \widehat{U} = U \times [0,1]$ and $j_1(x) = (x,1) \in U \times [0,1]$. Then for any $k \ge 1$ there exists a linear map $K : \Omega^k(\widehat{U}) \to \Omega^{k-1}(U)$ such that*

$$d \circ K + K \circ d = j_1^* - j_0^*,$$

i.e. for each differential k-form $\alpha \in \Omega^k(\widehat{U})$ one has

$$dK(\alpha) + K(d\alpha) = j_1^*\alpha - j_0^*\alpha.$$

Remark 13.9. Note that the first d in the above formula denotes the exterior differential $\Omega^k(U) \to \Omega^{k+1}(U)$, while the second one is the exterior differential $\Omega^k(\widehat{U}) \to \Omega^{k+1}(\widehat{U})$.

Proof. Let us write a point in $\widehat{U} = U \times [0,1]$ as (x,t), $x \in U, t \in [0,1]$. Given a differential k-form $\alpha \in \Omega^k(\widehat{U})$ we first contract it with the vector field $\frac{\partial}{\partial t}$, and then integrate the resultant form with respect to the t-coordinate. More precisely, note that any k-form α on $U \times [0,1]$ can be written as $\alpha = \beta(t) + dt \wedge \gamma(t)$, $t \in [0,1]$, where for each $t \in [0,1]$

$$\beta(t) \in \Omega^k(U), \gamma(t) \in \Omega^{k-1}(U).$$

Then $\frac{\partial}{\partial t} \lrcorner\, \alpha = \gamma(t)$ and we define $K(\alpha) = \int\limits_0^1 \gamma(t)dt$.

If $(u_1, \dots, u_m)$ are coordinates on U then $\gamma(t)$ can be written as $\gamma(t) = \sum\limits_{1 \le i_1 < \dots < i_{\le} m} h_{i_1 \dots i_k}(t) du_{i_1} \wedge \dots \wedge du_{i_k}$, and hence

$$K(\alpha) = \int\limits_0^1 \gamma(t)dt = \sum_{1 \le i_1 < \dots < i_{\le} m} \left(\int\limits_0^1 h_{i_1 \dots i_k}(t)dt \right) du_{i_1} \wedge \dots \wedge du_{i_k}.$$

Clearly, K is a linear operator $\Omega^k(\widehat{U}) \to \Omega^{k-1}(U)$.

Note that if $\alpha = \beta(t) + dt \wedge \gamma(t) \in \Omega^k(\widehat{U})$ then

$$j_0^*\alpha = \beta(0), \quad j_1^*\alpha = \beta(1).$$

We further have

$$K(\alpha) = \int_0^1 \gamma(t)dt;$$

$$d\alpha = d_{\widehat{U}}(\beta(t) + dt \wedge \gamma(t)) = d_U\beta(t) + dt \wedge \dot{\beta}(t) - dt \wedge d_U\gamma(t) = d_U\beta(t) + dt \wedge (\dot{\beta}(t) - d_U\gamma(t)),$$

where we denoted $\dot{\beta}(t) := \frac{\partial\beta(t)}{\partial t}$ and $I = [0,1]$. Here the notation $d_{\widehat{U}}$ stands for exterior differential on $\Omega(\widehat{U})$ and d_U denotes the exterior differential on $\Omega(U)$. In other words, when we write $d_U\beta(t)$ we view $\beta(t)$ as a form on U depending on t as a parameter. We do not write any subscript for d when there could not be any misunderstanding.

Hence,

$$K(d\alpha) = \int_0^1 (\dot{\beta}(t) - d_U\gamma(t))dt = \beta(1) - \beta(0) - \int_0^1 d_U\gamma(t)dt;$$

$$d(K(\alpha)) = \int_0^1 d_U\gamma(t)dt.$$

Therefore,

$$\begin{aligned} d(K(\alpha)) + K(d(\alpha)) &= \beta(1) - \beta(0) - \int_0^1 d_U\gamma(t)dt + \int_0^1 d_U\gamma(t)dt \\ &= \beta(1) - \beta(0) = j_1^*(\alpha) - j_0^*(\alpha). \end{aligned}$$

■

Proof of Theorem 13.6. Let us assume that U is star-shaped with respect to $0 \in V$. Cosider a map $f : \widehat{U} = U \times [0,1] \to U$ given by the formula

$$f(x,t) = tx, \quad x \in U, t \in [0,1].$$

Consider the pull-back differential k-form $\widehat{\alpha} = f^*\alpha$ on $\widehat{U}$ and note that is is closed as the pull-back of a closed form. Applying to it Lemma 13.8:

$$j_1^*\widehat{\alpha} - j_0^*\widehat{\alpha} = dK(\widehat{\alpha}) + K(d\widehat{\alpha}) = dK(\widehat{\alpha}).$$

But $j^*\widehat{\alpha} = \alpha$ and $j^*\widehat{\alpha} = 0$. Hence,

$$\alpha = d\beta, \text{ where } \beta = K(\widehat{\alpha}).$$

■

13.5 More about vector fields

Similarly to the case of linear coordinates, given any curvilinear coordinate system $(u_1, \dots, u_n)$ in U, one denotes by

$$\frac{\partial}{\partial u_1}, \dots, \frac{\partial}{\partial u_n}$$

the vector fields which correspond to the partial derivatives with respect to the coordinates $u_1, \dots, u_n$. In other words, the vector field $\frac{\partial}{\partial u_i}$ is tangent to the u_i-coordinate lines and represents the velocity vector of the curve

$$\begin{aligned} &u_1 = c_1, \\ &\dots \\ &u_{i-1} = c_{i-1}, \\ &u_{i+1} = c_{i+1}, \\ &\dots \\ &u_n = c_n\,, \end{aligned}$$

parameterized by the coordinate u_i.

For instance, suppose we are given spherical coordinates (r, θ, φ), $r \geq 0, \phi \in [0, \pi], \theta \in [0, 2\pi)$ in $\mathbb{R}^3$. The spherical coordinates are related to the Cartesian coordinates (x, y, z) by the formulas

$$x = r \sin\varphi \cos\theta, \tag{13.5.1}$$

$$y = r \sin\varphi \sin\theta, \tag{13.5.2}$$

$$z = r \cos\varphi\,. \tag{13.5.3}$$

Then the vector fields

$$\frac{\partial}{\partial r}, \quad \frac{\partial}{\partial \varphi}, \quad \text{and} \quad \frac{\partial}{\partial \theta}$$

are mutually orthogonal and define the same orientation of the space as the standard Cartesian coordinates in $\mathbb{R}^3$. We also have $||\frac{\partial}{\partial r}|| = 1$. However the length of vector fields $\frac{\partial}{\partial \varphi}$ and $\frac{\partial}{\partial \theta}$ vary. When r and θ are fixed and φ varies, then the corresponding point (r, θ, φ) is moving along a meridian of radius r with a constant angular speed 1. Hence,

$$\left\| \frac{\partial}{\partial \varphi} \right\| = r \, .$$

When r and φ are fixed and θ varies, then the point (r, φ, θ) is moving along a latitude of radius $r \sin \varphi$ with a constant angular speed 1. Hence,

$$\left\| \frac{\partial}{\partial \theta} \right\| = r \sin \varphi \, .$$

Note that it is customary in Physics to introduce unit vector fields in the direction of $\frac{\partial}{\partial r}$, $\frac{\partial}{\partial \theta}$ and $\frac{\partial}{\partial \varphi}$:

$$\mathbf{e}_r := \frac{\partial}{\partial r}, \ \mathbf{e}_\varphi := \frac{1}{r} \frac{\partial}{\partial \varphi}, \ \ \mathbf{e}_\theta = \frac{1}{r \sin \varphi} \frac{\partial}{\partial \theta} \ ,$$

which form an orthonormal basis at every point. The vector field $\mathbf{e}_r$ is not defined at the origin, while $\mathbf{e}_\theta$ and $\mathbf{e}_\varphi$ are not defined along the z-axis.

Given any curvilinear coordinate system $(u_1, \dots, u_n)$, the chain rule allows us to express the vector fields $\frac{\partial}{\partial u_1}, \dots, \frac{\partial}{\partial u_n}$ through the vector fields $\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n}$. Indeed, for any function $f : U \to \mathbb{R}$ we have

$$\frac{\partial f}{\partial u_i} = \sum_{j=1}^{n} \frac{\partial f}{\partial x_j} \frac{\partial x_j}{\partial u_i} \, ,$$

and, therefore,

$$\frac{\partial}{\partial u_i} = \sum_{j=1}^{n} \frac{\partial x_j}{\partial u_i} \frac{\partial}{\partial x_j} \, ,$$

For instance, spherical coordinates (r, θ, ϕ) are related to the Cartesian coordinates (x, y, z) by the formulas (13.5.1), and hence we derive the following expression of the vector fields $\frac{\partial}{\partial r}, \frac{\partial}{\partial \theta}, \frac{\partial}{\partial \varphi}$

through the vector fields $\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}$:

$$\begin{aligned}
\frac{\partial}{\partial r} &= \sin\varphi\cos\theta\frac{\partial}{\partial x} + \sin\varphi\sin\theta\frac{\partial}{\partial y} + \cos\varphi\frac{\partial}{\partial z}\,, \\
\frac{\partial}{\partial \theta} &= -r\sin\varphi\sin\theta\frac{\partial}{\partial x} + r\sin\varphi\cos\theta\frac{\partial}{\partial y}\,, \\
\frac{\partial}{\partial \varphi} &= r\cos\varphi\cos\theta\frac{\partial}{\partial x} + r\cos\varphi\sin\theta\frac{\partial}{\partial y} - r\sin\varphi\frac{\partial}{\partial z}\,.
\end{aligned}$$

Respectively, we get

$$\begin{aligned}
\mathbf{e}_r &= \sin\varphi\cos\theta\frac{\partial}{\partial x} + \sin\varphi\sin\theta\frac{\partial}{\partial y} + \cos\varphi\frac{\partial}{\partial z}\,, \\
\mathbf{e}_\theta &= -\sin\theta\frac{\partial}{\partial x} + \cos\theta\frac{\partial}{\partial y}\,, \\
\mathbf{e}_\varphi &= \cos\varphi\cos\theta\frac{\partial}{\partial x} + \cos\varphi\sin\theta\frac{\partial}{\partial y} - \sin\varphi\frac{\partial}{\partial z}\,.
\end{aligned}$$

13.6 Summary of isomorphisms

Let U be a domain in a space V of dimension n. Consider spaces $\Omega^k(U)$, $k = 0, \dots, n$, of differential k-forms on U, and the space $\mathrm{Vect}(U)$ of vector fields on U.

Let us fix a volume form $\omega \in \Omega^n(U)$ that is any nowhere vanishing differential n-form. In coordinates ω can be written as

$$\omega = f(x)dx_1 \wedge dx_2 \wedge dx_n,$$

where the function $f : U \to \mathbb{R}$ is never equal to 0. The choice of the form ω allows us to define the following isomorphisms.

- $\Lambda_\omega : C^\infty(U) = \Omega_0(U) \to \Omega^n(U), \quad \Lambda_\omega(h) = h\omega$ for any function $h \in C^\infty(U)$.

- $\lrcorner_\omega : \mathrm{Vect}(U) \to \Omega^{n-1}(U) \quad \lrcorner_\omega(v) = v \lrcorner \omega$.

Sometimes we will omit the subscript ω an write just Λ and $\lrcorner$.

Our next isomorphisms depends on a choice of a scalar product $\langle\cdot,\cdot\rangle$ in V. Fixing a scalar product allows us to define isomorphisms

- $\mathcal{D}$: Vect$(U) \to \Omega^1(U)$, which associates with a vector field v on U a differential 1-form $\mathcal{D}(v) = \langle v, \cdot \rangle$.

Finally we recall the Hodge star operator

- $\star : \Omega_k(U) \to \Omega^{n-k}(U)$

which depends on the scalar product and a *choice of an orientation.*

Let us write down the coordinate expressions for all these isomorphisms in the 3-dimensional case. Fix a Cartesian coordinate system (x_1, x_2, x_3) in V which defines its given orientation. Suppose also that $\omega = dx_1 \wedge dx_2 \wedge dx_3$. Then

$$\Lambda(h) = \star h = h dx_1 \wedge dx_2 \wedge dx_3$$

for $h : U \to \mathbb{R}$.

For a vector field $v = v_1 \frac{\partial}{\partial x_1} + v_2 \frac{\partial}{\partial x_2} + v_3 \frac{\partial}{\partial x_3}$ we have

$$\mathcal{D}(v) = v_1 dx_1 + v_2 dx_2 + v_3 dx_3 \text{ and}$$

$$\lrcorner(v) = \star\mathcal{D}(v) = v_1 dx_2 \wedge dx_3 + v_2 dx_3 \wedge dx_1 + v_3 dx_1 \wedge dx_2.$$

Furthermore,

$$\star (a_1 dx_1 + a_2 dx_2 + a_3 dx_3) = a_1 dx_2 \wedge dx_3 + a_2 dx_3 \wedge dx_1 + a_3 dx_1 \wedge dx_2,$$

$$\star (a_1 dx_2 \wedge dx_3 + a_2 dx_3 \wedge dx_1 + a_3 dx_1 \wedge dx_2) = a_1 dx_1 + a_2 dx_2 + a_3 dx_3.$$

13.7 Gradient, curl and divergence of a vector field

The above isomorphism, combined with the operation of exterior differentiation, allows us to define the following operations on the vector fields. First recall that for a function $f \in C^\infty(U)$, its gradient vector field is defined as

$$\text{grad} f = \mathcal{D}^{-1}(df).$$

Now let $v \in$ Vect (U) be a vector field. Then its *divergence* $\operatorname{div} v$ is the function defined by the formula

$$\operatorname{div} v = \Lambda^{-1}(d(\lrcorner v)) = \Lambda^{-1}(d(v \lrcorner \omega))$$

In other words, we take the 2-form $v\lrcorner\omega$ (ω is the volume form) and compute its exterior differential $d(v\lrcorner\omega)$. The result is a 3-form, and, therefore, it is proportional to the volume form ω, i.e. $d(v\lrcorner\omega) = h\omega$. This proportionality coefficient (which is a function; it varies from point to point) is the divergence: div $v = h$.

Given a vector field v, its *curl* is another vector field defined by the formula

$$\text{curl } v := \lrcorner^{-1} d(\mathcal{D}v) = \mathcal{D}^{-1} * d(\mathcal{D}v).$$

If one fixes a Cartesian coordinate system in V defining its orientation, and if $\omega = dx_1 \wedge dx_2 \wedge dx_3$ then we get the following formulas

$$\begin{aligned}
\text{grad } f &= \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \frac{\partial f}{\partial x_3}\right) \\
\text{div } v &= \frac{\partial v_1}{\partial x_1} + \frac{\partial v_2}{\partial x_2} + \frac{\partial v_3}{\partial x_3} \\
\text{curl } v &= \left(\frac{\partial v_3}{\partial x_2} - \frac{\partial v_2}{\partial x_3}, \frac{\partial v_1}{\partial x_3} - \frac{\partial v_3}{\partial x_1}, \frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2}\right)
\end{aligned}$$

where $v = (v_1, v_2, v_3)$.

We will discuss the geometric meaning of these operations later in Section 17.3.

13.8 Example: expressing vector analysis operations in spherical coordinates

Sometimes in applications it is useful to be able to express vector calculus operations in curvilinear coordinates. In this section we show how this can be done in spherical cordinates.

Given spherical coordinates (r, θ, ϕ) in $\mathbb{R}^3$, consider orthonormal vector fields

$$\mathbf{e}_r = \frac{\partial}{\partial r}, \ \mathbf{e}_\varphi = \frac{1}{r}\frac{\partial}{\partial \varphi} \ \text{ and } \ \mathbf{e}_\theta = \frac{1}{r\sin\varphi}\frac{\partial}{\partial\theta}$$

tangent to the coordinate lines and which form orthonormal basis at every point.

Operator $\mathcal{D}$.

Given a vector field $\mathbf{v} = a\mathbf{e}_r + b\mathbf{e}_\varphi + c\mathbf{e}_\theta$ we compute $\mathcal{D}\,\mathbf{v}$. Given a vector field $h = h_1\mathbf{e}_r + h_2\mathbf{e}_\theta + h_3\mathbf{e}_\varphi$ let us write $\mathcal{D}\,\mathbf{v} = c_1dr + c_2d\varphi + c_3d\theta$. Then

$$\begin{aligned}
\mathcal{D}\,\mathbf{v}(h) &= \langle\,\mathbf{v}, h\rangle = ah_1 + bh_2 + ch_3 \\
&= (c_1dr + c_2d\varphi + c_3d\theta)(h_1\mathbf{e}_r + h_2\mathbf{e}_\varphi + h_3\mathbf{e}_\theta) \\
&= c_1h_1dr(\mathbf{e}_r) + c_2h_2d\varphi(\mathbf{e}_\varphi) + c_3h_3d\theta(\mathbf{e}_\theta) = c_1h_1 + \frac{c_2}{r}h_2 + \frac{c_3}{r\sin\varphi}h_3,
\end{aligned}$$

because

$$dr\left(\frac{\partial}{\partial r}\right) = d\varphi\left(\frac{\partial}{\partial\varphi}\right) = d\theta\left(\frac{\partial}{\partial\theta}\right) = 1.$$

Hence, $c_1 = a,\ c_2 = rb,\ c_3 = cr\sin\varphi$, i.e.

$$\mathcal{D}\,\mathbf{v} = adr + brd\varphi + cr\sin\varphi d\theta.$$

Given a differential form $\alpha = Adr + Bd\varphi + Cd\theta$ we get

$$\mathcal{D}^{-1}(\alpha) = A\mathbf{e}_r + \frac{B}{r}\mathbf{e}_\varphi + \frac{C}{r\sin\varphi}\mathbf{e}_\theta.$$

Volume form.

Let us express the volume form $\Omega = dx\wedge dy\wedge dz$ in spherical coordinates. One way to do that is just to plug into the form the expression of x, y, z through spherical coordinates. But we can also argue as follows. We have $\Omega = cdr\wedge d\varphi\wedge d\theta$. Let us evaluate both sides of this equality on vectors $\mathbf{e}_r, \mathbf{e}_\varphi, \mathbf{e}_\theta$. Then $\Omega(\mathbf{e}_r, \mathbf{e}_\varphi, \mathbf{e}_\theta) = 1$ because these vectors fields are at every point orthonormal and define the standard orientation of the space. On the other hand,

$$\begin{aligned}
&(dr\wedge d\varphi\wedge d\theta)(\mathbf{e}_r, \mathbf{e}_\varphi, \mathbf{e}_\theta) \\
&= \frac{1}{r^2\sin\varphi}(dr\wedge d\varphi\wedge d\theta)(\frac{\partial}{\partial r}, \frac{\partial}{\partial\varphi}, \frac{\partial}{\partial\theta}) = \frac{1}{r^2\sin\varphi},
\end{aligned}$$

and therefore $c = r^2\sin\varphi$, i.e.

$$\Omega = r^2\sin\varphi\, dr\wedge d\varphi\wedge d\theta.$$

In particular, given a 3-form $\eta = adr\wedge d\varphi\wedge d\theta$ we get $\star\eta = \Lambda_\Omega^{-1}(\eta) = \frac{a}{r^2\sin\varphi}$.

For a vector field $\mathbf{v} = a\mathbf{e}_r + b\mathbf{e}_\varphi + c\mathbf{e}_\theta$ We get

$$\begin{aligned} \lrcorner_\Omega(\mathbf{v}) &= \mathbf{v} \lrcorner \Omega \\ &= ar^2 \sin\varphi \, d\phi \wedge d\theta + br \sin\varphi \, d\theta \wedge dr + cr dr \wedge d\varphi. \end{aligned}$$

and given a differential 2-form

$$\lambda = A d\varphi \wedge d\theta + B d\theta \wedge dr + C dr \wedge d\varphi$$

we get

$$\lrcorner^{-1}(\lambda) = \frac{A}{r^2 \sin\varphi}\mathbf{e}_r + \frac{B}{r\sin\varphi}\mathbf{e}_\varphi + \frac{C}{r}\mathbf{e}_\theta.$$

Now we are ready to express the vector analysis operations in spherical coordinates.

Exercise 13.10. (Gradient) Given a function $f(r, \theta, \varphi)$ expressed in spherical coordinates, compute ∇f.

Solution. We have

$$\nabla f = D^{-1}(f_r dr + f_\varphi d\varphi + f_\theta d\theta) = f_r \mathbf{e}_r + \frac{f_\varphi}{r}\mathbf{e}_\varphi + \frac{f_\varphi}{r\sin\varphi}\mathbf{e}_\theta,$$

where we denoted by f_r, f_φ and f_θ the respective partial derivatives of the function f. ■

Exercise 13.11. Given a vector field $\mathbf{v} = a\mathbf{e}_r + b\mathbf{e}_\varphi + c\mathbf{e}_\theta$ compute $\operatorname{div}\mathbf{v}$ and $\operatorname{curl}\mathbf{v}$.

Solution.

a) **Divergence**. We have

$$\begin{aligned} \mathbf{v} \lrcorner \Omega &= (a\mathbf{e}_r + b\mathbf{e}_\varphi + c\mathbf{e}_\theta) \lrcorner r^2 \sin\varphi dr \wedge d\varphi \wedge d\theta \\ &= ar^2 \sin\varphi \, d\varphi \wedge d\theta + br \sin\varphi \, d\theta \wedge dr + cr \, dr \wedge d\varphi \end{aligned}$$

and

$$\begin{aligned} d(\mathbf{v} \lrcorner \Omega) &= \big((r^2 a_r + 2ra)\sin\varphi + rb_\varphi \sin\varphi + rb\cos\varphi + rc_\theta\big)\, dr \wedge d\varphi \wedge d\theta \\ &= \left(a_r + \frac{2a}{r} + \frac{b_\varphi}{r} + \frac{b}{r}\cot\varphi + \frac{c_\theta}{r\sin\varphi}\right)\Omega. \end{aligned}$$

Hence,

$$\operatorname{div} \mathbf{v} = a_r + \frac{2a}{r} + \frac{b_\varphi}{r} + \frac{b}{r}\cot\varphi + \frac{c_\theta}{r\sin\varphi}\,.$$

b) **Curl**. We have

$\mathcal{D}\,\mathbf{v} = adr + brd\varphi + cr\sin\varphi d\theta$;

$d(\mathcal{D}\,\mathbf{v}) = (c_\varphi r\sin\phi + cr\cos\phi - b_\theta r)\,d\varphi\wedge d\theta + (a_\theta - rc_r\sin\varphi - c_\varphi)\,d\theta\wedge dr + (rb_r + b - a_\varphi)\,dr\wedge d\varphi$;

$\operatorname{curl}\mathbf{v} = \lrcorner^{-1}\left(d(\mathcal{D}(\,\mathbf{v}))\right) =$

$\left(\frac{c_\varphi}{r} + \frac{c\cot\varphi}{r} - \frac{b_\theta}{r\sin\varphi}\right)\mathbf{e}_r + \left(\frac{a_\theta}{r\sin\varphi} - c_r - \frac{c_\varphi}{r\sin\varphi}\right)\mathbf{e}_\varphi + \left(b_r + \frac{b}{r} - \frac{a_\varphi}{r}\right)\mathbf{e}_\theta.$

■

13.9 Complex-valued differential k-forms

One can consider complex-valued differential k-forms. A $\mathbb{C}$-valued differential 1-form is a field of $\mathbb{C}$-valued k-forms, or simply it is an expression $\alpha + i\beta$, where α, β are usual real-valued differential k-forms. All operations on complex valued k-forms (exterior multiplication, pull-back and exterior differential) are defined in a natural way:

$$(\alpha_1 + i\beta_1)\wedge(\alpha_2 + i\beta_2) = \alpha_1\wedge\alpha_2 - \beta_1\wedge\beta_2 + i(\alpha_1\wedge\beta_2 + \beta_1\wedge\alpha_2),$$

$$f^*(\alpha + i\beta) = f^*\alpha + if^*\beta,\ d(\alpha + i\beta) = d\alpha + id\beta.$$

We will be in particular interested in complex valued differential forms on $\mathbb{C}$. Note that a complex-valued function (or 0-form) on a domain $U \subset \mathbb{C}$ is just a map $f = u + iv : U \to \mathbb{C}$. Its differential df is the same as the differential of this map, but it also can be viewed as a $\mathbb{C}$-valued differential 1-form $df = du + idv$.

Example 13.12.

$$dz = dx + idy, d\bar z = dx - idy, zdz = (x + iy)(dx + idy) = xdx - ydy + i(xdy + ydx),$$

$$dz\wedge d\bar z = (dx + idy)\wedge(dx - idy) = -2idx\wedge dy.$$

Exercise 13.13. Prove that $d(z^n) = nz^{n-1}dz$ for *any* integer $n \neq 0$.

Solution. Let us do the computation in polar coordinates. Then $z^n = r^n e^{in\phi}$ and assuming that $n \neq 0$ we have

$$d(z^n) = nr^{n-1}e^{in\phi}dr + inr^n e^{in\phi}d\phi = nr^{n-1}e^{in\phi}(dr + ird\phi).$$

On the other hand,

$$nz^{n-1}dz = nr^{n-1}e^{i(n-1)\phi}d(re^{i\phi}) = nr^{n-1}e^{i(n-1)\phi}\left(e^{i\phi}dr + ire^{i\phi}d\phi\right) = nr^{n-1}e^{in\phi}(dr + ird\phi).$$

Comparing the two expressions we conclude that $d(z^n) = nz^{n-1}dz$.

It follows that the 1-form $\frac{dz}{z^n}$ is exact on $\mathbb{C} \setminus 0$ for $n > 1$. Indeed,

$$\frac{dz}{z^n} = d\left(\frac{1}{(1-n)z^{n-1}}\right).$$

On the other hand the form $\frac{dz}{z}$ is closed on $\mathbb{C} \setminus 0$ but not exact. Indeed,

$$\frac{dz}{z} = \frac{d(re^{i\phi})}{re^{i\phi}} = \frac{e^{i\phi}dr + ire^{i\phi}d\phi}{re^{i\phi}} = \frac{dr}{r} + id\phi,$$

and hence $d\left(\frac{dz}{z}\right) = 0$.

On the other hand, we already had seen that the form $d\phi$, and hence $\frac{dz}{z}$, is not exact.

Chapter 14

Integration of differential forms and functions over domains in a vector space

14.1 Integration of functions over domains in high-dimensional spaces

We define here *Riemann integral* of a *bounded* function over a *bounded* domain in a vector space V. We fix a coordinate system, i.e. identify V in $\mathbb{R}^n$. Let η denote the volume form $x_1 \wedge \cdots \wedge x_n$. As it will be clear below, the definition of an integral will not depend on the choice of a coordinate system but only on the background volume form, or rather its absolute value because the orientation of V will be irrelevant.

We will need a special class of parallelepipeds in V, namely those which are generated by vectors proportional to basic vectors, or in other words, parallelepipeds with edges parallel to the coordinate axes. We will also allow these parallelepipeds to be parallel transported anywhere in the space. Let us denote

$$P(a_1, b_1; a_2, b_2; \dots; a_n, b_n) := \{a_i \le x_i \le b_i;\ i = 1, \dots, n\} \subset \mathbb{R}^n.$$

We will refer to $P(a_1, b_1; a_2, b_2; \dots; a_n, b_n)$ as a *special parallelepiped*, or *rectangle*.

Let us fix one rectangle $P := P(a_1, b_1; a_2, b_2; \dots; a_n, b_n)$. Following the same scheme as we used in the 1-dimensional case, we define a *partition* $\mathcal{P}$ of P as a product of partitions $a_1 = t_0^1 < \dots < t_{N_1}^1 = b_1, \ \dots, a_n = t_0^n < \dots < t_{N_n}^n = b_n$, of intervals $[a_1, b_1], \dots, [a_n, b_n]$. For simplicity of notation we will always assume that each of the coordinate intervals is partitioned into the same number of intervals, i.e. $N_1 = \dots = N_n = N$. This defines a partition of P into N^n smaller rectangles $P_{\mathbf{j}} = \{t_{j_1}^1 \leq x_1 \leq t_{j_1+1}^1, \dots, t_{j_n}^n \leq x_n \leq t_{j_n+1}^n\}$, where $\mathbf{j} = (j_1, \dots, j_n)$ and each index j_k takes values between 0 and $N-1$. Let us define

$$\operatorname{Vol}(P_{\mathbf{j}}) := \prod_{k=1}^{n} (t_{j_k+1}^k - t_{j_k}^k). \tag{14.1.1}$$

This agrees with the definition of the volume of a parallelepiped which we introduced earlier (see formula (9.3.1) in Section 9.3). We will also denote $\delta_{\mathbf{j}} := \max\limits_{k=1,\dots,n} (t_{j_k+1}^k - t_{j_k}^k)$ and $\delta(\mathcal{P}) := \max\limits_{\mathbf{j}}(\delta_{\mathbf{j}})$. Let us fix a point $c_{\mathbf{j}} \in P_{\mathbf{j}}$ and denote by C the set of all such $c_{\mathbf{j}}$. Given a function $f : P \to \mathbb{R}$ we form an integral sum

$$I(f; \mathcal{P}, C) = \sum_{\mathbf{j}} f(c_{\mathbf{j}}) \operatorname{Vol}(P_{\mathbf{j}}) \tag{14.1.2}$$

where the sum is taken over all elements of the partition. If there exists a limit $\lim\limits_{\delta(\mathcal{P}) \to 0} I(f; \mathcal{P}, C)$ then the function $f : P \to \mathbb{R}$ is called *integrable* (in the sense of Riemann) over P, and this limit is called the *integral of f over P*.

There exist several different notations for this integral: $\int\limits_P f$, $\int\limits_P f dV$, $\int\limits_P f d\operatorname{Vol}$, etc. In the particular case of $n = 2$ one often uses notation $\int\limits_P f dA$, or $\iint\limits_P f dA$. Sometimes the functions we integrate may depend on a parameter, and in these cases it is important to indicate with respect to which variable we integrate. Hence, one also uses the notation like $\int\limits_P f(x, y) dx^n$, where the index n refers to the dimension of the space over which we integrate. One also use the notation $\underbrace{\int \cdots \int}_{P} f(x_1, \dots x_n) dx_1 \dots dx_n$, which is reminiscent both of the integral $\int_P f(x_1, \dots x_n) dx_1 \wedge \dots \wedge dx_n$ which will be defined later in Section 14.6 and the notation $\int\limits_{a_n}^{b_n} \cdots \int\limits_{a_1}^{b_1} f(x_1, \dots x_n) dx_1 \dots dx_n$ for n-iterated integral which will be discussed in Section 14.5.

Alternatively and equivalently the integrability can be defined via upper and lower integral sum, similar to the 1-dimensional case. Namely, we define

$$U(f;\mathcal{P}) = \sum_{\mathbf{j}} M_{\mathbf{j}}(f)\mathrm{Vol}(P_{\mathbf{j}}), \quad L(f;\mathcal{P}) = \sum_{\mathbf{j}} m_{\mathbf{j}}(f)\mathrm{Vol}(P_{\mathbf{j}}),$$

where $M_{\mathbf{j}}(f) = \sup\limits_{P_{\mathbf{j}}} f$, $m_{\mathbf{j}}(f) = \inf\limits_{P_{\mathbf{j}}} f$, and say that the function f is integrable over P if

$$\inf_{\mathcal{P}} U(f;\mathcal{P}) = \sup_{\mathcal{P}} L(f,\mathcal{P}).$$

Note that $\inf\limits_{\mathcal{P}} U(f;\mathcal{P})$ and $\sup\limits_{\mathcal{P}} L(f,\mathcal{P})$ are sometimes called *upper* and *lower* integrals, respectively, and denoted by $\overline{\int\limits_P} f$ and $\underline{\int\limits_P} f$. Thus a function $f : P \to \mathbb{R}$ is integrable iff $\overline{\int\limits_P} f = \underline{\int\limits_P} f$.

Let us list some properties of Riemann integrable functions and integrals.

Proposition 14.1. *Let $f, g : P \to \mathbb{R}$ be integrable functions. Then*

1. *$af + bg$, where $a, b \in \mathbb{R}$, is integrable and $\int\limits_P af + bg = a \int\limits_P f + b \int\limits_P g$;*
2. *If $f \le g$ then $\int\limits_P f \le \int\limits_P g$;*
3. *$h = \max(f, g)$ is integrable; in particular the functions $f_+ := \max(f, 0)$ and $f_- := \max(-f, 0)$ and $|f| = f_+ + f_-$ are integrable;*
4. *fg is integrable.*

Proof. Parts 1 and 2 are straightforward and we leave them to the reader as an exercise. Let us check properties 3 and 4.

3. Take any partition $\mathcal{P}$ of P. Note that

$$M_{\mathbf{j}}(h) - m_{\mathbf{j}}(h) \le \max(M_{\mathbf{j}}(f) - m_{\mathbf{j}}(f), M_{\mathbf{j}}(g) - m_{\mathbf{j}}(g)) \,. \tag{14.1.3}$$

Indeed, we have $M_{\mathbf{j}}(h) = \max(M_{\mathbf{j}}(f), M_{\mathbf{j}}(g))$ and $m_{\mathbf{j}}(h) \ge \max(m_{\mathbf{j}}(f), m_{\mathbf{j}}(g))$. Suppose for determinacy that $\max(M_{\mathbf{j}}(f), M_{\mathbf{j}}(g)) = M_{\mathbf{j}}(f)$. We also have $m_{\mathbf{j}}(h) \ge m_{\mathbf{j}}(f)$. Thus

$$M_{\mathbf{j}}(h) - m_{\mathbf{j}}(h) \le M_{\mathbf{j}}(f) - m_{\mathbf{j}}(f) \le \max(M_{\mathbf{j}}(f) - m_{\mathbf{j}}(f), M_{\mathbf{j}}(g) - m_{\mathbf{j}}(g)).$$

Then using (14.1.3) we have

$$
\begin{aligned}
U(h;\mathcal{P}) - L(h;\mathcal{P}) = &\sum_{\mathbf{j}} (M_{\mathbf{j}}(h) - m_{\mathbf{j}}(h))\mathrm{Vol}(P_{\mathbf{j}}) \le \\
&\sum_{\mathbf{j}} \max\left(M_{\mathbf{j}}(f) - m_{\mathbf{j}}(f), M_{\mathbf{j}}(g) - m_{\mathbf{j}}(g)\right)\mathrm{Vol}(P_{\mathbf{j}}) = \\
&\max\left(U(f;\mathcal{P}) - L(f;\mathcal{P}), U(f;\mathcal{P}) - L(f;\mathcal{P})\right).
\end{aligned}
$$

By assumption the right-hand side can be made arbitrarily small for an appropriate choice of the partition $\mathcal{P}$, and hence h is integrable.

4. We have $f = f_+ - f_-$, $g = g_+ - g_-$ and $fg = f_+g_+ + f_-g_- - f_+g_- - f_-g_+$. Hence, using 1 and 3 we can assume that the functions f, g are non-negative. Let us recall that the functions f, g are by assumption bounded, i.e. there exists a constant $C > 0$ such that $f, g \le C$. We also have $M_{\mathbf{j}}(fg) \le M_{\mathbf{j}}(f)M_{\mathbf{j}}(g)$ and $m_{\mathbf{j}}(fg) \ge m_{\mathbf{j}}(f)m_{\mathbf{j}}(g)$. Hence

$$
\begin{aligned}
&U(fg;\mathcal{P}) - L(fg;\mathcal{P}) = \sum_{\mathbf{j}} \left(M_{\mathbf{j}}(fg) - m_{\mathbf{j}}(fg)\right)\mathrm{Vol}(P_{\mathbf{j}}) \le \\
&\sum_{\mathbf{j}} (M_{\mathbf{j}}(f)M_{\mathbf{j}}(g) - m_{\mathbf{j}}(f)m_{\mathbf{j}}(g))\mathrm{Vol}(P_{\mathbf{j}}) = \\
&\sum_{\mathbf{j}} \left(M_{\mathbf{j}}(f)M_{\mathbf{j}}(g) - m_{\mathbf{j}}(f)M_{\mathbf{j}}(g) + m_{\mathbf{j}}(f)M_{\mathbf{j}}(g) - m_{\mathbf{j}}(f)m_{\mathbf{j}}(g)\right)\mathrm{Vol}(P_{\mathbf{j}}) \le \\
&\sum_{\mathbf{j}} \left((M_{\mathbf{j}}(f) - m_{\mathbf{j}}(f))M_{\mathbf{j}}(g) + m_{\mathbf{j}}(f)(M_{\mathbf{j}}(g) - m_{\mathbf{j}}(g))\right)\mathrm{Vol}(P_{\mathbf{j}}) \le \\
&C(U(f;\mathcal{P}) - L(f;\mathcal{P}) + U(g;\mathcal{P}) - L(g;\mathcal{P})).
\end{aligned}
$$

By assumption the right-hand side can be made arbitrarily small for an appropriate choice of the partition $\mathcal{P}$, and hence fg is integrable. ■

Consider now a bounded subset $K \subset \mathbb{R}^n$ and choose a rectangle $P \supset K$. Given any function $f : K \to \mathbb{R}$ one can always extend it to P as equal to 0. A function $f : K \to \mathbb{R}$ is called *integrable over* K if this trivial extension $\overline{f}$ is integrable over P, and we define $\int\limits_K f dV := \int\limits_P \overline{f} dV$. When this will not be confusing we will usually keep the notation f for the above extension.

14.2 Volume

We further define the volume

$$\mathrm{Vol}(K) = \int\limits_K 1 dV = \int\limits_P \chi_K dV,$$

provided that this integral exists. In this case we call the set K *measurable in the sense of Riemann*, or just measurable.[1] Here χ_K is the *characteristic* or *indicator* function of K, i.e. the function which is equal to 1 on K and 0 elsewhere. In the 2-dimensional case the volume is called the area, and in the 1-dimensional case the length.

Remark 14.2. For any bounded set A there is defined a *lower* and *upper* volumes,

$$\underline{\mathrm{Vol}}(A) = \underline{\int} \chi_A dV \leq \overline{\mathrm{Vol}}(A) = \overline{\int} \chi_A dV.$$

The set is measurable iff $\underline{\mathrm{Vol}}(A) = \overline{\mathrm{Vol}}(A)$. If $\overline{\mathrm{Vol}}(A) = 0$ then $\underline{\mathrm{Vol}}(A) = 0$, and hence A is measurable and $\mathrm{Vol}(A) = 0$.

Exercise 14.3. *Prove that for the rectangles this definition of the volume coincides with the one given by the formula* (14.1.1).

The next proposition lists some properties of the volume.

Proposition 14.4. 1. *Volume is monotone, i.e. if $A, B \subset P$ are measurable and $A \subset B$ then* $\mathrm{Vol}(A) \leq \mathrm{Vol}(B)$.

2. *If sets $A, B \subset P$ are measurable then $A \cap B$, $A \setminus B$ and $A \cup B$ are measurable as well and we have*

$$\mathrm{Vol}(A \cup B) = \mathrm{Vol}(A) + \mathrm{Vol}(B) - \mathrm{Vol}(A \cap B).$$

3. *If A can be covered by a measurable set of arbitrarily small total volume then* $\mathrm{Vol}(A) = 0$. *Conversely, if* $\mathrm{Vol}(A) = 0$ *then for any $\epsilon > 0$ there exists $\delta > 0$ such that for any partition*

[1] There exists a more general and more common notion of measurability in the sense of Lebesgue. Any Riemann measurable set is also measurable in the sense of Lebesgue, but not the other way around. Historically an attribution of this notion to Riemann is incorrect. It was defined by Camille Jordan and Giuseppe Peano before Riemann integral was introduced. What we call in these notes *volume* is also known by the name *Jordan content.*

$\mathcal{P}$ with $\delta(\mathcal{P}) < \delta$ the elements of the partition which intersect A have arbitrarily small total volume.

4. *A is measurable iff* $\mathrm{Vol}\,(\partial A) = 0$.

Proof. The first statement is obvious. To prove the second one, we observe that $\chi_{A\cup B} = \max(\chi_A, \chi_B)$, $\chi_{A\cap B} = \chi_A\chi_B$, $\max(\chi_A, \chi_B) = \chi_A + \chi_B - \chi_A\chi_B$, $\chi_{A\setminus B} = \chi_A - \chi_{A\cap B}$ and then apply Proposition 14.1. To prove 14.4.3 we first observe that if a set B is measurable and $\mathrm{Vol}B < \epsilon$ then then for a sufficiently fine partition $\mathcal{P}$ we have $U(\chi_B;\mathcal{P}) < \mathrm{Vol}B + \epsilon < 2\epsilon$. Since $A \subset B$ then $\chi_A \leq \chi_B$, and therefore $U(\chi_A, \mathcal{P}) \leq U(\chi_B, \mathcal{P}) < 2\epsilon$. Thus, $\inf_{\mathcal{P}} U(\chi_A, \mathcal{P}) = 0$ and therefore A is measurable and $\mathrm{Vol}(A) = 0$. Conversely, if $\mathrm{Vol}(A) = 0$ then for any $\epsilon > 0$ for a sufficiently fine partition $\mathcal{P}$ we have $U(\chi_A;\mathcal{P}) < \epsilon$. But $U(\chi_A;\mathcal{P})$ is equal to the sum of volumes of elements of the partition which have non-empty intersection with A.

Finally, let us prove 14.4.4. Consider any partition $\mathcal{P}$ of P and form lower and upper integral sums for χ_A. Denote $M_{\mathbf{j}} := M_{\mathbf{j}}(\chi_A)$ and $m_{\mathbf{j}} = m_{\mathbf{j}}(\chi_A)$. Then all numbers $M_{\mathbf{j}}, m_{\mathbf{j}}$ are equal to either 0 or 1. We have $M_{\mathbf{j}} = m_{\mathbf{j}} = 1$ if $P_{\mathbf{j}} \subset A$; $M_{\mathbf{j}} = m_{\mathbf{j}} = 0$ if $P_{\mathbf{j}} \cap A = \varnothing$ and $M_{\mathbf{j}} = 1, m_{\mathbf{j}} = 0$ if $P_{\mathbf{j}}$ has non-empty intersection with both A and $P \setminus A$. In particular,

$$B(\mathcal{P}) := \bigcup_{\mathbf{j};M_{\mathbf{j}}-m_{\mathbf{j}}=1} P_{\mathbf{j}} \supset \partial A.$$

Hence, we have

$$U(\chi_A;\mathcal{P}) - L(\chi_A;\mathcal{P}) = \sum_{\mathbf{j}} (M_{\mathbf{j}} - m_{\mathbf{j}})\mathrm{Vol}(P_{\mathbf{j}}) = \mathrm{Vol}B(\mathcal{P}).$$

Suppose that A is measurable. Then there exists a partition such that $U(\chi_A;\mathcal{P}) - L(\chi_A;\mathcal{P}) < \epsilon$, and hence ∂A is can be covered by the set $B(\mathcal{P})$ of volume $< \epsilon$. Thus applying part 3 we conclude that $\mathrm{Vol}(\partial A) = 0$. Conversely, we had seen below that if $\mathrm{Vol}(\partial A) = 0$ then there exists a partition such that the total volume of the elements intersecting ∂A is $< \epsilon$. Hence, for this partition we have $L(\chi_A;\mathcal{P}) \leq U(\chi_A;\mathcal{P}) < \epsilon$, which implies the integrability of χ_A, and hence measurability of A. ■

Corollary 14.5. *If a bounded set $A \subset V$ is measurable then its interior* $\mathrm{Int}\,A$ *and its closure* $\overline{A}$ *are also measurable and we have in this case*

$$\mathrm{Vol}A = \mathrm{Vol}\,\mathrm{Int}\,A = \mathrm{Vol}\overline{A}.$$

Proof. 1. We have $\partial\overline{A} \subset \partial A$ and $\partial(\operatorname{Int} A) \subset \partial A$. Therefore, $\operatorname{Vol}\partial\overline{A} = \operatorname{Vol}\partial\operatorname{Int} A = 0$, and therefore the sets $\overline{A}$ and $\operatorname{Int} A$ are measurable. Also $\operatorname{Int} A \cup \partial A = \overline{A}$ and $\operatorname{Int} A \cap \partial A = \varnothing$. Hence, the additivity of the volume implies that $\operatorname{Vol}\overline{A} = \operatorname{Vol}\partial\operatorname{Int} A + \operatorname{Vol}\partial A = \operatorname{Vol}\partial\operatorname{Int} A$. On the other hand, $\operatorname{Int} A \subset A \subset \overline{A}$. and hence the monotonicity of the volume implies that $\operatorname{Vol}\operatorname{Int} A \leq \operatorname{Vol} A \leq \operatorname{Vol}\overline{A}$. Hence, $\operatorname{Vol} A = \operatorname{Vol}\operatorname{Int} A = \operatorname{Vol}\overline{A}$. ■

Exercise 14.6. If $\operatorname{Int} A$ or $\overline{A}$ are measurable then this does not imply that A is measurable. For instance, if A is the set of rational points in interval $I = [0,1] \subset \mathbb{R}$ then $\operatorname{Int} A = \varnothing$ and $\overline{A} = I$. However, show that A is not Riemann measurable.
2. A closed set A is called *nowhere dense* if $\operatorname{Int} A = \varnothing$. Prove that if A is nowhere dense then either $\operatorname{Vol} A = 0$, or A is not measurable in the sense of Riemann. Find an example of a non-measurable nowhere dense set.

14.3 Smooth maps and volume

14.3.1 Lipshitz maps

Let V, W be two Euclidean spaces. Recall the definition of the norm of a linear operator $\mathcal{A} : V \to W$:

$$||\mathcal{A}|| = \max_{||x||=1} \frac{||\mathcal{A}(x)||}{||x||}.$$

Thus we have $\mathcal{A}(B_R(0)) \subset B_{aR}(0) \subset W$, where we denoted $a := ||\mathcal{A}||$.

Given a subset $A \subset V$ a map $f : A \to W$ is called *Lipshitz* if there exists a constant $C > 0$ such that for any $x, y \in A$ we have

$$||f(y) - f(x)|| \leq C||y - x||.$$

Lemma 14.7. *Let $A \subset V$ be a compact set. Then any C^1-smooth map $A \to W$ is Lipshitz.*

Let us recall that given a compact set $C \subset V$, we say that a map $f : C \to W$ is smooth if it extends to a smooth map defined on an open neighborhood $U \supset C$.

Proof. Let $K : A \to \mathbb{R}$ be the function defined by $K(x) = ||d_x f||, x \in A$. The function K is continuous because f is C^1-smooth. Hence it is bounded: there exists a constant $E > 0$ such that $K(x) \leq E$ for all $x \in A$.

Let us first consider the case when A is a convex set, i.e. with any two points $x, y \in A$ the interval connecting them is also contained in A. Given two points $x, y \in A$ at a distance $d = ||x - y|| > 0$ consider a path $\phi(s) = x + \frac{s}{d}(y - x),\ s \in [0, d]$ which connects them. Note that the velocity vector $\phi'(s) = \frac{y-x}{d}$ has the unit length. Denote $\widetilde{f}(s) = f(\phi(s)), s \in [0, d]$. Note that $\widetilde{f}'(s) = df_{\phi(s)}(\phi'(s))$ by the chain rule. In particular,

$$||\widetilde{f}'(s)|| \leq ||df_{\phi(s)}|| ||\phi'(s)|| \leq E.$$

We also have $f(y) - f(x) = \widetilde{f}(d) - \widetilde{f}(0)$. Let $\widetilde{f} = (\widetilde{f}_1, \dots, \widetilde{f}_m)$ be the coordinate functions of $\widetilde{f}$ in some Cartesian coordinates in W. By the intermediate value theorem, for each $k = 1, \dots, n$ we have $\widetilde{f}_k(d) - \widetilde{f}_k(0) = \widetilde{f}_k'(c_k)d$ for some $c_k \in [0, d]$. Hence, $|\widetilde{f}_k(d) - \widetilde{f}_k(0)| \leq Cd$ and therefore

$$||f(y) - f(x)|| = ||\widetilde{f}(d) - \widetilde{f}(0)|| = \sqrt{\sum_1^n (\widetilde{f}_k(d) - \widetilde{f}_k(0))^2} \leq E\sqrt{n}d = \widetilde{E}||y - x||, \tag{14.3.1}$$

where we denoted $\widetilde{E} := E\sqrt{n}$.

For a general bounded set A choose an open neighborhood $U \supset A$ to which the map f extends C^1-smoothly. Let $U' \subset U$ be a smaller open neighborhood of A such that $\overline{U}' \subset U$. We will also assume that $\overline{U}'$ is bounded, and hence compact.

For every point $x \in A$ there is $\epsilon(x) > 0$ such that the closed ball $\overline{B_{\epsilon(x)}(x)}$ is contained in U'. We have $\bigcup\limits_{x \in A} B_{\frac{\epsilon(x)}{2}}(x) \supset A$, and hence by compactness of A there are finitely many balls $B_i := B_{\frac{\epsilon(x_i)}{2}}(x_i)$, $i = 1, \dots, N$, such that $\bigcup\limits_1^N B_i \supset A$. Denote $\epsilon := \min\limits_{i=1,\dots,N} \frac{\epsilon(x_i)}{2}$. Then for any two points $x, y \in A$ with $||y - x|| \leq \epsilon$ belong to one of the balls $\widehat{B}_i := B_{\epsilon(x_i)}(x_i)$ which is convex, and hence according to (14.3.1) we have $||f(y) - f(x)|| \leq \widetilde{E}||y - x||$.

Denote by D the diameter of the compact set $f(A)$, i.e. $D := \max\limits_{x,y \in A} ||f(y) - f(x)||$. Then if for the distance between two points $x, y \in A$ is $\geq \epsilon$ we have $||f(y) - f(x)|| \leq \frac{D}{\epsilon}||y - x||$. Finally, if we denote $C := \max(\widetilde{E}, \frac{D}{\epsilon})$ we get

$$||f(y) - f(x)|| \leq C||y - x|| \text{ for any } x, y \in A.$$

■

14.3.2 Sard's theorem

Lemma 14.8. *Let $A \subset V$ be a compact set of volume* 0 *and $f : V \to W$ a Lipshitz map, where* $\dim W \geq \dim V$. *Then* $\mathrm{Vol} f(A) = 0$.

Proof. According to Lemma 14.7 there is a constant $C > 0$ such that $||f(y) - f(x)|| \leq C||y - x||$ for any $x, y \in A$. In particular, the image $f(P)$ of a cube P of size δ in V_x, $x \in A$, centered at x, is contained in a cube of size $K\delta$ in $W_{f(x)}$ centered at $f(x)$.

The volume 0 assumption implies that for any $\epsilon > 0$ there exists a partition of some size $\delta > 0$ of a larger cube containing A such that the total volume $N\delta^n$ of cubes $P_1, \dots, P_N$ intersecting A is $< \epsilon$. But then $\bigcup\limits_1^N f(P_j) \supset f(A)$ while $\overline{\mathrm{Vol}}_m(f(P_j)) \leq K^M \delta^m$, and hence

$$\overline{\mathrm{Vol}}_m f(A) \leq N\delta^n K^m \delta^{m-n} \leq \epsilon K^m \delta^{m-n} \underset{\epsilon \to 0}{\to} 0.$$

■

Corollary 14.9. *Let $A \subset V$ be any compact set and $f : A \to W$ a C^1-smooth map. Suppose that* $n = \dim V < m = \dim W$. *Then* $\mathrm{Vol}(f(A)) = 0$.

Indeed, f can be extended to a smooth map defined on a neighborhood of $A \times 0$ in $V \times \mathbb{R}$ (e.g. as independent of the new coordinate $t \in \mathbb{R}$). But $\mathrm{Vol}_{n+1}(A \times 0) = 0$ and $m \geq n + 1$. Hence, the required statement follows from Lemma 14.8.

Remark 14.10. The statement of Corollary 14.9 is wrong for continuous maps. For instance, there exists a continuous map $h : [0, 1] \to \mathbb{R}^2$ such that $h([0, 1])$ is the square $\{0 \leq x_1, x_1 \leq 1\}$. (This is a famous Peano curve passing through every point of the square.)

Corollary 14.9 is a simplest special case of *Sard's theorem* which asserts that the set of critical values of a sufficiently smooth map has volume 0. More precisely,

Proposition 14.11. (A. Sard, 1942) *Given a C^k-smooth map $f : A \to W$ (where A is a compact subset of V,* $\dim V = n, \dim W = m$*) let us denote by*

$$\Sigma(f) : \{x \in A;\ \mathrm{rank}\, d_x f < m\}.$$

Then if $k \geq \max(n - m + 1, 1)$ *then* $\mathrm{Vol}_m(f(\Sigma(f)) = 0$.

If $m > n$ then $\Sigma(f) = A$, and hence the statement is equivalent to Corollary 14.9.

Proof. We prove the proposition only for the case $m = n$.

To clarify the main idea we first consider the case when $n = 1$ and $A = [0, 1]$, i.e. f is a function $f : A \to \mathbb{R}$ with a continuous derivative f'. According to Cantor's theorem f' is uniformly continuous and hence for any ϵ there exists $\delta > 0$ such that

$$|f'(x) - f'(y)| < \epsilon \text{ when } |x - y| < \delta. \tag{14.3.2}$$

Let us take a partition of the interval of the size $< \delta$. Let $I_1, \dots, I_N$ be the interval of the partition which contain critical points, i.e. points where the derivative is 0. Then for any point c in on of these intervals $|f'(c)| < \epsilon$, and hence by the intermidiate value theorem for any two points $x, y \in I_j,\ j = 1, \dots, N$ we have $|f(x) - f(y)| = |f'(c)||x - y| < \epsilon\delta$, i.e. the image $f(I_j)$ is contained in an interval of length $\epsilon\delta$. But total length of theintrvals I_j is ≤ 1, and thus $f(\Sigma)(f)$ is covered by the union of intervals of the total length $< \epsilon$. Hence $\mathrm{Vol}_1(f(\Sigma(f))) = 0$.

Consider now the case of a general n. Again the C^1-smoothness of f implies that $d_x f$ is uniformly continuous, i.e. for every $\epsilon > 0$ there exists $\delta > 0$ such that

$$||d_x f - d_y f|| < \epsilon \text{ when } ||x - y|| < \delta. \tag{14.3.3}$$

The inequality $||d_x f - d_y f|| < \epsilon$ means that for any unit vector h in V

$$||d_x f(h) - d_y f(h)|| < \epsilon, \tag{14.3.4}$$

where we parallel transport the vector h to points x and y.

Consider a partition of a cube Q containing A by cubes of size $< \frac{\delta}{2\sqrt{n}}$, so that the ball surrounding each of the cubes and centered at any point of the cube has radius $< \delta$.

Let $Q_1, \dots, Q_N$ be the cubes of the partition intersecting $\Sigma(f)$. The total volume $2^n n^{-\frac{n}{2}} N\delta^n$ of these cubes is bounded by $\mathrm{Vol}P$, where P is a fixed cube containing A.

Choose a point $c_j \in Q_j \cap \Sigma(f)$ for each $j = 1, \dots, N$ Let $B_1, \dots, B_N$ be balls of radius δ centered at c_j. As we already pointed out, $B_j \supset Q_j$ for each $j = 1, \dots, N$.

The differential $d_{c_j} f$ is degenerate, and hence the image $d_{c_j} f(V_{c_j}) \subset W_{f(c_j)}$ is contained in a codimension 1 subspace $L_j \subset W_{f(c_j)}$. Let us choose a Cartesian coordinate system $(y_1, \dots, y_n)$ in $W_{f(c_j)}$ such that $L_j = \{y_n = 0\}$ we can view y_j as coordinates in V with the origin shifted to

$f(c_j)$. Let $(f_1,\dots,f_n)$ be the coordinate functions of f with respect to these coordinates. Then $d_{c_j}f_n=0$, i.e. the directional derivatives of the function f at c_j at every direction are equal to 0. But then according to inequality (14.3.4) the (absolute value of the) directional derivatives of f_n at any point of B_j are $<\epsilon$. Hence, using our above 1-dimensional argument along each radius of B_j, we conclude that $|f_n(x)|<\epsilon\delta$, i.e. the image $f(B_j)$ is contained in an $\epsilon\delta$-neighborhood $U_{\epsilon\delta}$ of the hyperplane L_j viewed as an affine hyperplane in V. We also recall that the map f is Lipshitz, and hence the image $f(B_j)$ is contained in a ball $B_{K\delta}(f(c_j))\subset W$ of radius $K\delta$ centered at $f(c_j)$ for some constant $K>0$. Hence, $f(B_j)\subset U_{\epsilon\delta}\cap B_{K\delta}(f(c_j))$, so $f(B_j)$ is contained in a rectangular P_j with all sides equal to $2K\delta$ and one side of size $2\epsilon\delta$. In particular, $\mathrm{Vol}(P_j)=K^{n-1}2^n\delta^n\epsilon$.

Therefore, $f(\Sigma(f)$ can be covered by N such rectangular of total volume

$$K^{n-1}2^nN\delta^n\epsilon=(K^{n-1}n^{\frac{n}{2}}\mathrm{Vol}P)\epsilon\underset{\epsilon\to0}{\to}0.$$

Hence, $\mathrm{Vol}f(\Sigma(f))=0$. ■

Corollary 14.12. *Let $A\subset\mathbb{R}^n$ be a measurable set and $f:A\to\mathbb{R}^q$, where $q\geq n$, be a C^1-smooth map. Then the image $f(A)\subset\mathbb{R}^q$ is also measurable.*

Proof. If $q>n$ then $\mathrm{Vol}_qf(A)=0$ according to Corollary 14.9, and hence $f(A)$ is measurable. Suppose that $q=n$. Then for any interior point $a\in A$ such that $\mathrm{rank}d_af=n$ the image $f(a)$ is an interior point of $f(A)$ according to the implicit function theorem. Hence, the boundary $\partial(f(A))\subset f(\partial A)\cup f(\Sigma(f))$, where $\Sigma(f)$ is the set critical points of f (i.e. points $a\in A$ where $\mathrm{rank}d_af<n$). But $\mathrm{Vol}\partial A=0$ and hence, according to Lemma 14.8 $\mathrm{Vol}(f(\partial A))=0$. On the other hand, Sard's theorem 14.11 implies that $\mathrm{Vol}f(\Sigma(f))=0$, and therefore $\mathrm{Vol}(\partial f(A))=0$, which means that $f(A)$ is measurable.

14.4 Orthogonal invariance of volume

14.4.1 Properties which hold almost everywhere

We say that some property holds *almost everywhere* (we will abbreviate a.e.) if it holds in the complement of a set of volume 0. For instance, we say that a bounded function $f:P\to\mathbb{R}$ is almost everywhere continuous (or a.e. continuous) if it is continuous in the complement of a set $A \subset P$ of volume 0. For instance, a *characteristic function of any measurable set is a.e. continuous.* Indeed, it is constant away from the set ∂A which according to Proposition 14.4.4 has volume 0.

Proposition 14.13. *Suppose that the bounded functions $f, g : P \to \mathbb{R}$ coincide a.e. Then if f is integrable, then so is g and we have $\int\limits_P f = \int\limits_P g$.*

Proof. Denote $A = \{x \in P : f(x) \neq g(x)\}$. By our assumption, $\mathrm{Vol}A = 0$. Hence, for any ϵ there exists a $\delta > 0$ such that for every partition $\mathcal{P}$ with $\delta(\mathcal{P}) \leq \delta$ the union B_δ of all rectangles of the partition which have non-empty intersection with A has volume $< \epsilon$. The functions f, g are bounded, i.e. there exists $C > 0$ such $-C \leq |f(x)|, |g(x)| \leq C$ for all $x \in P$. Due to integrability of f we can choose δ small enough so that $|U(f, \mathcal{P}) - L(f, \mathcal{P})| \leq \epsilon$ when $\delta(\mathcal{P}) \leq \delta$. Then we have

$$|U(g, \mathcal{P}) - U(f, \mathcal{P})| = \left| \sum_{J:\, P_J \subset B_\delta} \sup_{P_J} g - \sup_{P_J} f) \right| \leq 2C\mathrm{Vol}B_\delta \leq 2C\epsilon.$$

Similarly, $|L(g, \mathcal{P}) - L(f, \mathcal{P})| \leq 2C\epsilon$, and hence

$$\begin{aligned} |U(g, \mathcal{P}) - L(g, \mathcal{P})| &\leq |U(g, \mathcal{P}) - U(f, \mathcal{P})| + |U(f, \mathcal{P}) - L(f, \mathcal{P})| + |L(f, \mathcal{P}) - L(g, \mathcal{P})| \\ &\leq \epsilon + 4C\epsilon \underset{\delta \to 0}{\longrightarrow} 0, \end{aligned}$$

and hence g is integrable and

$$\int\limits_P g = \lim_{\delta(\mathcal{P}) \to 0} U(g, \mathcal{P}) = \lim_{\delta(\mathcal{P}) \to 0} U(f, \mathcal{P}) = \int\limits_P f.$$

■

Proposition 14.14. *1. Suppose that a function $f : P \to \mathbb{R}$ is a.e. continuous. Then f is integrable.*

2. Let $A \subset V$ be compact and measurable, $f : U \to W$ a C^1-smooth map defined on a neighborhood $U \supset A$. Suppose that $\dim W = \dim V$. *Then $f(A)$ is measurable.*

Proof. 1. Let us begin with a

WARNING. One could think that in view of Proposition 14.13 it is sufficient to consider only the case when the function f is continuous. However, this is not the case, because for a given a.e. continuos function one cannot, in general, find a continuos function g which coincides with f a.e.

Let us proceed with the proof. Given a partition $\mathcal{P}$ we denote by J_A the set of multi-indices $\mathbf{j}$ such that $\operatorname{Int} P_{\mathbf{j}} \cap A \neq \varnothing$, and by $\overline{J}_A$ the complementary set of multi-indices, i.e. for each $\mathbf{j} \in \overline{J}_A$ we have $P_{\mathbf{j}} \cap A = \varnothing$. Let us denote $C := \bigcup\limits_{\mathbf{j} \in J_A} P_{\mathbf{j}}$. According to Proposition 14.4.3 for any $\epsilon > 0$ there exists a partition $\mathcal{P}$ such that $\operatorname{Vol}(C) = \sum\limits_{\mathbf{j} \in J_A} \operatorname{Vol}(P_{\mathbf{j}}) < \epsilon$. By assumption the function f is continuous over a compact set $B = \bigcup\limits_{\mathbf{j} \in \overline{J}_A} P_{\mathbf{j}}$, and hence it is uniformly continuous over it. Thus there exists $\delta > 0$ such that $|f(x) - f(x')| < \epsilon$ provided that $x, x' \in B$ and $||x - x'|| < \delta$. Thus we can further subdivide our partition, so that for the new finer partition $\mathcal{P}'$ we have $\delta(\mathcal{P}') < \delta$. By assumption the function f is bounded, i.e. there exists a constant $K > 0$ such that $M_{\mathbf{j}}(f) - m_{\mathbf{j}}(f) < K$ for all indices $\mathbf{j}$. Then we have

$$
\begin{aligned}
U(f;\mathcal{P}') - L(f,\mathcal{P}') = & \sum_{\mathbf{j}} (M_{\mathbf{j}}(f) - m_{\mathbf{j}}(f)) \operatorname{Vol}(P_{\mathbf{j}}) = \\
& \sum_{\mathbf{j}; P_{\mathbf{j}} \subset B} (M_{\mathbf{j}}(f) - m_{\mathbf{j}}(f)) \operatorname{Vol}(P_{\mathbf{j}}) + \sum_{\mathbf{j}; P_{\mathbf{j}} \subset C} (M_{\mathbf{j}}(f) - m_{\mathbf{j}}(f)) \operatorname{Vol}(P_{\mathbf{j}}) < \\
& \epsilon \operatorname{Vol} B + K \operatorname{Vol} C < \epsilon(\operatorname{Vol} P + K).
\end{aligned}
$$

Hence $\inf\limits_{\mathcal{P}} U(f;\mathcal{P}) = \sup\limits_{\mathcal{P}} L(f;\mathcal{P})$, i.e. the function f is integrable.

2. If x is an interior point of A and $detDf(x) \neq 0$ then the inverse function theorem implies that $f(x) \in \operatorname{Int} f(A)$. Denote $C = \{x \in A; \det Df(x) = 0\}$. Hence, $\partial f(A) \subset f(\partial A) \cup f(C)$. But $\operatorname{Vol}(\partial A) = 0$ because A is measurable and $\operatorname{Vol} f(C) = 0$ by Sard's theorem 14.11. Therefore, $\operatorname{Vol} \partial f(A) = 0$ and thus $f(A)$ is measurable. ■

14.4.2 Volume of a parallelepiped revisited

The following lemma provides a way of computing the volume via packing by balls rather then cubes. An *admissible set of balls in* A is any finite set of disjoint balls $B_1, \dots, B_K \subset A$

Lemma 14.15. *Let A be a measurable set. Then* $\operatorname{Vol} A$ *is the supremum of the total volume of admissible sets of balls in A. Here the supremum is taken over all admissible sets of balls in A.*

Proof. Let us denote this supremum by β. The monotonicity of volume implies that $\beta \leq \operatorname{Vol} A$. Suppose that $\beta < VolA$. Let us denote by μ_n the volume of an n-dimensional ball of radius 1 (we will compute this number later on). This ball is contained in a cube of volume 2^n. It follows then that the ratio of the volume of any ball to the volume of the cube to which it is inscribed is equal to $\frac{\mu_n}{2^n}$. Choose an $\epsilon < \frac{\mu_n}{2^n}(\mathrm{Vol}A - \beta)$. Then there exists a finite set of disjoint balls $B_1, \dots, B_K \subset A$ such that $\mathrm{Vol}(\bigcup\limits_1^K B_j) > \beta - \epsilon$. The volume of the complement $C = A \setminus \bigcup\limits_1^K B_j$ satisfies

$$\mathrm{Vol}C = \mathrm{Vol}A - \mathrm{Vol}\left(\bigcup_1^K B_j\right) > \mathrm{Vol}A - \beta.$$

Hence there exists a partition $\mathcal{P}$ of P by cubes such that the total volume of cubes $Q_1, \dots, Q_L$ contained in C is $> \mathrm{Vol}A - \beta$. Let us inscribe in each of the cubes Q_j a ball $\widetilde{B}_j$. Then $B_1, \dots, B_K, \widetilde{B}_1, \dots, \widetilde{B}_L$ is an admissible set of balls in A. Indeed, all these balls are disjoint and contained in A. The total volume of this admissible set is equal to

$$\sum_1^K \mathrm{Vol}B_j + \sum_1^L \mathrm{Vol}\widetilde{B}_i \geq \beta - \epsilon + \frac{\mu_n}{2^n}(\mathrm{Vol}A - \beta) > \beta,$$

in view of our choice of ϵ, but this contradicts to our assumption $\beta < \mathrm{Vol}A$. Hence, we have $\beta = \mathrm{Vol}A$. ∎

Lemma 14.16. *Let $A \subset V$ be any measurable set in a Euclidean space V. Then for any linear orthogonal transformation $F : V \to V$ the set $F(A)$ is also measurable and we have* $\mathrm{Vol}(F(A)) = \mathrm{Vol}(A)$.

Proof. First note that if $\mathrm{Vol}A = 0$ then the claim follows from Lemma 14.8. Indeed, an orthogonal transformation is, of course a smooth map.

Let now A be an arbitrary measurable set. Note that $\partial F(A) = F(\partial A)$. Measurability of A implies $\mathrm{Vol}(\partial A) = 0$. Hence, as we just have explained, $\mathrm{Vol}(\partial F(A)) = \mathrm{Vol}(F(\partial A)) = 0$, and hence $F(A)$ is measurable. According to Lemma 14.15 the volume of a measurable set can be computed as a supremum of the total volume of disjoint inscribed balls. But the orthogonal transformation F moves disjoint balls to disjoint balls of the same size, and hence $\mathrm{Vol}A = \mathrm{Vol}F(A)$. ∎

Next proposition shows that the volume of a parallelepiped can be computed by formula (9.3.1) from Section 9.3.

Proposition 14.17. *Let $v_1, \dots, v_n \in V$ be linearly independent vectors. Then*

$$\mathrm{Vol}\,P(v_1, \dots, v_n) = |x_1 \wedge \dots \wedge x_n(v_1 \dots, v_n)|. \tag{14.4.1}$$

Proof. The formula (14.4.1) holds for rectangles, i.e. when $v_j = c_j e_j$ for some non-zero numbers c_j, $j = 1, \dots n$. Using Lemma 14.16 we conclude that it also holds for any orthogonal basis. Indeed, any such basis can be moved by an orthogonal transformation to a basis of the above form $c_j e_j,\ j = 1, \dots n$. Lemma 14.16 ensures that the volume does not change under the orthogonal transformation, while Proposition 7.17 implies the same about $|x_1 \wedge \dots \wedge x_n(v_1 \dots, v_n)|$.

The Gram-Schmidt orthogonalization process shows that one can pass from any basis to an orthogonal basis by a sequence of following elementary operations:

- \- reordering of basic vectors, and
- *shears*, i.e. an addition to the last vector a linear combination of the other ones:

$$v_1, \dots, v_{n-1}, v_n \mapsto v_1, \dots, v_{n-1}, v_n + \sum_1^{n-1} \lambda_j v_j.$$

Note that the reordering of vectors $v_1, \dots, v_n$ changes neither $\mathrm{Vol} P(v_1, \dots, v_n)$, nor the absolute value $|x_1 \wedge \dots \wedge x_n(v_1 \dots, v_n)|$. On the other hand, a shear does not change

$$x_1 \wedge \dots \wedge x_n(v_1 \dots, v_n).$$

It remains to be shown that a shear does not change the volume of a parallelepiped. We will consider here only the case $n = 2$ and will leave to the reader the extension of the argument to the general case.

Let v_1, v_2 be two orthogonal vectors in $\mathbb{R}^2$. We can assume that $v_1 = (a, 0)$, $v_2 = (0, b)$ for $a, b > 0$, because we already proved the invariance of volume under orthogonal transformations. Let $v_2' = v_2 + \lambda v_1 = (a', b)$, where $a' = a + \lambda b$. Let us partition the rectangle $P = P(v_1, v_2)$ into N^2 smaller rectangles $P_{i,j}$, $i, j = 0, \dots, N-1$, of equal size. We number the rectangles in such a way that the first index corresponds to the first coordinate, so that the rectangles $P_{00}, \dots, P_{N-1,0}$ form the lower layer, $P_{01}, \dots, P_{N-1,1}$ the second layer, etc. Let us now shift the rectangles in k-th layer horizontally by the vector $(\frac{k\lambda b}{N}, 0)$. Then the total volume of the rectangles, denoted $\widetilde{P}_{ij}$ remains the same, while when $N \to \infty$ the volume of part of the parallelogram $P(v_1, v_2')$ that *is not covered* by rectangles $\widetilde{P}_{i,j}$, $i, j = 0, \dots, N-1$ converges to 0.

■

14.5 Fubini's Theorem

Let us consider $\mathbb{R}^n$ as a direct product of $\mathbb{R}^k$ and $\mathbb{R}^{n-k}$ for some $k = 1, \dots, n-1$. We will denote coordinates in $\mathbb{R}^k$ by $x = (x_1, \dots, x_k)$ and coordinates in $\mathbb{R}^{n-k}$ by $y = (y_1, \dots, y_{n-k})$, so the coordinates in $\mathbb{R}^n$ are denoted by $(x_1, \dots, x_k, y_1, \dots, y_{n-k})$. Given rectangles $P_1 \subset \mathbb{R}^k$ and $P_2 \subset \mathbb{R}^{n-k}$ their product $P = P_1 \times P_2$ is a rectangle in $\mathbb{R}^n$.

The following theorem provides us with a basic tool for computing multiple integrals.

Theorem 14.18 (Guido Fubini). *Suppose that a function $f : P \to \mathbb{R}$ is integrable over P. Given a point $x \in P_1$ let us define a function $f_x : P_2 \to \mathbb{R}$ by the formula $f_x(y) = f(x, y)$, $y \in P_2$. Then*

$$\int\limits_P f dV_n = \int\limits_{P_1} \left(\underline{\int\limits_{P_2}} f_x dV_{n-k} \right) dV_k = \int\limits_{P_1} \left(\overline{\int\limits_{P_2}} f_x dV_{n-k} \right) dV_k.$$

In particular, if the function f_x is integrable for all (or almost all) $x \in P_1$ then one has

$$\int\limits_P f dV_n = \int\limits_{P_1} \left(\int\limits_{P_2} f_x dV_{n-k} \right) dV_k.$$

Here by writing dV_k, dV_{n-k} and dV_n we emphasize the integration with respect to the k-, $(n-k)$- and n-dimensional volumes, respectively.

Proof. Choose any partition $\mathcal{P}_1$ of P_1 and $\mathcal{P}_2$ of P_2. We will denote elements of the partition $\mathcal{P}_1$ by $P_1^{\mathbf{j}}$ and elements of the partition $\mathcal{P}_2$ by $P_2^{\mathbf{i}}$. Then products of $P^{\mathbf{j},\mathbf{i}} = P_1^{\mathbf{j}} \times P_2^{\mathbf{i}}$ form a partition $\mathcal{P}$ of $P = P_1 \times P_2$. Let us denote

$$\overline{I}(x) := \overline{\int\limits_{P_2}} f_x, \; \underline{I}(x) := \underline{\int\limits_{P_2}} f_x, \; x \in P_1.$$

Let us show that

$$L(f, \mathcal{P}) \le L(\underline{I}, \mathcal{P}_1) \le U(\overline{I}, \mathcal{P}_1) \le U(f, \mathcal{P}). \tag{14.5.1}$$

Indeed, we have

$$L(f, \mathcal{P}) = \sum_{\mathbf{j}} \sum_{\mathbf{i}} m_{\mathbf{j},\mathbf{i}}(f) \mathrm{Vol}_n P^{\mathbf{j},\mathbf{i}}.$$

Here the first sum is taken over all multi-indices $\mathbf{j}$ of the partition $\mathcal{P}_1$, and the second sum is taken over all multi-indices $\mathbf{i}$ of the partition $\mathcal{P}_2$. On the other hand,

$$L(\underline{I}, \mathcal{P}_1) = \sum_{\mathbf{j}} \inf_{x \in P_1^{\mathbf{j}}} \left(\underline{\int_{P_2}} f_x dV_{n-k} \right) \mathrm{Vol}_k P_1^{\mathbf{j}}.$$

Note that for every $x \in P_1^{\mathbf{j}}$ we have

$$\underline{\int_{P_2}} f_x dV_{n-k} \geq L(f_x; \mathcal{P}_2) = \sum_{\mathbf{i}} m_{\mathbf{i}}(f_x) \mathrm{Vol}_{n-k}(P_2^{\mathbf{i}}) \geq \sum_{\mathbf{i}} m_{\mathbf{i},\mathbf{j}}(f) \mathrm{Vol}_{n-k}(P_2^{\mathbf{i}}),$$

and hence

$$\inf_{x \in P_1^{\mathbf{j}}} \underline{\int_{P_2}} f_x dV_{n-k} \geq \sum_{\mathbf{i}} m_{\mathbf{i},\mathbf{j}}(f) \mathrm{Vol}_{n-k}(P_2^{\mathbf{i}}).$$

Therefore,

$$L(\underline{I}, \mathcal{P}_1) \geq \sum_{\mathbf{j}} \sum_{\mathbf{i}} m_{\mathbf{i},\mathbf{j}}(f) \mathrm{Vol}_{n-k}(P_2^{\mathbf{i}}) \mathrm{Vol}_k(P_1^{\mathbf{j}}) = \sum_{\mathbf{j}} \sum_{\mathbf{i}} m_{\mathbf{j},\mathbf{i}}(f) \mathrm{Vol}_n(P^{\mathbf{j},\mathbf{i}}) = L(f, \mathcal{P}).$$

Similarly, one can check that $U(\overline{I}, \mathcal{P}_1) \leq U(f, \mathcal{P})$. Together with an obvious inequality $L(\underline{I}, \mathcal{P}_1) \leq U(\overline{I}, \mathcal{P}_1)$ this completes the proof of (14.5.1). Thus we have

$$\max(U(\overline{I}, \mathcal{P}_1) - L(\overline{I}, \mathcal{P}_1), U(\underline{I}, \mathcal{P}_1) - L(\underline{I}, \mathcal{P}_1)) \leq U(\overline{I}, \mathcal{P}_1) - L(\underline{I}, \mathcal{P}_1) \leq U(f, \mathcal{P}) - L(f, \mathcal{P}).$$

By assumption for appropriate choices of partitions, the right-hand side can be made $< \epsilon$ for any a priori given $\epsilon > 0$. This implies the integrability of the function $\underline{I}(x)$ and $\overline{I}(x)$ over P_1. But then we can write

$$\int_{P_1} \underline{I}(x) dV_{n-k} = \lim_{\delta(\mathcal{P}_1) \to 0} L(\underline{I}; \mathcal{P}_1)$$

and

$$\int_{P_1} \overline{I}(x) dV_{n-k} = \lim_{\delta(\mathcal{P}_1) \to 0} U(\overline{I}; \mathcal{P}_1).$$

We also have

$$\lim_{\delta(\mathcal{P}) \to 0} L(f; \mathcal{P}) = \lim_{\delta(\mathcal{P}) \to 0} U(f; \mathcal{P}) = \int_{P} f dV_n.$$

Hence, the inequality (14.5.1) implies that

$$\int_P f dV_n = \int_{P_1} \left(\underline{\int}_{P_2} f_x dV_{n-k} \right) dV_k = \int_{P_1} \left(\overline{\int}_{P_2} f_x dV_{n-k} \right) dV_k .$$

■

Corollary 14.19. *Suppose $f : P \to \mathbb{R}$ is a continuous function. Then*

$$\int_P f = \int_{P_1} \int_{P_2} f_x = \int_{P_2} \int_{P_1} f_y .$$

Thus if we switch back to the notation $x_1, \dots, x_n$ for coordinates in $\mathbb{R}^n$, and if $P = \{a_1 \le x_1 \le b_1, \dots, a_n \le x_n \le b_n\}$ then we can write

$$\int_P f = \int_{a_n}^{b_n} \left(\dots \left(\int_{a_1}^{b_1} f(x_1, \dots, x_n) dx_1 \right) \dots \right) dx_n . \qquad (14.5.2)$$

The integral in the right-hand side of (14.5.2) is called an *iterated integral*. Note that the order of integration is irrelevant there. In particular, for continuous functions one can change the order of integration in the iterated integrals.

■

14.6 Integration of n-forms over domains in an n-dimensional space

Differential forms are much better suited to be integrated than functions. For integrating a function, one needs a measure. To integrate a differential form, one needs nothing except an orientation of the domain of integration.

Let us start with the integration of a n-form over a domain in a n-dimensional space. Let ω be a n-form on a domain $U \subset V$, $\dim V = n$.

Let us fix now an orientation of the space V. Pick any coordinate system $(x_1 \dots x_n)$ that agrees with the chosen orientation.

We proceed similar to the way we defined an integral of a function. Let us fix a rectangle $P = P(a_1, b_1; a_2, b_2; \dots ; a_n, b_n) = \{a_i \le x_i \le b_i;\ i = 1, \dots, n\}$. Choose its partition $\mathcal{P}$ by N^n

smaller rectangles $P_{\mathbf{j}} = \{t^1_{j_n} \le x_1 \le t^1_{j_n+1}, \ldots, t^n_{j_1} \le x_1 \le t^n_{j_1+1}\}$, where $\mathbf{j} = (j_1, \ldots, j_n)$ and each index j_k takes values between 0 and $N-1$. Let us fix a point $c_{\mathbf{j}} \in P_{\mathbf{j}}$ and denote by C the set of all such $c_{\mathbf{j}}$. We also denote by $t_{\mathbf{j}}$ the point with coordinates $t^1_{j_1}, \ldots, t^n_{j_n}$ and by $T_{\mathbf{j},m} \in V_{c_{\mathbf{j}}}$, $m = 1, \ldots, n$ the vector $t_{\mathbf{j}+1_m} - t_{\mathbf{j}}$, parallel-transported to the point $c_{\mathbf{j}}$. Here we use the notation $\mathbf{j} + 1_m$ for the multi-index $j_1, \ldots, j_{m-1}, j_m + 1, j_{m+1}, \ldots, j_n$. Thus the vector $T_{\mathbf{j},m}$ is parallel to the m-th basic vector and has the length $|t_{j_m+1} - t_{j_m}|$.

Given a differential n-form α on P we form an integral sum

$$I(\alpha; \mathcal{P}, C) = \sum_{\mathbf{j}} \alpha_{c_{\mathbf{j}}}(T_{\mathbf{j},1}, T_{\mathbf{j},2}, \ldots, T_{\mathbf{j},n}), \tag{14.6.1}$$

where the sum is taken over all elements of the partition. We call an n-form α *integrable* if there exists a limit $\lim\limits_{\delta(\mathcal{P}) \to 0} I(\alpha; \mathcal{P}, C)$ which we denote by $\int\limits_P \alpha$ and call the *integral of α over P*. Note that if $\alpha = f(x)dx_1 \wedge \cdots \wedge dx_n$ then the integral sum $I(\alpha, \mathcal{P}, C)$ from (14.6.1) coincides with the integral sum $I(f; \mathcal{P}, C)$ from (14.1.2) for the function f. Thus the integrability of α is the same as integrability of f and we have

$$\int\limits_P f(x)dx_1 \wedge \cdots \wedge dx_n = \int\limits_P f dV. \tag{14.6.2}$$

Note, however, that the equality (14.6.2) holds *only if the coordinate system $(x_1, \ldots, x_n)$ defines the given orientation of the space V*. The integral $\int\limits_P f(x)dx_1 \wedge \cdots \wedge dx_n$ changes its sign with a change of the orientation while the integral $\int\limits_P f dV$ is not sensitive to the orientation of the space V.

It is not clear from the above definition whether the integral of a differential form depends on our choice of the coordinate system. It turns out that it does not, as the following theorem, which is the main result of this section, shows. Moreover, we will see that one even can use arbitrarty curvilinear coordinates.

In what follows we use the convention introduced at the end of Section 2.1. Namely by a diffeomorphism between two closed subsets of vector spaces we mean a diffeomorphism between their neighborhoods.

Theorem 14.20. *Let $A, B \subset \mathbb{R}^n$ be two measurable compact subsets. Let $f : A \to B$ be an orientation preserving diffeomorphism. Let η be a differential n-form defined on B. Then if η is* *integrable over B then $f^*\alpha$ is integrable over A and we have*

$$\int\limits_A f^*\eta = \int\limits_B \eta. \tag{14.6.3}$$

For an orientation reversing diffeomorphism f we have $\int\limits_A f^*\eta = -\int\limits_B \eta$.

Let $\alpha = g(x)dx_1 \wedge \dots \wedge dx_n$. Then $f^*\alpha = g \circ f \det Df dx_1 \wedge \dots \wedge dx_n$, and hence the formula (14.6.3) can be rewritten as

$$\int\limits_B g(x_1, \dots, x_n)dx_1 \wedge \dots \wedge dx_n = \int\limits_A g \circ f \det Df \, dx_1 \wedge \dots \wedge dx_n.$$

Here

$$\det Df = \begin{vmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \dots & \dots & \dots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{vmatrix}$$

is the determinant of the Jacobian matrix of $f = (f_1, \dots, f_n)$.

Hence, in view of formula (14.6.2) we get the following *change of variables* formula for multiple integrals of functions.

Corollary 14.21. [Change of variables in a multiple integral] *Let $g : B \to \mathbb{R}$ be an integrable function and $f : A \to B$ a diffeomorphism. Then the function $g \circ f$ is also integrable and*

$$\int\limits_B g dV = \int\limits_A g \circ f |\det Df| dV\,. \tag{14.6.4}$$

We begin the proof with the following special case of Theorem 14.20.

Proposition 14.22. *The statement of 14.20 holds when $\eta = dx_1 \wedge \dots \wedge dx_n$ and the set A is the unit cube $I = I^n$. In other words,*

$$\mathrm{Vol} f(I) = \left| \int\limits_I f^*\eta \right|.$$

We will use below the following notation.

For any set $A \subset V$ and any positive number $\lambda > 0$ we denote by λA the set $\{\lambda x, x \in A\}$. For any linear operator $F : V \to W$ between two Euclidean spaces V and W we define its *norm* $||F||$ by the formula

$$||F|| = \max_{||v||=1} ||F(v)|| = \max_{v \in V, v \neq 0} \frac{||F(v)||}{||v||}.$$

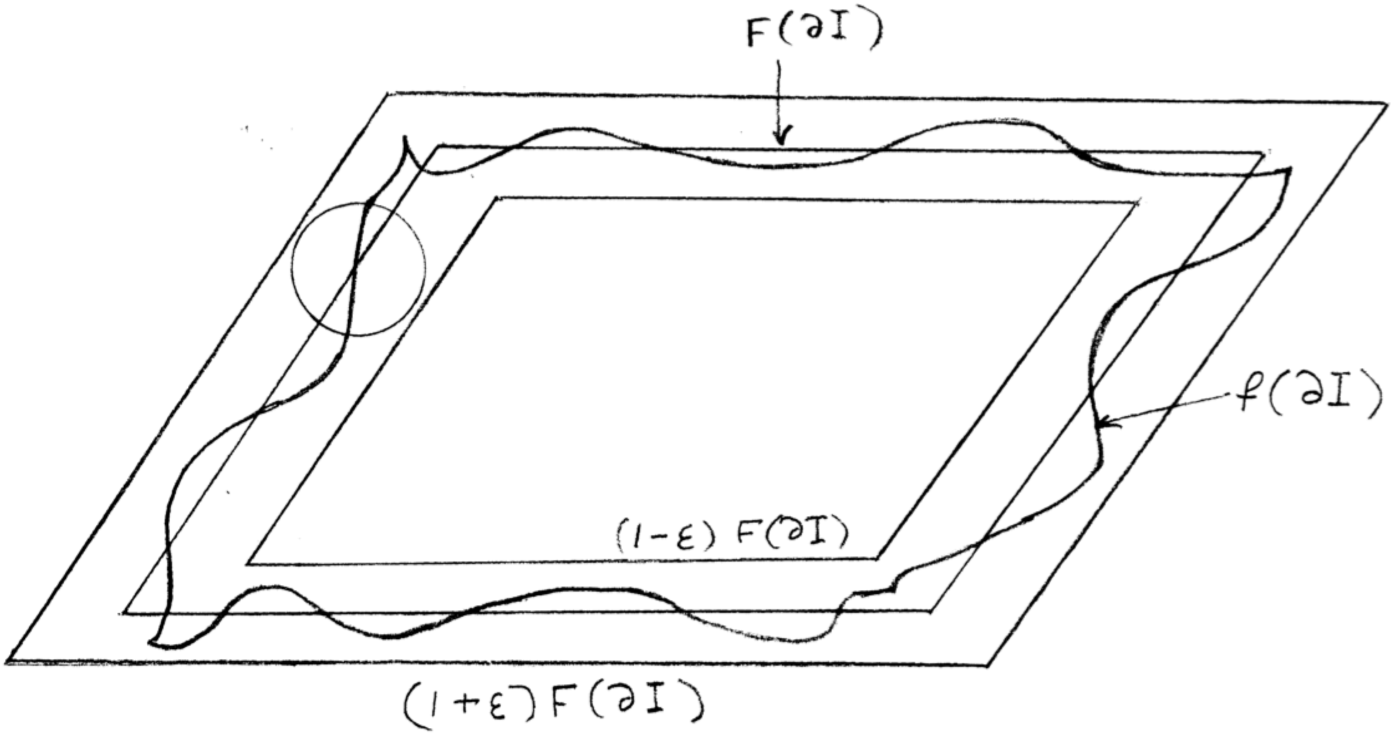


Figure 14.1: Image of a cube under a diffeomorphism and its linearization

Equivalently, we can define $||F||$ as follows. The linear map F maps the unit sphere in the space V onto an ellipsoid in the space W. Then $||F||$ is the biggest semi-axis of this ellipsoid.

Let us begin by observing the following geometric fact:

Lemma 14.23. *Let* $I = \{|x_i| \leq \frac{1}{2}, i = 1, \ldots, n\} \subset \mathbb{R}^n$ *be the unit cube centered at* 0 *and* $F : \mathbb{R}^n \to \mathbb{R}^n$ *a non-degenerate linear map. Take any* $\epsilon \in (0,1)$ *and set* $\sigma = \frac{\epsilon}{||F^{-1}||}$*. Then for any boundary point* $z \in \partial I$ *we have*

$$B_\sigma(F(z)) \subset (1+\epsilon)F(I) \setminus (1-\epsilon)F(I), \tag{14.6.5}$$

see Fig. 14.6

Proof. Inclusion (14.6.5) can be rewritten as

$$F^{-1}(B_\sigma(F(z)) \subset (1+\epsilon)I \setminus (1-\epsilon)I.$$

But the set $F^{-1}(B_\sigma(F(z))$ is an ellipsoid centered at z whose greatest semi-axis is equal to $\sigma||F^{-1}||$. Hence, if $\sigma||F^{-1}|| \leq \epsilon$ then $F^{-1}(B_\sigma(F(z)) \subset (1+\epsilon)I \setminus (1-\epsilon)I$. ■

Recall that we denote by δI the cube I scaled with the coefficient δ, i.e. $\delta I = \{|x_i| \le \frac{\delta}{2}, i = 1,\dots,n\} \subset \mathbb{R}^n$. We will also need

Lemma 14.24. *Let $U \subset \mathbb{R}^n$ be an open set, $f : U \to \mathbb{R}^n$ such that $f(0) = 0$. Suppose that f is differentiable at 0 and its differential $F = d_0 f : \mathbb{R}^n \to \mathbb{R}^n$ at 0 is non-degenerate. Then for any $\epsilon \in (0,1)$ there exists $\delta > 0$ such that*

$$(1-\epsilon)F(\delta I) \subset f(\delta I) \subset (1+\epsilon)F(\delta I), \tag{14.6.6}$$

see Fig. 14.6.

Proof. First, we note that inclusion (14.6.5) implies, using linearity of F, that for any $\delta > 0$ we have

$$B_{\delta\sigma}(F(z)) \subset (1+\epsilon)F(\delta I) \setminus (1-\epsilon)F(\delta I), \tag{14.6.7}$$

where $z \in \partial(\delta I)$ and, as in Lemma 14.23, we assume that $\epsilon = \sigma||F^{-1}||$.

According to the definition of differentiability we have

$$f(h) = F(h) + o(||h||).$$

Denote $\widetilde{\sigma} := \frac{\sigma}{\sqrt{n}} = \frac{\epsilon}{\sqrt{n}||F^{-1}||}$. There exists $\rho > 0$ such that if $||h|| \le \rho$ then

$$||f(h) - F(h)|| \le \widetilde{\sigma}||h|| \le \widetilde{\sigma}\rho.$$

Denote $\delta := \frac{\rho}{\sqrt{n}}$. Then $\delta I \subset B_\rho(0)$, and hence $||f(z) - F(z)|| \le \widetilde{\sigma}\rho$ for any $z \in \delta I$. In particular, for any point $z \in \partial(\delta I)$ we have

$$f(z) \in B_{\widetilde{\sigma}\rho}(F(z)) = B_{\sqrt{n}\widetilde{\sigma}\delta}(F(z)) = B_{\sigma\delta}(F(z)),$$

and therefore in view of (14.6.7)

$$f(\partial(\delta I)) \subset (1+\epsilon)F(\delta I) \setminus (1-\epsilon)F(\delta I).$$

But this is equivalent to inclusion (14.6.6). ■

Lemma 14.25. *Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be a non-degenerate orientation preserving linear map, $P = P(v_1,\dots,v_n)$ a parallelepiped, and $\eta = dx_1 \wedge \dots \wedge dx_n$. Then $\int\limits_{F(P)} \eta = \int\limits_P F^*\eta$. Here we assume that the orientation of P and $F(P)$ are given by the orientation of $\mathbb{R}^n$.*

Proof. We have $\int\limits_{F(P)} \eta = \int\limits_{F(P)} dx_1 \wedge \dots dx_n = \mathrm{Vol}F(P) = (\det F)\mathrm{Vol}P$. On the other hand, $F^*\eta = \det F\eta$, and hence $\int\limits_{P} F^*\eta = \det F \int\limits_{P} \eta = (\det F)\mathrm{Vol}P$. ■

Proof of Proposition 14.22. We have $f^*\eta = (\det Df)dx_1 \wedge \dots \wedge dx_n$, and hence the form $f^*\eta$ is integrable because f is C^1-smooth, and hence $\det Df$ is continuous.

Choose a partition $\mathcal{P}$ of the cube I by N^n small cubes I_K, $K = 1, \dots, N^n$, of the same size $\frac{1}{N}$. Let $c_K \in I_K$ be the center of the cube I_K. Then

$$\int\limits_I f^*\eta = \sum_{K=1}^{N^n} \int\limits_{I_K} f^*\eta.$$

Note that in view of the uniform continuity of the function $\det Df$, for any $\epsilon > 0$ the number N can be chosen so large that $|\det Df(x) - \det Df(x')| < \epsilon$ for any two points $x, x' \in I_K$ and any $K = 1, \dots, N^n$. Let η_K be the form $\det Df(c_k)dx_1 \wedge \dots \wedge dx_n$ on I_K. Then

$$\left|\int\limits_{I_K} f^*\eta - \int\limits_{I_K} \eta_K\right| \le \int\limits_{I_K} |\det Df(x) - \det Df(c_K)|\, dx_1 \wedge \dots \wedge dx_n \le \epsilon \mathrm{Vol}(I_K) = \frac{\epsilon}{N^n}.$$

Thus

$$\left|\int\limits_I f^*\eta - \sum_{K=1}^{N^n} \int\limits_{I_K} \eta_K\right| \le \epsilon. \tag{14.6.8}$$

Next, let us analyze the integral $\int\limits_{I_K} \eta_K$. Denote $F_K := d_{c_K}(f)$. We can assume without loss of generality that $c_K = 0$, and $f(c_K) = 0$, and hence F_K can be viewed just as a linear map $\mathbb{R}^n \to \mathbb{R}^n$.

Using Lemma 14.24 we have for a sufficiently large N

$$(1-\epsilon)F_K(I_K) \subset f(I_K) \to (1+\epsilon)F_K(I_K).$$

Again in view of compactness of I the number ϵ can be chosen the same for all cubes I_K. Hence

$$(1-\epsilon)^n \mathrm{Vol}(F_K(I_K)) \le \mathrm{Vol}f(I_K) \le (1+\epsilon)^n \mathrm{Vol}(F_K(I_K)). \tag{14.6.9}$$

Note that $\mathrm{Vol}f(I_K) = \int\limits_{f(I_K)} \eta$, and hence summing up inequality (14.6.9) over K we get

$$(1-\epsilon)^n \sum_{K=1}^{N^n} \mathrm{Vol}(F_K(I_K)) \le \sum_{K=1}^{N^n} \mathrm{Vol}f(I_K) = \sum_{K=1}^{N^n} \int\limits_{f(I_K)} \eta = \int\limits_{f(I)} \eta \le (1+\epsilon)^n \sum_{K=1}^{N^n} \mathrm{Vol}(F_K(I_K)). \tag{14.6.10}$$

Note that $\eta_K = F_K^*\eta$ and by Lemma 14.25 we have $\int\limits_{I_K} \eta_K = \int\limits_{I_K} F_K^*\eta = \int\limits_{F_K(I_K)} \eta = \mathrm{Vol}(F_K(I_K))$. Hence, it follows from (14.6.10) that

$$(1-\epsilon)^n \sum_{K=1}^{N^n} \int\limits_{I_K} \eta_K \leq \int\limits_{f(I)} \eta \leq (1+\epsilon)^n \sum_{K=1}^{N^n} \int\limits_{I_K} \eta_K. \tag{14.6.11}$$

Recall that from (14.6.8) we have

$$\int\limits_{I} f^*\eta - \epsilon \leq \sum_{K=1}^{N^n} \int\limits_{I_K} \leq \int\limits_{I} f^*\eta + \epsilon.$$

Combining with (14.6.11) we get

$$(1-\epsilon)^n \left(\int\limits_{I} f^*\eta - \epsilon \right) \leq \int\limits_{f(I)} \eta \leq (1+\epsilon)^n \left(\int\limits_{I} f^*\eta + \epsilon \right). \tag{14.6.12}$$

Passing to the limit when $\epsilon \to 0$ we get

$$\int\limits_{I} f^*\eta \leq \int\limits_{f(I)} \eta \leq \int\limits_{I} f^*\eta, \tag{14.6.13}$$

i.e. $\int\limits_{I} f^*\eta = \int\limits_{f(I)} \eta$. ∎

Corollary 14.26. *The statement of Theorem 14.20 holds for $\eta = dx_1 \wedge \dots \wedge dx_n$ and an arbitrary measurable A.*

Proof. Suppose for determinacy that f preserves the orientation. By assumption the diffeomorphism A extends to an open neighborhood $U \supset A$. Consider a cube $P \supset A$. Choose a $\delta > 0$ and consider a partition $\mathcal{P}$ of P by small cubes of size $< \delta$. Denote by A_δ^+ the union of elements of the partition which intersect A, and by A_δ^- the union of elements which are completely inside A. If δ is small enough then $A_\delta^+ \subset U$. We have

$$\int\limits_{A_\delta^-} f^*\eta \leq \int\limits_{A} f^*\eta \leq \int\limits_{A_\delta^+} f^*\eta$$

and

$$\int\limits_{A_\delta^+} f^*\eta - \int\limits_{A_\delta^-} f^*\eta \underset{\delta \to 0}{\to} 0.$$

On the other hand, each of the sets A_δ^+ is a union of cubes. Hence, Proposition 14.22 implies that

$$\int\limits_{A_\delta^\pm} f^*\eta = \int\limits_{f(A_\delta^\pm)} \eta = \mathrm{Vol}(A_\delta^\pm).$$

But $f(A_\delta^-) \subset f(A) \subset f(A_\delta^+)$, and hence $\mathrm{Vol}(f(A_\delta^-)) \le \mathrm{Vol} f(A) \le \mathrm{Vol}(f(A_\delta^+))$ which implies when $\delta \to 0$ that $\mathrm{Vol} f(A) = \int\limits_{f(A)} \eta = \int\limits_A f^*\eta$.

Proof of Theorem 14.20. Let us recall that the diffeomorphism f is defined as a diffeomorphism between open neighborhoods $U \supset A$ and $U' \supset B$. We also assume that the form η is extended to U' as equal to 0 outside B. The form η can be written as $hdx_1 \wedge \cdots \wedge dx_n$. Let us take a partition $\mathcal{P}$ of a rectangular containg U' by cubes $I_{\mathbf{j}}$ of the same size δ. Consider forms $\eta_{\mathbf{j}}^+ := M_{\mathbf{j}}(h)dx_1 \wedge \cdots \wedge dx_n$ and $\eta_{\mathbf{j}}^- := m_{\mathbf{j}}(h)dx_1 \wedge \cdots \wedge dx_n$ on $I_{\mathbf{j}}$, where $m_{\mathbf{j}}(h) = \inf_{I_{\mathbf{j}}} h,\ M_{\mathbf{j}}(h) = \sup_{I_{\mathbf{j}}}(h)$. Let $\eta^\pm$ be the form on U' equal to $\eta_{\mathbf{j}}^\pm$ on each cube $I_{\mathbf{j}}$. The assumption of integrability of η over B guarantees that for any $\epsilon > 0$ if δ is chosen small enough we have $\int_B \eta^+ - \int_B \eta \le \epsilon$. The forms $f^*\eta^\pm$ are a.e. continuous, and hence integrable over A and we have $\int\limits_A f^*\eta^- \le \int\limits_{\underline{A}} f^*\eta \le \int\limits_A f^*\eta^+$. Hence, if we prove that $\int\limits_A f^*\eta^\pm = \int\limits_B \eta^\pm$ then this will imply that η is integrable and $\int\limits_A \eta = \int\limits_B \eta$.

On the other hand, $\int\limits_B \eta^\pm = \sum\limits_{\mathbf{j}} \int\limits_{I_{\mathbf{j}}} \eta_{\mathbf{j}}^\pm$ and $\int\limits_A f^*\eta^\pm = \sum\limits_{\mathbf{j}} \int\limits_{B_{\mathbf{j}}} f^*\eta_{\mathbf{j}}^\pm$, where $B_{\mathbf{j}} = f^{-1}(I_{\mathbf{j}})$. But according to Corollary 14.26 we have $\int\limits_{B_{\mathbf{j}}} f^*\eta_{\mathbf{j}}^\pm = \int\limits_{I_{\mathbf{j}}} \eta_{\mathbf{j}}^\pm$, and hence $\int\limits_A f^*\eta^\pm = \int\limits_B \eta^\pm$. ∎

Chapter 15

Manifolds and submanifolds

15.1 Manifolds

Manifolds of dimension n are spaces which locally look like open subsets of $\mathbb{R}^n$ but globally could be much more complicated. We give a precise definition below.

A structure of an n-dimensional C^k*-smooth* (resp. topological if $k = 0$) *manifold* on a set M is is given by subsets $U_\lambda \subset X$, $\lambda \in \Lambda$, where Λ is a finite or countable set of indices, and for every $\lambda \in \Lambda$ a map $\Phi_\lambda : U_\lambda \to \mathbb{R}^n$ such that

M1. $M = \bigcup\limits_{\lambda\in\Lambda} U_\lambda$.

M2. The image $G_\lambda = \Phi_\lambda(U_\lambda)$ is an open set in $\mathbb{R}^n$.

M3. The map Φ_λ viewed as a map $U_\lambda \to G_\lambda$ is one-to-one.

M4. For any two sets $U_\lambda, U_\mu, \lambda, \mu \in \Lambda$ the images $\Phi_\lambda(U_\lambda \cap U_\mu), \Psi_\mu(U_\lambda \cap U_\mu) \subset \mathbb{R}^n$ are open and the map

$$h_{\lambda\mu} := \Phi_\mu \circ \Phi_\lambda^{-1} : \Phi_\lambda(U_\lambda \cap U_\mu) \to \Phi_\mu(U_\lambda \cap U_\mu) \subset \mathbb{R}^n$$

is a C^k-diffeomorphism (resp. homeomorphism).

Sets U_λ are called *coordinate neighborhoods* and maps $\Phi_\lambda : U_\lambda \to \mathbb{R}^n$ are called *coordinate maps*. The pairs $(U_\lambda, \Phi_\lambda)$ are also called *local coordinate charts*. The maps $h_{\lambda\mu}$ are called *transiton maps* between different coordinate charts. The inverse maps $\Psi_\lambda = \Phi_\lambda^{-1} : G_\lambda \to U_\lambda$ are called *(local) parametrization maps.* An *atlas* is a collection $\mathfrak{A} = \{U_\lambda, \Phi_\lambda\}_{\lambda \in \Lambda}$ of all coordinate charts.

One says that two atlases $\mathfrak{A} = \{U_\lambda, \Phi_\lambda\}_{\lambda \in \Lambda}$ and $\mathfrak{A}' = \{U'_\gamma, \Phi'_\gamma\}_{\gamma \in \Gamma}$ on the same manifold X are *equivalent*, or that they *define the same smooth structure on* X if their union $\mathfrak{A} \cup \mathfrak{A}' = \{(U_\lambda, \Phi_\lambda), (U'_\gamma, \Phi'_\gamma)\}_{\lambda \in \Lambda, \gamma \in \Gamma}$ is again an atlas on X. In other words, two atlases define the same smooth structure if transition maps from local coordinates in one of the atlases to the local coordinates in the other one are given by smooth functions.

A subset $G \subset M$ is called *open* if for every $\lambda \in \Lambda$ the image $\Phi_\lambda(G \cap U_\lambda) \subset \mathbb{R}^n$ is open. In particular, coordinate charts U_λ themselves are open, and we can equivalently say that a set G is open if its intersection with every coordinate chart is open. By a *neighborhood* of a point $a \in M$ we will mean any open subset $U \subset M$ such that $a \in U$.

It is useful to observe that if $\mathfrak{A} = \{U_\lambda, \Phi_\lambda\}_{\lambda \in \Lambda}$ is an atlas, then if we consider any covering $U_\lambda = \bigcup_\alpha U_{\lambda,\alpha}$ of U_λ by open subsets $U_{\lambda,\alpha} \subset U_\lambda, \alpha \in A_\lambda$, then

$$\widetilde{\mathfrak{A}} := \{U_{\lambda,\alpha}, \Phi_\lambda|_{U_{\lambda,\alpha}}\}_{\alpha \in A_\lambda, \lambda \in \Lambda}$$

is an atlas equivalent to $\mathfrak{A}$. In particular, one can always choose an atlas which consists of arbitrary small neighborhoods.

Given two smooth manifolds M and $\widetilde{M}$ of dimension m and n then a map $f : M \to \widetilde{M}$ is called *continuous* if if for every point $a \in M$ there exist local coordinate charts $(U_\lambda, \Phi_\lambda)$ in M and $(\widetilde{U}_\lambda, \widetilde{\Phi}_\lambda)$ in $\widetilde{M}_\lambda$, such that $a \in U_\lambda, f(U_\lambda) \subset \widetilde{U}_\lambda$ and the composition map

$$G_\lambda = \Phi_\lambda(U_\lambda) \xrightarrow{\Psi_\lambda} U_\lambda \xrightarrow{f} \widetilde{U}_\lambda \xrightarrow{\widetilde{\Phi}_\lambda} \mathbb{R}^n$$

is continuous.

Similarly, for $k = 1, \dots, \infty$ a map $f : M \to \widetilde{M}$ is called C^k*-smooth* if for every point $a \in M$ there exist local coordinate charts $(U_\lambda, \Phi_\lambda)$ in M and $(\widetilde{U}_\lambda, \widetilde{\Phi}_\lambda)$ in $\widetilde{M}_\lambda$, such that $a \in U_\lambda, f(U_\lambda) \subset \widetilde{U}_\lambda$ and the composition map

$$G_\lambda = \Phi_\lambda(U_\lambda) \xrightarrow{\Psi_\lambda} U_\lambda \xrightarrow{f} \widetilde{U}_\lambda \xrightarrow{\widetilde{\Phi}_\lambda} \mathbb{R}^n$$

is C^k-smooth. In other words, a map is continuous or smooth, if it is continuous or smooth when expressed in local coordinates.

A map $f : M \to N$ is called a *diffeomorphism* if it is smooth, one-to-one, and the inverse map is also smooth. One-to-one continuous maps with continuous inverses are called *homeomorphisms.*

Note that in view of the chain rule the C^k-smoothness is independent of the choice of local coordinate charts $(U_\lambda, \Phi_\lambda)$ and $(\widetilde{U}_\lambda, \widetilde{\Phi}_\lambda)$. Note that for C^k-smooth manifolds one can talk only about C^l-smooth maps for $l \leq k$. For topological manifolds one can talk only about continuous maps.

If one replaces condition M2 in the definition of a manifold by

M2^b. The image $G_\lambda = \Psi_\lambda(U_\lambda)$ is either an open set in $\mathbb{R}^n$ or an intersection of an open set in $\mathbb{R}^n$ with $\mathbb{R}^n_+ = \{x_1 \geq 0\}$

then one gets a definition of a *manifold with boundary.*

A slightly awkward nuance in the above definition is that a manifold with boundary is not a manifold! It would be, probably, less confusing to write this as a 1 word *manifold-with-boundary*, but of course nobody does that.

The points of a manifold M with boundary which are mapped by coordinate maps Ψ_λ to points in $\mathbb{R}^{n-1} = \partial\mathbb{R}^n_+$ are called the *boundary points* of M. The set of boundary points is called *the boundary* of M and denoted by ∂M. It is itself a manifold of dimension $n - 1$.

Note that any (interior) point a of an n-dimensional manifold M has a neighborhood B diffeomorphic to an open ball $B_1(0) \subset \mathbb{R}^n$, while any boundary point has a neighborhood diffeomorphic to a semi-ball $B_1(0) \cap \{x_1 \geq 0\} \subset \mathbb{R}^n$.

Exercise 15.1. Prove that a boundary point *does not have* a neighborhood diffeomorphic to an open ball. In other words, the notion of boundary and interior point of a manifold with boundary are well defined.

Next we want to introduce a notion of compactness for subsets in a manifold. Let us recall that for subsets in a Euclidean vector space we introduced three equivalent definition of compactness, see Section 2.1. The first definition, COMP1 is unapplicable because we cannot talk about bounded sets in a manifold. However, definitions COMP2 and COMP3 make perfect sense in an arbitrary manifold. For instance, we can say that a subset $A \subset M$ is compact if from any infinite sequence of points in A one can choose a subsequence converging to a point in A.

A compact manifold (without boundary) is called *closed*. Note that the word closed is used here in a different sense than a closed set. For instance, a closed interval is not a closed manifold because it has a boundary. An open interval or a real line $\mathbb{R}$ is not a closed manifold because it is not compact. On the other hand, a circle, or a sphere S^n of any dimension n is a closed manifold.

The notions of connected and path connected subsets of a manifold are defined in the same way as in an Euclidean space.

Exercise 15.2. Let A, B be two diffeomorphic manifolds. Prove that

1. if A is path-connected then so is B;

2. if A is compact then so is B;

3. if $\partial A = \varnothing$ then $\partial B = \varnothing$;

4. $\dim A = \dim B$; [1]

15.2 Gluing construction

The construction which is described in this section is called *gluing* or *quotient* construction. It provides a rich source of examples of manifolds. We discuss here only very special cases of this construction.

a) Let M be a manifold and U, U' its two open disjoint subsets. Let us moreover assume that each point $x \in M$ has a neighborhood which does not intersect at least one of the sets U and U'.[2] Consider a diffeomorphism $f : U \to U'$.

Let us denote by $M/\{f(x) \sim x\}$ the set obtained from M by identifying each point $x \in U$ with its image $f(x) \in U'$. In other words, a point of $M/\{f(x) \sim x\}$ is either a point from $x \in M \setminus (U \cup U')$, or a pair of points $(x, f(x))$, where $x \in U$. Note that there exists a canonical projection $\pi : M \to M/\{f(x) \sim x\}$. Namely $\pi(x) = x$ if $x \notin U \cup U'$, $\pi(x) = (x, f(x))$ if $x \in U$ and $\pi(x) = (f^{-1}(x), x)$ if $x \in U'$. By our assumption each point $x \in M$ has a neighborhood $U_x \ni x$

[1] In fact, we will prove later that even *homeomorphic* manifolds should have the same dimension.

[2] Here is an example when this condition *is not satisfied*: $M = (0, 2), U = (0, 1), U' = (1, 2)$. In this case any neighborhood of the point 1 intersect both sets, U and U'.

with a coordinate map $\Phi_x : U_x \to \mathbb{R}^n$ which intersects at most one of the sets U, U', and hence, $f(U_x \cap U) \cap U_x = \varnothing$, and therefore, the projection $\pi|_{U_x} : U_x \to \widetilde{U}_x := \pi(U_x)$ is one-to-one. We will declare by definition that $\widetilde{U}_x$ is a coordinate neighborhood of $\pi(x) \in M/\{f(x) \sim x\}$ and define a coordinate map $\widetilde{\Phi}_x : \widetilde{U}_x \to \mathbb{R}^n$ by the formula $\widetilde{\Phi}_x = \Phi_x \circ \pi^{-1}$. We will call the resulted manifold *the quotient manifold* of M, or say that $M/\{f(x) \sim x\}$ is obtained from M *by gluing U with U' with the diffeomorphism f*.

Exercise 15.3. *Work out the transition maps between two coordinate charts $(\widetilde{U}_x, \widetilde{\Phi}_x)$ and $(\widetilde{U}_y, \widetilde{\Phi}_y)$.*

Solution. We have $G_x := \Phi_x(U_x) = \widetilde{\Phi}_x(\widetilde{U}_x)$ and $G_y := \Phi_y(U_y) = \widetilde{\Phi}_y(\widetilde{U}_y)$. Suppose that $U_x \cap U' = \varnothing$ (this is not a constraint: by assumption either $U_x \cap U' = \varnothing$ or $U_x \cap U = \varnothing$). If $f(U_x \cap U) \cap U_y = \varnothing$ then $\pi|_{U_x \cap U_y}$ is a diffeomorphism $U_x \cap U_y \to \widetilde{U}_x \cap \widetilde{U}_y$, and the transition map

$$(\widetilde{\Phi}_y)^{-1} \circ \widetilde{\Phi}_x^{-1} : \widetilde{\Phi}_x(U_x \cap \widetilde{U}_y) \to \widetilde{P}hi_y(\widetilde{U}_x \cap \widetilde{U}_y)$$

is equal to the corresponding transition map

$$\Phi_y \circ \Phi_x^{-1} : \Phi_x(U_x \cap U_y) = \widetilde{\Phi}_x(U_x \cap \widetilde{U}_y) \to \Phi_y(U_x \cap U_y) = \widetilde{P}hi_y(\widetilde{U}_x \cap \widetilde{U}_y).$$

Suppose that $f(U_x \cap U) \cap U_y \neq \varnothing$. Then $\widetilde{U}_x \cap \widetilde{U}_y = \pi(U_x \cap U_y) \cup \pi(f(U_x) \cap U_y)$. Note that $\pi(U_x \cap U_y) \cap \pi(f(U_x \cap U) \cap U_y) = \varnothing$. Otherwise U_y would have to intersect both U and U' which by our assumption is not the case. The transition map $(\widetilde{\Phi}_y) \circ \widetilde{\Phi}_x^{-1}$ is equal to $\Phi_y \circ \Phi_x$ on $\Phi)x(\pi(U_x \cap U_y)$ and is equal to $\Phi_y \circ f^{-1} \circ \pi^{-1} \Phi_x^{-1} \circ f^{-1} \circ \pi^{-1}$ on $\Phi_x(\pi(f(U_x \cap U) \cap U_y)$.

Though the described above gluing construction always produce a manifold, the result could be quite pathological, if no additional care is taken. Here is an example of such pathology.

Example 15.4. Let $M = I \cup I'$ be the union of two disjoint open intervals $I = (0, 2)$ and $I' = (3, 5)$. Then M is a 1-dimensional manifold. Denote $U := (0, 1) \subset I, U' := (3, 4) \subset I'$. Consider a diffeomorphism $f : U \to U'$ given by the formula $f(t) = t + 3,\ t \in U$. Let $\widetilde{M} = M/\{f(x) \sim x\}$ be the corresponding quotient manifold. In other words, $\widetilde{M}$ is the result of gluing the intervals I and I' along their open sub-intervals U and U'. Note that the points $1 \in I$ and $4 \in I'$ are not identified, but $1 - \epsilon, 4 - \epsilon$ are identified for an arbitrary small $\epsilon > 0$. This means that any neighborhood of 1 and any neighborhood of 4 have non-empty intersection.

In order to avoid such pathological examples one usually (but not always) requires that manifolds satisfy an additional axiom, called *Hausdorff property*:

M5. Any two distinct points $x, y \in M$ have non-intersecting neighborhoods $U \ni x, G \ni y$.

In what follows **we always assume that the manifolds satisfy the Hausdorff property M5.**

Exercise 15.5. Show that if M is a Hausdorff manifold, $U, U' \subset M$ are two open sets as in a) (i.e. each point $x \in M$ has a neighborhood which does not intersect at least one of the sets U and U') and $f : U \to U'$ a diffeomorphism. Then $M/\{f(x) \sim x\}$ is Hausdorff if and only if the following condition is satisfied:

given a boundary (in the set-theoretic sense) point $x \in \partial U \subset M$ and any sequence of points $x_j \in U$, $x_j \to x$, then the sequence $f(x_j) \in U' \subset M$ goes to infinity in M, i.e. for any compact set $C \subset M$ there is and integer N such that for all $j > N$ we have $f(x_j) \notin C$.

Let us make the following general remark about diffeomorphisms $f : (a, b) \to (c, d)$ between two open intervals. Such diffeomorphism is simply a differentiable function whose derivative never vanishes and whose range is equal to the interval (c, d). If derivative is positive then the diffeomorphism is orientation preserving, and it is orientation reversing otherwise. The function f always extends to a *continuous* (but necessarily differentiable function $\overline{f} : [a, b] \to [c, d]$ such that $\overline{f}(a) = c$ and $\overline{f}(b) = d$ in the orientation preserving case, and $\overline{f}(a) = d$, $\overline{f}(b) = c$ in the orientation reversing case.

Lemma 15.6. *Given $a, b, a'b' \in (0, 1)$ such that $a < b$ and $a' < b'$ consider an orientation preserving diffeomorphisms $f : (0, a) \to (0, a')$ and $(b, 1) \to (b'1)$. Then for any $\widetilde{a} \in (0, a)$ and $\widetilde{b} \in (b, 1)$ there exists a diffeomorphism $F : (0, 1) \to (0, 1)$ which coincides with f on $(0, \widetilde{a})$ and coincides with g on $(\widetilde{b}, 1)$.*

Proof. Choose real numbers $c, \widetilde{c}, \widetilde{d}, d$ such that $a < c < \widetilde{c} < \widetilde{d} < d < b$. Consider a cut-off C^∞-function $\theta : (0, 1) \to (0, 1)$ which is equal to 1 on $(0, \widetilde{a}] \cup [\widetilde{c}, \widetilde{d}] \cup [\widetilde{b}, 1)$ and equal to 0 on $[a, c] \cup [d, b]$. For positive numbers $\epsilon > 0$ and $C > 0$ (which we will choose later) consider a function $h_{\epsilon,C}$ on

$(0,1)$ defined by the formula

$$h_{\epsilon,C}(x) = \begin{cases} \theta(x)f'(x) + (1-\theta(x))\epsilon, & x \in (0,a); \\ \epsilon, & x \in [a,c] \cup [d,b]; \\ C\theta(x) + (1-\theta(x))\epsilon, & x \in (c,d); \\ \theta(x)g'(x) + (1-\theta(x))\epsilon, & x \in (b,1). \end{cases}$$

Note that $h_{\epsilon,C}(x) = f'(x)$ on $(0,\widetilde{a}]$, $h_{\epsilon,C}(x) = g'(x)$ on $[\widetilde{b},1)$ and equal to C on $[\widetilde{c},\widetilde{d}]$. Define the function $F_{\epsilon,C} : (0,1) \to (0,1)$ by the formula

$$F_{\epsilon,C}(x) = \int\limits_0^x h(u)du.$$

Note that the derivative $F'_{\epsilon,C}$ is positive, and hence the function $F_{\epsilon,C}$ is strictly increasing. It coincides with f on $(0,\widetilde{a}]$ and coincides *up to a constant* with g on $(\widetilde{b},1)$. Note that when ϵ and C are small we have $F_{\epsilon,C}(\widetilde{b}) < b' < g(\widetilde{b})$, and $\lim\limits_{C\to\infty} F_{\epsilon,C}(\widetilde{b}) = \infty$. Hence, by continuity one can choose $\epsilon, C > 0$ in such a way that $F_{\epsilon,C}(\widetilde{b}) = g(\widetilde{b})$. Then the function $F = F_{\epsilon,C}$ is a diffeomorphism $(0,1) \to (0,1)$ with the required properties. ■

Lemma 15.7. *Suppose that a* 1*-dimensional manifold* M *(which satisfies the Hausdorff axiom M5) is covered by two coordinate charts,* $M = U \cup U'$*, with coordinate maps* $\Phi : U \to (0,1), \Phi' : U' \to (0,1)$ *such that* $\Phi(U \cap U') = (0,a) \cup (b,1), \Phi'(U \cap U') = (0,a') \cup (b'1)$ *for some* $a,a',b,b' \in (0,1)$ *with* $a < b, a' < b'$*. Then* M *is diffeomorphic to the circle* S^1*.*

Proof. Denote by Ψ and Ψ' the parameterization maps Φ^{-1} and $(\Phi')^{-1}$, and set $G := \Phi(U \cap U')$ and $G' := \Phi'(U \cap U')$. Let $h = \Phi' \circ \Psi : G \to G'$ be the transition diffeomorphism. There could be two cases: $h((0,a)) = (0,a'), h((b,1)) = (b',1)$ and $h((0,a)) = (b',1), h((b,1)) = (0,a)$. We will analyze the first case. The second one is similar.

Let $\overline{h}$ be the continuous extension of h to $[0,a] \cup [b,1]$. We claim that $\overline{h}(0) = a'$, $\overline{h}(a) = 0$, $\overline{h}(b) = 1$ and $\overline{h}(1) = b'$. Indeed, assuming otherwise we come to a contradiction with the Hausdorff property M5. Indeed, suppose $\overline{h}(a) = a'$. Note the points $A := \Psi(a), A' := \Psi'(a') \in M$ are disjoint. On the other hand, for any neighborhood $\Omega \ni A$ its image $\Phi(\Omega) \subset I$ contains an interval $(a-\epsilon, a)$, and similarly for any neighborhood $\Omega' \ni A'$ its image $\Phi'(\Omega) \subset I$ contains an interval $(a'-\epsilon, a')$. for a sufficiently small ϵ. But $h((a-\epsilon, a)) = (a'-\epsilon', a')$ for some $\epsilon' > 0$ and hence

$$\Omega \supset \Psi'((a', a'-\epsilon')) = \Psi' \circ h((a, a-\epsilon)) = \Psi' \circ \Phi' \circ \Phi((a, a-\epsilon)) = \Phi((a, a-\epsilon)) \subset \Omega,$$

i.e. $\Omega \cap \Omega' \neq \varnothing$. In other words, *any neighborhoods of the distinct points $A, A' \in M$ intersect*, which violates axiom M5. Similarly we can check that $h(b) = b'$.

Now take the unit circle $S^1 \subset \mathbb{R}^2$ and consider the polar coordinate ϕ on S^1. Let us define a map $g' : U' \to S^1$ by the formula $\phi = -\pi\Phi'(x)$. Thus g' is a diffeomorphism of U' onto an arc of S^1 given in polar coordinates by $-\pi < \phi < 0$. The points $A' = \Psi'(a')$ and $B' = \Psi'(b')$ are mapped to points with polar coordinates $\phi = -\pi a'$ and $\phi = -\pi b'$. On the intersection $U \cap U'$ we can describe the map g' in terms of the coordinate in U. Thus we get a map $f := g' \circ \Psi :$ $(0, a) \cup (b, 1) \to S^1$. We have $\overline{f}(0) = g'(A'), f(a) = g'(0), \overline{f}(1) = g'(B'), f(b) = g'(1)$. Here we denoted by $\overline{f}$ the continuous extension of f to $[0, a] \cup [b, 1]$. Thus $f((0, a)) = \{-\pi a' < \phi < 0\}$ and $f((b, 1) = \{\pi < \phi < 3\pi - \pi b'\}$. Note that the diffeomorphism f is orientation preserving assuming that the circle is oriented counter-clockwise. Using Lemma 15.6 we can find a diffeomorphism F from $(0, 1)$ to the arc $\{-\pi a' < \phi < 3\pi - \pi b'\} \subset S^1$ which coincides with f on $(0, \widetilde{a}) \cup (\widetilde{b}, 1)$ for any $\widetilde{a} \in (0, a)$ and $\widetilde{b} \in (b, 1)$. Denote $\widetilde{a}' := h(\widetilde{a}), \widetilde{b}' = h(\widetilde{b})$. Notice that the neighborhoods U and $\widetilde{U}' = \Psi'((\widetilde{a}, \widetilde{b}))$ cover M. Hence, the required diffeomorphism $\widetilde{F} : M \to S^1$ we can define by the formula

$$\widetilde{F}(x) = \begin{cases} g(x), & x \in \widetilde{U}'; \\ F \circ \Phi(x), & x \in U. \end{cases}$$

■

Similarly (and even simpler), one can prove

Lemma 15.8. *Suppose that a 1-dimensional manifold M (which satisfies the Hausdorff axiom M5) is covered by two coordinate charts, $M = U \cup U'$, with coordinate maps $\Phi : U \to (0, 1), \Phi' : U' \to (0, 1)$ such that $U \cap U'$ is connected. Then M is diffeomorphic to the open interval $(0, 1)$.*

Theorem 15.9. *Any (Hausdorff) connected closed 1-dimensional manifold is diffeomorphic to the circle S^1.*

Exercise 15.10. Show that the statement of the above theorem is not true without the axiom M5, i.e. the assumption that the manifold has the Hausdorff property.

Proof. Let M be a connected closed 1-dimensional manifold. Each point $x \in M$ has a coordinate neighborhood U_x diffeomorphic to an open interval. All open intervals are diffeomorphic, and hence we can assume that each neighborhood G_x is parameterized by the interval $I = (0,1)$. Let $\Psi_x : I \to G_x$ be the corresponding parameterization map. We have $\bigcup\limits_{x \in M} U_x = M$, and due to compactness of M we can choose finitely many $U_{x_1}, \dots, U_{x_k}$ such that $\bigcup\limits_{i=1}^{k} U_{x_i} = M$. We can further assume that none of these neighborhoods is completrely contained inside another one. Denote $U_1 := U_{x_1}, \Psi_1 := \Psi_{x_1}$. Note that $U_1 \cap \bigcup\limits_{2}^{k} U_{x_k} \neq \varnothing$. Indeed, if this were the case then due to connectedness of M we would have $\bigcup\limits_{2}^{k} U_{x_i} = \varnothing$ and hence $M = U_1$, but this is impossible because M is compact. Thus, there exists $i = 2, \dots, k$ such that $U_{x_i} \cap U_1 \neq \varnothing$. We set $U_2 := U_{x_i}, \Psi_2 = \Psi_{x_i}$. Consider open sets $G_{1,2} := \Psi_1^{-1}(U_1 \cap U_2), G_{2,1} = \Psi_2^{-1}(U_1 \cap U_2) \subset I$. The transition map $h_{1,2} := \Psi_2^{-1} \circ \Psi_1|_{G_{1,2}} : G_{1,2} \to G_{2,1}$ is a diffeomorphism.

Let us show that the set $G_{1,2}$ (and hence $G_{2,1}$) cannot contain more that two connected components. Indeed, in that case one of the components of $G_{1,2}$ has to be a subinterval $I' = (a,b) \subset I = (0,1)$ where $0 < a < b < 1$. Denote $I'' := h_{1,2}(I')$. Then at least of of the boundary values of the transition diffeomorphism $h_{1,2}|_{I'}$, say $\overline{h}_{1,2}(a)$, which is one of the end points of I'', has to be an interior point $c \in I = (0,1)$. We will assume for determinacy that $I'' = (c,d) \subset I$. But this contradicts the Hausdorff property M5. The argument repeats a similar argument in the proof of Lemma 15.7.

Indeed, note that $\Psi_1(a) \neq \Psi_2(c)$. Indeed, $\Psi_1(a)$ belongs to $U_1 \setminus U_2$ and $\Psi_2(c)$ is in $U_2 \setminus U_1$. Take any neighborhood $\Omega \ni \Psi_1(a)$ in M. Then $\Psi^{-1}(\Omega)$ is an open subset of I which contains the point a. Hence $\Psi_1((a, a+\epsilon)) \subset \Omega$, and similarly, for any neighborhood $\Omega' \ni \Psi_2(c)$ in M we have $\Psi_2((c, c+\epsilon)) \subset \Omega'$ for a sufficiently small $\epsilon > 0$. But $\Psi_1((a, a+\epsilon)) = \Psi_2(h_{1,2}((a, a_\epsilon))) = \Psi_2(c, c+\epsilon')$, where $c + \epsilon' = h_{1,2}(a+\epsilon)$. Hence $\Omega \cap \Omega' \neq \varnothing$, i.e. any two neighborhoods of two distict points $\Psi_1(a)$ and $\Psi_2(c)$ have a non-empty intersection, which violates the Hausdorff axiom M5.

If $G_{1,2} \subset (0,1)$ consists of two components then the above argument shows that each of these components must be adjacent to one of the ends of the interval I, and the same is true about the components of the set $G_{2,1} \subset I$. Hence, we can apply Lemma 15.7 to conclude that the union $U_1 \cup U_2$ is diffeomorphic to S^1. We also notice that in this case all the remaining neighborhoods U_{x_j} must contain in $U_1 \cup U_2$. Indeed, each U_{x_i} which intersects the circle $U_1 \cup U_2$ must be completely contained in it, because otherwise we would again get a contradiction with the Hausdorff property. Hence, we can eleiminate all neighborhoods which intersect $U_1 \cup U_2$. But then no other neighborhoods could be left because otherwise we would have $M = (U_1 \cup U_2) \cup \bigcup\limits_{U_{x_j} \cap (U_1 \cup U_2) = \varnothing} U_{x_j}$, i.e. the manifold M could be presented as a union of two disjoint non-empty open sets which is impossible due to connectedness of M. Thus we conclude that in this case $M = U_1 \cup U_2$ is diffeomorphic to S^1.

Finally in the case when $G_{1,2}$ consists of 1 component, i.e. when it is connected, one can Use Lemma 15.8 to show that $U_1 \cup U_2$ is diffeomorphic to an open interval. Hence, we get a covering of M by $k-1$ neighborhood diffeomorphic to S^1. Continuing inductively this process we will either find at some step two neighborhoods which intersect each other along two components, or continue to reduce the number of neighborhoods. However, at some moment the first situation should occur because otherwise we would get that M is diffeomorphic to an interval which is impossible because by assumption M is compact. ■

b) Let M be a manifold, $f : M \to M$ be a diffeomorphism. Suppose that f satisfies the following property: *There exists a positive integer p such that for any point $x \in M$ we have $f^p(x) = \underbrace{f \circ f \circ \dots \circ f}_{p}(x) = x$, but the points $x, f(x) \dots, f^{p-1}(x)$ are all disjoint.* The set $\{x, f(x) \dots, f^{p-1}(x)\} \subset M$ is called the *trajectory* of the point x under the action of f. It is clear that trajectory of two different points either coincide or disjoint. Then one can consider the *quotient space X/f*, whose points are trajectories of points of M under the action of f. Similarly to how it was done in a) one can define on M/f a structure of an n-dimensional manifold.

c) Here is a version of the construction in a) for the case when trajectory of points are infinite. Let $f : M \to M$ be a diffeomorphism which satisfies the following property: *for each point $x \in M$ there exists a neighborhood $U_x \ni x$ such that all sets*

$$\dots, f^{-2}(U_x), f^{-1}(U_x), U_x, f(U_x), f^2(U_x), \dots$$

are mutually disjoint. In this case the *trajectory* $\{\dots, f^{-2}(x), f^{-1}(x), x, f(x), f^2(x), \dots\}$ of each point is infinite. As in the case b), the trajectories of two different points either coincide or disjoint.

The set M/f of all trajectories can again be endowed with a structure of a manifold of the same dimension as M.

15.2.1 Examples of manifolds

1. **n-dimensional sphere S^n.** Consider the unit sphere $S^n = \{||x|| = \sqrt{\sum\limits_1^{n+1} x_j^2} = 1\} \subset \mathbb{R}^{n+1}$. Let introduce on S^n the structure of an n-dimensional manifold. Let $N = (0, \dots, 1)$ and $S = (0, \dots, -1)$ be the North and South poles of S^n, respectively.

Denote $U_- = S^n \setminus S, U_+ = S^n \setminus N$ and consider the maps $p_\pm : U_\pm \to \mathbb{R}^n$ given by the formula

$$p_\pm(x_1, \dots, x_{n+1}) = \frac{1}{1 \mp x_{n+1}}(x_1, \dots, x_n). \tag{15.2.1}$$

The maps $p_+ : U_+ \to \mathbb{R}_n$ and $p_- : U_- \to \mathbb{R}^n$ are called *stereographic projections* from the North and South poles, respectively. It is easy to see that stereographic projections are one-to one maps. Note that $U_+ \cap U_- = S^n \setminus \{S, N\}$ and both images, $p_+(U_+ \cap U_-)$ and $p_-(U_+ \cap U_-)$ coincide with $\mathbb{R}^n \setminus 0$. The map $p_- \circ p_+^{-1} : \mathbb{R}^n \setminus 0 \to \mathbb{R}^n \setminus 0$ is given by the formula

$$p_- \circ p_+^{-1}(x) = \frac{x}{||x||^2}, \tag{15.2.2}$$

and therefore it is a diffeomorphism $\mathbb{R}^n \setminus 0 \to \mathbb{R}^n \setminus 0$.

Thus, the atlas which consists of two coordinate charts (U_+, p_+) and (U_-, p_-) defines on S^n a structure of an n-dimensional manifold. One can equivalently defines the manifold S^n as follows. Take two disjoint copies of $\mathbb{R}^n$, let call them $\mathbb{R}^n_1$ and $\mathbb{R}^n_2$. Denote $M = \mathbb{R}^n_1 \cup \mathbb{R}^n_2$, $U = \mathbb{R}^n_1 \setminus 0$ and $U' = \mathbb{R}^n_2 \setminus 0$. Let $f : U \to U'$ be a difeomorphism defined by the formula $f(x) = \frac{x}{||x||^2}$, as in (15.2.2). Then S^n can be equivalently described as the quotient manifold M/f.

Note that the 1-dimensional sphere is the circle S^1. It can be d as follows. Consider the map $T : \mathbb{R} \to \mathbb{R}$ given by the formula $T(x) = x + 1$, $x \in \mathbb{R}$. It satisfies the condition from 15.2 and hence, one can define the manifold $\mathbb{R}/T$. This manifold is diffeomorphic to S^1.

2. **Real projective space.** The real projective space $\mathbb{R}P^n$ is the set of all lines in $\mathbb{R}^{n+1}$ passing through the origin. One introduces on $\mathbb{R}P^n$ a structure of an n-dimensional manifold as follows. For each $j = 1, \dots, n+1$ let us denote by U_j the set of lines which are not parallel to the affine subspace $\Pi_j = \{x_j = 1\}$. Clearly, $\bigcup\limits_1^{n+1} U_j = \mathbb{R}P^n$. There is a natural one-to one map $\pi_j : U_j \to \Pi_j$

which associates with each line $\mu \in U_j$ the unique intersection point of μ with Π_j. Furthermore, each Π_j can be identified with $\mathbb{R}^n$, and hence pairs (U_j, π_j), $j = 1, \dots, n+1$ can be chosen as an atlas of coordinate charts. We leave it to the reader to check that this atlas indeed define on $\mathbb{R}P^n$ a structure of a smooth manifold, i.e. that the transition maps between different coordinate charts are smooth.

Exercise 15.11. Let us view S^n as the unit sphere in $\mathbb{R}^{n+1}$. Consider a map $p : S^n \to \mathbb{R}P^n$ which associates to a point of S^n the line passing through this point and the origin. Prove that this two-to-one map is smooth, and moreover a *local diffeomorphism*, i.e. that the restriction of p to a sufficiently small neighborhood of each point is a diffeomorphism. Use it to show that $\mathbb{R}P^n$ is diffeomorphic to the quotient space S^n/f, where $f : S^n \to S^n$ is the antipodal map $f(x) = -x$.

3. **Products of manifolds and n-dimensional tori.** Given two manifolds, M and N of dimension m and n, respectively, one can naturally endow the *direct product*

$$M \times N = \{(x, y);\ x \in M, y \in N\}$$

with a structure of a manifold of dimension $m+n$. Let $\{(U_\lambda, \Phi_\lambda)\}_{\lambda \in \Lambda}$ and $\{(V_\gamma, \Psi_\gamma)\}_{\gamma \in \Gamma}$ be atlases for M and N, so that $\Phi_\lambda : U_\lambda \to U'_\lambda \subset \mathbb{R}^m, \Psi_\mu : V_\mu \to V'_\mu \subset \mathbb{R}^n$ are diffeomorphisms on open subsets of $\mathbb{R}^m$ and $\mathbb{R}^n$. Then the smooth structure on $M \times N$ can be defined by an atlas

$$\{(U_\lambda \times V_\gamma, \Phi_\lambda \times \Psi_\gamma)\}_{\lambda \in \Lambda, \gamma \in \Gamma},$$

where we denote by $\Phi_\lambda \times \Psi_\gamma : U_\lambda \times V_\gamma \to U'_\lambda \times V'_\gamma \subset \mathbb{R}^m \times \mathbb{R}^n = \mathbb{R}^{m+n}$ are diffeomorphisms defined by the formula $(x, y) \mapsto (\Phi_\lambda(x)\Psi_\mu(y))$ for $x \in U_\lambda$ and $y \in V_\mu$.

One can similarly define the direct product of any finite number of smooth manifolds. In particular the *n-dimensional torus* T^n is defined as the product of n circles: $T^n = \underbrace{S^1 \times \dots \times S^1}_{n}$. Let us recall, that the circle S^1 is diffeomorphic to $\mathbb{R}/T$, i.e. a point of S^1 is a real number up to adding any integer. Hence, the points of the torus T^n can be viewed as the points of $\mathbb{R}^n$ up to adding any vector with all integer coordinates.

15.3 Tangent spaces, tangent bundle and differential

Given an n-dimensional manifold M we want to introduce a notion of a *tangent vector* at a point $a \in M$, the *tangent space* T_aM to M at the point a as the set of all tangent vectors at a, and the *tangent bundle* TM to M as the collection $\{T_aM\}_{a\in M}$ of all its tangent spaces.

If $M = U \subset V$ is a domain in a vector space V then we were already using the term "tangent vector" for a vector $v \in V_a$ originated at $a \in U$, the notation T_aU for V_a calling it tangent space to U and were calling the collection $\{T_aU\}_{a\in U}$ the tangent bundle TU to U. The parallel transport identifies TU with the product $U \times V$. This definition is not well suited to be transported to a general manifold, so we will search for another equivalent and more suitable for our purpose definition.

Note that given two paths $\gamma, \widetilde{\gamma} : (-\epsilon, \epsilon) \to U$ in U such that $\gamma(0) = \widetilde{\gamma}(0) = a$ and $\gamma'(0) = \widetilde{\gamma}'(0) = v \in T_aU$ then

$$\gamma(t) - \widetilde{\gamma}(t) = o(t) \tag{15.3.1}$$

and the converse is also true: if condition (15.3.1) holds then $\gamma(0) = \widetilde{\gamma}(0)$ and $\gamma'(0) = \widetilde{\gamma}'(0)$. We call two paths γ and $\widetilde{\gamma}$ 1-*equivalent* if (15.3.1) holds. Hence, *the 1-equivalence class* $[\gamma]$ of all paths $\gamma : (-\epsilon, \epsilon) \to U$ can be identified with the tangent vector $\gamma'(0) \in T_aU$, the tangent bundle TU can be viewed as the set of all such equivalence classes. The natural projection $\pi : TU \to U$ given by $\pi([\gamma]) = \gamma(0)$ and $\pi^{-1}(a)$ is the tangent space T_aU.

In this form the notions of tangent vectors, tangent spaces and tangent bundles could be extended to general manifolds.

Exercise 15.12. Suppose that paths γ and $\widetilde{\gamma}$ are 1-equivalent and $f : U \to \widetilde{U}$ be any diffeomorphism. Show that the paths $f \circ \gamma$ and $f \circ \widetilde{\gamma}$ are 1-equivalent as well.

Thus, given any n-dimensional manifold M, and a point $a \in M$ the notion of the 1-equivalence class $[\gamma]$ of a path $\gamma : (-\epsilon, \epsilon) \to M$ with $\gamma(0) = a$ makes sense, because it is independent of the choice of a local coordinate system. Hence we call this class $[\gamma]$ a tangent vector to M at a and denote it by $\gamma'(0)$. All equivalence classes $[\gamma]$ with $\gamma(0) = a$ form an n-dimensional vector space, called the *tangent space* to M at the point a and is denoted by T_aM.

Given a smooth map $f : M \to N$ between two manifolds, the *differential* $d_af : T_aM \to T_{f(a)}N$

is defined as follows. Given a tangent vector $v = [\gamma] \in T_aM$ we define $d_af(v) := [f \circ \gamma] \in T_{f(a)}N$. If M and N are domains in vector spaces this definition coincides with the one we were using so far.

The collection $TM = \{T_aM\}_{a\in M}$ is called the *tangent bundle* of M. The tangent bundle has itself a natural structure of a $2n$-dimensional manifold. Indeed, consider an atlas $\{U_\lambda, \Phi_\lambda\}_{\lambda\in\Lambda}$ for M. Denote $G_\lambda = \Phi_\lambda(U_\lambda) \subset \mathbb{R}^n$. Then the atlas for TM is given by $\{TU_\lambda, d\Phi_\lambda\}_{\lambda\in\Lambda}$, where $d\Phi_\lambda : TU_\lambda \to TG_\lambda = G_\lambda \times \mathbb{R}^n$ is the differential of Φ_λ.

15.4 Submanifolds

Let V be an n-dimensional manifold. A subset $A \subset V$ is called a *k-dimensional submanifold* of V, or simply a *k-submanifold* of V, $0 \le k \le n$, if for any points $a \in A$ there exists a local coordinate chart $(U_a, u = (u_1, \dots, u_n) \to \mathbb{R}^n)$ such that $u(a) = 0$ (i.e. the point a is the origin of this coordinate system) and

$$A \cap U_a = \{u = (u_1, \dots, u_n) \in U_a;\ u_{k+1} = \dots = u_n = 0\}. \tag{15.4.1}$$

We will always assume the local coordinates at least as smooth as necessary for our purposes (but at least C^1-smooth), but more precisely, one can talk of C^m- submanifolds if the implied coordinate systems are at least C^m-smooth.

Note that in the above we can replace the vector space V by any n-dimensional manifold, and thus will get a notion of a k-dimesional submanifold of an n-dimensional manifold V.

Example 15.13. *Suppose a subset $A \subset U \subset V$ is given by equations $F_1 = \dots = F_{n-k} = 0$ for some C^m-smooth functions $F_1, \dots, F_{n-k}$ on U. Suppose that for any point $a \in A$ the differentials $d_aF_1, \dots, d_aF_{n-k} \in V_a^*$ are linearly independent. Then $A \subset U$ is a C^m-smooth submanifold.*

Indeed, for each $a \in A$ one can choose a linear functions $l_1, \dots, l_k \in V_a^*$ such that together with $d_aF_1, \dots, d_aF_{n-k} \in V_a^*$ they form a basis of V^*. Consider functions $L_1, \dots, L_k : V \to \mathbb{R}$, defined by $L_j(x) = l_j(x - a)$ so that $d_a(L_j) = l_j$, $j = 1, \dots, k$. Then the Jacobian $\det D_aF$ of the map $F : (L_1, \dots, L_k, F_1, \dots, F_{n-k}) : U \to \mathbb{R}^n$ does not vanish at a, and hence the inverse function theorem implies that this map is invertible in a smaller neighborhood $U_a \subset U$ of the point $a \in A$. Hence, the functions $u_1 = L_1, \dots, u_k = L_k, u_{k+1} = F_1, \dots, u_n = F_{n-k}$ can be chosen as a local coordinate system in U_a, and thus $A \cap U_a = \{u_{k+1} = \dots = u_n = 0\}$.

Note that the map $u' = (u_1, \dots, u_k)$ maps $U_a^A = U_a \cap A$ onto an open neighborhood $\widetilde{U} = u(U_a) \cap \mathbb{R}^k$ of the origin in $\mathbb{R}^k \subset \mathbb{R}^n$, and therefore $u' = (u_1, \dots u_k)$ defines a local coordinates, so that the pair (U_a^A, u') is a *coordinate chart*. The restriction $\widetilde{\phi} = \phi|_{\widetilde{U}}$ of the parameterization map $\phi = u^{-1}$ maps $\widetilde{U}$ onto U_a^A. Thus $\widetilde{\phi}$ a *parameterization* map for the neighborhood U_a^A. The atlas $\{(U_a^A, u')\}a \in A$ defines on a a structure of a k-dimensional manifold. The complementary dimension $n - k$ is called the *codimension* of the submanifold A. We will denote dimension and codimension of A by $\dim A$ and $\mathrm{codim} A$, respectively.

As we already mentioned above in Section 5.2 1-dimensional submanifolds are usually called curves. We will also call 2-dimensional submanifolds *surfaces* and codimension 1 submanifolds *hypersurfaces*. Sometimes k-dimensional submanifolds are called k-surfaces. Submanifolds of codimension 0 are open domains in V.

In the case when V is a vector space, an important class form *graphical* k-submanifolds. Let us recall that given a map $f : B \to \mathbb{R}^{n-k}$, where B is a subset $B \subset \mathbb{R}^k$, then *graph* is the set

$$\Gamma_f = \{(x, y) \in \mathbb{R}^k \times \mathbb{R}^{n-k} = \mathbb{R}^n;\ x \in B, y = f(x)\}.$$

A (C^m)-submanifold $A \subset V$ is called *graphical* with respect to a splitting $\Phi : \mathbb{R}^k \times \mathbb{R}^{n-k} \to V$, if there exist an open set $U \subset \mathbb{R}^k$ and a (C^m)-smooth map $f : U \to \mathbb{R}^{n-k}$ such that

$$A = \Phi(\Gamma_f).$$

In other words, A is graphical if there exists a coordinate system in V such that

$$A = \{x = (x_1, \dots, x_n); (x_1, \dots x_k) \in U, x_j = f_j(x_1, \dots, x_k),\ j = k+1, \dots, n\}.$$

for some open set $U \subset \mathbb{R}^k$ and smooth functions, $f_{k+1}, \dots, f_n : U \to \mathbb{R}$.

For a graphical submanifold there is a global coordinate system given by the projection of the submanifold to $\mathbb{R}^k$.

It turns out that that *any submanifold locally is graphical.*

Proposition 15.14. *Let $A \subset V$ be a submanifold. Then for any point $a \in A$ there is a neighborhood $U_a \ni a$ such that $U_a \cap A$ is graphical with respect to a splitting of V. (The splitting may depend on the point $a \in A$).*

We leave it to the reader to prove this proposition using the implicit function theorem.

A map $f : M \to Q$ is called an *embedding* of a manifold M into another manifold Q if it is a diffeomorphism of M onto a submanifold $A \subset Q$. In other words, f is an embedding if the image $A = f(M)$ is a submanifold of Q and the map f viewed as a map $M \to A$ is a diffeomorphism.

One can prove that any n-dimensional manifold can be embedded into $\mathbb{R}^N$ with a sufficiently large N (in fact $N = 2n + 1$ is always sufficient).

Hence, one can think of any *manifold* as a *submanifold of some* $\mathbb{R}^N$, given up to a diffeomorphism, i.e. ignoring how this submanifold is embedded in the ambient space.

In the exposition below we mostly restrict our discussion to submanifolds of $\mathbb{R}^n$ rather than general abstract manifolds.

15.4.1 Submanifolds with boundary

A slightly different notion is of a *submanifold with boundary*. A subset $A \subset V$ is called a *k-dimensional submanifold with boundary*, or simply a *k-submanifold of V with boundary*, $0 \le k < n$, if for any points $a \in A$ there is a neighborhood $U_a \ni a$ in V and local (curvi-linear) coordinates $(u_1, \dots, u_n)$ in U_a with the origin at a if *one of two conditions* is satisfied: condition (15.4.1), or the following condition

$$A \cap U_a = \{u = (u_1, \dots, u_n) \in U_a;\ u_1 \ge 0, u_{k+1} = \dots = u_n = 0\}. \tag{15.4.2}$$

In the latter case the point a is called a *boundary* point of A, and the set of all boundary points is called the *boundary* of A and is denoted by ∂A.

As in the case of submanifolds without boundary, any submanifold with boundary has a structure of a manifold with boundary.

Exercise 15.15. *Prove that if A is k-submanifold with boundary then ∂A is a $(k-1)$-dimensional submanifold (without boundary).*

Remark 15.16. 1. As we already pointed out when we discussed manifolds with boundary, a submanifold with boundary is not a submanifold!

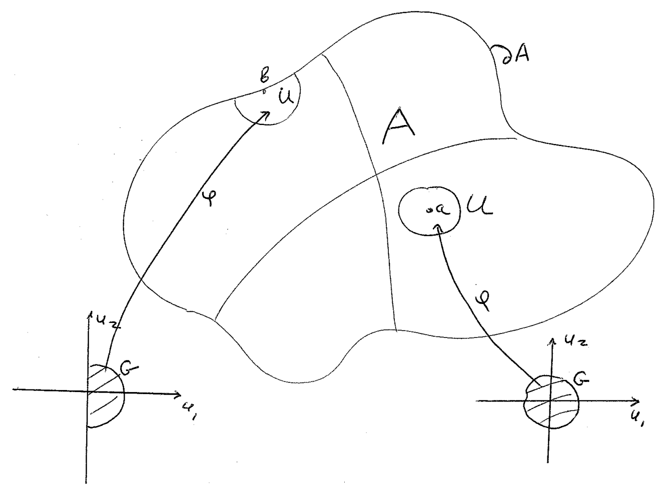


Figure 15.1: The parameterization ϕ introducing local coordinates near an interior point a and a boundary point b.

2. As it was already pointed out when we discussed 1-dimensional submanifolds with boundary, the boundary of a k-submanifold with boundary is not the same as its set-theoretic boundary, though traditionally the same notation ∂A is used. Usually this should be clear from the context, what the notation ∂A stands for in each concrete case. We will explicitly point this difference out when it could be confusing.

A compact manifold (without boundary) is called *closed*. The boundary of any compact manifold with boundary is closed, i.e. $\partial(\partial A) = \varnothing$.

Example 15.17. An open ball $B_r^n = B_R^n(0) = \{\sum\limits_1^n x_j^2 < 1\} \subset \mathbb{R}^n$ is a codimension 0 submanifold, A closed ball $D_r^n = D_R^n(0) = \{\sum\limits_1^n x_j^2 \leq 1\} \subset \mathbb{R}^n$ ia codimension 0 submanifold with boundary. Its boundary ∂D_R^n is an $(n-1)$-dimensional sphere $S_R^{n-1} = \{\sum\limits_1^n x_j^2 = 1\} \subset \mathbb{R}^n$. It is a closed hypersurface. For $k = 0, 1 \dots n-1$ let us denote by L^k the subspace $L_k = \{x_{k+1} = \dots = x_n = 0\} \subset \mathbb{R}^n$. Then the intersections

$$B_R^k = B_R^n \cap L^k, D_R^k = D_R^n \cap L^k, \text{ and } S_R^{k-1} = S_R^{n-1} \cap L^k \subset \mathbb{R}^n$$

are, respectively a k-dimensional submanifold, a k-dimensional submanifold with boundary and a closed $(k-1)$-dimensional submanifold of $\mathbb{R}^n$. Among all above examples there is only one (which one?) for which the manifold boundary is the same as the set-theoretic boundary.

A neighborhood of a boundary point $a \in \partial A$ can be always locally parameterized by the semi-open upper-half ball

$$B_+(0) = \{x = (x_1, \dots, x_k) \in \mathbb{R}^k;\ x_1 \geq 0, \sum_1^n x_j^2 < 1\}.$$

We will finish this section by defining submanifolds with *piece-wise smooth boundary*. A subset $A \subset V$ is called a k-dimensional submanifold of V with *piecewise smooth boundary* or with *boundary with corners*, $0 \leq k < n$, if for any points $a \in A$ there is a neighborhood $U_a \ni a$ in V and local (curvi-linear) coordinates $(u_1, \dots, u_n)$ in U_a with the origin at a if one of three condittions satisfied: conditions (15.4.1), (15.4.2) or the following condition

$$A \cap U_a = \{u = (u_1, \dots, u_n) \in U_a;\ l_1(u) \geq 0, \dots, l_m(u) \geq 0, u_{k+1} = \dots = u_n = 0\}, \qquad (15.4.3)$$

where $m > 1$ and $l_1, \dots, l_m \in (\mathbb{R}^k)^*$ are linear functions In the latter case the point a is called a *corner* point of ∂A.

Note that the system of linear inequalities $l_1(u) \geq 0, \dots, l_m(u) \geq 0$ defines a *convex cone* in $\mathbb{R}^k$. Hence, near a corner point of its boundary the manifold is diffeomorphic to a convex cone. Thus convex polyhedra and their diffeomorphic images are important examples of submanifolds with boundary with corners.

15.4.2 Tangent spaces, tangent bundle and differential in the case of a submanifold

Let $N \subset V$ be a k-dimensional submanifold of an n-dimensional manifold M, $U \subset N$ a local coordinate chart and $\phi : G \to U$ its parametrization. Suppose that $0 \in G$ and $\phi(0) = a \in U$. The parametrization map ϕ can be viewed as an embedding $G \to U \hookrightarrow M$ and its differential $d_0\phi$ as a linear map $d_0\phi : T_0G \to T_aM$. As ϕ is a diffeomorphism of G onto U, the differential has the maximal rank k, so that $d_0\phi(T_0G)$ is a k-dimensional subspace of T_aM. If $\widetilde{\phi} : \widetilde{G} \to U$ is another parametrization of U with $\widetilde{\phi}(0) = a$. Then we can write $\widetilde{\phi} = \phi \circ (\phi^{-1} \circ \widetilde{\phi})$, so by the chain rule $d_0\widetilde{\phi} = d\phi \circ d_0(\phi^{-1} \circ \widetilde{\phi})$, and hence, $d_0\phi(T_0G) = d_0\widetilde{\phi}(T_0\widetilde{G})$. In other words, the subspace $d_0\phi(T_0G) \subset T_aM$ is independent of the choice of a parametrization. We call $T_aN := d_0\phi(T_0G)$ the tangent space to the submanifold $N \subset M$.

If N is a submanifold with boundary and $a \in \partial N$ then there are defined both the k-dimensional tangent space $T_aN \subset T_aM$ and its $(k-1)$-dimensional subspace $T_a(\partial N) \subset T_aN \subset T_aM$.

Example 15.18. 1. Suppose a submanifold $N \subset M$ is globally parameterized by an embedding $\phi : G \to N \hookrightarrow V$, $G \subset \mathbb{R}^k$. Suppose the coordinates in $\mathbb{R}^k$ are denoted by $(u_1, \ldots, u_k)$. Then the tangent space T_aN at a point $a = \phi(b), b \in G$ is equal to the span

$$\operatorname{Span}\left(\frac{\partial\phi}{\partial u_1}(a), \ldots, \frac{\partial\phi}{\partial u_k}(a)\right).$$

2. In particular, suppose $M = \mathbb{R}^n$, a submanifold N is graphical and given by equations

$$x_{k+1} = g_1(x_1, \ldots, x_k), \ldots, x_n = g_{n-k}(x_1, \ldots, x_k), \quad (x_1, \ldots, x_k) \in G \subset \mathbb{R}^k.$$

Take points $b = (b_1, \ldots b_k) \in G$ and $a = (b_1, \ldots, b_k, g_1(b), \ldots, g_{n-k}(b)) \in A$. Then $T_aN = \operatorname{Span}(T_1, \ldots T_k)$, where

$$T_1 = \left(\underbrace{1, 0, \ldots, 0}_{k}, \frac{\partial g_1}{\partial x_1}(b), \ldots, \frac{\partial g_{n-k}}{\partial x_1}(b)\right),$$

$$T_1 = \left(\underbrace{0, 1, \ldots, 0}_{k}, \frac{\partial g_1}{\partial x_2}(b), \ldots, \frac{\partial g_{n-k}}{\partial x_2}(b)\right),$$

$$\ldots$$

$$T_1 = \left(\underbrace{0, 0, \ldots, 1}_{k}, \frac{\partial g_1}{\partial x_k}(b), \ldots, \frac{\partial g_{n-k}}{\partial x_k}(b)\right).$$

3. Suppose a hypersurface $\Sigma \subset \mathbb{R}^n$ is given by an equation $\Sigma = \{F = 0\}$ for a smooth function F defined on an neighborhood of Σ and such that $d_aF \neq 0$ for any $a \in \Sigma$. In other words, the function F has no critical points on Σ. Take a point $a \in \Sigma$. Then $T_a\Sigma \subset \mathbb{R}^n_a$ is given by a linear equation

$$\sum_1^n \frac{\partial F}{\partial x_j}(a)h_j = 0, \quad h = (h_1, \ldots, h_n) \in \mathbb{R}^n_a.$$

Note that sometimes one is interested to define $T_a\Sigma$ as an affine subspace of $\mathbb{R}^n = \mathbb{R}^n_0$ and not as a linear subspace of $\mathbb{R}^n_a$. We get the required equation by shifting the origin:

$$T_a\Sigma = \{x = (x_1, \dots, x_n) \subset \mathbb{R}^n;\ \sum_1^n \frac{\partial F}{\partial x_j}(a)(x_j - a_j) = 0\}.$$

If for some parameterization $\phi : G \to A$ with $\phi(0) = a$ the composition $f \circ \phi$ is differentiable at 0, and the linear map

$$d_0(f \circ \phi) \circ (d_0\phi)^{-1} : T_aA \to W_{f(a)}$$

is called the *differential* of f at the point a and denoted, as usual, by d_af. Similarly one can define C^m-smooth maps $A \to W$.

Exercise 15.19. Let $N \subset M$ a submanifold Show that a map $f : N \to V$ is differentiable at a point $a \in N$ iff for some neighborhood U of a in M there exists a map $F : U \to V$ that is differentiable at a and such that $F|_{U\cap N} = f|_{U\cap N}$, and we have $dF|_{T_aN} = d_af$. As it follows from the above discussion the map $dF|_{T_aN}$ is independent of this extension. Similarly, any C^m-smooth map of N locally extends to a C^m-smooth map of a neighborhood of N in M.

15.5 Vector bundles and their homomorphisms

Let us put the above discussion in a bit more global and general setup.

A collection of all tangent spaces $\{T_aA\}_{a\in A}$ to a submanifold A is called its *tangent bundle* and denoted by TA or $T(A)$. This is a special case of a more general notion of a *vector bundle of rank* r over a set $A \subset V$. One understands by this a family of r-dimensional vector subspaces $L_a \subset V_a$, parameterized by points of A and *continuously* (or C^m-smoothly) depending on a. More precisely one requires that each point $a \in A$ has a neighborhood $U \subset A$ such that there exist linear independent vector fields $v_1(a), \dots, v_r(a) \in L_a$ which continuously (smoothly, etc.) depend on a.

Besides the tangent bundle $T(A)$ over a k-submanifold A an important example of a vector bundle over a submanifold A is its *normal bundle* $NA = N(A)$, which is a vector bundle of rank $n - k$ formed by orthogonal complements $N_aA = T_a^{\perp}A \subset V_a$ of the tangent spaces T_aA of A. We assume here that V is Euclidean space.

A vector bundle L of rank k over A is called *trivial* if one can find k continuous linearly independent vector fields $v_1(a), \dots, v_k(a) \in L_a$ defined for *all* $a \in A$. The set A is called the *base* of the bundle L.

An important example of a trivial bundle is the bundle $TU = \{T_aU\}_{a\in U}$ for a domain U in a vector space V..

Exercise 15.20. Prove that the tangent bundle to the unit circle $S^1 \subset \mathbb{R}^2$ is trivial. Prove that the tangent bundle to $S^2 \subset \mathbb{R}^3$ is not trivial, but the tangent bundle to the unit sphere $S^3 \subset \mathbb{R}^4$ is trivial. (The case of S^1 is easy, of S^3 is a bit more difficult, and of S^2 even more difficult. It turns out that the tangent bundle TS^{n-1} to the unit sphere $S^{n-1} \subset \mathbb{R}^n$ is trivial if and only if $n = 2, 4$ and 8. The *only if* part is a very deep topological fact which was proved by F. Adams in 1960.

Suppose we are given two vector bundles, L over A and $\widetilde{L}$ over $\widetilde{A}$ and a continuous (resp. smooth) map $\phi : A \to \widetilde{A}$. By a continuous (resp. smooth) *homomorphism* $\Phi : L \to \widetilde{L}$ which *covers the map* $\phi : A \to \widetilde{A}$ we understand a continuous (resp. smooth) family of linear maps $\Phi_a : L_a \to \widetilde{L}_{\phi(a)}$. For instance, a C^m-smooth map $f : A \to B$ defines a C^{m-1}-smooth homomorphism $df : TA \to TB$ which covers $f : A \to B$. Here $df = \{d_af\}_{a\in A}$ is the family of linear maps $d_af : T_aA \to T_{f(a)}B$, $a \in A$.

15.6 Orientation

By an *orientation* of a vector bundle $L = \{L_a\}_{a\in A}$ over A we understand continuously depending on a orientation of all vector spaces L_a. An orientation of a submanifold k is the same as an orientation of its tangent bundle $T(A)$. A *co-orientation* of a k- submanifold A is an orientation of its normal bundle $N(A) = T^{\perp}A$ in V. Note that *not all bundles are orientable*, i.e. some bundles admit no orientation. But if L is orientable and the base A is connected, then L admits exactly two orientations. Here is a simplest example of a non-orientable rank 1 bundle over the circle $S^1 \subset \mathbb{R}^2$. Let us identify a point $a \in S^1$ with a complex number $a = e^{i\phi}$, and consider a line $l_a \in \mathbb{R}^2_a$ directed by the vector $e^{\frac{i\phi}{2}}$. Hence, when the point completes a turn around S^1 the line l_a rotates by the angle π. We leave it to the reader to make a precise argument why this bundle is not orientable . In fact, rank 1 bundles are orientable if and only if they are trivial.

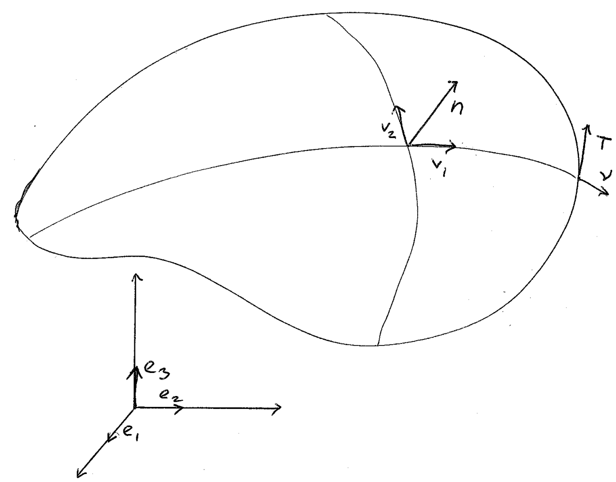


Figure 15.2: The orientation of the surface is induced by its co-orientation by the normal vector $\mathbf{n}$. The orientation of the boundary us induced by the orientation of the surface.

If the ambient space V is oriented then co-orientation and orientation of a submanifold A determine each other according to the following rule. For each point a, let us choose any basis $v_1, \dots, v_k$ of $T_a(A)$ and any basis $w_1, \dots, w_{n-k}$ of $N_a(A)$. Then $w_1, \dots, w_{n-k}, v_1, \dots, v_k$ is a basis of $V_a = V$. Suppose one of the bundles, say $N(A)$, is oriented. Let us assume that the basis $w_1, \dots, w_{n-k}$ defines this orientation. Then we orient T_aA by the basis $v_1, \dots, v_k$ if the basis $w_1, \dots, w_{n-k}, v_1, \dots, v_k$ defines the given orientation of V, and we pick the opposite orientation of T_aA otherwise.

Example 15.21. (Induced orientation of the boundary of a submanifold.) Suppose A is an oriented manifold with boundary. Let us co-orient the boundary ∂A by orienting the rank 1 normal bundle to $T(\partial A)$ in $T(A)$ by the unit ourtward normal to $T(\partial A)$ in $T(A)$ vector field. Then the above rule determine an orientation of $T(\partial A)$, and hence of ∂A.

15.7 Useful technical tools: partition of unity and cut-off functions

Let us recall that the *support of a function* θ is the closure of the set of points where it is not equal to 0. We denote the support by $\mathrm{Supp}\,(\theta)$. We say that θ is *supported* in an open set U if $\mathrm{Supp}\,(\theta) \subset U$.

Lemma 15.22. *There exists a C^∞ function $\rho : \mathbb{R} \to [0, \infty)$ with the following properties:*

- $\rho(x) \equiv 0,\ |x| \geq 1$;
- $\rho(x) = \rho(-x)$;
- $\rho(x) > 0$ *for* $|x| < 1$.

Proof. There are a lot of functions with this property. For instance, one of them can be constructed as follows. Take the function

$$h(x) = \begin{cases} e^{-\frac{1}{x^2}} & , \quad x > 0 \\ 0 & , \quad x \leq 0. \end{cases} \tag{15.7.1}$$

The function $e^{-\frac{1}{x^2}}$ has the property that all its derivatives at 0 are equal to 0, and hence the function h is C^∞-smooth. Then the function $\rho(x) := h(1+x)h(1-x)$ has the required properties. ■

Lemma 15.23. Existence of cut-off functions. *Let $C \subset V$ be compact set and $U \supset C$ its open neighborhood. Then there exists a C^∞-smooth function $\sigma_{C,U} : V \to [0, \infty)$ with its support in U which is equal to 1 on C*

Proof. As C is compact we can assume U to be bounded. Let us fix a Euclidean structure in V and a Cartesian coordinate system. Thus we can identify V with $\mathbb{R}^n$ with the standard dot-product. Given a point $a \in V$ and $\delta > 0$ let us denote by $\psi_{a,\delta}$ the *bump* function on V defined by

$$\psi_{a,\delta}(x) := \rho\left(\frac{||x-a||^2}{\delta^2}\right), \tag{15.7.2}$$

where $\rho : \mathbb{R} \to [0, \infty)$ is the function constructed in Lemma 15.22. Note that $\psi_{a,\delta}(x)$ is a C^∞-function with Supp $(\psi_{a,\delta}) = D_\delta := \overline{B_\delta(a)}$ and such that $\psi_{a,\delta}(x) > 0$ for $x \in B_\delta(a)$.

Let us denote by $U_\epsilon(C)$ the ϵ-neighborhood of C, i.e.

$$U_\epsilon(C) = \{x \in V; \exists y \in C, ||y - x|| < \epsilon.\}$$

There exists $\epsilon > 0$ such that $U_\epsilon(C) \subset U$. Using compactness of C we can find finitely many points $z_1, \dots, z_N \in C$ such that the balls $B_\epsilon(z_1), \dots, B_\epsilon(z_N) \subset U$ cover $\overline{U_{\frac{\epsilon}{2}}(C)}$, i.e. $\overline{U_{\frac{\epsilon}{2}}}(C) \subset \bigcup\limits_1^N B_\epsilon(z_j)$. Consider a function

$$\sigma_1 := \sum_1^N \psi_{z_i,\frac{\epsilon}{2}} : V \to \mathbb{R}.$$

The function ψ_1 is positive on $\overline{U}_{\frac{\epsilon}{2}}(C)$ and has Supp $(\psi_1) \subset U$.

The complement $E = V \setminus U_{\frac{\epsilon}{2}}(C)$ is a closed but unbounded set. Take a large $R > 0$ such that $B_R(0) \supset \overline{U}$. Then $E_R = D_R(0) \setminus U_{\frac{\epsilon}{2}}(C)$ is compact. Choose finitely many points $x_1, \dots, x_M \in E_R$ such that $\bigcup\limits_1^M B_{\frac{\epsilon}{4}}(x_i) \supset E_R$. Notice that $\bigcup\limits_1^M B_{\frac{\epsilon}{4}}(x_i) \cap C = \varnothing$. Denote

$$\sigma_2 := \sum_1^M \psi_{x_i,\frac{\epsilon}{4}}.$$

Then the function σ_2 is positive on V_R and vanishes on C. Note that the function $\sigma_1 + \sigma_2$ is positive on $B_R(0)$ and it coincides with σ_1 on C. Finally, define the function $\sigma_{C,U}$ by the formula

$$\sigma_{C,U} := \frac{\sigma_1}{\sigma_1 + \sigma_2}$$

on $B_(R)(0)$ and extend it to the whole space V as equal to 0 outside the ball $B_R(0)$. Then $\sigma_{C,U} = 1$ on C and Supp $(\sigma_{C,U}) \subset U$, as required. ■.

Let $C \subset V$ be a compact set. Consider its finite covering by open sets $U_1, \dots, U_N$, i.e.

$$\bigcup_1^N U_j \supset C.$$

We say that a finite sequence $\theta_1, \dots, \theta_K$ of C^∞-functions defined on some open neighborhood U of C in V forms a *partition of unity* over C subordinated to the covering $\{U_j\}_{j=1,\dots,N}$ if

- $\sum\limits_1^K \theta_j(x) = 1$ for all $x \in C$;
- Each function θ_j, $j = 1, \dots, K$ is supported in one of the sets U_i, $i = 1, \dots, N$.

Lemma 15.24. *For any compact set C and its open covering $\{U_j\}_{j=1,\dots,N}$ there exists a partition of unity over C subordinated to this covering.*

Proof. In view of compactness of C there exists $\epsilon > 0$ and finitely many balls $B_\epsilon(z_j)$ centered at points $z_j \in C$, $j = 1, \dots, K$, such that $\bigcup\limits_1^K B_\epsilon(z_j) \supset C$ and each of these balls is contained in one of the open sets U_j, $j = 1, \dots, N$. Consider the functions $\psi_{z_j,\epsilon}$ defined in (15.7.2). We have $\sum\limits_1^K \psi_{z_j,\epsilon} > 0$ on some neighborhood $U \supset C$. Let $\sigma_{C,U}$ be the cut-off function constructed in Lemma 15.23. For $j = 1, \dots, K$ we define

$$\theta_j(x) = \begin{cases} \frac{\psi_{z_j,\epsilon}(x)\sigma_{C,U}(x)}{\sum\limits_1^K \psi_{z_j,\epsilon}(x)}, & \text{if } x \in U, \\ \\ 0, & \text{otherwise} \end{cases}.$$

Each of the functions is supported in one of the open sets U_j, $j = 1, \dots, N$, and we have for every $x \in C$

$$\sum_1^K \theta_j(x) = \frac{\sum\limits_1^K \psi_{z_j,\epsilon}(x)}{\sum\limits_1^K \psi_{z_j,\epsilon}(x)} = 1.$$

■

15.8 Tubular neighborhood

It is convenient to consider a convenient class of neighborhoods of submanifolds of $\mathbb{R}^n$, called *tubular neighborhhoods*. We restrict below the discussion to the case closed submanifolds.

Let $A \subset \mathbb{R}^n$ be a closed k-dimensional submanifold of $\mathbb{R}^n$. A compact domain $N = N(A) \subset \mathbb{R}^n$ (with smooth boundary if A is closed, and with boundary with corners if $\partial A \neq \varnothing$) is called a *tubular neighborhhood* of A if and there is a projection $\pi : N \to A$ and $\epsilon > 0$ with the following properties:

- $\pi|_A = \mathrm{Id}$;
- for every point $a \in A$ the pre-image $N_a := \pi^{-1}(a) \subset N$ (called a *fiber* of the tubular neighborhood) is an $(n-k)$-dimensional normal disc of radius ϵ, i.e.

$$N_a := \{a + y; y \perp T_a A, ||y|| \leq \epsilon\}.$$

The number ϵ is called the *radius* of the tubular neighborhood N.

Theorem 15.25 (Tubular neighborhood theorem)**.** *Any compact k-dimensional submanifold $A \subset \mathbb{R}^n$ admits a tubular neighborhood of a sufficiently small radius $\epsilon > 0$.*

Proof. Let $\mathrm{Norm}(A)$ be the normal bundle of A, i.e. the set of pairs $\{(a, v); a \in A, v \in T_a^{\perp}(A) \subset \mathbb{R}^n_a$. As $\mathbb{R}^n_a$ can be identified with $\mathbb{R}^n$ via a parallel transport, we can view $\mathrm{Norm}(A)$ as a $2k$-dimensional submanifold of $A \times \mathbb{R}^n \subset \mathbb{R}^n \times \mathbb{R}^n = \mathbb{R}^{2n}$. Denote by p the projection $\mathrm{Norm}(A) \to A$ given by the formula $p(a, v) = a$, and denote $\mathrm{Norm}_\epsilon(A) := \{(a, v) \in \mathrm{Norm}(A); ||v|| \leq \epsilon\}$. Note that if $\partial A \neq \varnothing$ then $\mathrm{Norm}(A)$ has boundary $\partial(\mathrm{Norm}(A)) = p^{-1}(\partial A)$. For a closed manifold A the manifold $\mathrm{Norm}_\epsilon(A)$ has boundary $\partial(\mathrm{Norm}_\epsilon(A)) = \{(a, v); ||v|| = \epsilon\}$ and if $\partial A \neq \varnothing$ then $\mathrm{Norm}_\epsilon(A)$ has boundary $\partial(\mathrm{Norm}_\epsilon(A)) = \{(a, v); ||v|| = \epsilon\} \cup \{(a, v); a \in \partial A, ||v|| \leq \epsilon\}$ with corners along $\{(a, v); ||v|| = \epsilon\}$.

Consider a map $\nu : \mathrm{Norm}(A) \to \mathbb{R}^n$ given by the formula

$$\nu(a, v) = a + v.$$

We will show that for a sufficiently small $\epsilon > 0$ the map $\nu|_{\mathrm{Norm}_\epsilon}$ is a diffeomorphism of Norm_ϵ onto its image. First, we note that for each point $(a, 0) \in \mathrm{Norm}(N)$ the differential $d_{(a,0)}\nu : T_{(a,0)}\mathrm{Norm}(N) \to \mathbb{R}^n_a$ is an isomorphism (why?). Hence, by the inverse function theorem ν is a diffeomorphism on a sufficiently small neighborhood of each point $(a, 0)$ onto its image. Hence, there is $\sigma > 0$ and for each point $a \in A$, a neighborhood $U_a \ni a, U_a \subset A$ such that $\nu|_{p^{-1}(U_a) \cap \mathrm{Norm}_\sigma}$ is a diffeomorphism onto its image. Furthermore, for each $a \in A$ we can choose a smaller neighborhood $\widetilde{U}_a \ni a, \widetilde{U}_a \subset U_a$ such that if $\widetilde{U}_a \cap \widetilde{U}_b \neq \varnothing$ then $\widetilde{U}_b \subset U_a$. In view of compactness we can choose finitely many neighbohoods $\widetilde{U}_{a_1}, \dots, \widetilde{U}_{a_N}$ such that $\bigcup\limits_1^N \widetilde{U}_{a_j} = A$. Furthermore, we can find a small $\epsilon < \sigma$ such that if

$$\nu(p^{-1}(\widetilde{U}_{a_j}) \cap \mathrm{Norm}_\epsilon) \cap \nu(p^{-1}(\widetilde{U}_{a_i}) \cap \mathrm{Norm}_\epsilon) \neq \varnothing, \;\; i, j \in \{1, \dots, N\},$$

then $\widetilde{U}_{a_i} \subset U_{a_j}$. This implies that $\nu|_{\mathrm{Norm}_\epsilon}$ is a diffeomorphism onto its image. Indeed, first of all the map ν is an immersion, i.e. each point has a neighborhood which is diffeomorphically mapped onto its image. So to verify that it is a diffeomorphism it is sufficient to check that $\nu|_{\mathrm{Norm}_\epsilon}$ is injective. Take any two distinct points $(a, v), (a', v') \in \mathrm{Norm}_\epsilon$ if a, a' belong to the same neighborhood U_{a_j} then $\nu(a, v) \neq \nu(a', v')$ because the restriction $\nu|_{p^{-1}(U_{a_j}) \cap \mathrm{Norm}_\epsilon}$ is injective by construction. If $a \in \widetilde{U}_{a_j}$ and $a' \notin U_{a_j}$ then $a' \in U_{a_i}$ such that $\nu(p^{-1}(\widetilde{U}_{a_j}) \cap \mathrm{Norm}_\epsilon) \cap \nu(p^{-1}(\widetilde{U}_{a_i}) \cap \mathrm{Norm}_\epsilon) = \varnothing$, and therefore again we have $\nu(a, v) \neq \nu(a', v')$.

Thus we can define the tubular neighborhood N as the image $\nu(\mathrm{Norm}_\epsilon)$, and define the projection $\pi : N \to A$ by the formula $\pi = \nu \circ p \circ \nu^{-1}$. We note that, as Norm_ϵ, N is manifold with boundary $\partial N = \nu(\{(a, v);\, ||v|| = \epsilon\})$ if A is closed, and has boundary $\partial N = \nu(\{(a, v);\, ||v|| = \epsilon\} \cup \{(a, v);\, a \in \partial A, ||v|| \leq \epsilon\})$ with corners along $\nu(\{(a, v);\, ||v|| = \epsilon\})$ if A itself has non-empty boundary ∂A. ■

In the case $\partial A \neq \varnothing$ we will sometimes refer to the part $\nu(\{(a, v);\, ||v|| = \epsilon\})$ as *vertical part* of the boundary ∂N, and to $\nu(\{(a, v);\, a \in \partial A, ||v|| \leq \epsilon\})$ as the *horizontal part*.

It will also useful to consider a vector field $\mathbf{r}$ on the tubular neighborhood N as follows. Viewing $\mathrm{Norm}A$ a submanifold of $\mathbb{R}^{2n} = \mathbb{R}^n \times \mathbb{R}^n$ and denoting the corresponding coordinates $(x_1, \dots, x_n, y_1, \dots, y_n)$ consider a vector field $Y := \sum\limits_1^n y_j \frac{\partial}{\partial y_j}$. It vanishes along $\mathbb{R}^n \times 0$ and its radial vector field tangent to fibers $x \times \mathbb{R}^n$. The vector field Y is also tangent to $\mathrm{Norm}A \subset \mathbb{R}^{2n}$. We define the vector field $\mathbf{r}$ on N as $d\nu(Y|_{\mathrm{Norm}_\epsilon})$. The vector field $\mathbf{r}$ vanishes along A and its is radial vector field tangent to normal fibers $N_a = \pi^{-1}(a), a \in A$. We also note that if A is closed, then $\mathbf{r}$ is outward transverse to the boundary ∂N, and if $\partial A \neq \varnothing$ then $\mathbf{r}$ is outward transverse to the vertical part of the boundary ∂N and tangent to the horizontal part.

Part IV

Stokes' theorem and its applications

Chapter 16

Stokes' theorem

16.1 Differential k-forms on manifolds and their integration

Let us first generalize the notion of differential k-forms for an arbitrary n-dimensional manifold M. As in the case when M is a domain in a vector space we say that a differential k-form α on M is a field of exterior k-forms α_a on tangent spaces T_aM, $a \in M$. In a local coordinate system on a neighborhood $U \subset M$ the k-form can be written in the usual was as $\sum\limits_{i_1<\dots<i_k} f_{i_1\dots i_k}(x)dx_{i_1} \wedge \dots \wedge dx_{i_k}$, and we require the coefficients $f_{i_1\dots i_k}(x)$ to be smooth functions. Changing to another local coordinate system results to the pull-back of α by a transition map g, and as it is smooth as well, the smoothness condition for coefficients is independent of the a choice of a coordinate system.

If $N \subset M$ is a submanifold, then any differential k-form α on M can be pulled back to N by the inclusion map $i : N \hookrightarrow M$. The resulted k-form $i^*\alpha$ on N is called the *restriction* of the form α to N and is often denoted by $\alpha|_N$. Thus for any k tangent to N vectors $T_1, \dots, T_k \in T_aN \subset T_aM$ we have

$$\alpha|_N(T_1, \dots, T_k) = \alpha(T_1, \dots, T_k).$$

Remark 16.1. Conversely, for any differential k-form α on a submanifold $N \subset M$ there exists a differential k-form $\widehat{\alpha}$ on a neighborhood $U \supset N$, $U \subset M$, such that $\widehat{\alpha}|_N = \alpha$. We will not be using this fact.

Let M be any compact oriented n-dimensional manifold with boundary and α a differential n-form. Suppose first that α is supported in a coordinate neighborhood $U \subset M$, so that there exists an orientation preserving diffeomorphism $\psi : G \to U$, where G is an open bounded subset of $\mathbb{R}^n_+ = \{x_n \geq 0\} \subset \mathbb{R}^n$. We define

$$\int\limits_M \alpha := \int\limits_G \psi^* \alpha.$$

Note that if $\widetilde{\psi} : \widetilde{G} \to U$ be another orientation preserving parametrization of U then $\widetilde{\psi} = \psi \circ h$, where $h = \psi^{-1} \circ \widetilde{\psi} : \widetilde{G} \to G$ is a diffeomorphism. Hence, by Theorem 14.20 we have

$$\int\limits_{\widetilde{G}} \widetilde{\psi}^* \alpha = \int\limits_{\widetilde{G}} h^*(\psi^* \alpha) = \int\limits_G \psi^* \alpha.$$

$\int\limits_{\widetilde{G}} \widetilde{\psi}^* \alpha = \int\limits_{\widetilde{G}} \psi \circ \psi^{-1} \circ \widetilde{\psi}(^{-1} \circ \widetilde{\psi}) \psi^* \alpha$, hence $\int\limits_M \alpha$ *is independent of a choice of parametrization, provided that the orientation is preserved.*

To define the integral of an arbitrary differential n-form α we choose a partition of unity $1 = \sum\limits_1^K \theta_j$ of M such that each function θ_j is supported in some coordinate neighborhood of M. Denote $\alpha_j = \theta_j \alpha$. Then $\alpha = \sum\limits_1^K \alpha_j$, where each form α_j is supported in one of coordinate neighborhoods. We define

$$\int\limits_M \alpha := \sum_1^K \int\limits_M \alpha_j.$$

Lemma 16.2. *The above definition of* $\int\limits_M \alpha$ *is independent of a choice of a partition of unity.*

Proof. Consider two partitions of unity $1 = \sum\limits_1^K \theta_j$ and $1 = \sum\limits_1^{\widetilde{K}} \widetilde{\theta}_j$ subordinated to coverings $U_1, \dots, U_K$ and $\widetilde{U}_1, \dots, \widetilde{U}_{\widetilde{K}}$, respectively. Taking the product of two partitions we get another partition $1 = \sum\limits_{i=1}^K \sum\limits_{j=1}^{\widetilde{K}} \theta_{ij}$, where $\theta_{ij} := \theta_i \widetilde{\theta}_j$, which is subordinated to the covering by intersections $U_i \cap \widetilde{U}_j$, $i = 1, \dots, K,\ j = 1, \dots, \widetilde{K}$. Denote $\alpha_i := \theta_i \alpha,\ \widetilde{\alpha}_j := \widetilde{\theta}_j \alpha$ and $\alpha_{ij} = \theta_{ij} \alpha$. Then

$$\sum_{i=1}^K \alpha_{ij} = \widetilde{\alpha}_j,$$

$$\sum_{j=1}^{\widetilde{K}} \alpha_{ij} = \alpha_i, \quad \text{and}$$

$$\alpha = \sum_1^K \alpha_i = \sum_1^{\widetilde{K}} \widetilde{\alpha}_j.$$

Then we get

$$\sum_{1}^{K}\int\limits_{M}\alpha_i=\sum_{i=1}^{K}\sum_{j=1}^{\widetilde{K}}\int\limits_{M}\alpha_{ij}=\sum_{1}^{\widetilde{K}}\int\limits_{M}\widetilde{\alpha}_j.$$

■

Given a compact oriented k-dimensional submanifold $N\subset M$ and a differential k-form α on a neighborhood $U\supset N, U\subset M$ we define

$$\int\limits_{N}\alpha:=\int\limits_{N}\alpha|_N.$$

When $k=1$ the above definition of the integral coincides with the definition of the integral of a 1-form over an oriented curve which was given above in Section 5.1.

Let us extend the definition of integration of differential forms to an important case of integration of 0-form over oriented 0-dimensional submanifolds. Let us recall a compact oriented 0-dimensional submanifold of M is just a finite set of points $a_1,\dots,a_m\in M$, and their orientation is just an assignment of a sign to every point. In view of the additivity of the integral it is sufficient to define integration over 1 point with a sign. On the other hand, a 0-form is just a function $f:V\to\mathbb{R}$. So we define

$$\int\limits_{\pm a}f:=\pm f(a).$$

A partition of unity is a convenient tool for studying integrals, but not so convenient for practical computations. The following proposition provides a more practical method for computations.

Proposition 16.3. *Let A be a compact oriented submanifold of V and α a differential k-form given on a neighborhood of A. Suppose that A presented as a union $A=\bigcup\limits_{1}^{N}A_j$, where A_j are codimension 0 submanifolds of A with boundary with corners. Suppose that A_i and A_j for any $i\neq j$ intersect only along pieces of their boundaries. Then*

$$\int\limits_{A}\alpha=\sum_{1}^{N}\int\limits_{A_j}\alpha.$$

In particular, if each A_j is parameterized by an orientation preserving embedding $\phi_j : G_j \to A$, where $G_j \subset \mathbb{R}^k$ is a domain with boundary with corners. Then

$$\int\limits_A \alpha = \sum_1^N \int\limits_{G_j} \phi_j^* \alpha.$$

We leave the proof to the reader as an exercise.

Exercise 16.4. *Compute the integral*

$$\int\limits_S 1/3(x_1 dx_2 \wedge dx_3 + x_2 dx_3 \wedge dx_1 + x_3 dx_1 \wedge dx_2),$$

where S is the sphere

$$\{x_1^2 + x_2^2 + x_3^2 = 1\},$$

cooriented by its exterior normal vector.

Solution. Let us present the sphere as the union of northern and southern hemispheres:

$$S = S_- \cup S_+, \ \text{ where } S_- = S \cap \{x_3 \le 0\},\ S_+ = S \cap \{x_3 \ge 0\}.$$

Then $\int\limits_S \omega = \int\limits_{S+} \omega + \int\limits_{S_-} \omega$. Let us first compute $\int\limits_{S+} \omega$.

We can parametrize S_+ by the map $(u, v) \to (u, v, \sqrt{R^2 - u^2 - v^2})$, $(u, v) \in \{u^2 + v^2 \le R^2\} = \mathcal{D}_R$. One can check that this parametrization agrees with the prescribed coorientation of S. Thus, we have

$$\int\limits_{S_+} \omega = 1/3 \int\limits_{\mathcal{D}_R} \left(u dv \wedge d\sqrt{R^2 - u^2 - v^2} + v d\sqrt{R^2 - u^2 - v^2} \wedge du + \sqrt{R^2 - u^2 - v^2} du \wedge dv \right).$$

Passing to polar coordinates (r, φ) in the plane (u, v) we get

$$\int\limits_{S_+} \omega = 1/3 \int\limits_P r\cos\varphi d(r\sin\varphi) \wedge d\sqrt{R^2 - r^2} + r\sin\varphi d\sqrt{R^2 - r^2} \wedge d(r\cos\varphi)$$

$$+\sqrt{R^2 - r^2} d(r\cos\varphi) \wedge d(r\sin\varphi),$$

where $P = \{0 \le r \le R, 0 \le \varphi \le 2\pi\}$. Computing this integral we get

$$\begin{aligned}
\int\limits_{S_+} \omega &= \frac{1}{3}\int_P -\frac{r^3\cos^2\varphi d\varphi \wedge dr}{\sqrt{R^2-r^2}} + \frac{r^3\sin^2\varphi dr\wedge d\varphi}{\sqrt{R^2-r^2}} + \sqrt{R^2-r^2}dr\wedge d\varphi \\
&= \frac{1}{3}\int_P \left(\frac{r^3}{\sqrt{R^2-r^2}} + r\sqrt{R^2-r^2}\right) dr\wedge d\varphi \\
&= \frac{2\pi}{3}\int_0^R \frac{rR^2}{\sqrt{R^2-r^2}}dr = -\frac{2\pi R^2}{3}\sqrt{R^2-r^2}\,\Big|_0^R = \frac{2\pi R^3}{3}
\end{aligned}$$

Similarly, one can compute that

$$\int\limits_{S_-} \omega = \frac{2\pi R^3}{3}.$$

Computing this last integral, one should notice the fact that the parametrization

$$(u,v) \mapsto (u, v, -\sqrt{R^2-u^2-v^2})$$

defines the *wrong* orientation of S_-. Thus one should use instead the parametrization

$$(u,v) \mapsto (v, u, -\sqrt{R^2-u^2-v^2}),$$

and we get the answer

$$\int\limits_{S} \omega = \frac{4\pi R^3}{3}.$$

This is just the volume of the ball bounded by the sphere. The reason for such an answer will be clear below from Stokes' theorem.

16.2 Statement of Stokes' theorem

Theorem 16.5. *Let M be a compact oriented k-dimensional manifold with boundary (and possibly with corners). Let ω be a C^1-smooth differential $(k-1)$-form defined on M. Then*

$$\int\limits_{\partial M} \omega = \int\limits_{M} d\omega\,.$$

Here $d\omega$ is the exterior differential of the form ω and ∂M is the boundary of M co-oriented by its outward normal vector field.

As a corollary we get

Corollary 16.6. *Let M be a compact n-dimensional manifold and $N \subset M$ its k-dimensional oriented submanifolds with boundary (and possibly with corners). Let α be a C^1-smooth differential $(k-1)$-form defined on a neighborhood $U \supset N$. Then*

$$\int\limits_{\partial N} \alpha = \int\limits_{N} d\alpha\,.$$

We will discuss below what exactly Stokes' theorem means for the case $k \leq 3$ and $n = \dim V \leq 3$.

Let us begin with the case $k = 1$, $n = 2$. Thus $V = \mathbb{R}^2$. Let x_1, x_2 be coordinates in $\mathbb{R}^2$ and U a domain in $\mathbb{R}^2$ bounded by a smooth curve $\Gamma = \partial U$. Let us co-orient Γ with the outward normal ν to the boundary of U. This defines a counter-clockwise orientation of Γ.

Let $\omega = P_1(x_1, x_2)dx_1 + P_2(x_1, x_2)dx_2$ be a differential 1-form. Then the above Stokes' formula asserts

$$\int\limits_{U} d\omega = \int\limits_{\Gamma} \omega\,,$$

or

$$\int\limits_{U} \left(\frac{\partial P_2}{\partial x_1} - \frac{\partial P_1}{\partial x_2}\right) dx_1 \wedge dx_2 = \int_{\Gamma} P_1 dx_1 + P_2 dx_2.$$

This is called *Green's formula.* In particular, when $d\omega = dx_1 \wedge dx_2$, e.g. $\omega = x_d x_2$ or $\omega = \frac{1}{2}(x_1 dx_2 - x_2 dx_1)$, the integral $\int\limits_{\Gamma} \omega$ computes the area of the domain U.

Consider now the case $n = 3$, $k = 2$. Thus

$$V = \mathbb{R}^3,\ \omega = P_1 dx_2 \wedge dx_3 + P_2 dx_3 \wedge dx_1 + P_3 dx_1 \wedge dx_2.$$

Let $U \subset \mathbb{R}^3$ be a domain bounded by a smooth surface S. We co-orient S with the exterior normal ν. Then

$$d\omega = \left(\frac{\partial P_1}{\partial x_1} + \frac{\partial P_2}{\partial x_2} + \frac{\partial P_3}{\partial x_3}\right) dx_1 \wedge dx_2 \wedge dx_3.$$

Thus, Stokes' formula

$$\int\limits_{S} \omega = \int\limits_{U} d\omega$$

gives in this case

$$\int\limits_S P_1 dx_2 \wedge dx_3 + P_2 dx_3 \wedge dx_1 + P_3 dx_1 \wedge dx_2 = \int\limits_U \left(\frac{\partial P_1}{\partial x_1} + \frac{\partial P_2}{\partial x_2} + \frac{\partial P_3}{\partial x_3}\right) dx_1 \wedge dx_2 \wedge dx_3$$

This is called the *divergence theorem* or Gauss-Ostrogradski's formula.

Consider the case $k = 0$, $n = 1$. Thus ω is just a function f on an interval $I = [a, b]$. The boundary ∂I consists of 2 points: $\partial I = \{a, b\}$. One should orient the point a with the sign $-$ and the point b with the sign $+$.

Thus, Stokes' formula in this case gives

$$\int\limits_{[a,b]} df = \int\limits_{\{-a,+b\}} f,$$

or

$$\int_a^b f'(x)dx = f(b) - f(a).$$

This is Newton-Leibnitz' formula. More generally, for a 1-dimensional oriented connected curve $\Gamma \subset \mathbb{R}^3$ with boundary $\partial\Gamma = B \cup (-A)$ and any smooth function f we get the formula

$$\int\limits_\Gamma df = \int\limits_{B\cup(-A)} f = f(B) - f(A),$$

which we already proved earlier, see Theorem 5.8 .

Consider now the case $n = 3$, $k = 1$.

Thus $V = \mathbb{R}^3$ and $\omega = P_1 dx_1 + P_2 dx_2 + P_3 dx_3$. Let $S \subset \mathbb{R}^3$ be an oriented surface with boundary Γ. We orient Γ in the same way, as in Green's theorem. Then Stokes' formula

$$\int\limits_S d\omega = \int\limits_\Gamma \omega$$

gives in this case

$$\int\limits_S \left(\frac{\partial P_3}{\partial x_2} - \frac{\partial P_2}{\partial x_3}\right) dx_2 \wedge dx_3 + \left(\frac{\partial P_1}{\partial x_3} - \frac{\partial P_3}{\partial x_1}\right) dx_3 \wedge dx_1 + \left(\frac{\partial P_2}{\partial x_1} - \frac{\partial P_1}{\partial x_2}\right) dx_1 \wedge dx_2$$

$$= \int_{\Gamma} P_1 dx_1 + P_2 dx_2 + P_3 dx_3.$$

This is the original Stokes' theorem.

Stokes' theorem allows one to clarify the geometric meaning of the exterior differential.

Lemma 16.7. *Let β be a differential k-form in a domain $U \subset V$. Take any point $a \in U$ and vectors $X_1, \dots, X_{k+1} \in V_a$. Given $\epsilon > 0$ let us consider the parallelepiped $P(\epsilon X_1, \dots, \epsilon X_{k+1})$ as a subset of V with vertices at points $a_{i_1 \dots i_{k+1}} = a + \epsilon \sum\limits_1^{k+1} i_j X_j$, where each index i_j takes values $0, 1$. Then*

$$d\beta_a(X_1, \dots X_{k+1}) = \lim_{\epsilon \to 0} \frac{1}{\epsilon^{k+1}} \int\limits_{\partial P(\epsilon X_1, \dots, \epsilon X_{k+1})} \beta.$$

Proof. First, it follows from the definition of integral of a differential form that

$$d\beta_a(X_1, \dots, X_{k+1}) = \lim_{\epsilon \to 0} \frac{1}{\epsilon^{k+1}} \int\limits_{P(\epsilon X_1, \dots, \epsilon X_{k+1})} d\beta. \tag{16.2.1}$$

Then we can continue using Stokes' formula

$$d\beta_a(X_1, \dots, X_{k+1}) = \lim_{\epsilon \to 0} \frac{1}{\epsilon^{k+1}} \int\limits_{P(\epsilon X_1, \dots, \epsilon X_{k+1})} d\beta = \lim_{\epsilon \to 0} \frac{1}{\epsilon^{k+1}} \int\limits_{\partial P(\epsilon X_1, \dots, \epsilon X_{k+1})} \beta. \tag{16.2.2}$$

■

16.3 Integration of closed and exact forms

Let us recall that a differential k-form ω is called *closed* if $d\omega = 0$, and that it is called *exact* if there exists a $(k-1)$-form α, called *primitive* of ω, such that $\omega = d\alpha$.

Any exact form is closed, because $d(d\alpha) = 0$. Any n-form in a n-dimensional space is closed.

Proposition 16.8. *a) For a closed k-form ω on a compact $(k+1)$-dimensional manifold Σ with boundary $\partial\Sigma$ we have*

$$\int\limits_{\partial\Sigma} \omega = 0\,.$$

b) If ω is an exact k-form defined on a closed k-dimensional manifold S then

$$\int_S \omega = 0\,.$$

The proof immediately follows from Stokes' formula. Indeed, in case a) we have

$$\int_{\partial\Sigma} \omega = \int_{\Sigma} d\omega = 0\,.$$

In case b) we have $\omega = d\alpha$ and $\partial S = \varnothing$. Thus

$$\int_S d\alpha = \int_{\emptyset} \alpha = 0\,.$$

Proposition 16.8b) gives a necessary condition for a closed form to be exact.

We already had seen that the differential 1-form $\alpha = \frac{1}{x^2+y^2}(xdy - ydx)$ defined on the punctured plane $\mathbb{R}^2 \setminus 0$ is closed but not exact.

Lemma 16.9. *The $(n-1)$-form*

$$\theta_n = \sum_{i=1}^{n} (-1)^{i-1} \frac{x_i}{r^n} dx_1 \wedge \overset{i}{\check{\dots}} \wedge dx_n \quad (dx_i \ \ \textit{is missing})$$

is closed in $\mathbb{R}^n \setminus 0$ but not exact.

Proof. It is straightforward to check that θ_n is closed. To show that it is not exact we we observe $\int\limits_{S^{n-1}} \theta_n \neq 0$, where S^{n-1} is the unit sphere oriented as the boundary of the unit ball. Indeed, consider the form

$$\mu_n := \sum_{i=1}^{n} (-1)^{i-1} x_i dx_1 \wedge \overset{i}{\check{\dots}} \wedge dx_n$$

and observe that $\theta_n|_{S^{n-1}} = \mu_n|_{S^{n-1}}$. But the form μ_n is well defined on the whole space $\mathbb{R}^n$. Hence, we can apply Stokes' formula to the unit ball B_n. Note that

$$d\mu_n = n dx_1 \wedge dx_n.$$

Thus, we get

$$\begin{aligned} \int_{S^{n-1}} \theta_n &= \int_{S^{n-1}} \mu_n = \int_{B^n} d\mu_n \\ &= n \int_{B^n} dx_1 \wedge \dots dx_n = n\mathrm{Vol}(B^n) > 0. \end{aligned}$$

■

16.4 Proof of Stokes' theorem

We prove in this section Theorem 16.5. We will consider only the case when M is a manifold with boundary *without corners* and leave the corner case as an exercise to the reader.

Let us cover M by coordinate neighborhoods. First, using a partition of unity we can reduce to the case of a form supported in one of these coordinate neighborhoods. Assuming that, consider two cases, when the coordinate neighborhood U intersects ∂M or not.

Consider the former case. Then $\omega|_{\partial M} = 0$, and hence, $\int\limits_{\partial M} \omega = 0$. We can assume that U is parameterized by an open cube $I = \{|x_1|, \dots, |x_k| < 1\}$, i.e. we are given a diffeomorphism $\psi : I \to U$. By definition

$$\int\limits_M d\omega = \int\limits_I \psi^* d\omega = \int\limits_I \psi^* d\omega = \int\limits_I d(\psi^*\omega).$$

Let us write

$$\psi^*\omega = \sum_{j=1}^k (-1)^{j-1} f_j dx_1 \wedge \dots \overset{j}{\vee} \dots \wedge dx_k, \text{ and} \tag{16.4.1}$$

$$d(\psi^*\omega) = \left(\sum_{j=1}^k \frac{\partial f_j}{\partial x_j}\right) dx_1 \wedge \dots \wedge dx_k. \tag{16.4.2}$$

Note that the functions f_j are equal to 0 on ∂I. Using Fubini's theorem we get

$$\int\limits_I d(\psi^*\omega) = \underbrace{\int \cdots \int}_{k} \left(\sum_{j=1}^k \frac{\partial f_j}{\partial x_j}\right) dx_1 \dots dx_k$$

$$\sum_1^k \underbrace{\int \cdots \int}_{k-1} (f(x_1, \cdots x_{j-1}, 1, \dots, x_k) - f(x_1, \cdots x_{j-1}, -1, \dots, x_k)) dx_1 \dots \overset{j}{\vee} \dots dx_k = 0.$$

Consider now the case when the coordinate neighborhood U supporting ω intersects ∂M. We can arrange that U is parameterized by an half-cube $I_- = \{-1 < x_1 \le 0; |x_2| < 1, \dots, |x_k| < 1\}$, i.e. we are given a diffeomorphism $\psi : I_- \to U$. We have $\partial I_- = \{x_1 = 0, \dots, |x_2| < 1, \dots, |x_k| < 1\}$. The orientation of ∂I_- is induced by the outward normal vector fields $(1, 0, \dots, 0)$, and hence, it coincides with the orientation given by coordinates $(x_2, \dots, x_k)$ (we have chosen the half-cube I_-

rather that I_+ to ensure this property). Thus,

$$\begin{aligned}\int\limits_{\partial M} \omega = \int\limits_{\partial I_-} \psi^*\omega &= \int\limits_{|x_j|<1, j=2,\dots,k} f_1(0, x_2, \dots, x_k) dx_2 \wedge \dots \wedge dx_k \\ &= \underbrace{\int\limits_{-1}^{1} \cdots \int\limits_{-1}^{1}}_{k-1} f_1(0, x_2, \dots, x_k) dx_2 \dots dx_k.\end{aligned} \tag{16.4.3}$$

On the other hand,

$$\begin{aligned}\int\limits_M d\omega = \int\limits_{I_-} d(\psi^*\omega) &= \underbrace{\int \cdots \int}_{k-1} \left(\int\limits_0^1 \frac{\partial f}{\partial x_1}(x_1, \dots, x_k) dx_1 \right) dx_2 \dots dx_k \\ &+ \sum_{j=2}^{k} \underbrace{\int \cdots \int}_{k-2} \left(\int\limits_{-1}^{0} \left(\int\limits_{-1}^{1} \frac{\partial f}{\partial x_j}(x_1, \dots, x_k) dx_j \right) dx_1 \right) dx_2 \dots \overset{j}{\vee} \dots dx_k \\ &= \underbrace{\int\limits_{-1}^{1} \cdots \int\limits_{-1}^{1}}_{k-1} f_1(0, x_2, \dots, x_k) dx_2 \dots dx_k.\end{aligned} \tag{16.4.4}$$

Comparing (16.4.3) and (16.4.3) we get Stokes' formula $\int\limits_M d\omega = \int\limits_{\partial M} \omega$. ■

Chapter 17

Volume of submanifolds in $\mathbb{R}^n$ and integral formulas of vector analysis

17.1 Integration of functions over submanifolds in a Euclidean space

In order to integrate functions over a submanifold we need a notion of volume form of a submanifold in a Euclidean space V.

Let $A \subset V$ be an oriented k-dimensional submanifold a Euclidean space V, $0 \leq k \leq n$. By definition, the *volume form* $\sigma = \sigma_A$ of A (or the *area form* if $k = 2$, or the *length form* if $k = 1$) is a differential k-form on A whose value on any k tangent vectors $v_1, \dots, v_k \in T_x A$ equals the *oriented* volume of the parallelepiped generated by these vectors.

Given a function $f : A \to \mathbb{R}$ we define its integral over A by the formula

$$\int_A f dV = \int_A f \sigma_A \,, \tag{17.1.1}$$

and, in particular,

$$\operatorname{Vol} A = \int_A \sigma_A \,.$$

Notice that the integral $\int_A f dV$ *is independent of the orientation of* A. Indeed, changing the orientation we also change the sign of the form σ_A, and hence the integral remains unchanged. This allows us to define the integral $\int\limits_A f dV$ even for a non-orientable A. Indeed, we can cover A by coordinate charts, find a subordinated partition of unity and split correspondingly the function $f = \sum\limits_1^N f_j$ in such a way that each function f_j is supported in a coordinate neighborhood (which is, therefore, orientable). By orienting in arbitrary ways each of the coordinate neighborhoods we can compute each of the integrals $\int\limits_A f_j dV$, $j = 1, \dots, N$. It is straightforward to see that the integral $\int\limits_A f dV = \sum\limits_j \int\limits_A f_j dV$ is independent of the choice of the partition of unity.

Let us study in some examples how the form σ_A can be effectively computed.

Example 17.1. *Volume form of a hypersurface.* Let us fix a Cartesian coordiantes in V. Let $A \subset V$ is given by the equation

$$A = \{F = 0\}$$

for some function $F : V \to \mathbb{R}$ which has no critical points on A. The vector field ∇F is orthogonal to A, and

$$\mathbf{n} = \frac{\nabla F}{||\nabla F||}$$

is the unit normal vector field to A. Assuming A to be co-oriented by $\mathbf{n}$ we can write down the volume form of A as the contraction of $\mathbf{n}$ with the volume form $\Omega = dx_1 \wedge \dots \wedge dx_n$ of $\mathbb{R}^n$, i.e.

$$\sigma_A = \mathbf{n} \lrcorner \Omega = \frac{1}{||\nabla F||} \sum_1^n (-1)^{i-1} \frac{\partial F}{\partial x_i} dx_1 \wedge \overset{i}{\dots\vee\dots} \wedge dx_n \,.$$

In particular, if $n = 3$ we get the following formula for the area form of an implicitely given 2-dimensional surface $A = \{F = 0\} \subset \mathbb{R}^3$:

$$\sigma_A = \frac{1}{\sqrt{\left(\frac{\partial F}{\partial x_1}\right)^2 + \left(\frac{\partial F}{\partial x_2}\right)^2 + \left(\frac{\partial F}{\partial x_3}\right)^2}} \left(\frac{\partial F}{\partial x_1} dx_2 \wedge dx_3 + \frac{\partial F}{\partial x_2} dx_3 \wedge dx_1 + \frac{\partial F}{\partial x_3} dx_1 \wedge dx_2 \right) . \quad (17.1.2)$$

Example 17.2. *Length form of a curve.*

Let $\Gamma \subset \mathbb{R}^n$ be an oriented curve given parametrically by a map $\gamma : [a, b] \to \mathbb{R}^n$. Let $\sigma = \sigma_\Gamma$ be the length form. Let us compute the form $\gamma^* \sigma_\Gamma$. Denoting the coordinate in $[a, b]$ by t and the unit vector field on $[a, b]$ by e we have

$$\gamma^* \sigma_\Gamma = f(t) dt,$$

where

$$f(t) = \gamma^* \sigma_\Gamma(e) = \sigma_\Gamma\left(\gamma'(t)\right) = ||\gamma'(t)||\,.$$

In particular the length of Γ is equal to

$$\int\limits_\Gamma \sigma_\Gamma = \int\limits_a^b ||\gamma'(t)||dt = \int\limits_a^b \sqrt{\sum_{i=1}^n (x_i'(t))^2}\, dt\,,$$

where

$$\gamma(t) = (x_1(t), \dots, x_n(t))\,.$$

Similarly, given any function $f : \Gamma \to \mathbb{R}$ we have

$$\int\limits_\Gamma f ds = \int\limits_a^b f\left(\gamma(t)\right) ||\gamma'(t)||dt\,.$$

Example 17.3. *Area form of a surface given parametrically.*

Suppose a surface $S \subset \mathbb{R}^n$ is given parametrically by a map $\Phi : U \to \mathbb{R}^n$ where U in the plane $\mathbb{R}^2$ with coordinates (u, v).

Let us compute the pull-back form $\Phi^*\sigma_S$. In other words, we want to express σ_S in coordinates u, v. We have

$$\Phi^*\sigma_S = f(u, v) du \wedge dv\,.$$

To determine $f(u, v)$ take a point $z = (u, v) \in \mathbb{R}^2$ and the standard basis $e_1, e_2 \in \mathbb{R}^2_z$. Then

$$(\Phi^*\sigma_S)_z(e_1, e_2) = f(u, v) du \wedge dv(e_1, e_2)\,. \tag{17.1.3}$$

On the other hand, by the definition of the pull-back form we have

$$(\Phi^*\sigma_S)_z(e_1, e_2) = (\sigma_S)_{\Phi(z)}(d_z\Phi(e_1), d_z\Phi(e_2))\,. \tag{17.1.4}$$

But $d_z\Phi(e_1) = \frac{\partial\Phi}{\partial u}(z) = \Phi_u(z)$ and $d_z\Phi(e_2) = \frac{\partial\Phi}{\partial v}(z) = \Phi_v(z)$. Hence from (17.1.3) and (17.1.4) we get

$$f(u, v) = \sigma_S(\Phi_u, \Phi_v)\,. \tag{17.1.5}$$

The value of the form σ_S on the vectors Φ_u, Φ_v is equal to the area of the parallelogram generated by these vectors, because the surface is assumed to be oriented by these vectors, and hence

$\sigma_S(\Phi_u, \Phi_v) > 0$. Denoting the angle between Φ_u and Φ_v by α we get[1] $\sigma_S(\Phi_u, \Phi_v) = ||\Phi_u||\,||\Phi_v||\sin\alpha$. Hence

$$\sigma_S(\Phi_u, \Phi_v)^2 = ||\Phi_u||^2\,||\Phi_v||^2\sin^2\alpha = ||\Phi_u||^2\,||\Phi_v||^2(1-\cos^2\alpha) = ||\Phi_u||^2\,||\Phi_v||^2 - (\Phi_u\cdot\Phi_v)^2\,,$$

and therefore,

$$f(u,v) = \sigma_S(\Phi_u, \Phi_v) = \sqrt{||\Phi_u||^2\,||\Phi_v||^2 - (\Phi_u\cdot\Phi_v)^2}\,.$$

It is traditional to introduce the notation

$$E = ||\Phi_u||^2,\ F = \Phi_u\cdot\Phi_v,\ G = ||\Phi_v||^2\,,$$

so that we get

$$\Phi^*\sigma_S = \sqrt{EG-F^2}du\wedge dv\,,$$

and hence we get for any function $f : S \to \mathbb{R}$

$$\int\limits_S f dS = \int\limits_S f\sigma_S = \int\limits_U f(\Phi(u,v))\sqrt{EG-F^2}du\wedge dv = \iint\limits_U f(\Phi(u,v))\sqrt{EG-F^2}dudv\,. \quad (17.1.6)$$

Consider a special case when the surface S defined as a graph of a function ϕ over a domain $D \subset \mathbb{R}^2$. Namely, suppose

$$S = \{z = \phi(x,y),\ \ (x,y)\in D\subset\mathbb{R}^2\}\,.$$

The surface S as parametrized by the map

$$(x,y)\overset{\Phi}{\mapsto}(x,y,\phi(x,y)).$$

Then

$$E = ||\Phi_x||^2 = 1+\phi_x^2,\ G = ||\Phi_y||^2 = 1+\phi_y^2,\ F = \Phi_x\cdot\Phi_y = \phi_x\phi_y\,,$$

and hence

$$EG - F^2 = (1+\phi_x^2)(1+\phi_y^2) - \phi_x^2\phi_y^2 = 1+\phi_x^2+\phi_y^2\,.$$

Therefore, the formula (17.1.6) takes the form

$$\int\limits_S f dS = \iint\limits_D f(\Phi(x,y))\sqrt{EG-F^2}dx\wedge dy =$$
$$\iint\limits_D f\left(x,y,\phi(x,y)\right)\sqrt{1+\phi_x^2+\phi_y^2}\,dxdy\,. \quad (17.1.7)$$

[1] See a computation in a more general case below in Example 17.4.

Note that the formula (17.1.7) can be also deduced from (17.1.2). Indeed, the surface

$$S = \{z = \phi(x,y), \ \ (x,y) \in D \subset \mathbb{R}^2\},$$

can also be defined implicitly by the equation

$$F(x,y,z) = z - \phi(x,y) = 0, \ \ (x,y) \in D\,.$$

We have

$$\nabla F = (-\frac{\partial \phi}{\partial x}, -\frac{\partial \phi}{\partial y}, 1),$$

and, therefore,

$$\begin{aligned}
\int\limits_S f dS &= \int\limits_S \frac{f(x,y,z)}{||\nabla F||}\left(\frac{\partial F}{\partial x} dy \wedge dz + \frac{\partial F}{\partial y} dz \wedge dx + \frac{\partial F}{\partial z} dx \wedge dy\right) \\
&= \int\!\!\int\limits_D \frac{f(x,y,\phi(x,y))}{\sqrt{1 + \left(\frac{\partial \phi}{\partial x}\right)^2 + \left(\frac{\partial \phi}{\partial y}\right)^2}} \left(-\frac{\partial \phi}{\partial x} dy \wedge d\phi - \frac{\partial \phi}{\partial y} d\phi \wedge dx + dx \wedge dy\right) \\
&= \int\!\!\int\limits_D \frac{f(x,y,\phi(x,y))}{\sqrt{1 + \left(\frac{\partial \phi}{\partial x}\right)^2 + \left(\frac{\partial \phi}{\partial y}\right)^2}} \left(1 + \left(\frac{\partial \phi}{\partial x}\right)^2 + \left(\frac{\partial \phi}{\partial y}\right)^2\right) dxdy \\
&= \int\!\!\int\limits_D f(x,y,\phi(x,y)) \sqrt{1 + \left(\frac{\partial \phi}{\partial x}\right)^2 + \left(\frac{\partial \phi}{\partial y}\right)^2}\, dxdy\,.
\end{aligned}$$

Example 17.4. *Integration over a parametrically given k-dimensional submanifold.*

Consider now a more general case of a parametrically given k-submanifold A in an n-dimensional Euclidean space V. We fix a Cartersian coordinate system in V and thus identify V with $\mathbb{R}^n$ with the standard dot-product.

Let $U \subset \mathbb{R}^k$ be a compact domain with boundary and $\phi : U \to \mathbb{R}^n$ be an embedding. We assume that the submanifold with boundary $A = \phi(U)$ is oriented by this parameterization. Let σ_A be the volume form of A. We will find an explicit expression for $\phi^*\sigma_A$. Namely. denoting coordinates in $\mathbb{R}^k$ by $(u_1, \dots, u_k)$ we have $\phi^*\sigma_A = f(u) du_1 \wedge \dots \wedge du_k$, and our goal is to compute the function f.

By definition, we have

$$\begin{aligned}
f(u) =& \phi^*(\sigma_A)_u(e_1, \dots, e_k) = (\sigma_A)_u(d_u\phi(e_1), \dots, d_u\phi(e_k)) = \\
& (\sigma_A)_u\left(\frac{\partial \phi}{\partial u_1}(u), \dots, \frac{\partial \phi}{\partial u_k}(u)\right) = \mathrm{Vol}_k\left(P\left(\frac{\partial \phi}{\partial u_1}(u), \dots, \frac{\partial \phi}{\partial u_k}(u)\right)\right) \qquad (17.1.8)
\end{aligned}$$

In Section 10.2 we proved two formulas for the volume of a parallelepiped. Using formula (10.2.1) we get

$$\mathrm{Vol}_k\left(P\left(\frac{\partial\phi}{\partial u_1}(u),\dots,\frac{\partial\phi}{\partial u_k}(u)\right)\right)=\sqrt{\sum_{1\le i_1<\dots<i_k\le n} Z^2_{i_1\dots i_k}}. \tag{17.1.9}$$

where

$$Z_{i_1\dots i_k}=\begin{vmatrix}\frac{\partial\phi_{i_1}}{\partial u_1}(u) & \dots & \frac{\partial\phi_{i_1}}{\partial u_k}(u)\\ \dots & \dots & \dots\\ \frac{\partial\phi_{i_k}}{\partial u_1}(u) & \dots & \frac{\partial\phi_{i_k}}{\partial u_k}(u)\end{vmatrix}. \tag{17.1.10}$$

Thus

$$\int\limits_A f dV=\int\limits_A f\sigma_A=\int\limits_U f(\phi(u))\sqrt{\sum_{1\le i_1<\dots<i_k\le n} Z^2_{i_1\dots i_k}}du_1\wedge\dots\wedge du_k.$$

Rewriting this formula for $k=2$ we get

$$\int\limits_A f dV=\int\limits_U f(\phi(u))\sqrt{\sum_{1\le i<j\le n}\begin{vmatrix}\frac{\partial\phi_i}{\partial u_1} & \frac{\partial\phi_i}{\partial u_2}\\ \frac{\partial\phi_j}{\partial u_1} & \frac{\partial\phi_j}{\partial u_2}\end{vmatrix}^2}du_1\wedge du_2. \tag{17.1.11}$$

Alternatively we can use formula (9.4.2). Then we get

$$\mathrm{Vol}_k\left(P\left(\frac{\partial\phi}{\partial u_1}(u),\dots,\frac{\partial\phi}{\partial u_k}(u)\right)\right)=\sqrt{\det\left((D\phi)^T D\phi\right)}, \tag{17.1.12}$$

where

$$D\phi=\begin{pmatrix}\frac{\partial\phi_1}{\partial u_1} & \dots & \frac{\partial\phi_1}{\partial u_k}\\ \dots & \dots & \dots\\ \frac{\partial\phi_n}{\partial u_1} & \dots & \frac{\partial\phi_n}{\partial u_k}\end{pmatrix}$$

is the Jacobi matrix of ϕ.[2] Thus using this expression we get

$$\int\limits_A f dV=\int\limits_A f\sigma_A=\int\limits_U f(\phi(u))\sqrt{\det\left((D\phi)^T D\phi\right)}\,dV. \tag{17.1.13}$$

Exercise 17.5. *In case $n=3$, $k=2$ show explicitely equivalence of formulas* (17.1.6), (17.1.11) *and* (17.1.13).

[2]The symmetric matrix $(D\phi)^T D\phi$ is the Gram matrix of vectors $\frac{\partial\phi}{\partial u_1}(u),\dots,\frac{\partial\phi}{\partial u_k}(u)$, and its entrees are pairwise scalar products of these vectors..

Exercise 17.6. *Integration over an implicitly defined k-dimensional submanfold.* Suppose that $A = \{F_1 = \cdots = F_{n-k} = 0\}$ and the differentials of defining functions are linearly independent at points of A. Show that

$$\sigma_A = \frac{*(dF_1 \wedge \cdots \wedge dF_{n-k})}{||\, dF_1 \wedge \cdots \wedge dF_{n-k}\, ||}.$$

Example 17.7. Let us compute the volume of the unit 3-sphere $S^3 = \{x_1^2 + x_2^2 + x_3^2 + x_4^2 = 1\}$.

By definition, $\mathrm{Vol}(S^3) = \int\limits_{S^3} \mathbf{n} \lrcorner \Omega$, where $\Omega = dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4$, and $\mathbf{n}$ is the outward unit normal vector to the unit ball B_4. Here S^3 should be co-oriented by the vector field $\mathbf{n}$. Then using Stokes' theorem we have

$$\int\limits_{S^3} \mathbf{n} \lrcorner \Omega = \int\limits_{S^3} (x_1 dx_2 \wedge dx_3 \wedge dx_4 - x_2 dx_1 \wedge dx_3 \wedge dx_4 + x_3 dx_1 \wedge dx_2 \wedge dx_4 - x_4 dx_1 \wedge dx_2 \wedge dx_3) =$$
$$4 \int\limits_{B^4} dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4 = 4 \int\limits_{B^4} dV. \tag{17.1.14}$$

Introducing polar coordinates (r, ϕ) and (ρ, θ) in the coordinate planes (x_1, x_2) and (x_3, x_4) and using Fubini's theorem we get

$$\int\limits_{B^4} = \int\limits_0^{2\pi} \int\limits_0^{2\pi} \int\limits_0^{1} \int\limits_0^{\sqrt{1-r^2}} r\rho d\rho dr d\theta d\phi =$$
$$2\pi^2 \int\limits_0^1 (r - r^3) dr = \frac{\pi^2}{2}. \tag{17.1.15}$$

Hence, $\mathrm{Vol}(S^3) = 2\pi^2$.

Exercise 17.8. Find the ratio $\frac{\mathrm{Vol}_n(B_R^n)}{\mathrm{Vol}_{n-1}(S_R^{n-1})}$.

17.2 Work and Flux

We introduce in this section two fundamental notions of vector analysis: a *work of a vector field along a curve*, and a *flux of a vector field through a surface.*

Let Γ be an oriented smooth curve in a Euclidean space V and T the unit tangent vector field to Γ. Let $\mathbf{v}$ be another vector field, defined along Γ. The function $\langle \mathbf{v}, \mathbf{T} \rangle$ equals the projection of the vector field $\mathbf{v}$ to the tangent directions to the curve. If the vector field $\mathbf{v}$ is viewed as a force field, then the integral $\int\limits_{\Gamma}\langle \mathbf{v}, \mathbf{T}\rangle ds$ has the meaning of a *work* $\mathrm{Work}_{\Gamma}(\mathbf{v})$ performed by the field $\mathbf{v}$ to transport a particle of mass 1 along the curve Γ in the direction determined by the orientation. If the curve Γ is closed then this integral is sometimes called the *circulation* of the vector field $\mathbf{v}$ along Γ and denoted by $\oint\limits_{\Gamma}\langle \mathbf{v}, \mathbf{T}\rangle ds$. As we already indicated earlier, the sign $\oint$ in this case has precisely the same meaning as $\int$, and it is used only to stress the point that we are integrating along a *closed* curve.

Consider now a co-oriented hypersurface $\Sigma \subset V$ and denote by $\mathbf{n}$ the unit normal vector field to Σ which determines the given co-orientation of Σ. Given a vector field $\mathbf{v}$ along Σ we will view it as the velocity vector field of a flow of a fluid in the space. Then we can interpret the integral

$$\int\limits_{\Sigma}\langle \mathbf{v}, \mathbf{n}\rangle dV$$

as the *flux* $\mathrm{Flux}_{\Sigma}(\mathbf{v})$ of $\mathbf{v}$ through Σ, i.e. the volume of fluid passing through Σ in the direction of $\mathbf{n}$ in time 1.

Lemma 17.9. *1. For any co-oriented hypersurface Σ and a vector field $\mathbf{v}$ given in its neighborhood we have*

$$\langle \mathbf{v}, \mathbf{n}\rangle \sigma_{\Sigma} = (\mathbf{v} \lrcorner \Omega)_{\Sigma},$$

where Ω is the volume form in V.

2. For any oriented curve Γ and a vector field $\mathbf{v}$ near Γ we have

$$\langle \mathbf{v}, \mathbf{T}\rangle \sigma_{\Gamma} = \mathcal{D}(\mathbf{v})|_{\Gamma}.$$

Proof. 1. For any $n-1$ vectors $T_1, \dots, T_{n-1} \in T_x\Sigma$ we have

$$\mathbf{v} \lrcorner \Omega(T_1, \dots, T_{n-1}) = \Omega(\mathbf{v}, T_1, \dots, T_{n-1}) = \mathrm{Vol} P(\mathbf{v}, T_1, \dots, T_{n-1}).$$

Using (9.3.1) we get

$$\begin{aligned}&\mathrm{Vol} P(\mathbf{v}, T_1, \dots, T_{n-1}) = \langle \mathbf{v}, \mathbf{n}\rangle \mathrm{Vol}_{n-1} P(T_1, \dots, T_{n-1}) = \\ &\langle \mathbf{v}, \mathbf{n}\rangle \mathrm{Vol} P(\mathbf{n}, T_1, \dots, T_{n-1}) = \langle \mathbf{v}, \mathbf{n}\rangle \sigma_{\Sigma}(T_1, \dots, T_{n-1}).\end{aligned} \tag{17.2.1}$$

2. The tangent space $T_x\Gamma$ is generated by the vector $\mathbf{T}$, and hence we just need to check that $\langle \mathbf{v}, \mathbf{T}\rangle \sigma_\Gamma(\mathbf{T}) = \mathcal{D}(\mathbf{v})(\mathbf{T})$. But $\sigma_\Gamma(\mathbf{T}) = \mathrm{Vol}(\mathbf{n}, \mathbf{T}) = 1$, and hence

$$\mathcal{D}(\mathbf{v})(\mathbf{T}) = \langle \mathbf{v}, \mathbf{T}\rangle = \langle \mathbf{v}, \mathbf{T}\rangle \sigma_\Gamma(\mathbf{T}).$$

■

Note that if we are given a Cartesian coordinate system in V and $\mathbf{v} = \sum\limits_1^n a_j \frac{\partial}{\partial x_j}$, then

$$\mathbf{v} \lrcorner \Omega = \sum_1^{n-1} (-1)^{i-1} a_i dx_1 \overset{i}{\ldots\vee\ldots} dx_n, \quad \mathcal{D}(\mathbf{v}) = \sum_1^n a_i dx_i\,.$$

Thus, we have

Corollary 17.10.

$$\mathrm{Flux}_\Sigma(\mathbf{v}) = \int\limits_\Sigma \langle \mathbf{v}, \mathbf{n}\rangle dV = \int\limits_\Sigma (\mathbf{v} \lrcorner \Omega) = \int\limits_\Sigma \sum_1^n (-1)^{i-1} a_i dx_1 \overset{i}{\ldots\vee\ldots} dx_n;$$

$$\mathrm{Work}_\Gamma(\mathbf{v}) = \int\limits_\Gamma \langle \mathbf{v}, \mathbf{T}\rangle = \int\limits_\Gamma \mathcal{D}(\mathbf{v}) = \int\limits_\Gamma \sum_1^n a_i dx_i.$$

In particular if $n = 3$ we have

$$\mathrm{Flux}_\Sigma(\mathbf{v}) = \int\limits_\Sigma a_1 dx_2 \wedge dx_3 + a_2 dx_3 \wedge dx_1 + a_3 dx_1 \wedge dx_2.$$

Let us also recall that in a Euclidean space V we have $v \lrcorner \Omega = *\mathcal{D}(v)$. Hence, the equation $\omega = v \lrcorner \Omega$ is equivalent to the equation

$$v = \mathcal{D}^1\left(*^{-1}\omega\right) = (-1)^{n-1}\mathcal{D}^{-1}(*\omega).$$

In particular, when $n = 3$ we get $\mathbf{v} = \mathcal{D}^{-1}(*\omega)$. Thus we get

Corollary 17.11. *For any differential $(n-1)$-form ω and an oriented compact hypersurface Σ we have*

$$\int\limits_\Sigma \omega = \mathrm{Flux}_\Sigma \mathbf{v},$$

where $\mathbf{v} = (-1)^{n-1}\mathcal{D}^{-1}(*\omega)$.

Integration of functions along curves and surfaces can be interpreted as the work and the flux of appropriate vector fields. Indeed, suppose we need to compute an integral $\int\limits_\Gamma f ds$. Consider the tangent vector field $\mathbf{v}(x) = f(x)\mathbf{T}(x),\ \ x \in \Gamma$, along Γ. Then $\langle\, \mathbf{v}, \mathbf{T}\rangle = f$ and hence the integral $\int_\Gamma f ds$ can be interpreted as the work $\mathrm{Work}_\Gamma(\,\mathbf{v})$. Therefore, we have

$$\int\limits_\Gamma f ds = \mathrm{Work}_\Gamma(\,\mathbf{v}) = \int\limits_\Gamma \mathcal{D}(\,\mathbf{v})$$

.

Note that we can also express $\mathbf{v}$ through ω by the formula $\mathbf{v} = \mathcal{D}^{-1} \star \omega$, see Section 13.7.

Similarly, to compute an integral $\int\limits_\Sigma f dS$ let us co-orient the surface Σ with a unit normal to Σ vector field $\mathbf{n}(x),\ x \in \Sigma$ and set $\mathbf{v}(x) = f(x)\,\mathbf{n}(x)$. Then $\langle\, \mathbf{v},\, \mathbf{n}\rangle = f$, and hence

$$\int\limits_\Gamma f dS = \int\limits_\Gamma \langle\, \mathbf{v},\, \mathbf{n}\rangle dS = \mathrm{Flux}_\Sigma(\,\mathbf{v}) = \int\limits_\Gamma \omega,$$

where $\omega = \ \mathbf{v} \lrcorner \Omega$.

17.3 Integral formulas of vector analysis

We interpret in this section Stokes' formula in terms of integrals of functions and operations on vector fields. Let us consider again differential forms, which one can associate with a vector field $\mathbf{v}$ in an Euclidean 3-space. Namely, this is a differential 1-form $\alpha = \mathcal{D}(\,\mathbf{v})$ and a differential 2-form $\omega = \ \mathbf{v} \lrcorner \Omega$, where $\Omega = dx \wedge dy \wedge dz$ is the volume form.

Using Corollary 17.10 we can reformulate Stokes' theorem for domains in a $\mathbb{R}^3$ as follows.

Theorem 17.12. *Let $\mathbf{v}$ be a smooth vector field in a domain $U \subset \mathbb{R}^3$ with a smooth (or piece-wise) smooth boundary Σ. Suppose that Σ is co-oriented by an outward normal vector field. Then we have*

$$\mathrm{Flux}_\Sigma\, \mathbf{v} = \int\int\limits_U\int \mathrm{div}\, \mathbf{v} dxdydz\,.$$

Indeed, $\mathrm{div}\, \mathbf{v} = *d\omega$. Hence we have

$$\int\limits_U \mathrm{div}\, \mathbf{v} dV = \int\limits_U (\star d\omega) dx \wedge dy \wedge dz = \int\limits_U d\omega = \int\limits_\Sigma \omega = \int\limits_\Sigma \mathbf{v} \lrcorner \Omega = \mathrm{Flux}_\Sigma\, \mathbf{v}\,.$$

This theorem clarifies the meaning of div $\mathbf{v}$:

Let $B_r(x)$ be the ball of radius r centered at a point $x \in \mathbb{R}^3$, and $S_r(x) = \partial B_r(x)$ be its boundary sphere co-oriented by the outward normal vector field. Then

$$\operatorname{div} \mathbf{v}(x) = \lim_{r \to 0} \frac{\mathrm{Flux}_{S_r(x)} \mathbf{v}}{\mathrm{Vol}(B_r(x))} .$$

Theorem 17.13. *Let Σ be a piece-wise smooth compact oriented surface in $\mathbb{R}^3$ with a piece-wise smooth boundary $\Gamma = \partial\Sigma$ oriented respectively. Let $\mathbf{v}$ be a smooth vector field defined near Σ. Then*

$$\mathrm{Flux}_\Sigma(\operatorname{curl} \mathbf{v}) = \int\limits_\Sigma \langle \operatorname{curl} \mathbf{v},\, \mathbf{n} \rangle dV = \oint\limits_\Gamma \mathbf{v} \cdot \mathbf{T} ds = \mathrm{Work}_\Gamma \mathbf{v} .$$

To prove the theorem we again use Stokes' theorem and the connection between integrals of functions and differential forms. Set $\alpha = \mathcal{D}(\mathbf{v})$. Then $\operatorname{curl} \mathbf{v} = \mathcal{D}^{-1} \star d\alpha$. We have

$$\oint\limits_\Gamma \mathbf{v} \cdot \mathbf{T} ds = \int\limits_\Gamma \alpha = \int\limits_\Sigma d\alpha = \mathrm{Flux}_\Sigma(\mathcal{D}^{-1} \star (d\alpha)) = \mathrm{Flux}_\Sigma(\operatorname{curl} \mathbf{v}) .$$

Again, similar to the previous case, this theorem clarifies the meaning of curl. Indeed, let us denote by $D_r(x,\, \mathbf{w})$ the 2-dimensional disc of radius r in $\mathbb{R}^3$ centered at a point $x \in \mathbb{R}^3$ and orthogonal to a unit vector $\mathbf{w} \in \mathbb{R}^3_x$. Set

$$c(x,\, \mathbf{w}) = \lim_{r \to 0} \frac{\mathrm{Work}_{\partial D_r(x,\, \mathbf{w})} \mathbf{v}}{\pi r^2}.$$

Then

$$c(x,\, \mathbf{w}) = \lim_{r \to 0} \frac{\mathrm{Flux}_{D_r(x,\, \mathbf{w})}(\operatorname{curl} \mathbf{v})}{\pi r^2} = \lim_{r \to 0} \frac{\int\limits_{D_r(x,\, \mathbf{w})} \langle \operatorname{curl} \mathbf{v},\, \mathbf{w} \rangle}{\pi r^2} = \langle \operatorname{curl} \mathbf{v},\, \mathbf{w} \rangle.$$

Hence, $||\operatorname{curl} \mathbf{v}(x)|| = \max\limits_{\mathbf{w}} c(x,\, \mathbf{w})$ and direction of $\operatorname{curl} \mathbf{v}(x)$ coincides with the direction of the vector $\mathbf{w}$ for which the maximum value of $c(x,\, \mathbf{w})$ is achieved.

17.4 From integral measurements to PDE

Integral formulas of vector calculus is a tool to formulate experimental or integral measurements in Physics as partial differential equations. As an example, we consider here the *Maxwell equations* in the theory of electro-magnetism. There are the following fundamental quantities in this theory:

- Electric field $\mathbf{E}$;
- Magnetic field $\mathbf{B}$;
- Current density $\mathbf{J}$;
- Charge density ρ;

$\mathbf{E}$, $\mathbf{B}$ and $\mathbf{J}$ are vector fields [3] while ρ is a (scalar) function.

There are four fundamental physical laws which are observes experimental facts, and hence have a macroscopic or integral character. We use below so-called *Gaussian* measurement units which absorbs some constants. We also use the time unit making the speed of light c equal to 1.

Consider a domain $\Omega \subset \mathbb{R}^3$ with the boundary $\Sigma = \partial\Omega$ and a surface $S \subset \mathbb{R}^3$ with the boundary $\Gamma = \partial S$.

a) *Conservation of charge or continuity equation.*

$$\mathrm{Flux}_\Sigma \mathbf{J} = -\frac{d}{dt}\int\limits_\Omega \rho dV,$$

i.e. the rate of change of the total charge of Ω is equal to the negative of the flux of $\mathbf{J}$ through its boundary Σ.

b) *Gauss law for the electric field.*

$$\mathrm{Flux}_\Sigma \mathbf{E} = 4\pi\int\limits_\Omega \rho dV,$$

i.e. the flux of the electric field through the boundary of a domain is equal to the total electric charge.

c) *Gauss law for the magnetic field.*

$$\mathrm{Flux}_\Sigma \mathbf{B} = 0,$$

which is usually interpreted that *there are no magnetic charges.*

d) *Faraday's law of induction,*

$$\underset{\Gamma}{\mathrm{Work}}\,\mathbf{E} = -\frac{d}{dt}\mathrm{Flux}_S \mathbf{B},$$

[3] $\mathbf{B}$ is sometimes called a *pseudo-vector* field, because in reality it should be considered as a differential two-form η, so that $\mathbf{B} = \star\eta$, and as we know the Hodge star operator *depends on the choice of an orientation of the space.*

i.e. the circulation of an electric field along a boundary of a surface is equal to the negative of the rate of change of the flux of the magnetic field through the surface.

e) *Ampère-Maxwell law.*

$$\mathrm{Work}_\Gamma \mathbf{B} = \mathrm{Flux}_S \left(4\pi\mathbf{J} + \frac{d}{dt}\mathbf{E}\right).$$

The second term in the right-hand side is Maxwell's correction to the law of Ampère in the case when $\mathbf{E}$ may depend on time.

Summarizing, we get

$$\begin{aligned}
&\iint\limits_{\Sigma} \langle \mathbf{J}, \mathbf{n}\rangle dS = -\iiint\limits_{\Omega} \frac{d\rho}{dt} dV;\\
&\iint\limits_{\Sigma} \langle \mathbf{E}, \mathbf{n}\rangle dS = 4\pi \iiint\limits_{\Omega} \rho dV;\\
&\iint\limits_{\Sigma} \langle \mathbf{B}, \mathbf{n}\rangle dS = 0;\\
&\int\limits_{\Gamma} \langle \mathbf{E}, \tau\rangle d\ell = -\iint\limits_{S} \langle \frac{d\mathbf{B}}{dt}, \mathbf{n}\rangle dS;\\
&\int\limits_{\Gamma} \langle \mathbf{B}, \tau\rangle d\ell = \iint\limits_{S} \langle 4\pi\mathbf{J} + \frac{d\mathbf{E}}{dt}, \mathbf{n}\rangle dS.
\end{aligned} \tag{17.4.1}$$

Here we denoted by $\mathbf{n}$ the unit normal vector field to surfaces Σ and S, and by τ the unit tangent vector field to Γ, These are the Maxwell equations in the macroscopic, or integral form. Sometimes the first equation (continuity) is not included in the list of the Maxwell equations, but it must be added to get a self-contained system of equations relating $\mathbf{E}, \mathbf{B}, J$ and ρ.

On the other hand, Stokes' theorem (or rather its reinterpretation in the language of vector calculus) yields:

$$\begin{aligned}
&\iint\limits_{\Sigma} \langle \mathbf{J}, \mathbf{n}\rangle dS = \iiint\limits_{\Omega} \mathrm{div}\mathbf{J} dV;\\
&\iint\limits_{\Sigma} \langle \mathbf{E}, \mathbf{n}\rangle dS = \iiint\limits_{\Omega} \mathrm{div}\mathbf{E} dV;\\
&\iint\limits_{\Sigma} \langle \mathbf{B}, \mathbf{n}\rangle dS = \iiint\limits_{\Omega} \mathrm{div}\mathbf{B} dV\\
&\int\limits_{\Gamma} \langle \mathbf{E}, \tau\rangle d\ell = \iint\limits_{S} \langle \mathrm{curl}\mathbf{E}, \mathbf{n}\rangle dS;\\
&\int\limits_{\Gamma} \langle \mathbf{B}, \tau\rangle d\ell = \iint\limits_{S} \langle \mathrm{curl}\mathbf{B}, \mathbf{n}\rangle dS.
\end{aligned} \tag{17.4.2}$$

To deduce an infinitesimal version of the Maxwell equations (17.4.1) we will need the following simple lemma.

Lemma 17.14. *1. Let f, g be two continuous bounded functions on a domain $U \subset \mathbb{R}^n$. Suppose that for any domain $\Omega \subset U$ we have $\int\limits_{\Omega} f dV = \int\limits_{\Omega} g dV$. Then $f = g$ on U.*

2. Let X, Y be two continuous vector fields on a domain $U \subset \mathbb{R}^n$. Suppose that for any compact cooriented submanifold $\Sigma \subset U$ with boundary we have $\mathrm{Flux}_\Sigma X = \mathrm{Flux}_\Sigma Y$. Then $X = Y$ everywhere in U.

For the proof of the first statement we test the condition on balls $B_\epsilon(a), a \in U$ and pass to the limit $\epsilon \to 0$ in the equality

$$\frac{\int\limits_{B_\epsilon(a)} f dV}{\mathrm{Vol}_n B_\epsilon(a)} = \frac{\int\limits_{B_\epsilon(a)} g dV}{\mathrm{Vol}_n B_\epsilon(a)}.$$

Similarly, for the second statement we test the condition on $(n-1)$-dimensional discs $D_\epsilon(a, \mathbf{n})$ of radius ϵ orthogonal to a unit vector $\mathbf{n} \in T_a U$, $a \in U$. and similarly, pass to the limit $\epsilon \to 0$ in the equality

$$\frac{\int\limits_{D_\epsilon(a,\mathbf{n})} \langle X, \mathbf{n}\rangle dV_{n-1}}{\mathrm{Vol}_{n-1} D_\epsilon(a, \mathbf{n})} = \frac{\int\limits_{D_\epsilon(a,\mathbf{n})} \langle Y, \mathbf{n}\rangle dV_{n-1}}{\mathrm{Vol}_{n-1} D_\epsilon(a, \mathbf{n})}.$$

Using Lemma 17.14 and comparing right-hand sides of (17.4.1) and (17.4.2) we get

Summarizing, we get

$$
\begin{aligned}
&\mathrm{div}\mathbf{J} = -\frac{d\rho}{dt}\\
&\mathrm{div}\mathbf{E} = 4\pi\rho;\\
&\mathrm{div}\mathbf{B} = 0;\\
&\mathrm{curl}\mathbf{E} = -\frac{d\mathbf{B}}{dt};\\
&\mathrm{curl}\mathbf{B} = 4\pi\mathbf{J} + \frac{d\mathbf{E}}{dt}.
\end{aligned}
\tag{17.4.3}
$$

These are the Maxwell equations written in the equivalent form as a system of PDEs.

17.5 Expressing div and curl in curvilinear coordinates

Let us show how to compute $\mathrm{div}\,\mathbf{v}$ and $\mathrm{curl}\,\mathbf{v}$ of a vector field $\mathbf{v}$ in $\mathbb{R}^3$ given in a curvilinear coordinates u_1, u_2, u_3, i.e. expressed through the coordinate vector fields $\frac{\partial}{\partial u_1}, \frac{\partial}{\partial u_2}$ and $\frac{\partial}{\partial u_3}$. Let

$$\Omega = f(u_1, u_2, u_3) du_1 \wedge du_2 \wedge du_3$$

be the volume form $dx_1 \wedge dx_2 \wedge dx_3$ expressed in coordinates u_1, u_2, u_3.

Let us first compute $\star(du_1 \wedge du_2 \wedge du_3)$. We have

$$\star(du_1 \wedge du_2 \wedge du_3) = \star\left(\frac{1}{f} dx_1 \wedge dx_2 \wedge dx_3\right) = \frac{1}{f}.$$

Let

$$\mathbf{v} = a_1 \frac{\partial}{\partial u_1} + a_2 \frac{\partial}{\partial u_2} + a_3 \frac{\partial}{\partial u_3}.$$

Then we have

$$
\begin{aligned}
\mathrm{div}\,\mathbf{v} &= \star d(\mathbf{v} \lrcorner \Omega)\\
&= \star d\left(\left(\sum_1^3 a_i \frac{\partial}{\partial u_i}\right) \lrcorner f du_1 \wedge du_2 \wedge du_3\right)\\
&= \star d\left(f a_1 du_2 \wedge du_3 + f a_2 du_3 \wedge du_1 + f a_3 du_1 \wedge du_2\right)\\
&= \star\left(\left(\frac{\partial(f a_1)}{\partial u_1} + \frac{\partial(f a_2)}{\partial u_2} + \frac{\partial(f a_3)}{\partial u_3}\right) du_1 \wedge du_2 \wedge du_3\right)\\
&= \frac{1}{f}\left(\frac{\partial(f a_1)}{\partial u_1} + \frac{\partial(f a_2)}{\partial u_2} + \frac{\partial(f a_3)}{\partial u_3}\right)\\
&= \frac{\partial a_1}{\partial u_1} + \frac{\partial a_2}{\partial u_2} + \frac{\partial a_3}{\partial u_3} + \frac{1}{f}\left(\frac{\partial f}{\partial u_1} a_1 + \frac{\partial f}{\partial u_2} a_2 + \frac{\partial f}{\partial u_3} a_3\right).
\end{aligned}
$$

In particular, we see that the divergence of a vector field is expressed by the same formulas as in the Cartesian case *if and only if* the volume form is proportional to the form $du_1 \wedge du_2 \wedge du_3$ with a *constant* coefficient.

For instance, in the spherical coordinates the volume form can be written as

$$\Omega = r^2 \sin\varphi dr \wedge d\varphi \wedge d\theta \,,$$[4]

and hence the divergence of a vector field

$$\mathbf{v} = a\frac{\partial}{\partial r} + b\frac{\partial}{\partial \theta} + c\frac{\partial}{\partial \varphi}$$

can be computed by the formula

$$\operatorname{div} \mathbf{v} = \frac{\partial a}{\partial r} + \frac{\partial b}{\partial \theta} + \frac{\partial c}{\partial \varphi} + \frac{2a}{r} + c\cot\varphi \,.$$

The general formula for $\operatorname{curl}\mathbf{v}$ in curvilinear coordinates looks pretty complicated. So instead of deriving the formula we will explain here how it can be obtained in the general case, and then illustrate this procedure for the spherical coordinates.

By the definition we have

$$\operatorname{curl}\mathbf{v} = D^{-1}\left(\star\left(d\left(D(\mathbf{v})\right)\right)\right) \,.$$

Hence we first need to compute $D(\mathbf{v})$.

To do this we need to introduce a symmetric matrix

$$G = \begin{pmatrix} g_{11} & g_{12} & g_{13} \\ g_{21} & g_{22} & g_{23} \\ g_{31} & g_{32} & g_{33} \end{pmatrix} ,$$

where

$$g_{ij} = \langle \frac{\partial}{\partial u_i}, \frac{\partial}{\partial u_j} \rangle, \quad i,j = 1,2,3 \,.$$

The matrix G is called the *Gram matrix*.

[4] Note that the spherical coordinates ordered as (r, ϕ, θ) determine the same orientation of $\mathbb{R}^3$ as the Cartesian coordinates (x, y, z).

Notice, that if $D(\mathbf{v}) = A_1 du_1 + B du_2 + C du_3$ then for any vector $h = h_1 \frac{\partial}{\partial u_1} + h_2 \frac{\partial}{\partial u_2} + h_3 \frac{\partial}{\partial u_3}$. we have

$$D(\mathbf{v})(h) = A_1 h_1 + A_2 h_2 + A_3 h_3 = \begin{pmatrix} A_1 & A_2 & A_3 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \\ h_3 \end{pmatrix} = \langle \mathbf{v}, h \rangle .$$

But

$$\langle \mathbf{v}, h \rangle = \begin{pmatrix} a_1 & a_2 & a_3 \end{pmatrix} G \begin{pmatrix} h_1 \\ h_2 \\ h_3 \end{pmatrix} .$$

Hence

$$\begin{pmatrix} A_1 & A_2 & A_3 \end{pmatrix} = \begin{pmatrix} a_1 & a_2 & a_3 \end{pmatrix} G ,$$

or, equivalently, because the Gram matrix G is symmetric we can write

$$\begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix} = G \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} ,$$

and therefore,

$$A_i = \sum_{i=1}^{n} g_{ij} a_j , \quad i = 1, 2, 3 .$$

After computing

$$\omega = d(D(\mathbf{v})) = B_1 du_2 \wedge du_3 + B_2 du_3 \wedge du_1 + B_3 du_1 \wedge du_2$$

we compute $\operatorname{curl} \mathbf{v}$ by the formula $= \operatorname{curl} \mathbf{v} = D^{-1} (\star \omega)$. Let us recall (see Proposition 10.15 above) that for any vector field $\mathbf{w}$ the equality $D\,\mathbf{w} = \star\omega$ is equivalent to the equality $\mathbf{w} \lrcorner \Omega = \omega$, where $\Omega = f du_1 \wedge du_2 \wedge du_3$ is the volume form. Hence, if

$$\operatorname{curl} \mathbf{v} = c_1 \frac{\partial}{\partial u_1} + c_2 \frac{\partial}{\partial u_2} + c_3 \frac{\partial}{\partial u_3}$$

then we have

$$\mathbf{w} \lrcorner \Omega = f c_1 du_2 \wedge du_3 + f c_2 du_3 \wedge du_1 + f c_3 du_1 \wedge du_2 ,$$

and therefore,

$$\operatorname{curl} \mathbf{v} = \frac{B_1}{f} \frac{\partial}{\partial u_1} + \frac{B_2}{f} \frac{\partial}{\partial u_2} + \frac{B_3}{f} \frac{\partial}{\partial u_3} .$$

Let us use the above procedure to compute $\operatorname{curl}\mathbf{v}$ of the vector field

$$\mathbf{v} = a\frac{\partial}{\partial r} + b\frac{\partial}{\partial \phi} + c\frac{\partial}{\partial \theta}$$

given in the spherical coordinates. The Gram matrix in this case is the diagonal matrix

$$G = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2\sin^2\varphi \end{pmatrix}.$$

Hence,

$$D\,\mathbf{v} = adr + br^2 d\varphi + cr^2\sin^2\varphi d\theta\,,$$

and

$$\begin{aligned}
\omega = d(D\,\mathbf{v}) &= da\wedge dr + d(br^2)\wedge d\varphi + d(cr^2\sin^2\varphi)\wedge d\theta \\
&= \left(-r\frac{\partial b}{\partial \theta} + r^2 c\sin 2\varphi + r^2\sin^2\varphi\frac{\partial c}{\partial \varphi}\right) d\varphi\wedge d\theta \\
&+ \left(-\frac{\partial a}{\partial \varphi} + r^2\frac{\partial b}{\partial r} + 2br\right) dr\wedge d\varphi + \left(-2rc\sin^2\varphi - r^2\sin^2\varphi\frac{\partial c}{\partial r} + \frac{\partial a}{\partial \theta}\right) d\theta\wedge dr\,.
\end{aligned}$$

Finally, we get the following expression for $\operatorname{curl}\mathbf{v}$:

$$\operatorname{curl}\mathbf{v} = \frac{-r\frac{\partial b}{\partial \theta} + r^2 c\sin 2\varphi + r^2\sin^2\varphi\frac{\partial c}{\partial \varphi}}{r^2\cos\phi}\frac{\partial}{\partial r} + \frac{-\frac{\partial a}{\partial \varphi} + r^2\frac{\partial b}{\partial r} + 2br}{r^2\cos\varphi}\frac{\partial}{\partial \varphi} + \frac{-2rc\sin^2\varphi - r^2\sin^2\varphi\frac{\partial c}{\partial r} + \frac{\partial a}{\partial \theta}}{r^2\cos\varphi}\frac{\partial}{\partial \theta}\,.$$

Chapter 18

Phase flow of a vector field and Lie derivative

18.1 Action of a diffeomorphism on a vector field

Let us denote by $\mathrm{Vect}(U)$ and $\mathrm{Vect}(V)$ the spaces of vector fields on domains U and V of the same dimension. Given a diffeomorphism $f : U \to V$ one can define the *push-forward map* $f_* : \mathrm{Vect}(U) \to \mathrm{Vect}(V)$ as follows. Let $X \in \mathrm{Vect}(U)$ be a vector field on U. Define the vector field $Y = f_* X$ by the formula

$$Y(v) = d_x f(X(u)), \text{ where } u = f^{-1}(v).$$

Let us point out that unlike the pull-back operator f^* on differential forms which defined for *any smooth map* and not, necessarily for a diffeomorphism, the push-forward operator f_* on vector fields is defined *only for diffeomorphisms* (why?).

Let $(u_1, \dots, u_n)$ be the Cartesian coordinates in $U \subset \mathbb{R}^n$ and $(x_1, \dots, x_n)$ Cartesian coordinates in $V \subset \mathbb{R}^n$. Then a diffeomorphism $f : U \to V$, $U \subset \mathbb{R}^n$ can be viewed as a parameterization map introducing in V new curvilinear coordinates $(u_1, \dots, u_n)$. Then the vector fields $f_*\left(\frac{\partial}{\partial u_j}\right)$ are vector fields tangent to the corresponding coordinate lines in V, which are also usually denoted by $\frac{\partial}{\partial u_j}$.

For instance, consider the map $P(r, \phi) = (r\cos\phi, r\sin\phi)$, $r > 0, \phi \in (-\pi, \pi)$, introducing polar coordinates in $\mathbb{R}^2 \setminus \{x_2 = 0, x_1 \geq 0\}$. By definition, $P_*\left(\frac{\partial}{\partial r}\right)$ and $P_*\left(\frac{\partial}{\partial r}\right)$ directly using the definition.

The Jacobi matrix $J(P)$ is equal to

$$J(P) = \begin{pmatrix} \cos\phi & -r\sin\phi \\ \sin\phi & r\cos\phi \end{pmatrix}.$$

Hence,

$$P_*\left(\frac{\partial}{\partial r}\right) = J(P)\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \cos\phi \\ \sin\phi \end{pmatrix} = \begin{pmatrix} \frac{x}{\sqrt{x^2+y^2}} \\ \frac{y}{\sqrt{x^2+y^2}} \end{pmatrix} \quad \text{and}$$

$$P_*\left(\frac{\partial}{\partial \phi}\right) = J(P)\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -r\sin\phi \\ r\cos\phi \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}.$$

But the computed vector fields are exactly what we call $\frac{\partial}{\partial r}$ and $\frac{\partial}{\partial \phi}$ in polar coordinates.

We can similarly define the push-forward operator on *line* fields. If X is a vector field and $\lambda = \mathrm{Span}(X)$ the line field which it generates then $f_*\lambda = \mathrm{Span}(f_*v)$.

Exercise 18.1. Suppose $n = 2$ and a line field λ on U is defined by a Pfaffian equation $\alpha = 0$, where α is a 1-form on U. Show that given a diffeomorphism $f : U \to V$ the line field $f_*\lambda$ on V can be defined by a Pfaffian equation $\beta = 0$, where

$$\beta := \left(f^{-1}\right)^* \alpha = (f^*)^{-1}\alpha.$$

18.2 Isotopy and diffeotopy

Let us denote by $\Delta \subset \mathbb{R}$ an interval in $\mathbb{R}$. This interval can be closed, open, semi-open, and even concide with the whole $\mathbb{R}$ or rays (a, ∞) and $(-\infty, a)$.

Let us recall that a homotopy $f_t : U \to V$, $t \in \Delta$, is just a continuous family of continuous maps $U \to V$, which depends continuously on the parameter Δ. Equivalently, one can think of a homotopy as a continuous map $F : U \times \Delta \to V$. The relation to the first definition is given by the formula

$$F(x,t) = f_t(x), \quad \text{for } x \in U, \ \ t \in \Delta.$$

We will also need two special cases of a homotopy, called *isotopy* and *diffeotopy*.

A homotopy $f_t : U \to V$, $t \in \Delta$, is called a *diffeotopy* if $f_t : U \to V$ is a diffeomorphism for each $t \in U$. A homotopy $f_t : U \to V$, $t \in \Delta$, is called an *isotopy* if for each $t \in U$ the map $f_t : U \to V$

is an *embedding*, i.e. a diffeomorphism onto its image $f_t(U)$. Thus, an embedding need not to be *onto*, and the image $f_t(U)$ can move during an isotopy. Of course, a diffeotopy is a special case of an isotopy.

As in the case of a homotopy, an isotopy $f_t : U \to V, t \in \Delta$, can also be viewed as a continuous map $F : U \times \Delta \to V$ defined by the formula $F(x,t) = f_t(x)$, $x \in U, t \in \Delta$. The restriction of F to the slices, $F|_{U\times t}$ are required to be diffeomorphisms on their images.

It what follows we will always assume that the map F is *smooth.* In Section 19.1 below it will be explained how this property could be achieved.

18.3 Phase flow

The following theorem is a foundational fact of the theory of ordinary differential equations.

Theorem 18.2 (E. Picard)**.** *Let X be a C^1-smooth vector field in a domain $U \subset \mathbb{R}^n$. Then for any point $x_0 \in U$ there exists $\epsilon > 0$, a neighborhood $\Omega \ni x_0, \Omega \subset U$, and a uniquely defined isotopy $f_t : \Omega \to U$, $t \in (-\epsilon, \epsilon)$ such that*

$$f_0(x) = x \quad \textit{for all}\, x \in \Omega; \tag{18.3.1}$$

$$\frac{df_t(x)}{dt} = X(f_t(x)). \tag{18.3.2}$$

Note that for each $x \in U$ the path $t \in (-\epsilon, \epsilon) \mapsto u = f_t(x)$ is the solution of the system of differential equations

$$\dot{u} = X(u), \tag{18.3.3}$$

where we denoted $\dot{u} := \frac{du}{dt}$. Hence, Theorem 18.2 implies, in particular, existence of a solution of a system (18.3.3) for any initial data $u(t_0) = x_0$ on an interval $(t_0 - \epsilon, t_0 + \epsilon)$, provided that the vector field X is C^1-smooth. It also implies the uniqueness of solution with given initial data and its smooth dependence on the initial data.

The isotopy f_t is called the *local phase flow* of the vector field X. Sometimes the isotopy f_t extends to a diffeotopy $f_t : U \to U$ which is defined for all $t \in \mathbb{R}$. In that case we call the diffeotopy f_t the *global phase flow* of X.

Exercise 18.3. 1. Suppose that in Theorem 18.2 there exists the same $\epsilon > 0$ which works *for all* points $x_0 \in U$. Prove that the vector field X in that case admits a global phase flow.

2. Prove that if $f_t : U \to U$ is a global phase flow for a vector field X then $f_{t_1+t_2} = f_{t_1} \circ f_{t_2}$ for any $t_1, t_2 \in \mathbb{R}$.

3. Suppose X integrates to a global phase flow $f_t : U \to U$. Show that for any $t \in \mathbb{R}$ we have $(f_t)_* X = X$.

The next theorem shows that two *non-vanishing* smooth vector fields are *locally diffeomorphic.* More precisely,

Theorem 18.4. *Let X be a C^1-smooth vector field in a domain $U \subset \mathbb{R}^n$. Suppose that $X(a) \neq 0$ for some point $a \in U$. Then there exists a local coordinate system $(y_1, \dots, y_n)$ on a neighborhood $U \ni a$, $\Omega \subset U$, centered at the point a such that the vector field X on U is equal to $\frac{\partial}{\partial y_1}$.*

In particular,

Corollary 18.5. *Let λ be a C^1-smooth line field in a domain $U \subset \mathbb{R}^n$. Then for any point $a \in U$ there exists a neighborhood $\Omega \ni a, \Omega \subset U$ and a local coordinate system $(y_1, \dots, y_n)$ in Ω, centered at the point a such that the line field Y on Ω is spanned by the vector field $\frac{\partial}{\partial y_1}$, or, equivalently, λ is given by a system of Pfaffian equations $dy_2 = \dots = dy_n = 0$.*

Proof of Theorem 18.4. We can assume without loss of generality that a is the origin of the Cartesian coordinate system, and the vector $X(a)$ coincides with the vector $\frac{\partial}{\partial x_1}$ at the point a. This could be achieved by rotating and scaling the original Cartesian system of coordinates. Denote $D_\delta^{n-1} := \{x_1 = 0; \sum\limits_2^n x_j^2 \leq \delta^2\}$. Suppose that ϵ is chosen so small that $D_\delta^{n-1} \subset \Omega$, where Ω is the neighborhood provided by Theorem 18.2. Let $f_t : \Omega \to U, t \in (-\epsilon, \epsilon)$ be the local phase flow constructed in Theorem 18.2. Denote

$$H := \{|x_1| \leq \epsilon, \sum_2^n x_j^2 \leq \delta^2\}$$

and define a map $F : H \to \Omega$ given by the formula $F(x_1, x_2, \dots, x_n) = f_{x_1}(0, x_2, \dots, x_n))$.

The map F is an embedding, provided that ϵ, δ are small enough. Indeed, the differential of F at the origin is the identity map (why?), and hence by the implicit function theorem it is an embedding in a sufficiently small neighborhood of 0. But $F_*(\frac{\partial}{\partial x_1}) = X$, and hence, assuming that ϵ, δ are small enough, the coordinate system introduced on the neighborhood $U' = F(H)$ by the diffeomorphism $F^{-1} : U' \to H$ is the required one. ∎

It is customary to denote the phase flow of a vector field X (local or global) by X^t. Below we will usually use this notation. Equation (18.3.3) equivalently can be rewritten as

$$X^s(x) = x + sX(x) + o(s) \tag{18.3.4}$$

Examples and exercises.

1. Consider a vector field $X = x\frac{\partial}{\partial x}$. It corresponds to the differential equations $\dot{x} = x$. Its solution with an initial data $x(0) = x_0$ is $x(t) = e^s x_0$. Hence it admits a global phase flow $X^t(x) = e^t x$, which consists of scaling with the coefficient e^t.
2. Show that the vector field $X = x^2\frac{\partial}{\partial x}$ on $\mathbb{R}$ does not admit a global flow, and compute the maximal time for which exists a solution with the initial data $x(0) = x_0$.
3. Consider the vector field $Z = -y\frac{\partial}{\partial x} + x\frac{\partial}{\partial y}$ on $\mathbb{R}^2$. Show that it admits a global flow consisting of rotations of $\mathbb{R}^2$.

Proposition 18.6. *Suppose that a vector field X on U integrates to a flow $X^t : U \to U$, $t \in \mathbb{R}$, and $f : U \to \widetilde{U}$ a diffeomorphism. Denote $\widetilde{X} := f_* X$. Then the vector field $\widetilde{X}$ integrates to a flow $\widetilde{X}^t$, $t \in \mathbb{R}$, on $\widetilde{U}$ and*

$$\widetilde{X}^t = f \circ X^t \circ f^{-1},\ t \in \mathbb{R}.$$

Proof. For any point $y = f(x) \in \widetilde{U}$ we have

$$\begin{aligned}
\frac{d}{dt}(\widetilde{X}^t(y))\big|_{t=0} &= \frac{d}{dt}(f \circ X^t \circ f^{-1}(y))\big|_{t=0} \\
&= \frac{d}{dt}(f \circ X^t(x))\big|_{t=0} = d_x f(\frac{d}{dt}(X^t(x)\big|_{t=0}) = d_x f(X(x)) \\
&= f_* X(y) = \widetilde{X}(y).
\end{aligned}$$

∎

18.4 Directional derivative revisited

Let X be a smooth vector field defined on a domain $U \subset \mathbb{R}^n$ (more generally we can assume that U is any n-dimensional manifold). Recall that given a differentiable function $f : U \to \mathbb{R}$ its *directional derivative* $L_X f = df(X)$ is defined by the formula:

$$L_X f = \lim_{s \to 0} \frac{f(x + sX) - f(x)}{s}. \tag{18.4.1}$$

Assume that X is C^1-smooth. Then there is defined its local phase flow $X^t : U' \to U$, $t \in (-\epsilon, \epsilon)$, where $U' \subset U$ is a neighborhood of a point $a \in U$, and the directional derivative can be also defined by the formula

$$L_X f(a) = \frac{d}{ds} f \circ X^s \Big|_{s=0}(a). \tag{18.4.2}$$

Indeed, in view of (18.3.4) we have

$$\begin{aligned} &\frac{d}{ds} f \circ X^s \Big|_{s=0}(a) = \lim_{s \to 0} \frac{f(X^s(a)) - f(a)}{s} \\ &= \lim_{s \to 0} \frac{f(a + sX(a) + o(s)) - f(a)}{s} = \lim_{s \to 0} \frac{s d_a f(X(a)) + o(s)}{s} = d_a f(X(a)) = L_X f(a). \end{aligned}$$

It turns out that formula (18.4.2) can be generalized to define an analog of directional derivatives for differential forms and vector fields, which is the *Lie derivative*.

18.5 Lie derivative of a differential form

Let ω be a differential k-form. We define the Lie derivative $L_X \omega$ of ω along a vector field X as

$$L_X \omega = \frac{d}{ds} (X^s)^* \omega \Big|_{s=0}. \tag{18.5.1}$$

Note that if ω is a 0-form, i.e. a function f, then $(X^s)^* f = f \circ X^s$, and hence, in this case definitions (18.4.2) and (18.5.1) coincide, i.e. for functions the Lie derivative is the same as the directional derivative.

Proposition 18.7. *The following identities hold*

1. $L_X(\omega_1 \wedge \omega_2) = (L_X \omega_1) \wedge \omega_2 + \omega_1 \wedge L_X \omega_2$.

2. $L_X(d\omega) = d(L_X \omega)$.

Proof.

1. $$L_X(\omega_1 \wedge \omega_2) = \frac{d}{ds}(X^s)^*(\omega_1 \wedge \omega_2)\Big|_{s=0} = \frac{d}{ds}\big((X^s)^*\omega_1 \wedge (X^s)^*\omega_2\big)\Big|_{s=0}$$
$$= \frac{d}{ds}\big((X^s)^*\omega_1\big)\Big|_{s=0} \wedge \omega_2 + \omega_1 \wedge \frac{d}{ds}\big((X^s)^*\omega_2\big)\Big|_{s=0} = (L_X\omega_1) \wedge \omega_2 + \omega_1 \wedge L_X\omega_2.$$

2. $$L_X(d\omega) = \frac{d}{ds}\big((X^s)^*d\omega\big)\Big|_{s=0} = \frac{d}{ds}\big(d(X^s)^*\omega\big)\Big|_{s=0} = d\Big(\frac{d}{ds}(X^s)^*\omega\Big|_{s=0}\Big) = d(L_X\omega).$$

■

The following formula of Élie Cartan provides an effective way for computing the Lie derivative of a differential form.

Theorem 18.8. *Let X be a vector field and ω a differential k-form. Then*

$$L_X\omega = d(X \lrcorner\, \omega) + X \lrcorner\, d\omega. \tag{18.5.2}$$

Proof. Suppose first that $\omega = f$ is a 0-form. Then $L_X f = df(X) = X \lrcorner\, df$, which is equivalent to formula (18.5.2), because in this case the first term in the formula is equal to 0. Then, using Proposition 18.72) we get

$$L_X df = dL_X f = d(df(X)) = d(X \lrcorner\, df),$$

which is again equivalent to (18.5.2) because in this case $ddf = 0$. Next we note that if the formula (18.4.1) holds for ω_1 and ω_2 then it holds also for $\omega_1 \wedge \omega_2$. Indeed, we have

$$(\star)\quad L_X(\omega_1 \wedge \omega_2) = (L_X\omega_1) \wedge \omega_2 + \omega_1 \wedge L_X\omega_2$$
$$= (X \lrcorner\, d\omega_1 + d(X \lrcorner\, \omega_1)) \wedge \omega_2 + \omega_1 \wedge (X \lrcorner\, d\omega_2 + d(X \lrcorner\, \omega_2))$$
$$= (X \lrcorner\, d\omega_1) \wedge \omega_2 + \omega_1 \wedge (X \lrcorner\, d\omega_2) + d(X \lrcorner\, \omega_1) \wedge \omega_2 + \omega_1 \wedge d(X \lrcorner\, \omega_2)$$

On the other hand, denoting by d_1 and d_2 the degrees of ω_1 and ω_2, we get

$$(\star\star)\quad X \lrcorner\, d(\omega_1 \wedge \omega_2) + d(X \lrcorner\, (\omega_1 \wedge \omega_2))$$
$$= X \lrcorner\, (d\omega_1 \wedge \omega_2 + (-1)^{d_1}\omega_1 \wedge d\omega_2) + d((X \lrcorner\, \omega_1) \wedge \omega_2 + (-1)^{d_1}\omega_1 \wedge (X \lrcorner\, \omega_2))$$
$$= (X \lrcorner\, d\omega_1) \wedge \omega_2 + (-1)^{d_1+1}d\omega_1 \wedge (X \lrcorner\, \omega_2) + (-1)^{d_1}(X \lrcorner\, \omega_1) \wedge d\omega_2 + \omega_1 \wedge (X \lrcorner\, d\omega_2)$$
$$+ d(X \lrcorner\, \omega_1) \wedge \omega_2 + (-1)^{d_1+1}X \lrcorner\, \omega_1 \wedge d\omega_2 + (-1)^{d_1}d\omega_1 \wedge (X \lrcorner\, \omega_2) + \omega_1 \wedge (d(X \lrcorner\, \omega_2))$$
$$= (X \lrcorner\, d\omega_1) \wedge \omega_2 + \omega_1 \wedge (X \lrcorner\, d\omega_2) + d(X \lrcorner\, \omega_1) \wedge \omega_2 + \omega_1 \wedge d(X \lrcorner\, \omega_2)\,.$$

Comparing the computation in ($\star$) and ($\star\star$) we conclude that

$$L_X(\omega_1 \wedge \omega_2) = X \lrcorner\, d(\omega_1 \wedge \omega_2) + d(X \lrcorner\, (\omega_1 \wedge \omega_2)).$$

By induction we can prove a similar formulas for an exterior product of any number of forms.

Finally we observe that any differential k-form ω can be written in coordinates as

$$\sum_{1 \le i_1 < \cdots < i_k \le n} f_{i_1 \ldots i_k}(x) dx_{i_1} \wedge \cdots \wedge dx_{i_k},$$

i.e. ω is a sum of exterior products of functions (0-forms) and exact 1-forms, and hence Cartan's formula follows. ∎

Proposition 18.9. *We have*

$$L_X \omega = 0 \iff (X^s)^* \omega = \omega \text{ for all } s \in \mathbb{R}.$$

Proof. If $(X^s)^* \omega \equiv \omega$ then $L_X\omega = \frac{d}{ds}(X^s)^*\omega\Big|_{s=0} = 0$. To prove the converse we note that

$$\begin{aligned}\frac{d}{ds}(X^s)^*\omega\Big|_{s=s_0} &= \lim_{t \to 0} \frac{(X^{s_0+t})^*\omega - (X^{s_0})^*\omega}{t} = (X^{s_0})^* \left(\lim_{t \to 0} \frac{(X^t)^*\omega - \omega}{t} \right) \\ &= (X^{s_0})^* (L_X \omega),\end{aligned}$$

and hence if $L_X\omega = 0$ then $(X^s)^* \omega = \omega$. ∎

18.6 Divergence free vector fields

Let X be a C^1-smooth vector field on a domain $U \subset \mathbb{R}^n$. Let η be the volume form on $\mathbb{R}^n$.

Proposition 18.10.

$$\text{div}X = 0 \iff L_X \eta = 0.$$

Proof. According to Cartan's formula $L_X\eta = d(X \lrcorner\, \eta) = (\text{div}X)\eta$, and the claim follows. ∎

Note that compactness of Ω guarantees that there exists $\epsilon > 0$ such that there is defined the phase flow $X^t : \Omega \to U$ for $|t| < \epsilon$ (why?).

This gives us the following interpretation of $\text{Flux}_{\partial\Omega} X$:

Exercise 18.11. Show that

$$\text{Flux}_{\partial\Omega} X = \frac{d}{ds}\left(\text{Vol}(X^s(\Omega))\right)\Big|_{s=0} = \lim_{t\to 0}\frac{\text{Vol}(X^s(\Omega) - \text{Vol}(\Omega)}{s}.$$

The equation $L_X\eta = d(X \lrcorner \eta) = (\text{div}X)\eta$ yields another proof of the divergence theorem. Let Ω be a compact n-dimensional manifold with boundary. According to Exercise 18.11 we have

$$\text{Flux}_{\partial\Omega} X = \frac{d}{ds}\left(\text{Vol}(X^s(\Omega))\right)\Big|_{s=0} = \frac{d}{ds}\left(\int\limits_{X^s(\Omega)} \eta\right)\Big|_{s=0}$$

$$= \frac{d}{ds}\left(\int\limits_{\Omega}(X^s)^*\eta\right)\Big|_{s=0} = \int\limits_{\Omega}\frac{d}{ds}\left((X^s)^*\eta\right)\Big|_{s=0} = \int\limits_{\Omega} L_X\eta$$

$$= \int\limits_{\Omega} d(X \lrcorner \eta) = \int\limits_{\Omega}\text{div}X.$$

18.7 Lie bracket of two vector fields

18.7.1 Commutator of two vector fields

In this section we assume that all objects we consider are C^∞-smooth (or at least as smooth as necessary to justify our considerations below). Any vector field X on a domain $U \subset \mathbb{R}^n$ can be viewed as a first order differential operator on the space of smooth functions on U, i.e. if $X = \sum\limits_1^n f_j \frac{\partial}{\partial x_j}$ then for a function $\phi : U \to \mathbb{R}$ we have

$$L_X(\phi)(x) = d_x\phi(X(x)) = \sum_1^n f_j \frac{\partial \phi}{\partial x_j}(x).$$

Given, two vector fields X and Y on U we can compute their *commutator* $L_X(L_Y\phi) - L_Y(L_X\phi)$. Though at first glance this looks like the second order operator, but as the next lemma shows, this is again a first order operator, which therefore could be identified with a vector field.

Lemma 18.12. *For any two vector fields X, Y there exists a unique vector field Z such that for any smooth function ϕ we hhave*

$$L_X(L_Y\phi) - L_Y(L_X\phi) = L_Z\phi,$$

i.e. $L_Z = L_XL_Y - L_YL_X$ is the commutator of of the operators L_X and L_Y.

Proof. Let $X = \sum\limits_1^n f_j \frac{\partial}{\partial x_j}$ and $Y = \sum\limits_1^n g_i \frac{\partial}{\partial x_i}$. Then

$$
\begin{aligned}
L_X(L_Y\phi) &= \sum_{j=1}^n f_j \frac{\partial}{\partial x_j}\left(\sum_{i=1}^n g_i \frac{\partial \phi}{\partial x_i}\right) \\
&= \sum_{i,j=1}^n f_j g_i \frac{\partial^2 \phi}{\partial x_j \partial x_i} + \sum_{i,j=1}^n f_j \frac{\partial g_i}{\partial x_j}\frac{\partial \phi}{\partial x_i} \\
&= \sum_{i,j=1}^n f_j g_i \frac{\partial^2 \phi}{\partial x_j \partial x_i} + \sum_{i=1}^n \left(\sum_{j=1}^n f_j \frac{\partial g_i}{\partial x_j}\right)\frac{\partial \phi}{\partial x_i}, \quad \text{and} \\
L_Y(L_X\phi) &= \sum_{i,j=1}^n f_j g_i \frac{\partial^2 \phi}{\partial x_i \partial x_j} + \sum_{j=1}^n \left(\sum_{i=1}^n f_i \frac{\partial g_j}{\partial x_i}\right)\frac{\partial \phi}{\partial x_j}.
\end{aligned}
$$

Hence,

$$
L_X(L_Y\phi) - L_Y(L_X\phi) = \sum_{i=1}^n \left(\sum_{j=1}^n g_j \frac{\partial f_i}{\partial x_j}\right)\frac{\partial \phi}{\partial x_i} - \sum_{j=1}^n \left(\sum_{i=1}^n f_i \frac{\partial g_j}{\partial x_i}\right)\frac{\partial \phi}{\partial x_j}
$$
$$
= \sum_{j=1}^n \left(\sum_{i=1}^n f_i \frac{\partial g_j}{\partial x_i} - g_i \frac{\partial f_j}{\partial x_i}\right)\frac{\partial \phi}{\partial x_j}
$$

Hence, if we denote $Z := \sum h_j \frac{\partial}{\partial x_j}$, where $h_j := \sum\limits_{i=1}^n \left(g_i \frac{\partial f_j}{\partial x_i} - f_i \frac{\partial g_j}{\partial x_i}\right)$ then we get

$$
L_Z\phi = \sum_{j=1}^n \left(\sum_{i=1}^n f_i \frac{\partial g_j}{\partial x_i} - g_i \frac{\partial f_j}{\partial x_i}\right)\frac{\partial \phi}{\partial x_j} = L_X(L_Y\phi) - L_Y(L_X\phi). \tag{18.7.1}
$$

To verify the uniqueness of Z we note that if for two vector fields Z, Z' we have $L_Z\phi = L_{Z'}\phi$ for any function ϕ then $Z = Z'$. Indeed, if $Z = \sum h_j \frac{\partial}{\partial x_j}$. Then $L_Z x_j = h_j$ for any $j = 1, \dots, n$. ■The vector field Z from Lemma 18.12 is called the *Lie bracket* of the vector fields X and Y and is denoted by $[X, Y]$. It is clear from the definition that $[X, Y] = -[Y, X]$. Other properties of the Lie brackets will be discussed later in this section. But meanwhile we will relate the notions of Lie bracket a Lie derivative.

18.7.2 Lie derivative of a vector field

We define the *Lie derivative* $L_X Y$ of the vector field Y along the vector field X in a similar way as we defined in Section 18.5 the Lie derivative of a differential form. Namely,

$$
L_X Y = \frac{d\,(X^s)^* Y}{ds}\Big|_{s=0}. \tag{18.7.2}
$$

More explicitly,

$$L_X Y(x) = \lim_{s \to 0} \frac{d_{X^s(x)} \left(X^{-s}\right) \left(Y(X^s(x)) - Y(x)\right.}{s}.$$

Similarly, to Proposition 18.9 we have

Proposition 18.13.

$$L_X Y = 0 \iff (X^s)^* Y \equiv Y \text{ for all } s \in \mathbb{R} \text{ where flow is defined.}$$

Proof. We have

$$\begin{aligned}
&\frac{d\,(X^s)^* Y}{ds}\Big|_{s=s_0} = \lim_{s \to 0} \frac{(X^{s+s_0})^* Y - (X^{s_0})^* Y}{s} \\
&= \lim_{s \to 0} (X^{s_0})^* \left(\frac{(X^s)^* Y - Y}{s} \right) = (X^{s_0})^* \left(\lim_{s \to 0} \frac{(X^s)^* Y - Y}{s} \right) \\
&= (X^{s_0})^* (L_X Y).
\end{aligned}$$

Hence, if $L_X Y = 0$ then $\frac{d(X^s)^* Y}{ds}$ for all s and hence $(X^s)^* Y = \left(X^0\right)^* Y = Y$. The converse is obvious. ■

Theorem 18.14. *For any two vector fields $X, Y \in \mathrm{Vect}(U)$*

$$L_X Y = [X, Y].$$

Proof. Note that $X^s(x) = x + sX(x) + o(s)$. Hence, we can write

$$d_y X^{-s} = \mathrm{Id} - s d_y X + o(s),$$

where we view here X as a map $\mathbb{R}^n \to \mathbb{R}^n$. Furthermore, plugging $y = X^s(x)$ we get

$$d_{X^s(x)} X^{-s} = \mathrm{Id} - s d_x X + o(s).$$

Indeed, as X is C^1-smooth, we have $d_{X^s(x)} X - d_x X \underset{s \to 0}{\to} 0$ and hence $s(d_y X - d_x X) = o(s)$. We also have

$$Y(X^s(x)) = Y(x + sX(x) + o(x)) = Y(x) + s d_x Y(X(x)) + o(s).$$

Thus, ignoring $o(s)$-terms we get

$$
\begin{aligned}
L_X Y &= \lim_{s\to 0} \frac{1}{s}\left(d_{X^s(x)}\left(X^{-s}\right)(Y(X^s(x))) - Y(x)\right) \\
&= \lim_{s\to 0} \frac{1}{s}\left((\mathrm{Id} - s d_x X)\right)(Y(x) + s d_x Y(X(x))) - Y(x)) \\
&= \lim_{s\to 0} \frac{1}{s}\left(Y(x) - s d_x X(Y) + s d_x Y(X) - Y(x)\right) = d_x Y(X) - d_x X(Y).
\end{aligned}
$$

The right-hand-side expression written in coordinates has the form

$$
d_x Y(X) - d_x X(Y) = \sum_{i=1}^{n} \left(\sum_{j=1}^{n} f_j \frac{\partial g_i}{\partial x_j} - g_j \frac{\partial f_i}{\partial x_j} \right) \frac{\partial}{\partial x_i}
$$

which coincides with the expression (18.7.1) for the Lie bracket. ■

Corollary 18.15. $L_X Y = -L_Y X$.

Indeed, $L_X Y = [X, Y] = -[Y, X] = -L_Y X$.

Exercise 18.16. Prove that for any smooth function ϕ one has

$$
L_{[X,Y]}\phi = \frac{\partial^2(\phi \circ X^s \circ Y^t)}{\partial s \partial t}.
$$

If $[X, Y] = 0$ then one says that the vector field X and Y *commute.*

Lemma 18.17. *Suppose two commuting vector fields X, Y on Ω can be integrated into phase flows X^t, Y^s. Then*

$$
X^t \circ Y^s = Y^s \circ X^t,
$$

$t, s \in \mathbb{R}$, i.e. the flows of commuting vector fields. Conversely, if two flows X^t, Y^s commute for all $t, s \in \mathbb{R}$ then $[X, Y] = 0$.

Proof. We have $[X, Y] = L_X Y$. Then according to Proposition 18.13 we have

$$
(X^s)^* Y = Y. \tag{18.7.3}
$$

Recall from Proposition 18.6 that for any diffeomorphism $f : \Omega \to \Omega$ if $f^* Y = Z$ then

$$
Z^t = f^{-1} \circ Y^t \circ f, \quad t \in \mathbb{R}.
$$

Applying this to $f = X^s$ and using (18.7.3) we conclude

$$Y^t = X^{-s} \circ Y^t \circ X^s,$$

or

$$X^s \circ Y^t = Y^t \circ X^s, \quad s, t \in \mathbb{R}.$$

$\blacksquare$

18.7.3 Jacobi identity

Let us now return to the discussion of properties of the Lie bracket operation.

Proposition 18.18. *1. (Bilinearity.)* $[aX_1 + bX_2, Y] = a[X_1, Y] + b[X_2, Y]$ *for any real* a, b.

2. (Skew-symmetricity.) $[X, Y] = -[Y, X]$*;*

3. (Jacobi identity.) $[[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y]$ *for any 3 vector fields* X, Y, Z*;*

4. (Leibnitz rule.) $[X, fY] = f[X, Y] + df(X)Y$

Proof. First two claims are straightforward. To prove Jacobi identity, the vector field $[[X, Y]Z$ corresponds to the double commutator

$$(L_X L_Y - L_Y L_X)L_Z - L_Z(L_X L_Y - L_Y L_X) = L_X L_Y L_Z - L_Y L_X L_Z - L_Z L_X L_Y + L_Z L_Y L_X.$$

Opening up in a similar way each of the summands in the Jacobi identity we will get 12 terms, which contains each triple composition term twice with opposite signs. Hence, all terms cancel.

Let us check the Leibnitz rule. We have

$$[X, fY] = L_X(fY) = \frac{d}{ds}((X^s)_*(fY))|_{s=0} = \frac{d}{ds}((f \circ X^s)(X^s)_*(Y))|_{s=0} = df(X)Y + fL_X Y.$$

$\blacksquare$

Remark 18.19. A vector space with a skew-symmetric bilinear operation $[X, Y]$ is called a *Lie algebra*. Besides the space of vector fields with a Lie bracket there are many other important examples of Lie algebras, both finite-, and infinite-dimensional.

Chapter 19

Topological applications of Stokes' formula

19.1 Approximation of continuous functions by smooth ones

Theorem 19.1. *Let $C \subset V$ be a compact domain with smooth boundary. Then any continuous function $f : C \to \mathbb{R}$ can be C^0-approximated by C^∞- smooth functions, i.e. for any $\epsilon > 0$ there exists a C^∞-smooth function $g : C \to \mathbb{R}$ such that $|f(x) - g(x)| < \epsilon$ for any $x \in C$. Moreover, if the function f is already C^∞-smooth in a neighborhood of a closed subset $B \subset \operatorname{Int} C$, then one can arrange that the function g coincides with f over B.*

Lemma 19.2. *There is a continuous extension of f to V.*

Proof. Let $\mathbf{n}$ be the outward normal vector field to the boundary ∂C. If the boundary is C^∞-smooth then so is the vector field $\mathbf{n}$. Given a point $a \in \partial C$ and $\sigma < 1$ denote $G_\sigma(a) := \{(z,t) \in \partial C \times [0,1];\ ||z - a|| < \sigma,\ 0 \leq t \leq \sigma\} \subset \partial C \times [0,1]$. Consider a map $\nu : \partial C \times [0,1] \to V$ given by the formula $\nu(x,t) = x + t\,\mathbf{n}, x \in \partial C, t \in [0,1]$. The differential of ν at the points of $\partial C \times 0$ has rank n. (*Exercise: prove this.*) Hence by the inverse function theorem (and in view of compactness of ∂C) for a sufficiently small $\delta > 0$ the restriction of the map ν to a neighborhood $G_\delta(a)$ of each point $a \in \partial C$ is a diffeomorphism onto its image. This implies that if $\epsilon \ll \delta$ is small enough then $\nu|_{\partial C \times [0,\epsilon)}$ is a diffeomorphism of $\partial C \times [0,\epsilon)$ onto a neighborhood $U \setminus \operatorname{Int} C$

in V. Indeed, it is sufficient to prove that the map is 1-1, because any bijective smooth map between open sets is a diffeomorphism, provided that it is a local diffeomorphism. Note that the set $D := \{(x, y) \in C; ||x - y|| \geq \delta\} \subset C \times C$ is compact (as a closed subset of a compact set). Hence, there exists $\sigma > 0$ such for any $(x, y) \in D$ we have $\nu(G_\sigma(x)) \cap \nu(G_\sigma(y)) = \varnothing$. Choose $\epsilon := \min(\delta, \sigma)$. Then given any two distinct points $(x, t), (x', t') \in \partial C \times [0, \epsilon)$ we have $\nu(x, t) \neq \nu(x', t')$. Indeed, if $||x - x'|| < \delta$ then $(x, t), (x', t') \in U_\delta(x)$ and $\nu(x, t) \neq \nu(x', t')$ because $\nu|_{U_\delta(x)}$ is injective. But if $||x - x'|| \geq \delta$ then $\nu(G_\sigma(x)) \cap \nu(G_\sigma(x')) = \varnothing$ and the inequality $\nu(x, t) \neq \nu(x', t')$ follows because we have $(x, t) \in G_\sigma(x), (x', t') \in G_\sigma(x'))$.

Consider a function $F : \partial C \times [0, \epsilon) \to \mathbb{R}$, defined by the formula

$$F(x, t) = \left(1 - \frac{2t}{\epsilon}\right) f(x)$$

if $t \in [0, \frac{\epsilon}{2}]$ and $F(x, t) = 0$ if $t \in (\frac{\epsilon}{2}, \epsilon)$. Now we can extend f to V by the formula $f(y) = F(\nu^{-1}(y))$ if $y \in U \setminus C$, and setting it to 0 outside U. ■

Consider the function

$$\Psi_\epsilon = \frac{1}{\int\limits_{D_\epsilon(0)} \psi_{0,\epsilon} dV} \psi_{0,\epsilon},$$

where $\psi_{0,\sigma}$ is a bump function defined above in (15.7.2). It is supported in the disc $D_\sigma(0)$, non-negative, and satisfies

$$\int\limits_{D_\sigma(0)} \Psi_\sigma dV = 1.$$

Given a continuous function $f : V \to \mathbb{R}$ we define a function $f_\sigma : V \to \mathbb{R}$ by the formula

$$f_\sigma(x) = \int\limits_{\mathbb{R}^n} f(x - y)\Psi_\sigma(y) d^n y. \tag{19.1.1}$$

Then

Lemma 19.3. *1. The function f_σ is C^∞-smooth.*

2. For any $\epsilon > 0$ there exists $\delta > 0$ such that for all $x \in C$ we have $|f(x) - f_\sigma(x)| < \epsilon$ provided that $\sigma < \delta$.

Proof. 1. By the change of variable formula we have, replacing the variable y by $u = y - x$:

$$f_\sigma(x) = \int\limits_{D_\epsilon(0)} f(x-y)\Psi_\sigma(y)d^n y = \int\limits_{D_\epsilon(-x)} f(-u)\Psi_\sigma(x+u)d^n u = \int\limits_V f(-u)\Psi_\sigma(x+u)d^n u.$$

But the expression under the latter integral depends C^∞-smoothly on x as a parameter. Hence, by the theorem about differentiating integral over a parameter, we conclude that the function f_ϵ in C^∞-smooth.

2. Fix some $\sigma_0 > 0$. The function f is uniformly continuous in $\overline{U_{\sigma_0}(C)}$. Hence there exists $\delta > 0$ such that $x, x' \in \overline{U_{\sigma_0}(C)}$ and $||x - x'|| < \delta$ we have $|f(x) - f(x')| < \epsilon$. Hence, for $\sigma < \min(\sigma_0, \delta)$ and for $x \in C$ we have

$$|f_\sigma(x) - f(x)| = |\int\limits_{D_\epsilon(0)} f(x-y)\Psi_\sigma(y)d^n y - \int\limits_{D_\epsilon(0)} f(x)\Psi_\sigma(y)d^n y| \leq$$

$$\int\limits_{D_\epsilon(0)} |f(x-y) - f(x)|\Psi_\sigma(y)d^n y \leq \epsilon \int\limits_{D_\epsilon(0)} \Psi_\sigma(y)d^n y = \epsilon. \tag{19.1.2}$$

Proof of Theorem 19.1. Lemma 19.3 implies that for a sufficiently small σ the function $g = f_\sigma$ is the required C^∞-smooth ϵ-approximation of the continuous function f. To prove the second part of the theorem let us assume that f is already C^∞-smooth on a neighborhood U, $B \subset U \subset C$. Let us choose a cut-off function $\sigma_{B,U}$ constructed in Lemma 15.23 and define the required approximation g by the formula $f_\sigma + (f - f_\sigma)\sigma_{B,U}$. ■

Theorem 19.1 implies a similar theorem form continuous maps $C \to \mathbb{R}^n$ by applying it to all coordinate functions.

19.2 Homotopy

Let A, B be any 2 subsets of vector spaces V and W, respectively. Two continuous maps $f_0, f_1 : A \to B$ are called *homotopic* if there exists a continuous map $F : A \times [0,1] \to B$ such that $F(x,0) = f_0(x)$ and $F(x,1) = f_1(x)$ for all $x \in A$. Notice that the family $f_t : A \to B$, $t \in [0,1]$, defined by the formula $f_t(x) = F(x,t)$ is a continuous deformation connecting f_0 and f_1. Conversely, any such continuous deformation $\{f_t\}_{t\in[0,1]}$ provides a homotopy between f_0 and f_1.

Given a subset $C \subset A$, we say that a homotopy $\{f_t\}_{t\in[0,1]}$ is fixed over C if $f_t(x) = f_0(x)$ for all $x \in C$ and all $t \in [0,1]$.

A set A is called *contractible* if there exists a point $a \in A$ and a homotopy $f_t : A \to A$, $t \in [0,1]$, such that $f_1 = \mathrm{Id}$ and f_0 is a constant map, i.e. $f_1(x) = x$ for all $x \in A$ and $f_0(x) = a \in A$ for all $x \in A$.

Example 19.4. *Any star-shaped domain A in V is contractible. Indeed, assuming that it is star-shaped with respect to the origin, the required homotopy $f_t : A \to A$, $t \in [0,1]$, can be defined by the formula $f_t(x) = tx$, $x \in A$.*

Remark 19.5. In what follows we will always assume all homotopies to be smooth. According to Theorem 19.1 this is not a serious constraint. Indeed, any continuous map can be C^0-approximated by smooth ones, and any homotopy between smooth maps can be C^0-approximated by a smooth homotopy between the same maps.

Lemma 19.6. *Let $U \subset V$ be an open set, A a compact oriented manifold (possibly with boundary) and α a smooth closed differential k-form on U. Let $f_0, f_1 : A \to U$ be two maps which are homotopic relative to the boundary ∂A. Then*

$$\int\limits_A f_0^*\alpha = \int\limits_A f_1^*\alpha.$$

Proof. Let $F : A \times [0,1] \to U$ be the homotopy map between f_0 and f_1. By assumption $d\alpha = 0$, and hence $\int\limits_{A\times[0,1]} F^* d\alpha = 0$. Then, using Stokes' theorem we have

$$0 = \int\limits_{A\times[0,1]} F^* d\alpha = \int\limits_{A\times[0,1]} dF^*\alpha = \int\limits_{\partial(A\times[0,1])} F^*\alpha = \int\limits_{\partial A\times[0,1]} F^*\alpha + \int\limits_{A\times 1} F^*\alpha + \int\limits_{A\times 0} F^*\alpha$$

where the boundary $\partial(A \times [0,1]) = (A \times 1) \cap (A \times 0) \cap (\partial A \times [0,1])$ is oriented by an outward normal vector field $\mathbf{n}$. Note that $\mathbf{n} = \frac{\partial}{\partial t}$ on $A \times 1$ and $\mathbf{n} = -\frac{\partial}{\partial t}$ on $A \times 0$, where we denote by t the coordinate corresponding to the factor $[0,1]$. First, we notice that $F^*\alpha|_{\partial A\times[0,1]} = 0$ because the map F is independent of the coordinate t, when restricted to $\partial A \times [0,1]$. Hence $\int\limits_{\partial A\times[0,1]} F^*\alpha = 0$. Consider the inclusion maps $A \to A\times[0,1]$ defined by the formulas $j_0(x) = (x,0)$ and $j_1(x) = (x,1)$. Note that j_0, j_1 are diffeomorphisms $A \to A \times 0$ and $A \to A \times 1$, respectively. Note that the map j_1 preserves the orientation while j_0 reverses it. We also have $F \circ j_0 = f_0$, $F \circ j_1 = f_1$. Hence,

$\int\limits_{A\times 1} F^*\alpha = \int\limits_A f_1^*\alpha$ and $\int\limits_{A\times 0} F^*\alpha = -\int\limits_A f_0^*\alpha$. Thus,

$$0 = \int\limits_{\partial(A\times[0,1])} F^*\alpha = \int\limits_{\partial A\times[0,1]} F^*\alpha + \int\limits_{A\times 1} F^*\alpha + \int\limits_{A\times 0} F^*\alpha = \int\limits_A f_1^*\alpha - \int\limits_A f_0^*\alpha.$$

■

Lemma 19.7. *Let A be an oriented m-dimensional manifold, possibly with boundary. Let $\Omega(A)$ denote the space of differential forms on A and $\Omega(A\times[0,1])$ denote the space of differential forms on the product $A\times[0,1]$. Let $j_0, j_1 : A \to A\times[0,1]$ be the inclusion maps $j_0(x) = (x,0) \in A\times[0,1]$ and $j_1(x) = (x,1) \in A\times[0,1]$. Then there exists a linear map $K : \Omega(A\times[0,1]) \to \Omega(A)$ such that*

- *If α is a k-form, $k = 1,\dots,m$ then $K(\alpha)$ is a $(k-1)$-form;*
- *$d\circ K + K\circ d = j_1^* - j_0^*$, i.e. for each differential k-form $\alpha \in \Omega^k(A\times[0,1])$ one has $dK(\alpha) + K(d\alpha) = j_1^*\alpha - j_0^*\alpha$.*

Remark 19.8. Note that the first d in the above formula denotes the exterior differential $\Omega^k(A) \to \Omega^{k+1}(A)$, while the second one is the exterior differential $\Omega^k(A\times[0,1]) \to \Omega^{k+1}(A\times[0,1])$.

Proof. Let us write a point in $A\times[0,1]$ as (x,t), $x\in A, t\in[0,1]$. To construct $K(\alpha)$ for a given $\alpha \in \Omega^k(A\times[0,1])$ we first contract α with the vector field $\frac{\partial}{\partial t}$ and then integrate the resultant form with respect to the t-coordinate. More precisely, note that any k-form α on $A\times[0,1]$ can be written as $\alpha = \beta(t) + dt\wedge\gamma(t)$, $t\in[0,1]$, where for each $t\in[0,1]$

$$\beta(t) \in \Omega^k(A), \gamma(t) \in \Omega^{k-1}(A).$$

Then $\frac{\partial}{\partial t} \lrcorner\, \alpha = \gamma(t)$ and we define $K(\alpha) = \int\limits_0^1 \gamma(t)dt$.

If we choose a local coordinate system $(u_1,\dots,u_m)$ on A then $\gamma(t)$ can be written as $\gamma(t) = \sum\limits_{1\le i_1<\dots<i_\le m} h_{i_1\dots i_k}(t)du_{i_1}\wedge\dots\wedge du_{i_k}$, and hence

$$K(\alpha) = \int\limits_0^1 \gamma(t)dt = \sum_{1\le i_1<\dots<i_\le m} \left(\int\limits_0^1 h_{i_1\dots i_k}(t)dt\right) du_{i_1}\wedge\dots\wedge du_{i_k}.$$

Clearly, K is a linear operator $\Omega^k(A\times I) \to \Omega^{k-1}(A)$.

Note that if $\alpha = \beta(t) + dt \wedge \gamma(t) \in \Omega^k(A \times [0,1])$ then

$$j_0^*\alpha = \beta(0), \;\; j_1^*\alpha = \beta(1).$$

We further have

$$\begin{aligned} K(\alpha) &= \int\limits_0^1 \gamma(t)dt; \\ d\alpha &= d_{A\times I}\alpha = d_U\beta(t) + dt \wedge \dot\beta(t) - dt \wedge d_U\gamma(t) = d_A\beta(t) + dt \wedge (\dot\beta(t) - d_A\gamma(t)), \end{aligned}$$

where we denoted $\dot\beta(t) := \frac{\partial\beta(t)}{\partial t}$ and $I = [0,1]$. Here the notation $d_{A\times I}$ stands for exterior differential on $\Omega(A\times I)$ and d_A denotes the exterior differential on $\Omega(A)$. In other words, when we write $d_A\beta(t)$ we view $\beta(t)$ as a form on A depending on t as a parameter. We do not write any subscript for d when there could not be any misunderstanding.

Hence,

$$\begin{aligned} K(d\alpha) &= \int\limits_0^1 (\dot\beta(t) - d_A\gamma(t))dt = \beta(1) - \beta(0) - \int\limits_0^1 d_A\gamma(t)dt; \\ d(K(\alpha)) &= \int\limits_0^1 d_A\gamma(t)dt. \end{aligned}$$

Therefore,

$$\begin{aligned} d(K(\alpha)) + K(d(\alpha)) &= \beta(1) - \beta(0) - \int\limits_0^1 d_A\gamma(t)dt + \int\limits_0^1 d_A\gamma(t)dt \\ &= \beta(1) - \beta(0) = j_1^*(\alpha) - j_0^*(\alpha). \end{aligned}$$

■

Theorem 19.9. (Poincaré's lemma) *Let U be a contractible domain in V. Then any closed form in U is exact. More precisely, let $F : U \times [0,1] \to U$ be the contraction homotopy to a point $a \in U$, i.e. $F(x,1) = x,\; F(x,0) = a$ for all $x \in U$. Then if ω a closed k-form in U then*

$$\omega = dK(F^*\omega),$$

where $K : \Omega^{k+1}(U \times [0,1]) \to \Omega^k(U)$ is an operator constructed in Lemma 19.7.

Proof. Consider a contraction homotopy $F : U \times [0,1] \to U$. Then $F \circ j_0(x) = a \in U$ and $F \circ j_1(x) = x$ for all $x \in U$. Consider an operator $K : \Omega(U) \to \Omega(U)$ constructed above. Thus

$$K \circ d + d \circ K = j_1^* - j_0^*.$$

Let ω be a closed k-form on U. Denote $\alpha := F^*\omega$. Thus α is a k-form on $U \times [0,1]$. Note that $d\alpha = dF^*\omega = F^*d\omega = 0$, $j_1^*\alpha = (F \circ j_1)^*\omega = \omega$ and $j_0^*\alpha = (F \circ j_0)^*\omega = 0$. Then, using Lemma 19.7 we have

$$K(d\alpha) + dK(\alpha) = dK(\alpha) = j_1^*\alpha - j_0^*\alpha = \omega, \tag{19.2.1}$$

i.e. $\omega = dK(F^*\omega)$. ∎

In particular, *any closed form is locally exact.*

Example 19.10. Let us work out explicitly the formula for a primitive of a closed 1-form in a star-shaped domain $U \subset \mathbb{R}^n$. We can assume that U is star-shaped with respect to the origin. Let $\alpha = \sum\limits_1^n f_i dx_i$ be a closed 1-form. Then according to Theorem 19.9 we have $\alpha = dF$, where $F = K(\Phi^*\alpha)$, where $\Phi : U \times [0,1] \to U$ is a contraction homotopy, i.e. $\Phi(x,1) = x, \Phi(x,1) = 0$ for $x \in U$. Φ can be defined by the formula $\Phi(x,t) = tx,\ x \in U, t \in [0,1]$. Then

$$\Phi^*\alpha = \sum_1^n f_i(tx)d(tx_i) = \sum_1^n tf_i(tx)dx_i + \sum_1^n x_i f_i(tx)dt.$$

Hence,

$$K(\alpha) = \int\limits_0^1 \frac{\partial}{\partial t} \lrcorner\, \Phi^*\alpha = \int\limits_0^1 \left(\sum_1^n x_i f_i(tx)\right) dt.$$

Note that this expression coincides with the expression in formula (5.3.1) in Section 5.3.

Exercise 19.11. Work out an explicit expression for a primitive of a closed 2-form $\alpha = Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy$ on a star-shaped domain $U \subset \mathbb{R}^3$.

Example 19.12. 1. $\mathbb{R}^n_+ = \{x_1 \geq 0\} \subset \mathbb{R}^n$ *is not diffeomorphic to* $\mathbb{R}^n$. Indeed, suppose there exists such a diffeomorphism $f : \mathbb{R}^n_+ \to \mathbb{R}^n$. Denote $a := f(0)$. Without loss of generality we can assume that $a = 0$. Then $\widetilde{f} = f|_{\mathbb{R}^n_+ \setminus 0}$ is a diffeomorphism $\mathbb{R}^n_+ \setminus 0 \to \mathbb{R}^n \setminus 0$. But $\mathbb{R}^n_+ \setminus 0$ is star-shaped with respect to any point with positive coordinate x_1, and hence it is contractible. In particular any closed form on $\mathbb{R}^n_+ \setminus 0$ is exact. On the other hand, we exhibited above in Lemma 16.9 a closed $(n-1)$-form on $\mathbb{R}^n \setminus 0$ which is not exact.

2. Borsuk's theorem: *There is no continuous map $D^n \to \partial D^n$ which is the identity on ∂D^n.* We denote here by D^n the unit disc in $\mathbb{R}^n$ and by ∂D^n its boundary $(n-1)$-sphere.

Proof. Suppose that there is such a map $f : D^n \to \partial D^n$. One can assume that f is smooth. Indeed, according to Theorem 19.1 one can approximate f by a smooth map, keeping it fixed on the boundary where it is the identity map, and hence smooth. Take the closed non-exact form θ_n from Lemma 16.9 on ∂D^n. Then $\Theta_n = f^*\theta_n$ is a closed $(n-1)$-form on D^n which coincides with θ_n on ∂D^n. D^n is star-shaped, and therefore Θ_n is exact, $\Theta_n = d\omega$. But then $\theta_n = d(\omega|_{\partial D^n})$ which is a contradiction. ■

3. Brouwer's fixed point theorem: *Any continuous map $f : D^n \to D^n$ has at least* 1 *fixed point.*

Proof. Suppose $f : D^n \to D^n$ has no fixed points. Let us define a map $F : D^n \to \partial D^n$ as follows. For each $x \in D^n$ take a ray r_x from the point $f(x)$ which goes through x till it intersects ∂D^n at a point which we will denote $F(x)$. The map is well defined because for any x the points x and $f(x)$ are distinct. Note also that if $x \in \partial D^n$ then the ray r_x intersects ∂D^n at the point x, and hence $F(x) = x$ in this case. But existence of such F is ruled out by Borsuk' theorem. ■

k-connected manifolds

A subset $A \subset V$ is called *k-connected*, $k = 0, 1, \dots$, if for any $m \leq k$ any two continuous maps of discs $f_0, f_1 : D^m \to A$ which coincide along ∂D^m are homotopic relative to ∂D^m. Thus, 0-connectedness is equivalent to path-connectedness. 1-connected submanifolds are also called *simply connected.*

Exercise 19.13. *Prove that k-connectedness can be equivalently defined as follows: A is k-connected if any continuous map $f : S^m \to A$, $m \leq k$ is homotopic to a constant map.*

Example 19.14. 1. *If A is contractible then it is k-connected for any k.* For some classes of subsets, e.g. submanifolds, the converse is also true (J.H.C Whitehead's theorem) but this is a quite deep and non-trivial fact.

2. *The n-sphere S^n is $(n-1)$-connected but not n-connected.* Indeed, to prove that S^{n-1} simply connected we will use the second definition. Consider a map $f : S^k \to S^n$. We first notice that according to Theorem 19.1 we can assume that the map f is smooth. Hence, according to Corollary 14.9 $\mathrm{Vol}_n f(S^k) = 0$ provided that $k < n$. In particular, there exists a point $p \in S^n \setminus f(S^k)$. But the complement of a point p in S^n is diffeomorphic to S^n vis the stereographic projection from the point p. But R^n is contractible, and hence f is homotopic to a constant map. On the other hand, the identity map $\mathrm{Id} : S^n \to S^n$ is not homotopic to a constant map. Indeed, we know that there exists a closed n-form on S^n, say the form θ_n from Lemma 16.9, such that $\int\limits_{S^n} \theta_n \neq 0$. Hence, $\int\limits_{S^n} \mathrm{Id}^*\theta_n \neq 0$. On the other hand if Id were homotopic to a constant map this integral would vanish.

Exercise 19.15. Prove that $\mathbb{R}^{n+1} \setminus 0$ is $(n-1)-$ *connected* but not n-connected.

We proved in Theorem 5.17 that in a simply-connected domain any closed 1-form is exact. It turns out that this results can be generalized to k-forms.

Proposition 19.16. *If a manifold A is m-connected then for any $k \leq m$, $k > 0$, any closed differential k-form α on A is exact.*

Proof. We will prove the theorem only for the case of the n-sphere S^n. Let us argue by induction. The case $n = 1$ is vacuous. Let is begin with $n = 2$. Let α be a closed 1-form. Denote $S_\pm = S \setminus p_\pm$, where p_+, p_- are the poles. Then $\alpha|_{S_\pm} = dh_\pm$ for some functions $h_\pm : S_\pm \to \mathbb{R}$ because $S_\pm$ is contractible. Note that the functiion $h_+ - h_-$ on $S_+ \cap S_-$ is a constant, because $dh_- = dh_+ = \alpha$ there. By adding this constant to h_- we can arrange that $h_+ = h_-$ on $S_+ \cap S_-$, and hence there is a function $h : S^2 \to \mathbb{R}$ such that $h_\pm = h|_{S_\pm}$. But then $dh = \alpha$.

Suppose now that $m > 2$ and the statement is already proven for all all $n < m$ and all $k < n$. Arguing as in the 2-dimensional case we can find $(k-1)$-forms $\beta_\pm$ on $S_\pm$ such that $d\beta_\pm = \alpha|_{S_\pm}$. Denote by S^{n-1} the equatorial $(n-1)$-sphere in S^n. Then $\gamma := \beta_+ - \beta_-$ is a closed $(k-1)$-form on $S_+ \cap S_-$. By the induction hypothesis $\widehat{\gamma} := \gamma|_S^{n-1}$ is exact, i.e. there is a form δ on S^{n-1} such that $d\delta = \widehat{\gamma}$. Consider a projection $\pi : S_+ \cap S_- = S^n \setminus \{p_+, p_-\} \to S^{n-1}$ along the meridians. Then $\pi^*\widehat{\gamma} = d(\pi^*\delta)$. Let $j : S_+ \cap S_- \to S^n$ be the inclusion map. Note that the maps $j,\ j \circ \pi : S_+ \cap S_- \to S^n$ are homotopic. Indeed the required homotopy moves along the meridians to their middle point so that to arrive there in exactly time 1. As we just stated the form $\pi^*\widehat{\gamma} = \pi^*(j^*\gamma)$ is exact. Hence so is the form $j^*\gamma = \gamma$, i.e. $\gamma = d\eta$ for a $(k-2)$-form η. Let us take a small disk D centered at the point p_- and consider a cut-off function $\theta : S^n \to \mathbb{R}$ which is equal to 1 on $S^n \setminus D$ and which is equal to 0 in a smaller disc $D' \subset D$ centered at p_-. Denote $\widehat{\eta} := \theta\eta$ and extend this form as 0 to the pole p_-. Set $\widehat{\beta}_+ := \beta_+ + d\widehat{\eta}$ on S_+. Then $d\widehat{\beta}_+ = d\beta + - = \alpha$ on S^+. Note that $\widehat{\beta}_+ = \beta_+\gamma = \beta_-$ on $S_- \setminus D$. Hence, the form

$$\beta := \begin{cases} \widehat{\beta}_+, & \text{on } D; \\ \beta_-, & \text{on } S^n \setminus D \end{cases}$$

is the required primitive of α. ■

19.3 Properties of k-forms on k-dimensional manifolds

A k-form α on k-dimensional manifold is always closed. Indeed, $d\alpha$ is a $(k+1)$-form and hence it is identically 0 on a k-dimensional manifold.

Remark 19.17. Given a k-dimensional submanifold $A \subset V$, and a k-form α on V, the differential $d_x\alpha$ does not need to vanish at a point $x \in A$. However, $d\alpha_x|_{T_x(A)}$ does vanish.

Theorem 19.18. *Let A be an oriented compact connected k-dimensional manifold, possibly with boundary, and α a differential k-form on A.*

1. *Suppose that $\partial A \neq \varnothing$. Then α is exact, i.e. there exists a $(k-1)$-form β on A such that $d\beta = \alpha$.*

2. *Suppose that $\overline{\operatorname{Supp}\alpha} \supset \operatorname{Int} A$ and $\int\limits_A \alpha = 0$. Then there exists a $(k-1)$-form β on A with $\overline{\operatorname{Supp}\beta} \subset \operatorname{Int} A$ such that $d\beta = \alpha$.*

Remark 19.19. 1. The condition $\int\limits_A \alpha = 0$ is necessary for existence β with compact support in $\operatorname{Int} A$ such that $d\beta = \alpha$. Indeed, by Stokes' theorem we have $0 = \int\limits_{\partial A} \beta = \int\limits_A d\beta = \int\limits_A \alpha$.
2. If A is closed, i.e. $\partial A = \varnothing$ then the compact support condition is empty, so the second part of the theorem just reads: a k-form α on a closed oriented connected k-manifold A is exact if and only if $\int_A \alpha = 0$.

We will first prove Theorem 19.18 in two special cases, and then sketch a proof of the general result.

Proposition 19.20. *Theorem 19.18 holds when A is the k-dimensional disc D^k or the k-dimensional sphere S^k.*

Proof. We will argue by induction.

1. *Case* $k = 1$. For the first part we have $A = D^1 = [-1,1]$ and $\alpha = f(x)dx$. Define $F(x) := \int\limits_0^x f(s)ds$ and $\beta := F(x)$. Then $d\beta = F'(x)dx = f(x)dx = \alpha$. For the second part, if $A = D^1 = [-1,1]$ then $\alpha = f(x)dx$, where $f = 0$ near the end points of the interval $[-1,1]$. Then the condition $\int\limits_A \alpha = \int\limits_{-1}^1 f(x)dx = 0$ implies that the function $F(x) := \int\limits_0^x F(s)ds$ also vanishes near the boundary of D^1, and hence $\beta = F$ is the required 0-form. Similarly, in the case of $A = S^1$ we can consider the map $\pi : \mathbb{R} \to S^1$, given by the formula $\pi(t) = (\cos \pi t, \sin \pi t)$, and pull back forms from S^1 to $\mathbb{R}$ by π. In this interpretation functions on S^1 are the same as 2-periodic functions on $\mathbb{R}$, and differential 1-forms on S^1 are the same as 1-forms $f(x)dx$ on $\mathbb{R}$, where f is a 2-periodic function. The condition that $\int\limits_S^1 \alpha = 0$ translates into the condition $\int\limits_{-1}^1 f(x)dx = 0$ for a 2-periodic function f. In turn, this implies that the function $F(x) := \int\limits_0^x f(s)ds$ is 2-periodic.

2. *Induction step. Case* $A = D^k$. Suppose now that $k \geq 2$, Theorem 19.18 already proven for disks and spheres of dimension $< k$ and deduce from this assumption the k-dimensional case.

The first statement of Theorem 19.18 is straightforward even without any induction. Indeed, D^k is contractible, and hence, α is exact, $\alpha = d\beta$. For the second part let α be a k-form which vanishes outside a disk $D^k_{1-2\epsilon} := \{r \leq 1 - 2\epsilon\}$ for a small $\epsilon > 0$ (e.g. $\epsilon < \frac{1}{6}$) and $\int\limits_A \alpha = 0$. Applying Part 1 Theorem 19.18 we find a $(k-1)$-form β_1 on $D^k_{3\epsilon}$ such that $d\beta = \alpha$ on $D^k_{3\epsilon}$. Choose a cut-off function $\theta_1 : [0,1] \to [0,1]$ such that $\theta_1(t) = 1$ for $t \leq 2\epsilon$ and $\theta_1(t) = 0$ for $t \geq 3\epsilon$. Denote $\widetilde{\beta}_1 := \theta_1(r)\beta_1$ on $D^k_{3\epsilon}$ and extend the form $\widetilde{\beta}_1$ to the whole disk D^k as equal to 0 outside $D^k_{3\epsilon}$. Denote $\alpha_1 := \alpha - d\widetilde{\beta}_1$. Then $\alpha_1 = \alpha$ on $D^k \setminus D^k_{3\epsilon}$, $\alpha_1 = 0$ on $D^k_{2\epsilon}$ and

$$\int\limits_{D^k} \alpha_1 = \int\limits_{D^k} \alpha - \int\limits_{D^k} d\widetilde{\beta}_1 = 0.$$

Consider a diffeomorphism $\phi : D^k \setminus D^k_\epsilon \to [\epsilon, 1] \times S^{k-1}$ given by the formula $\phi(x) = (r, y)$, where $r = ||x|| \in [\epsilon, 1]$, and $y = \frac{x}{r} \in S^{n-1}$. We consider the pull-back form $\phi^* \widetilde{\alpha}_1$ as just expressed in coordinates (r, y), and thus will keep the notation $\widetilde{\alpha}_1$. Let η be the volume form S^{n-1}. As any two n-forms are proportional, we can write $\widetilde{\alpha}_1 = g(r, y)dr \wedge \eta$ for some smooth function g which is equal to 0 when $r \in [\epsilon, 2\epsilon] \cup [1-2\epsilon, 1$. Consider the function $G(r,y) := \int\limits_{\epsilon}^{r} g(\rho, y) d\rho$. Note that $\frac{\partial G(r,y)}{\partial r} = g(r,y)$ and $G(r,y) = G(r,1)$ for $r \geq 1-2\epsilon$.

Define the $(k-1)$-form β_2 on D^k by the formula

$$\beta_2 := \begin{cases} G(r,y) d\rho \eta, & r > \epsilon; \\ 0, & r \leq \epsilon. \end{cases}$$

Then $\beta_2 = 0$ for $r < \epsilon$, $\beta_2 = G(1,y)\eta$ for and $r > 1-2\epsilon$ and $d\beta_2 = g(r,y) dr \wedge \eta = \widetilde{\alpha}_1$. We will now modify the form $\beta_2 = 0$ to a $(k-1)$-form $\widetilde{\beta}_2$ on D^k which is equal to 0 for $r > 1-\epsilon$ while still serves as the primitive of $\widetilde{\alpha}_1$, i.e. satisfies the condition $d\widetilde{\beta}_2 = \widetilde{\alpha}_1$.

Note that

$$\int\limits_{S^{n-1}} G(1,y)\eta = \int\limits_{S^{n-1}} \beta_2 = \int\limits_{D}^{2} d\beta_2 = \int\limits_{D^2} \widetilde{\alpha}_1 = 0.$$

Hence, by the induction hypothesis there exists a $(k-2)$-form μ on S^{n-1} such that $d\mu = G(1,y)\eta$. Choose another cut-off function $\theta_2 : [0,1] \to [0,1]$ such that $\theta_2(t) = 1$ for $t \leq 1-2\epsilon$, $\theta_2(t) = 0$ for $t \geq 1-\epsilon$. We define now the required form $\widetilde{\beta}_2$ by the formula

$$\widetilde{\beta}_2 := \begin{cases} \beta_2, & \text{on } D^k_{1-2\epsilon} \\ \theta_2(r)\beta_2 + \theta_2'(r) dr \wedge \mu, & \text{on } D_k \setminus D^k_{1-2\epsilon}. \end{cases}$$

We note that $\theta_2'(r) = 0$ unless $r \in [1-2\epsilon, 1-\epsilon]$, so the form $\widetilde{\beta}_2 = \beta_2$ on $D^k_{1-2\epsilon}$. We also recall that $\widetilde{\alpha}_1 = 0$ on $D^k \setminus D^k_{1-2\epsilon}$. Hence, $\widetilde{\beta}_2$ is a well-defined $(k-1)$-form on D^k which is equal to 0 on $D^k_{1-\epsilon}$. Furthermore,

$$d\widetilde{\beta}_2 := \begin{cases} d\beta_2 = \widetilde{\alpha}_1, & \text{on } D^k_{1-2\epsilon}; \\ \theta_2(r)\widetilde{\alpha}_1 + \theta_2'(r) dr \wedge \widetilde{\beta}_2 - \theta_2'(r) dr \wedge d\mu = 0, & \text{on } D_k \setminus D^k_{1-2\epsilon}. \end{cases}$$

In other words, $d\widetilde{\beta}_2 = \widetilde{\alpha}_1$ everywhere on D^k. Finally, we get

$$d(\widetilde{\beta}_2 + \widetilde{\beta}_1) = \alpha,$$

as required.

3. *Induction step. Case* $A = S^k$. For $a \in (-1,1)$ denote $S^k_{+,a} := S^k \cap \{x_k \geq a\}$ and $S^k_{-,a} := S^k \cap \{x_k \leq a\}$. Note that $S^k_{\pm,a}$ is diffeomorphic to the disc D^k for any $a \in (-1,1)$. Choose any

$a \in (0, \frac{1}{2})$. Applying the proven in Step 2 case of Proposition 19.20.1 to $\alpha|_{S^k_{-,2a}}$ we find a form β_1 on $S^k_{-,2a}$ such that $d\beta_1 = \alpha$. Choosing a cut-off function $\theta : S^k \to \mathbb{R}$ which is equal to 1 on $S^k_{-,a}$ and to 0 on $S^k_{+,2a}$ we define the form $\widetilde{\beta}_1$ equal to $\theta\beta_1$ on $S^k_{-,2a}$ and equal to 0 on $S^k_{+,2a}$. Consider the form $\alpha_1 := \alpha - d\widetilde{\beta}_1$. This form is equal to 0 on $S^k_{-,a}$ and

$$\int\limits_{S^k} \alpha_1 = \int\limits_{S^k} \alpha - \int\limits_{S^k} d\widetilde{\beta}_1 = 0.$$

Hence, we can apply already proven case in Step 2 case of Proposition 19.20 to the form $\alpha_1|_{S^k_{+,0}}$ we find a $(k-1)$-form β_2 on $S^k_{+,0}$ which is equal to 0 near $\partial S^k_{+,0}$ and such that $d\beta_2 = \alpha_1|_{S^k_{+,0}}$. We extending the form β_2 as equal to 0 on $S^k_{-,0}$ we get

$$d(\widetilde{\beta}_1 + \beta_2) = d\widetilde{\beta} + \alpha_1 = d\widetilde{\beta} + \alpha - d\widetilde{\beta} = 0.$$

■

Sketch of the proof of Theorem 19.18. We first consider Part 2 of the theorem. Let A be an oriented compact connected k-dimensional manifold, possibly with boundary, and α a differential k-form on A with $\overline{\operatorname{Supp}\alpha} \subset \operatorname{Int} A$.

Using a partition of unity, we can present α as a sum $\eta = \sum\limits_1^N \alpha_j$, where η_j is compactly supported in the domain $U_j \subset \operatorname{Int} M$ diffeomorphic to the k-ball $B^k = \int D^k$.

Choose disjoint points $a_j \in U_j$ for each $j = 1, \dots, N$ and surround them by small disjoint balls $D_j \ni a_j, D_j \subset U_j$. Choose for each j a k-form ω_j with its support strictly inside U_j and such that $\int\limits_{D_j} \omega_j = \int\limits_{U_j} \alpha_j$. We can assume that the forms ω_j defined on the whole A (as equal to 0 outside D_j). Set $\widehat{\alpha}_j := \alpha_j - \omega_j$. Then $\int\limits_{U_j} \widehat{\alpha}_j = 0$, and hence, by Proposition 19.20 we $\widehat{\alpha}_j = d\beta_j$ for some $(k-1)$-form β_j supported in U_j. Let us choose disjoint embedded paths γ_j, $j = 1, \dots, N-1$ connecting a point on ∂D_j with a point on ∂D_{j+1}. There is a domain B which is diffeomorphic to an open k-ball and which contains $D_1 \cup \gamma_1 \cup D_2 \cup \dots \cup D_{N-1} \cup \gamma_{N-1} \cup D_N$. By construction we have $\int\limits_B \sum\limits_1^N \omega_j = \int\limits_A \eta = 0$, and hence, there exists a $(k-1)$-form β supported in B such that $d\beta = \sum\limits_1^N \omega_j$. Therefore

$$d(\beta + \sum_1^N \beta_j) = \sum_1^N \omega_j + \sum_1^N (\alpha_j - \omega_j) = \eta.$$

This concludes the proof of the second part of the theorem.

Next, we will reduce Part 1 to Part 2. Suppose that $\partial A \neq \varnothing$ and α is any k-form. Arguing as in Lemma 19.2 we can find an embedding (i.e. a diffeomorphism on its image) $h : \partial[0,\epsilon) \times \partial A \to A$ such that $h(0,x) = x$ for any $x \in \partial A$. Denote $U := h([0,\epsilon) \times \partial A)$. Let η be a volume form on ∂A (i.e. any non-vanishing $(k-1)$-form). Then we can write form $h^*\alpha = f(t,x))dt \wedge \eta$. Denote by

$$C := \frac{\int\limits_A \alpha}{\int\limits_{\partial A} \eta}.$$

Define a function

$$F(t,x) := C + \int\limits_0^t f(s,x)ds,\ x \in \partial A, t \in (0,\epsilon)$$

and then cut it off to a function $\widetilde{F}(t,x)$ such that

$\widetilde{F}(t,x) = F(t,x)$ for $t \leq \frac{\epsilon}{2}$;

$\widetilde{F}(t,x) = 0$ for $t \geq \frac{3\epsilon}{4}$.

Then $h^*\alpha = d(\widetilde{F}(t,x)\eta)$ on $[0,\frac{\epsilon}{2}] \times \partial A$ and $h^*\alpha = 0$ on $[\frac{3\epsilon}{4},\epsilon] \times \partial A$. Applying Stokes' theorem we get

$$\int\limits_{[0,\frac{\epsilon}{2}]\times\partial A} d(\widetilde{F}(t,x)\eta) = \int\limits_{\partial A} C\eta = \int\limits_A \alpha.$$

Consider the $(k-1)$-form $\mu := \left(h^{-1}\right)^* (\widetilde{F}(t,x)\eta)$ on U and extend it as 0 to the whole A. Then the form $\alpha - d\mu$ is equal to 0 near ∂A and

$$\int\limits_A (\alpha - d\mu) = \int\limits_A \alpha - \int\limits_A \alpha = 0.$$

Hence, we can apply Part 2 to get a $(k-1)$-form β such that $d\beta = \alpha - d\mu$, i.e $\alpha = d(\mu+\beta)$. ■

19.4 Degree of a map

Consider two closed connected oriented manifolds A, B of the same dimension k. Let ω be an n-form on B such that $\int\limits_B \omega = 1$. Given a smooth map $f : A \to B$ the integer $\deg(f) := \int\limits_A f^*\omega$ is called the *degree* of the map f.

Proposition 19.21. *1. Given any two k-forms on B such* $\int_B \omega = \int_B \widetilde{\omega}$ *we have* $\int_A f^*\omega = \int_A f^*\widetilde{\omega}$, *for any smooth map* $f : A \to B$, *and thus* $\deg(f)$ *is independent of the choice of the form* ω *on* B *with the property* $\int_A \omega = 1$.

2. *If the maps* $f, g : A \to B$ *are homotopic then* $\deg(f) = \deg(g)$.

3. *Let* $b \in B$ *be a regular value of the map* f. *Let* $f^{-1}(b) = \{a_1, \dots, a_d\}$. *Then*

$$\deg(f) = \sum_1^d \operatorname{sign}(\det Df(a_j)).$$

In particular, $\deg(f)$ *is an integer number.*

Proof. The second part follows from Lemma 19.6. To prove the first part, let us write $\widetilde{\omega} = \omega + \eta$, where $\int_B \eta = 0$. Using Theorem 19.18.2 we conclude that $\eta = d\beta$ for some $(k-1)$-form β on B. Then

$$\int_A f^*\widetilde{\omega} = \int_A f^*\omega + \int_A f^*\eta = \int_A f^*\omega + \int_A df^*\beta = \int_A f^*\omega.$$

Let us prove the last statement of the theorem. By the inverse function theorem there exists a neighborhood $U \ni b$ in B and neighborhoods $U_1 \ni a_1, \dots, U_d \ni a_d$ in A such that the restrictions of the map f to the neighborhoods $U_1, \dots, U_d$ are diffeomorphisms $f|_{U_j} : U_j \to U$, $j = 1, \dots, d$. Let us consider a form ω on B such that $\operatorname{Supp}\omega \subset U$ and $\int_B \omega = \int_U \omega = 1$. Then

$$\deg(f) = \int_A f^*\omega = \sum_1^d \int_{U_j} f^*\omega = \sum_1^d \operatorname{sign}(\det Df(a_j)),$$

because according to Theorem 14.20 we have

$$\int_{U_j} f^*\omega = \operatorname{sign}(\det Df(a_j)) \int_U \omega = \operatorname{sign}(\det Df(a_j)).$$

for each $j = 1, \dots, d$. ■

Remark 19.22. Any continuous map $f : A \to B$ can be approximated by a homotopic to f smooth map $A \to B$, and any two such smooth approximations of f are homotopic. Hence this allows us to define the degree of any *continuous* map $f : A \to B$.

Exercise 19.23. 1. Let us view $\mathbb{R}^2$ as $\mathbb{C}$. In particular, we view the unit sphere $S^1 = S^1_1(0)$ as the set of complex numbers of modulus 1:

$$S^1 = \{z \in \mathbb{C}; |z| = 1\}.$$

Consider a map $h_n : S^1 \to S^1$ given by the formula $h_n(z) = z^n, z \in S^1$. Then $\deg(h_n) = n$.

2. Let $f : S^{n-1} \to S^{n-1}$ be a map of degree d. Let $p_\pm$ be the north and south poles of $S^n \subset \mathbb{R}^{n+1}$, i.e. $p_\pm = (0, \dots, 0, \pm 1)$. Given any point $x = (x_1, \dots, x_{n+1}) \in S^n \setminus \{p_+, p_-\}$ we denote by $\pi(x)$ the point

$$\frac{1}{\sqrt{\sum\limits_1^n x_j^2}}(x_1, \dots, x_n) \in S^{n-1}$$

and define a map $\Sigma f : S^n \to S^n$ by the formula

$$\Sigma f(x) = \begin{cases} p_\pm, & \text{if } x = p_\pm, \\ \left(\sqrt{\sum\limits_1^n x_j^2} f(\pi(x)), x_{n+1}\right), & \text{if } x \neq p_\pm. \end{cases}$$

Prove that $\deg(\Sigma(f)) = d$.[1]

3. Prove that two maps $f, g : S^n \to S^n$ are homotopic if and only if they have the same degree. In particular, any map of degree n is homotopic to the map h_n. (*Hint: For n=1 this follows from Proposition 19.24. For $n > 1$ first prove that any map is homotopic to a suspension.*)

4. Give an example of two non-homotopic orientation preserving diffeomorphisms $T^2 \to T^2$. Note that the degree of both these maps is 1. Hence, for manifolds, other than spheres, having the same degree is not sufficient for their homotopy.

5. Let $\gamma, \delta : S^1 \to \mathbb{R}^3$ be two disjoint loops in $\mathbb{R}^3$. Consider a map $\widetilde{F}_{\gamma,\delta} : T^2 \to S^2$ defined by the formula

$$\widetilde{F}_{\gamma,\delta}(s,t) = \frac{\gamma(s) - \delta(t)}{||\gamma(s) - \delta(t)||}, \ s, t \in S^1.$$

Prove that $l(\gamma, \delta) = \deg(\widetilde{F}_{\gamma,\delta})$. Use this to solve Exercise **??**.4 above.

[1] The map Σf is called the *suspension* of the map f.

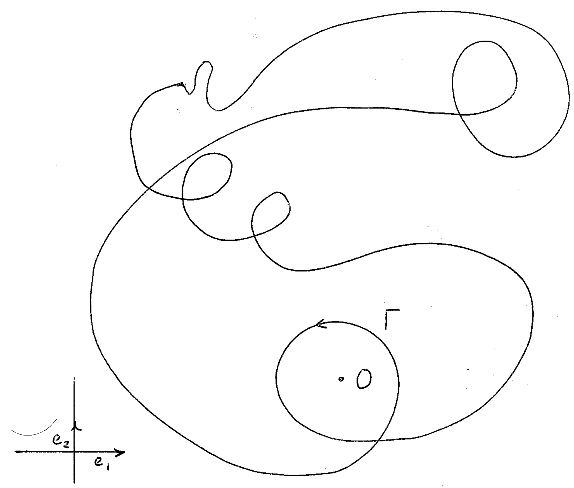


Figure 19.1: $w(\Gamma) = 2$.

19.5 Winding and linking numbers

Given a loop $\gamma : [a, b] \to \mathbb{R}^2 \setminus 0$ we define its *winding number* around 0 as the integral

$$w(\gamma) = \frac{1}{2\pi} \oint_\gamma \theta_2 = \frac{1}{2\pi} \oint_\gamma \frac{xdy - ydx}{x^2 + y^2}.$$

For instance, for the loop $\gamma_n(t) = (\cos nt, \sin nt), t \in [0, 2\pi]$ we have $w(\gamma_n) = n$.

Of course, we can always parametrize the loop γ by the interval $[0, 2\pi]$, and furthermore viewing t as the angular polar coordinate consider γ as a map of the map $S^1 \to \mathbb{R}^2 \setminus 0$. With this interpretation, $w(\gamma)$ is the degree of the maps

$$\frac{\gamma}{||\gamma||} : S^1 \to S^1.$$

In particular, it follows that $w(\gamma)$ is an integer number.

Proposition 19.24. *1. If loops $\gamma_0, \gamma_1 : S^1 \to \mathbb{R}^2 \setminus 0$ are homotopic in $\mathbb{R}^2 \setminus 0$ then $w(\gamma_0) = w(\gamma_1)$.*

2. $w(\gamma) = n$ then the loop γ is homotopic (as a loop in $\mathbb{R}^2 \setminus 0$) to the loop $\zeta_n : [0, 1] \to \mathbb{R}^2 \setminus 0$ given by the formula $\zeta_n(t) = (\cos 2\pi nt, \sin 2\pi nt)$.

Proof. 1. Let us define the loop γ parametrically in polar coordinates:

$$r = r(s), \phi = \phi(s),\ s \in [0,1],$$

where $r(0) = r(1)$ and $\phi(1) = \phi(0) + 2n\pi$. The form θ_2 in polar coordinates is equal to $d\phi$, and hence

$$\int\limits_{\gamma} \alpha = \frac{1}{2\pi} \int\limits_0^1 \phi'(s) ds = \frac{\phi(1) - \phi(0)}{2\pi} = n.$$

2. This is an immediate corollary of Proposition 19.6.

3. Let us write both loops γ and ζ_n in polar coordinates. Respectively,we have $r = r(t), \phi = \phi(s)$ for γ and $r = 1, \phi = 2\pi ns$ for ζ_n, $s \in [0,1]$. The condition $w(\gamma) = n$ implies, in view of part 1, that $\phi(1) = \phi(0) + 2n\pi$. Then the required homotopy γ_t, $t \in [0,1]$, connecting the loops $\gamma_0 = \gamma$ and $\gamma_1 = \zeta_n$ can be defined by the parametric equations $r = (1-t)r(s)+t$, $\phi = \phi_t(s) = (1-t)\phi(s)+2n\pi st$. Note that for all $t \in [0,1]$ we have $\phi_t(1) = \phi_t(0) + 2n\pi$. Therefore, γ_t is a loop for all $t \in [0,1]$. ■

Given two disjoint loops $\gamma, \delta : S^1 \to \mathbb{R}^3$ (i.e. $\gamma(s) \neq \delta(t)$ for any $s, t \in S^1$) consider a map $F_{\gamma,\delta} : T^2 \to \mathbb{R}^3 \setminus 0$, where $T^2 = S^1 \times S^1$ is the 2-torus, defined by the formula

$$F_{\gamma,\delta}(s,t) = \gamma(s) - \delta(t).$$

Then the number

$$l(\gamma,\delta) := \frac{1}{4\pi} \int\limits_T^2 F^*_{\gamma,\delta}\theta_3 = \frac{1}{4\pi} \int\limits_T^2 F^*_{\gamma,\delta} \left(\frac{xdy \wedge dz + ydz \wedge dx + zdx \wedge dy}{(x^2+y^2+z^2)^{\frac{3}{2}}} \right)$$

is called the *linking number* of loops γ, δ.[2] Equivalently, $l(\gamma,\delta)$ is the degree of the map

$$\frac{F_{\gamma,\delta}}{||F_{\gamma,\delta||}} : T^2 \to S^2.$$

Corollary 19.25. *1. The number $l(\gamma,\delta)$ remains unchanged if one continuously deforms the loops γ, δ keeping them disjoint;*

2. The number $l(\gamma,\delta)$ is an integer for any disjoint loops γ, δ;

3. $l(\gamma,\delta) = l(\delta,\gamma)$;

Exercise 19.26. *Let $\gamma(s) = (\cos s, \sin s, 0)$, $s \in [0,2\pi]$ and $\delta(t) = (-1+\frac{1}{2}\cos t, 0, \frac{1}{2}\sin t)$, $t \in [0,2\pi]$. Show that $l(\gamma,\delta) = 1$.*

[2]This definition of the linking number is due to Carl Friedrich Gauss.

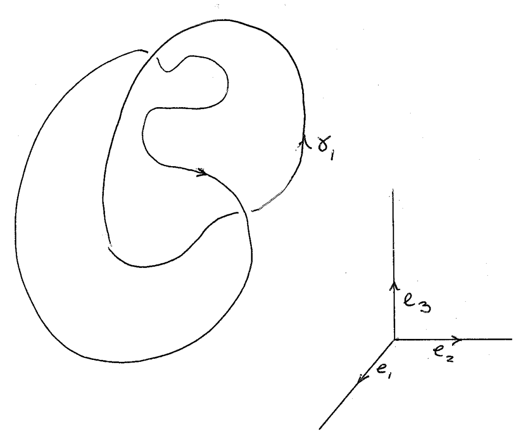


Figure 19.2: $l(\gamma_1, \gamma_2) = 1$.

19.6 Zeroes of a vector field

Let A be a compact n-dimensional oriented manifold with (possibly empty) boundary. Let X be a vector field on A. A point $a \in \operatorname{Int} A$ is called an *isolated zero* of X if $X(a) = 0$ and there exists a neighborhood $U \ni a, U \subset \operatorname{Int} A$ such that $X(x) \neq 0$ for any $x \in U \setminus a$.

Let $a \in \operatorname{Int} A$ be an isolated zero of X. Choose a local coordinate system on a neighborhood $U \ni a$, thus identifying U with a domain in $\mathbb{R}^n$. For a sufficiently small $\epsilon > 0$ take a sphere S_ϵ, centered at a and define a map $\phi_a : S_\epsilon \to S^{n-1}$, where S^{n-1} is the unit sphere in $\mathbb{R}^n$, by the formula:

$$\phi_a(x) = \frac{X(x)}{||X(x)||}.$$

The degree $\deg \phi_a$ is called the *index* of the isolated zero a and denoted $\operatorname{ind}_a X$.

Exercise 19.27. 1. Prove that the definition of the index is independent of a chooice of local coordinates.

2.Consider a linear vector field $X = \sum\limits_{i,j=1}^{n} a_{ij} x_i \frac{\partial}{\partial x_j}$. Show that 0 is an isolated zero if and only if det $A \neq 0$ and $\operatorname{ind}_0 X = \operatorname{sign} \det A$. For instance, for the vector field $x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}$ the origin has index

$+1$ and for the field $x\frac{\partial}{\partial x} - y\frac{\partial}{\partial y}$ the origin has index -1.

Theorem 19.28. *Let A be any compact manifold with boundary (which can be empty). Let Y be a vector field which has only isolated zeroes $z_1, \dots, z_N \in \operatorname{Int} A$, and which is outwardly transverse to ∂A. Then the sum of indices $\sum\limits_1^N \operatorname{ind}(z_j)$ is independent of the choice of the vector field and depends only on the topology of the manifold.*

This number is called the *Euler characteristic* of A, and denoted by $\chi(A)$. We have $\chi(S^1) = 0$, as S^1 has a tangent non-vanishing vector field, $\chi(S^2) = 2$, as the vector field tangent to S^2 has two zeroes of index $+1$. For the 2-torus T^2 we have $\chi(T^2) = 0$, because it admits a non-vanishing vector field. For an n-dimensional disc D^n we have $\chi(D^n) = 1$, because the radial vector field has a unique zero of index one at the center of the disc.

We split the proof of Theorem 19.28 into several steps. Each step has an independent interest and is formulated as a separate theorem.

Given a co-oriented closed hypersurface $\Sigma \subset \mathbb{R}^n$, its *normal degree* $\deg \operatorname{norm}(\Sigma)$ is defined as follows. Let ν be the unit normal vector field which define the co-orientation. By parallel transporting vectors $\nu(x), x \in \Sigma$, to the origin we define a map $G_\Sigma : \Sigma \to S^{n-1}$. Then

$$\deg \operatorname{norm}(\Sigma) := \deg G_\Sigma.$$

Theorem 19.29. *Let A be a compact n-dimensional domain in $\mathbb{R}^n$ with smooth boundary ∂A. Let us co-orient ∂A by the outward normal vector field ν. Let X be any vector field on U with isolated zeroes $z_1, \dots, z_N$, and which coincides with ν on ∂A. Then*

$$\sum_1^N \operatorname{ind}_{z_j} X = \deg \operatorname{norm}(\partial A).$$

In particular, the sum of indices of isolated zeroes for any vector field X extending to A the normal field ν is a constant which depends only on the domain A.

Proof. Choose a small $\epsilon > 0$ such that the balls $B_\epsilon(z_j)$, $j = 1, \dots, N$ are all disjoint and are contained in A. Denote $A_\epsilon := A \setminus \bigcup\limits_1^N B_\epsilon(z_j)$. Then A_ϵ is a compact manifold with boundary, and the vector X on A_ϵ has no zeroes. By parallel transporting vectors $\frac{X(x)}{||X(x)||}$, $x \in A_\epsilon$ to the origin we define a map $F : A_\epsilon \to S^{n-1}$. Let η be the $(n-1)$-form on S^{n-1} with $\int\limits_{S^{n-1}} \eta = 0$. Note that

$$\int\limits_{\partial A_\epsilon} F^*\eta = \int\limits_{\partial A} F^*\eta - \sum_1^N \int\limits_{\partial B_\epsilon(z_j)} F^*\eta = \deg \mathrm{norm}(\partial A) - \sum_1^N \mathrm{ind}_{z_j}(X).$$

But by Stokes theorem

$$\int\limits_{\partial A_\epsilon} F^*\eta = \int\limits_{A_\epsilon} dF^*\eta = 0,$$

because the form η is closed. Hence, $\deg \mathrm{norm}(\partial A) = \sum\limits_1^N \mathrm{ind}_{z_j}(X)$. ∎

Recall that a k-dimensional submanifold $A \subset \mathbb{R}^n$ has a *tubular neighborhood* $N(A)$, see Section 15.8. It is a manifold with boundary if A is closed and a manifold with boundary with corners, if A has non-empty boundary. It is also endowed by a vector field $\mathbf{r}$ which vanishes along A, and outward transverse to ∂A if A is closed, and it is outward transverse to the vertical part of ∂A and tangent to the horizontal part of ∂A if $\partial A \neq \varnothing$.

Theorem 19.30. *Let A be a k-dimensional compact submanifold of $\mathbb{R}^n$, $k < n$. Let $N(A)$ be its tubular neighborhood of A Then*

$$\sum_1^N \mathrm{ind}_{z_j} X = \deg \mathrm{norm}(\partial N).$$

Remark 19.31. If $\partial A \neq \varnothing$ then the boundary of N has corners. One needs to smooth them before computing $\deg \mathrm{norm}(\partial N)$.

Proof. Let X be any vector field on A with isolated zeroes $z_1, \dots, z_k$. Recall the vector field $\mathbf{r}$ on N which is tangent to the normal fibers N_a end restricts to each fiber as a radial vector field, see Section 15.8. It is also outward normal to the (vertical part of the) boundary ∂N of the tubular neighborhood. Define a vector field $\widetilde{X}$ on $N(A)$ by parallel transporting for each point $a \in A$ the vector $X(a)$ to all points of the fiber N_a. Consider a vector field $Y := \widetilde{X} + C\mathbf{r}$ on N, where C is a large constant. It has the same zeroes as X. If A is closed and C is large enough, then Y is outwardly transverse to ∂N, and if $\partial A \neq \varnothing$ then Y it is transverse to the smoothing of N, see Fig. 19.6.

Exercise 19.32. Prove that $\mathrm{ind}_{z_j} X = \mathrm{ind}_{z_j} Y$.

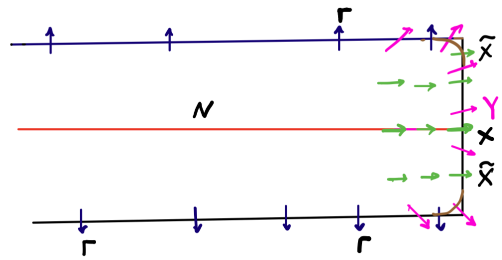


Figure 19.3: Vector field Y is transverse to the boundary ∂N after smoothing of corners.

HINT. Check that that the map defining the index of a zero z_j of Y is the suspension of the map which defines the index of the same zero z_j of X, and hence, have two maps have the same degrees.

Continuing the proof of Theorem 19.30 and using the above exercise we conclude that

$$\sum_1^N \mathrm{ind}_{z_j} X = \sum_1^N \mathrm{ind}_{z_j} Y.$$

On the other hand, Theorem 19.29 implies that

$$\sum_1^N \mathrm{ind}_{z_j} Y = \deg \mathrm{norm}(\partial N).$$

Hence,

$$\sum_1^N \mathrm{ind}_{z_j} X = \deg \mathrm{norm}(\partial N),$$

and thus the left-hand side is independent of a choice of the vector field X tangent to A. This finishes off the proof of Theorem 19.30, and with it, the proof of Theorem 19.28. ∎

Theorem 19.33. *Let $\Sigma \subset \mathbb{R}^n$ be a cooriented closed hypersurface in $\mathbb{R}^n$. Then*

$$\chi(\Sigma) = \begin{cases} 2 \deg \mathrm{norm}(\Sigma), & \text{if } n \text{ is odd;} \\ 0, & \text{if } n \text{ is even.} \end{cases}$$

Proof. Consider a tubular neighborhood $N \supset \Sigma$. Note that the boundary ∂N consists of two components $\partial N = \Sigma_+ \cup \Sigma_-$, see Fig. 19.6.

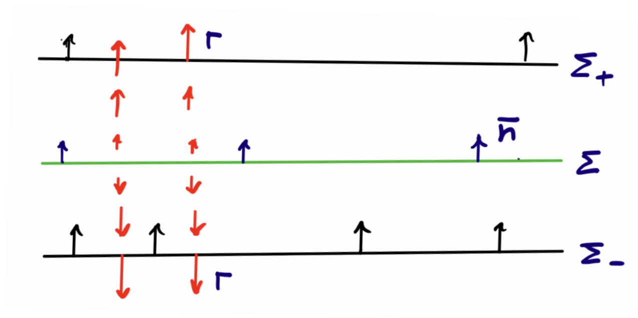


Figure 19.4: Comparing co-orientations of boundary components of N.

The chosen co-orientation of Σ by a normal vector field $\mathbf{n}$ defines co-orientations of components $\Sigma_\pm$. We denoted here by Σ_+ the component on which this co-orientation coincides with the co-orientation defined by the vector field $\mathbf{r}$, and by Σ_+ the component where the two co-orientations are opposite. Thus, we have

$$\deg \operatorname{norm}(\Sigma_+, \mathbf{r}) = \deg \operatorname{norm}(\Sigma, \mathbf{n}) = \deg \operatorname{norm}(\Sigma_-, -\mathbf{r}).$$

If n is odd then $\deg \operatorname{norm}(\Sigma_-, -\mathbf{r}) = \deg \operatorname{norm}(\Sigma_-, \mathbf{r})$, and if n is even then $\deg \operatorname{norm}(\Sigma_-, -\mathbf{r}) = -\deg \operatorname{norm}(\Sigma_-, \mathbf{r})$. Thus applying to the domain N Theorems 19.30 and 19.29 we conclude that

$$\chi(\Sigma) = \deg \operatorname{norm}(\partial N, \mathbf{r}) = \begin{cases} 2 \deg \operatorname{norm}(\Sigma), & \text{if } n \text{ is odd;} \\ 0, & \text{if } n \text{ is even.} \end{cases}$$

■

The above theorem implies, in particular, that the Euler characteristic of any closed hypersurface in an odd-dimensional space is equal to 0. It turns out that this can be generalized to *arbitrary* odd-dimensional closed manifolds:

Theorem 19.34. *For any closed odd-dimensional manifold A we have $\chi(A) = 0$.*

However, the proof of this result goes beyond the scope of these lecture notes.

Exercise 19.35. *Show that $\chi(M \times N) = \chi(M)\chi(N)$ for any 2 compact manifolds with or without boundary.*

19.7 De Rham Cohomology

Given a manifold A consider the vector space $\Omega^k(A)$ of differential k-forms on A. Denote by $\Omega^k_{\rm clo}(A)$ the subspace of $\Omega^k(A)$ which consists of closed k-form, and by $\Omega^k_{\rm exa}(A)$ the subspace of $\Omega^k_{\rm clo}(A)$ which consists of exact k-forms. Consider the quotient space $H^k(A;\mathbb{R}) := \Omega^k_{\rm clo}(A)/\Omega^k_{\rm exa}(A)$. Its elements are closed forms, up to adding an exact form. An equivalence class $[\omega]$ of a closed k-form ω is called its *de Rham cohomology class*, or simply *cohomology class.* The space $H^k(A;\mathbb{R})$ is called the k-th *de Rham cohomology space* (or simply the *cohomology space*) of the manifold A. Dimension $\dim H^k(A;\mathbb{R})$ is denoted by $b_k(A)$, and sometimes called the *k-th Betti number of A.*

Betti numbers of diffeomorphic manifolds coincide. Moreover, the next proposition shows that Betti numbers are preserved even by much more general class of transformations.

Two manifolds A, B (of possibly different dimension) are called *homotopy equivalent* if there are maps $f : A \to B, g : B \to A$ such that the compositions $f \circ g : B \to B$ and $g \circ f : A \to A$ are homotopic to the identity maps. The maps f, g are called homotopy equivalences.

Examples and Exercises.

1. Any contractible manifold is homotopy equivalent to a point.
2. $\mathbb{R}^n \setminus 0$ is homotopy equivalent to S^{n-1}.
3. For any manifold A the product $A \times \mathbb{R}^k$ is homotopy equivalent to A, for any k.
4. The group $GL(n)$ of non-degenerate matrices of order n is homotopy equivalent to its subgroup $O(n)$ of orthogonal matrices.

Lemma 19.36. *Any smooth map $f : A \to B$ induces a linear map $f^* : H^k(B;\mathbb{R}) \to H^k(B,\mathbb{R})$. If the maps $f, \widetilde{f} : A \to B$ are homotopic then $f^* = \widetilde{f}^*$. If $f : A \to B$ is a homotopy equivalence. Then the map $f^* : H^k(B;\mathbb{R}) \to H^k(B,\mathbb{R})$ is an isomorphism.*

Proof. The pull-back operator sends closed forms to closed and exact to exact. Hence it defines the map on cohomology. Suppose $F : A \times [0, 1] \to B$ is a homotopy between f and $\widetilde{f}$. Then according to Lemma 19.7 for any closed form α on B we have $f^*\alpha - \widetilde{f}^*\alpha = d(KF^*\alpha)$, i.e. the cohomology classes $[f^*\alpha]$ and $[\widetilde{f}^*\alpha]$ in $H^k(A;\mathbb{R})$ coincide.

Finally if $f \circ g$ and $g \circ f$ are homotopic to the identity maps. Then $f^* \circ g^* = \mathrm{Id}$ and $g^* \circ f^* = \mathrm{Id}$, i.e. f^* is an isomorphism. ■

Proposition 19.37. *Let A be a compact orientable n-dimensional manifold. Then*

a) $b_0(A) = \dim H^0(M;\mathbb{R})$ is the number of connected components of A;

b) if A is connected we have

$$b_n(A) = \dim H^n(A;\mathbb{R}) = \begin{cases} 0, & \text{if } \partial A \neq \varnothing; \\ 1, & \text{if } \partial A == \varnothing. \end{cases}$$

c) if A is k-connected we have $b_j(A) = \dim H^j(A;\mathbb{R}) = 0$ for $0 < j < k$.

Proof. To prove a) notice that any closed 0-form is a constant function, and if it is not 0 it is not exact. So the basis of $H^0(A)$ is formed by functions which are equal to 1 on one of the components and 0 on the others. Parts b) and c) are reformulation of Theorem 19.18 and Proposition 19.16, respectively.

Remark 19.38. 1. Cohomology can be defined for much more general spaces than manifolds. Moreover, the definition can be upgraded with many additional structures. This theory goes beyond the scope of the current lecture notes, though one of such structures is discussed below.
2. The spaces $\Omega^k_{\text{clo}}(A)$ and $\Omega^k_{\text{exa}}(A)$ are inifinite-dimensional. If A is compact than one can show that the cohomology spaces $H^k(A;\mathbb{R})$ are finite-dimensional. However, if A is not compact it could be infinite-dimensional.

Lemma 19.39. *If α is exact k-form and β as a closed ℓ-form that $\alpha \wedge \beta$ is exact.*

Indeed, if $\alpha = d\gamma$, then $d(\gamma \wedge \beta) = (d\gamma) \wedge \beta = \alpha \wedge \beta$.

Lemma 19.39 implies that the product operation can be defined for the cohomology classes of closed forms. i.e. if we can define

$$[\alpha] \wedge [\beta] := [\alpha \wedge \beta].$$

Given an n-dimensional manifold A define

$$H^*(A, \mathbb{R}) := \bigoplus_{k=0}^{n} H^k(A) = H^0(A) \oplus \cdots \oplus H^n(A)$$

Then the exterior product defines a product operation

$$a = [\alpha], b = [\beta] \in H^*(A) \mapsto a \wedge b := [\alpha \wedge \beta].$$

This operation transforms the vector space $H^*(A, \mathbb{R})$ into an *algebra*.

Exercise 19.40. Let $T = S^1 \times S^1$ be the 2-dimensional torus. Let us denote $\phi_1, \phi_2 \in S^1 = \mathbb{R}/2\pi\mathbb{Z}$ be cyclic coordinates corresponding to the 2 factors S^1. According to Proposition 19.37 we know that $H^0(T;\mathbb{R})$ is generated by the constant 1, $H^2(T;\mathbb{R})$ by the cohomology class $[\eta]$ of a 2-form with $\int\limits_T \eta = 1$. Prove that $H^1(T)$ is a 2-dimensional space generated by the cohomology classes of closed 1-forms $[d\phi_1]$ and $[d\phi_2]$, and that $[d\phi_1] \wedge [d\phi_2] = [\eta]$.

Theorem 19.41. *Let U be a compact n-dimensional manifold, possibly with boundary. Let $b_j(A) := \dim H^j(A;\mathbb{R})$ be the j-th Betti number, $j = 0, \dots, n$. Then*

$$\chi(U) = \sum_0^n (-1)^j b_j(A).$$

The proof of this theorem requires tools beyond the scope of these lecture notes.

Chapter 20

Main characters

Figure 20.1: Bernard Bolzano (1781-1848)

Figure 20.2: Karl Weierstrass (1815-1897)

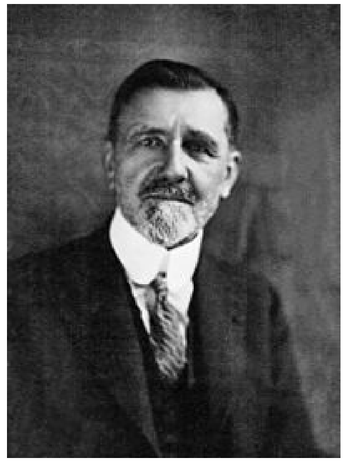


Figure 20.3: Émile Borel (1871-1956)

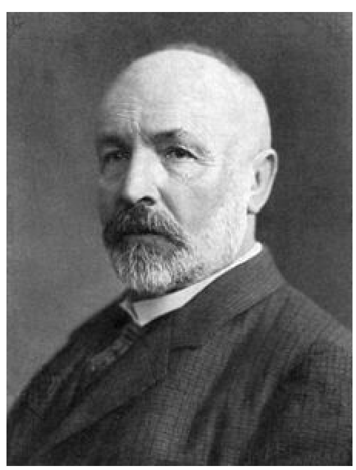


Figure 20.4: George Cantor (1845-1918)

Figure 20.5: Johann Friedrich Pfaff (1765–1825)

Figure 20.6: Bernhard Riemann (1826-1866)

Figure 20.7: Leonhard Euler (1707-1783)

Figure 20.8: Élie Cartan (1869–1951)

Figure 20.9: Henri Poincaré (1854–1912)

Figure 20.10: Guido Fubini (1879-1943)

Figure 20.11: Felix Hausdorff (1868-1942)

Figure 20.12: George Stokes (1819-1903)

Figure 20.13: George Green (1793-1841)

Figure 20.14: Carl Friedrich Gauss (1777-1855) Mikhail Ostrogradski (1801-1862)

Figure 20.15: Sofia Kovalevskaya (1850–1891)

Figure 20.16: Emmy Noether (1882-1935)